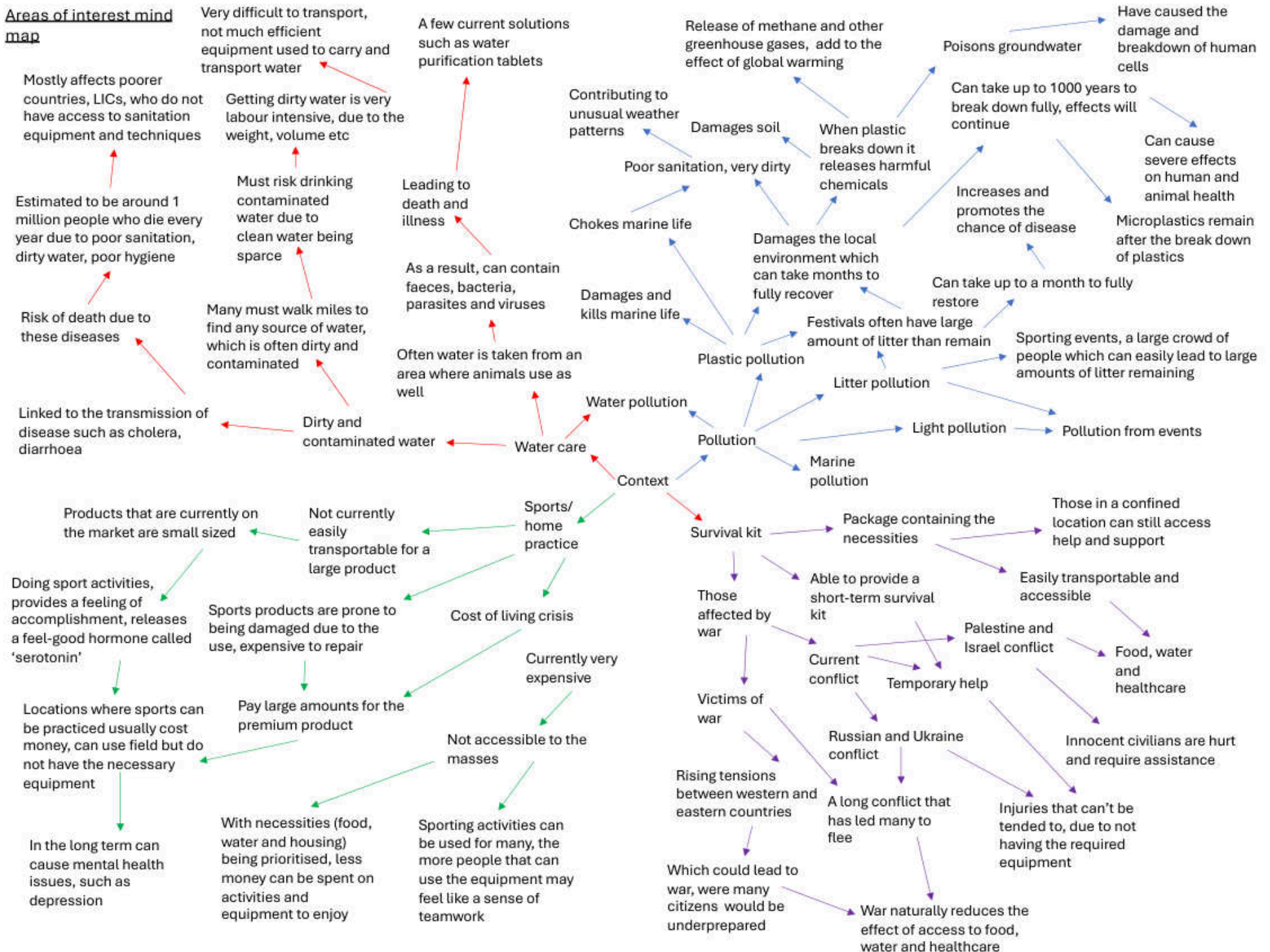


6408

Harry Castledine

Areas of interest mind map



Context of areas of interest

Whilst researching problems within current affairs and how these impact lives of many, it came clear that there were many problems that required a solution, these being pollution, survival kit/package for those affected by war, water care and the lack of reasonably priced home sports equipment.

Pollution has been an increasing problem that damages biodiversity, emits harmful gasses that contribute to global warming and helps to make an unpleasant environment. A solution is required to limit the effect that pollution has on our environment and communities. With the continuing rate of pollution, many animal lives will die, this could disrupt the natural food chain and cause serious problems. Pollution is unpleasant and attracts vermin and rodents which are a vector for disease, negatively affecting humans. There are areas of mass pollution, where pollution has built up to a devastating quantity, making areas unliveable and hazardous to human health, it is therefore vital that a solution is made to limit the effect of pollution so that human health will not be affected, animals will be less likely to die, and all areas are liveable and not hazardous or unpleasant due to pollution. Although products have been made to slow the effect down, more products should be developed or innovated to minimize pollution.



Poor water quality helps spread disease and leads to hundreds of thousands of people dying every year, many of these people do not have an option of clean water and only have the option of dirty, contaminated water which they have to risk to drink, or they will die of dehydration however they risk getting disease and infection as a result. Poor water mostly affects LICs (Low-income countries) as the government is unable to provide resources due to the lack of money, LICs often have to rely on charity money as a result, however corruption also plays a part, where the leaders sometimes use the money for corrupted use and not for the benefit. Water contamination affects many who are unable to get out of cycle of poverty, those who are unable to afford clean water and as a result are unable to escape it and often spend hours searching for clean water but can only find contaminated, dirty water and have to drink it. Everybody deserves the right to have clean water and those who can't suffer the consequences even though it's not their fault.



A survival kit/care package that could be easily accessible by the victims of war, this would provide aid and help to their current situation. Currently there are many countries at war such as Ukraine and Russia, Palestine and Israel with many civilians in danger are requiring help and assistance and the necessities to stay alive. Due to war food, water, sanitation, shelter and healthcare can be sparse and often civilians are affected by not having these necessities. Due to this many civilians can die or get the help they need and suffer severe consequences. It is a human necessity that the civilians have the requirements to live, especially as they are not participating in the war. Civilians are often unable to access survival kits and therefore can face the consequences. A new design needs to be developed so that all civilians can have the necessities that are required for life.



Sports practice equipment is currently expensive, poor quality and difficult to take to places for personal or group training due to the size and weight of these products, making it inaccessible to many, leaving people to have to use provided locations and equipment which are poor quality. This can heavily affect those who are in poverty, as they are unable to spend the money to access these locations that have the resources. Many sports clubs collapse as they can't facilitate the costs required to use a business' facilities. The NHS recommends at least 30 minutes of exercise on five or more days a week, although this exercise can be done in different ways, they may be unable to enjoy it, the sport maybe difficult to continue and the user may not complete it. Although high quality equipment and facilities will come with higher income, the necessities and basic to complete the activity should be facilitated, the costs should be low, at a high quality and transportable so that multiple people can use the benefit of them.



Problem area - pollution

Due to the major issues that arise because of pollution, I feel that it is necessary to try and take action in order to mitigate the effect. Pollution causes serious damage to the environment, wildlife and human; by enabling disease to prosper, pollution is not only dangerous to animals commonly by suffocation but to also humans via disease. Out of all the problems that I have investigated, pollution stands out as one that will remain within society for generations and centuries. Due to most plastics taking 20 to 500 years to decompose, and once it has decomposed it doesn't fully disappear, the plastic just gets into smaller micro plastic pieces, these pieces can have a dangerous effect on human and animal life. Although the other problems I looked at are also extremely serious, the longevity that pollution will last for is the major problem stands out to me, although other problems are serious as technology and way of life increases and improves, these problems should decrease. However, if action is not taken on pollution now, we will face the detrimental effects as plastic, which makes up a large percentage of pollution, decomposes it releases greenhouse gases, these contribute to global warming which is another problem that needs to be attended to otherwise it will be unrepairable damage. The plastics that decompose create micro plastics and degrade soil quality. As a result, I believe it to be necessary to focus on pollution as it is a problem that must be fixed or slowed as soon as possible, due to the major effects that it will have and currently has on our environment and lives.

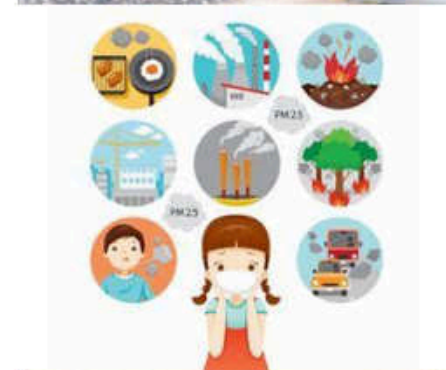


Due to pollution, micro plastics, which are small pieces of plastic, can be found within food, air and water, and can have many health affects on human and animal health when ingested. Some of these health issues can be weight gain, insulin resistance, decreased reproductive health, cancer and cardiovascular problems. These problems could also put a strain on the NHS which is already under stress, which could result in health issues not being examined for a long period of time or until the issue has worsened.

Pollution has many detrimental effects on the environment, wildlife and human life. The mass build up pollution can cause viruses and disease to emerge, as well as vermin and rodents which can carry these diseases, be difficult to contain and makes a place less appealing. With there being many violent ways to remove these pests, a simple solution is to remove the pollution within the area. Pollution can be categorised into many different areas, with the main three being air, water and land pollution.

Pollution can cause a variety of diseases, with most of them being a form of air pollution such as asthma, lung cancer, COPD. Additionally, vermin and rodents who can often be found in polluted areas due to there being food waste within the areas of litter and pollution. Rodents and vermin can carry disease that can spread to people, whilst also being vectors as well. These diseases may require treatment and therefore put a strain on the NHS. UNICEF stated that 8.1 million deaths in 2021 were due to air pollution, a number that must decrease

With animals, they are often to choke and end up dying due to suffocation of those plastics due to them thinking that it's food. We have already seen a change such as the United Kingdom using paper straws instead of plastic straws, however animals are still getting caught and being severely affected. Humans have a duty of care for these animals, and animals dying even if it was non-intentional such as an animal choking on a piece of plastic can cause affects to the food chain and can group to cause large sections of plastic to store in a specific area in the ocean, making living there for the animals not a solution.



Problem area - pollution effects on global warming

Air pollution is one of the main reasons for climate change, air pollution doesn't just occur from factories emitting large amount of violent greenhouse gases which is commonly thought, but instead many common methods used for resources also emit greenhouse gases as well, these **greenhouse gases contribute to global warming and climate change**. Examples of ways that produce greenhouse gases are vehicles, fracking, deforestation and power plants, with there being many others too, many of these contributors to pollution are already being mitigated or changed so that there is a reduced amount of greenhouse gases emitted, such as many coal power plants closing down in the UK, such as the UK closing down **its Ratcliffe-on-Soar power Station on the 30th of September 2024, meaning there will be no greenhouse gases or power produced by coal**. As well as deforestation reducing due to governmental police's and protected areas. Many of these other contributors to air pollution are being mitigated, however little has been currently done to reduce mass areas of litter, this is important to reduce as it is one of the contributors to climate change when the plastic biodegrades and releases harmful greenhouse gases.



Therefore, due to the many negative and detrimental effects that occur because of the release of greenhouse gases that contribute to the effect of climate change, it is essential to try and reduce pollution and litter. The litter is not only a contributor to climate change but also affects the biodiversity of many animals such as polar bear whose ice caps are melting due to the increasing temperatures, meaning they may not be able to survive, potentially causing extinction. **It is imperative that pollution is removed to only reduce that effects of climate change but to also reduce unpleasant areas, disease, damage to the environment and to human and animal life.**

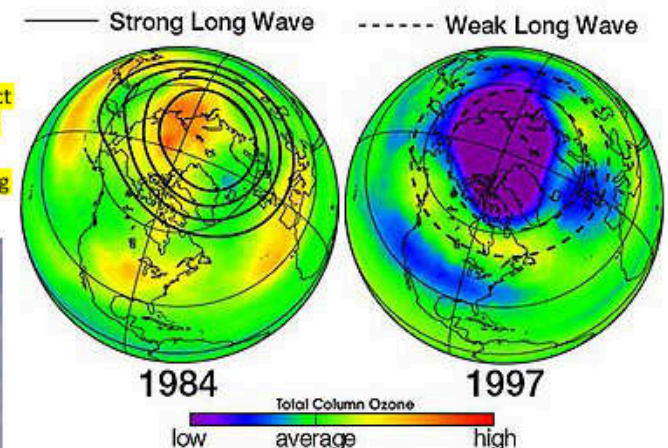


The United Nations states that there are **approximately 9 million deaths per year by non-infectious diseases like cancer and respiratory illness that a significantly contributed by environmental contamination by pollution, 7 million of which air pollution is responsible for**. Whilst contamination through the air, freshwater and the ocean severely affect humans, the chemical and biological pollutants that are released increase antimicrobial resistance making the diseases more difficult to stop, the toxic chemicals additionally are accumulated by pollution affecting humans, the food chain and animals.

It is clear to see that the effect of pollution doesn't just affect humans but also the environment, animals, biodiversity and the planet. If pollution is not stopped or mitigated, the damage will continue before we are able to contain it, having detrimental effects everywhere on the planet.



When plastic which is most of the pollution decomposes it releases harmful greenhouse gases such as carbon dioxide and methane, these gases are major contributors to climate change. Climate change is the long-term shifts in temperature and weather patterns, as a result there is a **significant variation of average weather patterns such as longer periods of rain and sunshine in the wrong seasons**. As well, there an increase in average temperature, this has numerous effects on way of life for many. Agriculture may change; hotter areas will be too hot to be able to produce certain foods. Places may become unliveable for many, if the area is too hot it may be unbearable to live in. Heat waves would increase due to the variation of weather patterns, this may increase the death rate for older people who often struggle in hotter areas. **Many ice caps will melt with the increase in temperatures, causing sea levels to rise, this will begin to flood certain areas of human land which may push people out and make them live elsewhere, as well as removing the homes and habitat for many animals, this can leave them in danger to extinction and a mass death of certain animals**. With the increase in greenhouse gases, the ozone layer, protecting humans from harmful UV rays emitted by the sun, will be more likely to become damaged, meaning Earth is exposed to more UV rays, which can lead to an increase in skin cancer, reduced crop yields, disruptions in the marine food chain and weakened immune systems. Reducing pollution would result in a stronger ozone layer, the stronger an ozone layer the safer humans will be from the effects of the UV rays.



What the UN are doing to limit pollution:

- Transition to a pollution-free planet
- Raise awareness of pollutions' impact and solutions
- Highlight the role communities can play in order to limit pollution
- Aim to remove all forms of pollution by 2030
- Working to boost investments in clean energy
- A sustainable use of natural resources
- Create policies outlawing the burning of solid waste

Mood board of pollution – to generate ideas to solve problems

Pollution can come into many forms, with all the forms having a negative effect on the environment, contributing to global warming, the damage of humans and animals. There are many ways that pollution can occur, such as litter on the street, over spilling bins, litter in the ocean or rivers as well as others. The longer the pollution and litter is affecting these area, the more damage that it will cause to the environment, humans and animals. With the some of the problems that I have identified, it will help me direct my product on to the problems that arise due to pollution.



This is pollution on beaches, here the litter is extremely compact and piled up, this is due to the litter being in the ocean and being carried onto the beach. This causes numerous problems such as not being able to use the beach to its fullest, creating an unpleasant atmosphere and damaging the lives of animals. The pollution makes the beaches inhabitable, dirty and an unpleasant place. The pollution carried by the water may be cause from careless and improper disposal of waste. Additionally, the litter has a huge effect on the health of humans, with the EPA estimating that up to 3.5 million people get ill from being in contact with pollution. Bacteria, animal waste and pollution may cause a wide range of illnesses for these beach goers.



Pollution and litter, which is most often plastic, in the water has numerous negative effects, such as the damage to marine life and the damage to animals' habitat. The litter often is carried by the wind and rainwater which leads to streams and rivers, grouping plastic together in large quantities. As well, drain systems often lead to the ocean too, making a litter dropped into the drains emerge in the ocean. Additionally, careless and improper waste disposal is a contributor to this too, the user maybe careless when disposing of the litter and this litter may be carried into waters. The plastic releases micro plastic when degrading in the water, further degrading the water quality. Furthermore, it creates an unpleasant environment and the animals that are in the water may think that the plastic is food, which may damage or even kills them, and as humans we have a right to care for the animals.



These are overfilled bins and bins that have litter around the base of the bin, this is commonly due to the bins being full so the user is unable to placed their litter in the bin, or if the opening for the user to put the litter in the bin is too small then it maybe difficult for the user to place litter inside and as a result the litter may not enter the bin fully and may spill out. Another reason for this could be laziness and selfishness of the user, due to this they may just placed the litter near the bin or placed it so that it is only in the opening of the bin and can easily spill out. With bins being extremely common in urban areas, the chances of rodents being around the overfilled bins are higher, creating an uncomfortable environment. A way to reduce this though would be to have more bins, so litter is evenly spread or to change the bins more often, however this may be unreliable due as the users may continue to be careless.



Festival pollution is a major problem that becomes noticeable once a festival has finished and people have left, leaving extensive amount of litter which can take weeks to clean up. This mass amount of litter causes serious harm to the local environment, degrading the soil quality and making the area unpleasant and unusable for weeks. Due to the enormous amount of litter, it can take a long time to clean up, and can be extremely expensive, with the Guardian reporting 'that it can cost £500,000 or more to return the site to its original state'. Often there are numerous clean up teams who have to clean which takes time to remove the quantity of litter and pollution. The extensive pollution causes the environment to suffer, with the pollution causes damage to the soil, plants and water nearby that animals may live off.

Festival pollution

Pollution that occurs at a festival is a problem that takes extensive periods of time to clean up. The mass quantities of waste that is produced due to the festival often takes long periods to remove, although the pollution can be removed, often the effects that the pollution has caused remain. The large amounts of pollution can cause severe amounts of damage to the local environment even in a short period of time. Not only does pollution affect the environment but the large amount of litter takes a long time to clean, the long-time taken can further damage the environment and cost more to clean up to hire more workers for the cleaning process, which a business may not want to do. Being able to develop a product that could reduce the amount of pollution that is left or decrease the time taken to remove the pollution would be beneficial and would result in less harm done to the environment.



Cleaning up process

The clean-up process of festivals can be extremely physically demanding, with the clean ups being completed by cleaners and volunteers to ensure that the process is done to return the festival environment back to its original setting. These clean ups can last from several days to up six weeks according to waste managed, the effect of this pollution will continue and will worsen and cause more damage to the environment the longer that it has been left, such as micro plastics and soil contamination. There is therefore a necessity to clean the festival and remove litter quickly before a large amount of damage can be done, however the clean-up can be physically demanding which slows down the clean-up. As well, finding a large crew of cleaners can be expensive so might not be wanted as those in charge may not want to spend the extra money, volunteers for this process may be difficult to find which means less cleaners, due to these factors and more, the process of the cleaning can vary and may be longer than anticipated. High quality tools can also be expensive and may not be bought, this may longer the process of cleaning. Developing a product that is more effective at the waste removal would decrease this time taken to clean up the festival and therefore create less of an impact on the environment.

The problem of the time taken in order to clean up and remove litter is something that I should keep in mind when developing and producing a product. Ensuring that it is effective in its use and not time consuming is essential for the user. The current products on the market used to clear litter are useful however the product that I will make, will have to be more effective at removing the litter. For example, commonly to clear litter from a festival a litter picker is used, however due to the large amount of litter it can take a long time to clear, if I was to produce a product that could pick up a larger amount of litter then it would reduce the time taken to clear the litter, and the environment could be cleared and cleaned at a faster rate.

<https://www.bbc.com/culture/article/20190627-the-people-fighting-the-war-on-waste-at-music-festivals>

Direction of development

With the development of my design, it is essential that it remains focused on the primary issue of helping to decrease the effects and quantity of pollution by producing a product that is simple, effective and comfortable for the user. It is important to stay realistic about my design and the capabilities that it can achieve. My design will not stop pollution from occurring and litter being left in the environment as it is just not realistic, which is why it is necessary to be effective on the area that I can change, being a product that can assist with the removal of litter to allow users to remove litter at a more effective rate in a wide range of circumstances. Although my product won't be able to stop the environmental damage completely, it may help reduce it which would help the battle against pollution.

Being able to create a product that isn't affected by the problems that the user faces will allow for a beneficial and good product that has been adapted to their use, allowing for the user to use the design optimally and efficiently which as a result would speed up the cleaning process for the user. Currently as of my investigation and research into the matter of pollution, it seems and feels that I should go down the route of a product that can be easily used such as a storage unit or a tool of some kind; a product that will simplify the current cleaning and removal of litter in local environments. This seems to be the case as targeting these areas is one that I can access with the local environment such as a woodland which is local to me, as well as volunteers who remove litter. With these resources nearby it would allow me to find the direct problems easier and find a solution in the form of a product to negate them and create a better experience for the user. Being able to tend to these problems would allow me to direct my product directly to the user to ensure that it is focused on their needs and solutions to their problems.

Additionally, it feels even more needed to focus my product on the problem of litter as reported by Keep Britain Tidy, 95% of people believe keeping the environment clean and tidy is important however only 58% are satisfied with the cleanliness of their local environment. This issue is another reason of why I believe a product should be produced to help with the removal of litter, to make the cleaning process easier and more effective, this would allow for less litter to be in the environment making it cleaner. A product should be in place to help limit the quantity of litter in the environment and is an issue I should investigate further into for a solution.

The investigation into the development of my design should be developed further with the help of a third party who cleans and removed litter from the environment and is something that I should investigate. Being able to hear and listen to someone else who is removing litter, hearing their problems will enable me to direct me onto the problem and find a solution to it, which as a result would allow me to produce a product designed to the needs of the user. I should also have a questionnaire from those who have removed litter which will help me direct my research onto the problems current users face, to produce a more appropriate product that is more effective. For this questionnaire, I will ask those who have removed litter, this being volunteers, cleaners and others who have removed litter such as friends. Furthermore, I think the Millenium Green area in my local area is essential to further developing my knowledge and understanding on how to produce a product to fit the needs of a user. With volunteers there, I can further develop my research and be given more information. I believe it is best to continue developing my product to meet the needs of the user; to create a user-centred product as this will allow for improvements to be made to ensure that the removal of litter can easily be done, allowing for more litter to be removed.

Research plan

To ensure that my project will be fulfilled to its best, I will need to create a plan that will cover all aspects of the research that I must conduct to make a product that is beneficial and useful. I will need to gather primary and secondary research from trusted sources, this will provide support and help guide me to develop a product that will negate the problems that are often faced. To develop my design and enable it to be effective on real-world problems which is removing pollution from streets and the environment, I will have to develop my design in a way that will be able to counter these problems directly. If I produce a product that doesn't meet the requirements, then it will not perform to the highest quality, and therefore could limit the quality of the user's experience. The user may then use a different product as it would not be worth the time and difficulty to use the product. I must be precise and accurate with the design of the product so that it is useful and accurate.

Primary research

To develop my design and allow it to be influenced by the real-world, I will have to research and gather data and information by myself. This data will allow me to be gain insight onto various things that I might not be able to achieve with secondary data. I will be able to witness firsthand the problem of pollution and be able to develop my product onto the problems that I have seen. To find this information about pollution I will have to venture out and explore, going around my local towns and even a major city such as London. To further develop my design ideas, I will visit a design museum, visiting a museum will allow me to develop my knowledge and understanding on the functions of the products which will enable me to design the product with more freedom. I will also be able to disassemble some products, enabling me to understand how the products function and see the pieces which are combined to produce a final product.

Secondary research

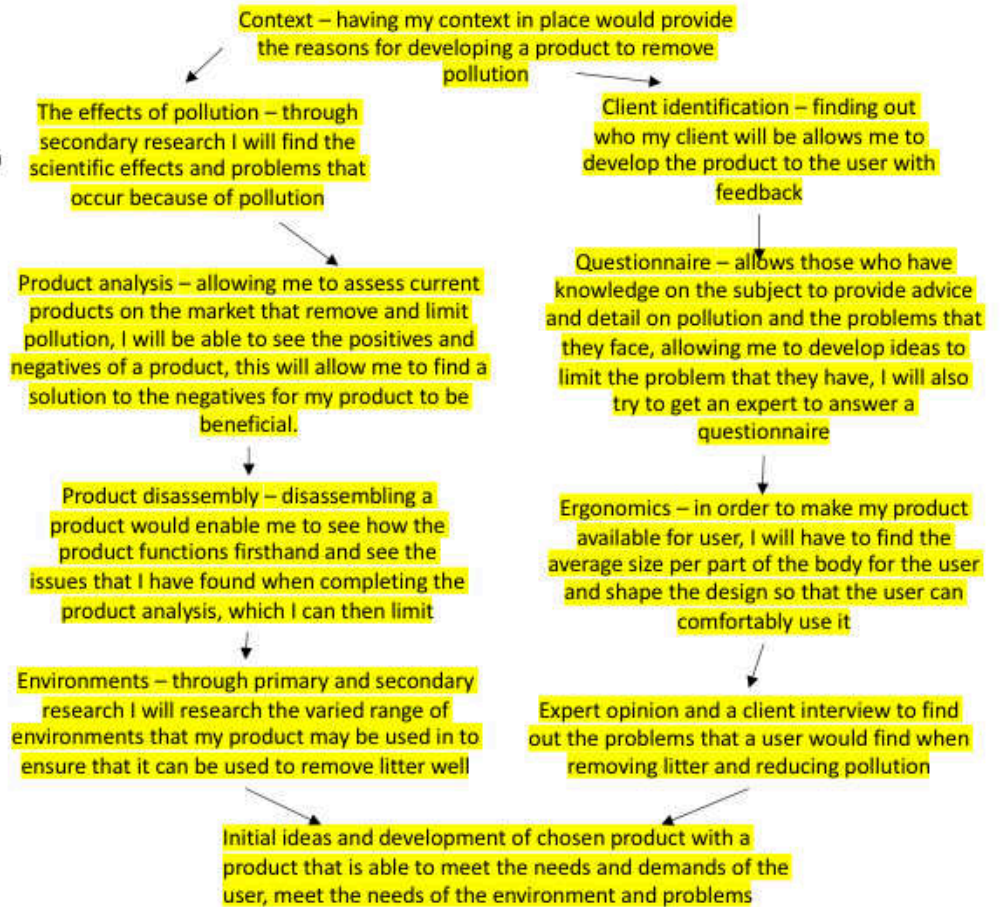
To develop my design, I will need to incorporate secondary research, this will be crucial to me as it will enable me to understand the effects of pollution that occurs elsewhere in the world, places that I am unable to visit and see for myself and the effect that pollution has on these places. Being able to see data and the effects that it has on people will enable me to develop a product to limit the effect. I will also be able to gain knowledge on the ergonomics and size of the user, this will allow me to develop and produce a product that will fit to the user and be comfortable for them to use. Secondary data will also provide support to the reasoning that I have found in my primary research.

Expert opinion

An expert opinion would be great to have, it would provide a different perspective compared to the research and opinion that I have, whilst also enabling me to explore the problems that they face. Being able to change and develop my design to counteract the problems that the expert often faces would allow for a product that is useful and negates the problem that they often find. Additionally, I will create a questionnaire for the expert to answer, this will allow for their opinion to be easy to come across whilst also providing their input on problems that occur, which I can use as input to help produce a product the doesn't allow these problems to occur.

Questionnaire

I will produce another questionnaire aimed at others who focus on removing pollution. This could be somebody who removes pollution and has some experience but isn't an expert. This will enable to develop my product based around the user and the experience that they have. The expert opinion could answer this questionnaire which will help reduce development time taken



Rough timeline of project, tasks overlap so all development can be done to improve the product:

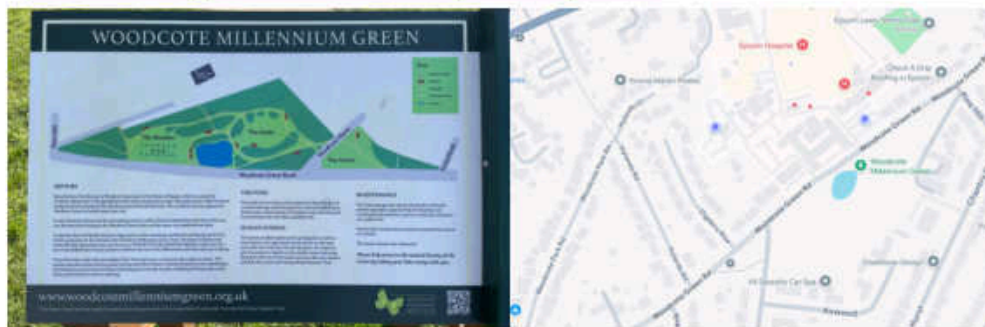
Task	Start date	End date	Duration (days)
Research investigation	18/06/2024	03/04/2025	300
Client interview and questionnaire	21/09/2024	23/09/2024	10
Product dissassemblies	25/09/2024	30/09/2024	10
Design brief and spec	02/10/2024	05/10/2024	10
Initial ideas	07/10/2024	16/10/2024	10
Developed ideas	17/10/2024	25/10/2024	10
Chosen product	26/10/2024	30/10/2024	10
Development of chosen product	31/10/2024	03/04/2025	200
Reseach and development	31/10/2014	03/04/2025	150
Testing and modelling	05/11/2024	10/01/2025	50
Computer Aided Design (CAD)	10/01/2025	30/01/2025	20
Cutting list, manufacturing table	15/01/2025	03/03/2025	20
Industrial manufacture	20/01/2025	30/01/2025	10
Manufacturing diary	01/02/2025	25/03/2025	80
Evaluation	25/03/2025	02/03/2025	10

Types of litter investigation and visit

The types of litter that can be found will vary my type of design and product that I will produce, for example if the average litter that can be found is large it wouldn't be suitable to produce a product that is focused on litter that is small. Focusing my product on the average litter that is available will allow for it to be easier and more suitable for the user's use of my product, providing a better and easier experience for the user which as a result will allow for a better product. Aswell, making the product based around the average litter found would allow for an easier experience as the tool would be based around the selected litter.

Primary research:

In order to analyse the types of litter that were most common, I went around the Millenium Green which is a woodland section with a pond in my local area and assessed the quantity of the types of litter that were in the environment. This would help me understand the most common types of litter that can be found within an environment so that the product can target the most common types of litter and fit solutions into my product. This would allow for my product to be most effective and focused so that my product can deal with the problem of pollution.



Count of types of litter results:

	Grass	Urban	Woodland	Watered	Sand/shore	
Food packaging	5	16	12	2	1	36
Tissue	2	3	4			9
Cardboard		1	3			4
Cigarette butts	106	179	198	3	2	488
Plastic bottle		5	1			6
Plastic pot			1			1
Gum wrapping	4		10		1	15
Glass bottles	1	3	2			6
Vape			1			1
General litter	10	26	57	5	2	100
Dog poo bag			2	1		3
Metal cans		1				1
Bottle caps	2					2
Other			3	1		3
	130	234	294	12	6	Total



The litter that I was able to count was all the litter that I was able to see, meaning the litter I counted is a minimum amount of litter in the environments, and there is a high chance that there would be more due to some sections of the Millenium Green being too dense with bushes and trees to reach and count, as well as human error were I have missed the litter or am unable to see due to looking at a wrong angle or direction, so that the litter is hidden.

The count for the watered and sand/shore environment are low compared to the others, which is most likely due to the shore/sand section being very small and mostly being a mix of grass and mud, so litter counted would be placed into the grass section. The watered environment also meant that the litter may have deteriorated already, sunk or be hidden by plants in the bottom of the pond, which meant that the litter was too difficult to count as it was unable to be seen.

Photos of some of the litter found:



Results from table analysis:

When analysing the results gathered on the table, the environments with the most litter was the grass, urban and woodland; this is due to factors stated previously such as the sand/shore being too small and the litter in the water being difficult to see meaning less litter could be counted.

The litter is mostly in these environments due to it being most accessible to the people, and as more people are able to use these areas, litter and pollution will increase with it. The litter that was found the most per count (every piece of litter was counted as one, volume or weight was not taken into consideration) was cigarette butts, with 488 being counted throughout the investigation which, per count of litter, was 72.3% of all found litter, this means that the cigarette butts are a problem that can cause problems, hurt animals and degrade the quality of the environment, which would make the location more unpleasant. This is a problem that I could develop my product around to help reduce the number of litter butts in the environment.

The reason for the quantity of litter butts is most likely due to the workers at Epsom Hospital when on their breaks, where they often visit the Millenium Green and smoke whilst sitting at the benches by the pond. Even though there are litter bins and cigarette bins, the number of cigarettes is most likely linked to human behaviour of negligence and not caring about dropping the cigarette, which could be something that I could investigate, however this would be more difficult, as human behaviour may not change for a product produced as there are already options provided to dispose of the cigarette butts, but these are not used, which could be linked to laziness and the person not bothering to properly dispose of their litter.

The next piece of litter that was most common, was general litter (wrappers ,bags , receipts ,etc) which resulted in 14.8% of all litter found per count. This litter was often small in size and would be the broken ends of sweets packets, small thin sheets of plastic or broken pieces of tools or toys. A reason for this could again be negligence of the person who has the litter or laziness to pick the litter up if they have dropped it due to it being 'small', however this would still have some impact on the environment.

Conclusion of primary investigation:

As the Millenium Green often has volunteers to remove litter with a litter picker to pick the litter up and a bin bag hoop to carry the litter, the data that I achieved with my investigation is appropriate to suggest that the litter I found the most was the most difficult to remove as it was smaller, being cigarette butts and small pieces of general litter, whereas the litter found the least would be the easiest to remove, being cans, pots and bottles, which is larger in size and easier to remove with a litter picker. This would suggest why it was most common to find smaller litter such as cigarette butts and small pieces of general litter, as these are the most difficult to remove with a litter picker due to there small size and not being able to grip the litter to remove it from the environment. As this is a problem found, I could direct my development onto this problem, smaller litter being more difficult to remove in order for my product to be directed onto the problem faced by those who volunteer to remove litter or cleaners who remove litter.

Further primary investigation



Having previously litter picked for the volunteering sections of my Bronze and Silver DofE where I removed litter from the Millenium Green, I have experience with removing litter from the environment. From my personal experience, removing most types of litter from the woodland, grass and urban environments was simple with a litter picker due to it being simple to use and effective when removing litter. However, when using it the problem that I found was that smaller sized objects such as cigarette butts and corners of wrapping where the most difficult to remove due to the extremely small size, meaning that there was a lack of surface for the litter picker to grip and secure the litter, due to this being a problem it would be appropriate to investigate further.

Secondary research:

In order to gain more knowledge on the average types of litter that can found, so that I can target my product to work in more conditions and provide a better experience for the user, I researched to find a litter composition report and found one from 'Keep Britain Tidy'. This report provided me with more information on the types of litter for me to keep in mind when developing my product for it to be most effective when using.

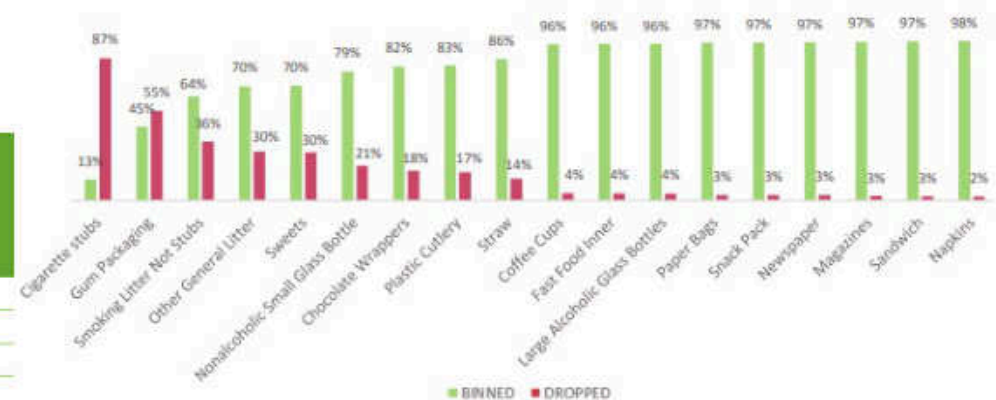
Litter types by count – Data from Keep Britain Tidy

Item	Count	% of dropped litter count	% of dropped litter count (without cigarette stubs) ²⁵
Cigarette stubs	50,088	66.3%	N/A
General litter – other	11,652	15.4%	45.8%
Paper – other	2,350	3.1%	9.2%
Smoking litter (not stubs)	1,511	2.0%	5.9%
Drinks: Cans – non-alcoholic	1,188	1.6%	4.7%
Sweets and mints packaging	1,177	1.6%	4.6%
Drinks: Plastic bottles – small, non-alcoholic	1,049	1.4%	4.1%
Chocolate wrappers	916	1.2%	3.6%
Unsure (litter type cannot be determined)	801	1.1%	3.1%
Fast food – inner packaging	613	0.8%	2.4%
Straws	537	0.7%	2.1%
Drinks: Cans – alcoholic	417	0.6%	1.6%
Gum packaging	379	0.5%	1.5%
Fast food drink container (not coffee)	307	0.4%	1.2%
Plastic cutlery	300	0.4%	1.2%

Litter types by volume

Item type	Count	% of dropped litter volume
Non-alcoholic small plastic bottle	1,049	24.4%
Non-alcoholic can	1,188	18.6%
Smoking litter not stubs	1,511	12.4%
Coffee cups	285	7.8%
Alcoholic can	417	6.8%
Fast food drink container (not coffee)	307	5.0%
Non-alcoholic large plastic bottle	71	4.4%
Other general litter	11,652	3.3%
Cold drink containers (not bottle)	98	3.0%
Small alcoholic glass bottles	146	2.4%
Newspaper	50	2.3%
Sandwich packaging	67	1.8%
Fast food inner	613	1.3%
Large alcoholic glass bottles	32	0.9%
Snack pack	264	0.7%

Proportion of total item count littered versus total item count binned at sites where litter bins are present



By looking at this data provided by 'Keep Britain Tidy', it can clearly be seen that the most common source of litter is cigarette stubs which account to 66.3% of all litter dropped by count. This means that there are large amounts of the stubs on the ground and because of their small size can often be a problem to clean and remove from the environment. Aswell, the tiny size of the stubs mean the tools which have been built to pick up litter of a larger size litter will find it difficult to pick up such a small size of litter, making it difficult and therefore more time consuming for the user as they have to spend more time being more accurate with their tool.

Opposingly another graph shows the litter types per volume, and it can clearly be seen that the litter with a higher volume have is drink packaging, with it mostly being from bottles, cans and cups. With this type of litter being often the largest, it may be difficult for a user who is picking up the litter to be able to carry such a large amount or even be able to pick up the large volume of the litter as it may be too large for the tool to pick up. This would be an inconvenience for the user, and they therefore might have to pick up the litter with their hands, something they may not be inclined to do as they would be touching litter which has been exposed to the environment and may be dirty or carrying disease etc.

Furthermore, when looking at the proportion of items littered verses binned, it is most common for the smaller types of litter, such as cigarette stubs or gum packaging to be the most common type of litter which is dropped on the floor and not binned. This creates a problem that would result in the removal of these types of litter taking longer to pick up as they require more precision to remove. This problem could also be a phycological issue where those who litter don't see the problem with dropping litter or have little care for the problems that it can cause.

When researching further, the problem with litter is also due to poverty and as quoted by Keep Britain Tidy : 'As would be expected based on the patterns shown in overall litter counts (Section 0), deprivation appears to have an impact on litter types, however some are more unevenly distributed than others. The most deprived 10% of sites contained nearly 12 times as many fast food drink containers (excluding coffee) as the 10% most affluent areas (59 to 5 respectively), 9 times as many non-alcoholic small plastic bottles (331 to 37), and nearly 8 times as many alcoholic cans (100 to 13)'. This problem may also be a phycological issue or a lack of care and could be something that I look into for when I am developing my product.

From this research, it is essential that when developing my product is is suitable for the most common types of litter as well as all litter so that the design is appropriate to limit pollution and the effects that it would cause on the environment, people and animals.

Expert opinion and client interview

To produce a product that will be used to reduce the effects of pollution, I will need to find the target market and identify a client and an expert opinion that faces the problems of pollution and is an expert in the field due to experience. My chosen area of investigation is pollution which is removed from my local environment by volunteers. Being able to investigate further into the users who remove pollution will enable me to develop my product around their issues for my product to be as beneficial to the user as possible, as the volunteer or any person such as cleaners, workers or those who want to help keep their community clean is the person that my product will be designed for.

To allow my product to be useful I talked one of the local volunteers, who volunteer by removing litter around my local area which is the Woodcote Millennium Green located in Epsom, who provided me more insight into their volunteering, helped me develop my knowledge as I already have some insight. Additionally, the expert and client is a scout leader, who organises, helps and directs removal of litter from the local environment of Epsom and Ewell.

The client/expert is part of the Woodcote Millennium Green Trust which has 3-4 major clean ups every year. The insight I have is due to me being a volunteer at the Woodcote Millennium Green to remove litter when participating for the volunteering section of my Bronze and Silver Duke of Edinburgh award, providing me with 9 months of experience.

My personal experience with removing litter and pollution from the environment whilst volunteering will help me to develop a product that is fitted directly to the user, as I am able to begin research on areas that I find challenging and find a solution, I will also be able to relate to the user of the product that I will produce, as I will be able to understand the problems that the client has faced, allowing me to produce a product fitted to the user's needs and solution to the problems that they have faced.

Before talking to the client/expert, I made some key points on the experience I had whilst volunteering to remove litter, whilst volunteering I was using a litter picker and a bin bag holder :

- Objects would be stuck in the ground or difficult to get pick up
- Small objects such as litter butts would be extremely difficult to pick up
- The holder would strain my arm due to the long period of litter picking
- The bin bag would sometimes break heavy objects being placed inside the bag

Client/expert profile:

- Age : 51 years old
- Occupation : Accountant
- Sex : female
- Takes part in communal clean ups 4 times a year, seasonal maintenance days for the local community
- Volunteers fortnightly removing litter
- A scout leader who volunteers with her patrol to remove litter from the local environment around twice a year
- Has volunteered for the local community for 8+ years, helping to keep the quality of the local environment and conditions to a high standard
- Lives locally to the Millenium Green so volunteers

Photo of client and expert volunteering which can be found on the Millenium Green website under gallery



After talking to the expert:

From talking with a expert, it provided me with more in-depth knowledge and understanding of the problems that they face whilst removing litter, the knowledge they gave me with will help me direct my product so that it is pinpointed on the problems faced.

The client/expert uses a bin bag holder to hold their bag and a litter picker also. They thought that combination of these two tools was imperative as the work together hand in hand, they thought that the litter picker was an excellent tool that was able to be used easily and, in many conditions, as well as the bin bag holder which helped them to not have to carry the litter uncomfortably or in a bag as it remains open throughout the whole process. However, the volunteer found many problems with the tools that would cause an inconvenience to them and make the process more difficult.

The main problem that I was able to gather from our conversation was the difficulty to obtain litter from different terrains, they found that picking up litter in flat conditions was accessible but when being in a different terrain became an issue, this being objects stuck or flat against the ground, objects behind bushes that was not accessible and objects in the pond that couldn't be captured due to them being slippery as they were in the water. Other problems was that carrying both tools for an extended period strained the volunteer which often led to them taking numerous breaks, which was an inconvenience for them especially in the colder seasons, where the low temperatures would affect them whilst on their break.

Furthermore, they found that when picking up a large quantity of litter it would be difficult to continue due to the strain that occurs for the user. Regarding the bin bag holder, the volunteer often found that the bag would often get in the way of their walking due to the large swinging of the bag, as well as the bag often tearing because of litter captured becoming too heavy or sharp. Additionally, the volunteer found that there was a lack of space for the range of litter, as they only had one bag to place the litter inside meaning the litter collected all had to go into the same bag, which left for them to have to either place the bag into a refuse bin to be sent to landfill or be incinerated, or either remove the litter from the bag and place the litter found into the specific bins, such as refuse, recycling, metal, glass or food waste.

When talking further about the problems with the expert, she said that the cigarettes were often too small for the litter picker to pick up effectively which then resulted in my client having to put on gloves and then remove the cigarette butts by hand which is a **laborious and straining process that would result in my client feeling large amounts of stress.**

She also said that **the product that I would make would have to be ergonomic, as the litter removal process can already be straining and damaging to the user as it is physically demanding** and having an ergonomic product which has been sized accurately would reduce the strain that the user would feel which would allow for a more enjoyable and simple experience when removing litter.

This conversation really helped to develop my understanding of the issues that take place when removing litter. I will keep this information in mind, as the issues that have taken place for the client/expert are ones that I will need to create a solution to as they are the ones that affect the users of the problem area I am investigating. The things that I will be taking forwards into my design due to the information that was given to me :

- A difficulty in litter being able to be picked up in a wide range of terrain
- Small objects were difficult to be picked up
- Tools are straining on the user when used for long periods of time
- The strain of picking up large quantities of litter due to it being straining on the user
- The difficulty to place the specific litter into its specific bin
- The equipment could be easily damaged with a bin bag holder due to the lack of support that is given

With talking further to the expert/client, she suggested some solutions to the problems they encounter:

-A tool that can pick smaller pieces of litter up

-A tool that is less straining and has more support on the user

-A tool that allows for the sorting of litter to be easy and simple

-A tool that is able to sort the litter collected

I will take this feedback the client/expert has given me and the problems and solutions into my designs as the user is directly impacted by these problems. Creating a solution to these problems will allow for a product that is better fitted to the user and one that allows for the process of removing litter and pollution to be easier, which as a result would allow for the user to be more effective.

When looking at my **litter count investigation and my conversation with an expert opinion (who has agreed to be my client, and I will update her on the findings of my investigation and development as it continues so I will remain focused on the client's needs), cigarette butts were a problem to remove.** This would be an area of interest and one that would help target and limit the problem that my users faces, which should help reduce the quantity of cigarette butts I found on my primary litter count investigation.



User questionnaire

To gather more information about the problems and difficulties that people face when removing litter, I created a questionnaire for those who remove or have removed litter such as volunteers to answer and provide me more information on their experience. By not limiting the questions to multiple choice **this information will enable me to focus more on the problem areas that the users have faced for me to target these areas, find solutions and develop a product that will stop these problems and provide a better and easier experience that can solve the problems that they face.**

How often do you volunteer or go out to remove litter from the environment?

6 Responses

Once a month

2

Once a week

Once a year

Once a quarter

Do it roughly once a year with scouts

What terrains/environments do you remove litter in (Woodland, urban, watered etc)?

6 Responses

Woodland and urban

3

Urban

Woodland, urban

Woodland, urban and watered

With each of the terrain/environment that you remove litter in, what are the problems do you have with removing litter and your tools?

6 Responses

With the woodlands there is a lot of mud so it is difficult to separate recyclables from non recyclables as I cannot see what the litter is.

In a woodland environment it can be hard to pick items up using the standard picker particularly when the litter is near tree roots, underneath shrubs. In an urban environment bottles and cans can roll away easily on tarmac/pavement and litter more likely to be blown by the wind so can be harder to pick up

It can be hard to grip certain litter when the weather conditions have been poor, lack of grip on the tool does not help.

Items get stuck against the floor which is difficult to pick up

Most commonly it is litter being stuck to the ground, in an urban environment the litter may be stuck to the floor but in a woodland environment the litter may be stuck in mud, which is difficult to remove

Litter and urban, litter can get stuck to the ground which is difficult to remove and in a watered, the litter is wet so is difficult to grip

What do you use to store the litter you picked up and what are the problems with these tools?

6 Responses

Bin bag clipped to my waist

I put the litter in a plastic bin bag. It can be tricky to put the items in the bag sometimes particularly where they are hard to pick up. You can release it too early and then the litter falls out of the bag. It would also be good to have one had free sometimes.

I put the litter in a bin bag

Bin bag, can sometimes tear

Place it in a bin bag which is in a bin bag hoop, the litter can sometimes tear the bag

A bag

What types of litter do you pick up and what is the most difficult to remove from the environment?

6 Responses

Mostly plastics and the most difficult to remove are cigarette butts as they sometimes amalgamate within the pavement/road.

Bottles, cans, crisp packets and cigarette butts. Cigarette butts are the most difficult as they are small and get caught easily in the soil/ground

Many plastic items but glass is the hardest to remove

Plastic bottles or litter butts

Bottles or cigarettes, larger and heavier bottles are difficult to pick up with a litter picker due to the large size which makes the holding of it uneven. As well cigarettes are very small and can get stuck to the ground very easily making it very difficult to remove with a litter picker. Removing litter that is stuck in a pond can also be difficult to capture due to it being wet.

All types of litter, wet plastic or cigarettes butts as these are small

What tools do you use to pick up litter and what are the problems with these tools?

6 Responses

Litter picker/ hands with gloves

Litter picker and bag secured on a hoop. It can be tricky to pick up smaller pieces of litter such as cigarette butts, small pieces of plastic.

A litter picker - can be straining for back when tool isn't long enough to grab the litter and I'm having to arch my back.

Litter picker and a bin bag holder

Litter picker and bin bag in a hoop

Litter picker and a bag, it can be difficult to pick up small pieces of the litter and litter that is stuck to the ground

How do you get to the location and how do you transport your tools?

6 Responses

Walk

Walk and carry them

I walk and carry them.

Walk

Walk, carry them to the location, litter pick whilst walking to my location

Given to me by the scout leaders

How long would you spend removing litter from the environment in one session and do the tools cause you strain?

6 Responses

I spend about 2 hours, sometimes the tools cause me strain as I have a weak lower back, clinging the garbage bag onto my belt imbalances me and causes me difficulty.

1 hour. I don't think they cause strain unless item is heavy but they can be awkward to use

1 hour and as time goes on it causes more strain on my body.

1.5 hours

Around 1-2 hours, or however long it takes to do a sweep of the Millenium Green

2 hours, can get tiring carrying the tools around for a while

Summary of results:

- All users use a litter picker to pick up their litter and a bag or bin bag in a hoop to store their litter
- Many users found a difficulty in picking up litter that was stuck to the ground, this litter may have been stuck in mud or stuck due to being wet and made the removal process difficult for the user
- Most users walked to their location of choosing to remove litter, meaning that their location was nearby and could be easily reached
- Nearly all users said that cigarettes were the most difficult type of litter to remove due to its small size which users found difficult to capture. 2 user said that objects such as bottles and glass were difficult to remove, due to the large size and sometimes being heavy which could strain on them
- All users spent a maximum of two hours removing litter, before they had either captured all the litter, ran out of time or were too strained to continue removing litter
- One user found litter difficult to remove from a watered environment due the litter being wet and difficult to grip

With the results from the users, I will be able to focus my design on the problem areas that they faced when removing litter, with one of the major problems being that small pieces of litter such as cigarettes were difficult to grip and remove from an environment.

Product analysis of a litter picker

What is a litter picker and where would it be used?

This tool is a litter picker and is used to pick up pieces of litter on the ground so that the litter doesn't cause damage to the environment, this could be damage to animal life or contamination/pollution. It can be used by squeezing the trigger and the claw will close on the litter, picking it up and able to release when the user stops squeezing. It would often be used through areas like a park, woods or streets where this won't be mass piles of litter, only small piles or individual pieces of plastic that need picking up due to the litter picker only having a singular claw, which results in small amounts of litter being picked up at a time, large amounts of litter would therefore take a long period of time to clear up, which may also be physically demanding on the user, so they may use another tool which could pick up larger amounts of pollution which each use.

How does the litter picker function?

The litter picker functions by pulling a trigger which is connected to a string, this string is attached around a pulley to keep it firm and straight against the frame, the string continues against the frame and is attached to a spring which is connected to the pulley of the litter picker. When the trigger is not being contracted, the spring is at its extended length and the hook of the picker is open. When the trigger is being contracted, the spring is at its shortened length and the hook of the picker is closed. This movement can be performed on a piece of litter for it to be picked up and then the pressure on the trigger can then be released which opens the hook and can drop the litter. The user can place the litter in a bin bag to keep the litter organised and contained by releases the pressure on the handle with the litter over or inside the bin bag.

Cost and value of product

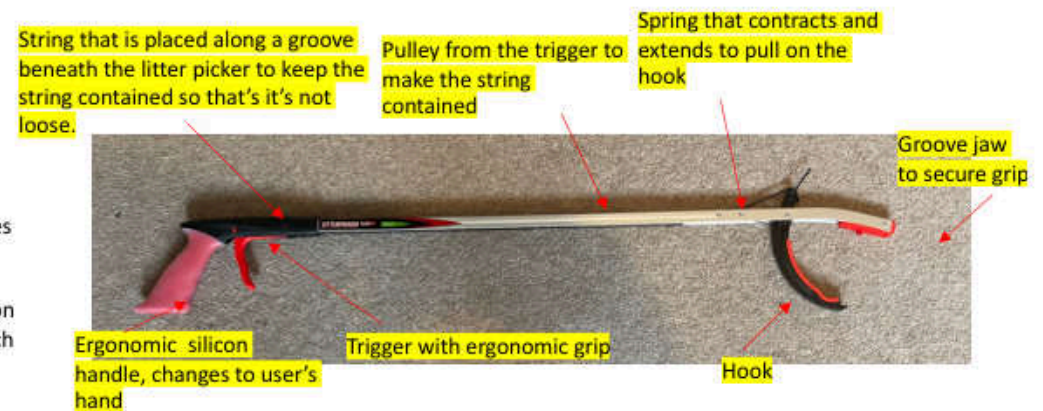
Most litter pickers range from £5-£20, making it accessible for many to purchase. They can be bought from stores or online from sellers such as Amazon. It appears to be a good value for money, it will be a long-lasting product if used safely and correctly, also allowing the user to avoid disease as they will pick up litter using the tool and not their hands which could be contaminated whilst also helping keep an area with less litter.

Ergonomics of the design

The ergonomic handle of the litter picker has been shaped so that it will fit the shape of the user's hand comfortably, this will help for accuracy when picking up litter, overall comfort when collecting litter over long periods, reduce strain and fatigue and therefore reduce the risk of injury on the user. The trigger is completed with curves that allow the finger to rest in when squeezing, allowing for a stronger grip so that the use of the tool is more accurate, and the user can remain squeezing the trigger when collecting an item to place into a storage area.

The quality and materials of the litter picker

Litter pickers are commonly made from polypropylene and aluminium, with the polypropylene that is often used to make the handle, trigger and hook which is 100% recycled. Both materials are recyclable, lightweight and durable, making it easy to carry around for extended periods of time whilst litter picking. The toughness and hardness of the aluminium allows for the litter picker to be durable and maintain shape, providing a long-lasting product that doesn't need to be repaired. With both commonly used materials being recyclable and the main frame which is the aluminium being durable, the product will maintain a high quality whilst being used.



User

The user of a litter picker could vary but would be from cleaners/workers who are trying to clean an area or members of a community who are trying to keep their area clean of litter and pollution, appropriate for both circumstances the lightweight tool allows for easy use of the product whilst also being accurate, durable which makes it appropriate for use.

Environmental effects

The manufacture of the litter picker use polypropylene and aluminium to produce the product, producing these materials is harmful to the environment, releasing large amounts of greenhouse gases and harmful emissions that contribute to global warming. However, these materials are commonly recycled to produce new products ensuring that the environment isn't affected by the deterioration of materials, the only problem that can arise from recycling is that the materials can downgrade in quality.

A common problem that I found when using the litter picker was the difficulty to pick up objects that are stuck to the ground or flat against the ground. The jaws of the design couldn't grip the litter easily and as a result made the process of litter removal difficult.

Product analysis of litter picker continued

After analysing the litter picker, I found the advantages and disadvantages of the product that improve the product and make it beneficial and some areas that could be improved:

Advantages:

- Ergonomic handle that allows for a comfortable grip, provides stability for the user when picking up litter, allows to be comfortable when using the tool for extensive periods of time
- Ergonomic trigger that allows for a comfortable grip when squeezing, easier to squeeze as the grip is shaped to fingers of the average sized hand, distance between handle and trigger is a comfortable reaching distance and shaped to meet the size of the average user
- The string that pulls the hook upwards is protected between a wall of plastic or metal, keeping it protected from damage, increasing the life of the product and reducing the risk of breakage
- A groove jaw that secures the grip of the litter picked up, ensuring that litter is less likely to fall out the grip, so picking is more efficient, saving time and energy
- Relatively cheap, meaning that it is accessible to many
- Relatively easy to use, a simple squeeze of the trigger, whilst also being lightweight allowing for long periods of use

Disadvantages:

- Only one, thin hook that can pick up litter, meaning it takes time to pick up large quantities of litter which may be physically demanding on the user
- Beginning of the hook is outside of the protection and frame of the product, meaning it is more likely to get damaged if not used carefully, which may result in a broken product
- Basic colours, raw colour of material, meaning it is not an attractive piece, children may not want to use it as it isn't appealing to them, should be appealed to children so that they understand the importance of removing litter

Photographic sequence of using a litter picker:



How to use the litter picker:

- 1- find a piece of litter
 - 2- place the hook of the litter picker above the piece of litter that you have found
 - 3- Squeeze the trigger, the hook will close, and you would have captured the litter
 - 4- Whilst remaining to squeeze the litter, place the litter in a suitable area, such as a bin or a bin bag
- The litter picker can also be used as a reacher/ grapple to bring items towards the user and can be functioned the same, the only difference is the user will collect the item they have captured instead of placing it in a bin, by releasing their pressure on the handle

Other variations of a litter picker

There are multiple variations of a litter picker, the tool that I investigated is the most common one and is composed of one thin hook that closes when the trigger is pulled. The other variations can have different handles, triggers and hooks, which can vary the quantity of litter picked up, the types of litter picked up and the environment of the litter picked up.

For example, the litter picker that I investigated had one small hook, this would only be able to pick up one piece of litter easily, would be able to pick up pieces of litter up such as plastic bottles, packets and small boxes, the small hook enabled for the picker to be placed inside some litter so there is more grip and would be able to pick up litter in numerous environments such as woods, forest, towns and cityscape. The only environment that the picker would struggle would be in a watered environment.



This variation of a litter picker is different in comparison to the one that I analysed, it can function as both a litter picker and a grabber. The picker has two hooks squeeze and close together into the middle with a grooved inside face on both hooks to provide grip picking up litter. As well the litter picker is extendable, enabling a user to increase the range that the litter picker can pick up from, this would be extremely useful for an elder user who has problems reaching to higher places, and this tool would enable them to reach the height of the object, squeeze and bring the object towards them, reaching an object they would be unable to reach without the litter picker.

This picker is very similar to the one I investigated, with the same trigger and grip the design would function with a squeeze of the trigger to close the hooks of the design, however this design is replaced with 'scoops' rather than having hooks. The hooks have a larger surface area enabling for the scoops to pick up a larger amount of litter in one squeeze, this would be very beneficial when cleaning large amounts of litter, decreasing the time taken to clean. This would enable the user to clean and remove more litter especially if the litter is in larger quantities, however due to the larger size of the scoops it may be more difficult to pick up smaller amounts of litter.



This picker is completely different to the pickers that I have investigated, there is no trigger, meaning there is no string attached to it to pull open the hooks, instead the user can squeeze the handle which will squeeze the two ends together, enabling it to pick up litter. The design is like that of chopsticks, with two rods that squeeze together to hold an object in place. The simple and easy design of the tool with only two rods, enables for the tool to be open to a wide audience. Due to the user having to squeeze the two connected rods together, the user may feel a large amount of strain when using, the handle as well may be uncomfortable as it isn't ergonomic to fit the shape of the user's hand.

Product analysis of a bin bag holder

What is a bin bag holder and where would it be used?

A bin bag holder is used to maintain an opening to the bin bag so that placing litter inside the bin bag is efficient. The frame provides a circular opening that easily enables for litter to be placed inside to be kept organised area. It is commonly used when using a litter picker due to the litter picker easily being able to go through the centre of the hoop and drop litter into the bin bag.

How does the bin bag holder function?

The hoop is consisted of five hooks, that open and close with a pull and a push. When a bin bag is placed in the middle and wrapped around the hoop, the five hooks can be pulled open and then pushed closed onto the bag to secure it tightly. The bin bag will then remain open, and the user can easily place litter inside where the litter can be stored. The user can easily hold the holder by the foamed grip which is shaped to fit the hand, providing comfort and helping the hand to not strain whilst holding it.

Cost and value of the product

A bin bag holder is commonly priced at around £5-£10, making it accessible for many to purchase. I believe that this is a good value for money due to the service that it can provide, which is being able to keep a bin bag open so litter can easily be placed inside of the bag rather than the user having to manually open the bag which may be difficult if using with a litter picker, where the user may have to place the bag on the floor, open it and then place the litter inside, which is very time consuming. The product can be bought from online shops like Amazon and in store shops such as B&Q.

Ergonomics and design of the product

The holder is a circular shape, which easily allows the user to place the litter caught by them inside. There is also a foam grip which is shaped so that the foam grip is comfortable around the palm of the user's hand so that it is less likely to strain which is more likely on the thinner plastic that the foam is wrapped around. As well the hooks are shaped so that the bin bag can fit comfortably and stay secure inside the product.

The quality and materials of the product

The material that is used to make the litter picker is most commonly HDPE for the frame and clips, a hard and robust material that is chosen due for this product to its hardness, toughness and bending strength that allow the frame to support and carry the weight without breaking or becoming disfigured. The clips allow to be tightly fitted around the bag, securing it and ensuring the bag won't be released, the clips also bend backwards against itself, the flexibility of the HDPE allow for this to happen. The frame also has a foam handle on it, this is shaped to fit the hand of the user, the soft foam provides comfort for the user and helps to reduce strain when using.

User

The user of a litter picker would commonly use this product, the user is therefore very similar being cleaners or members of a community who want to clean an area and remove litter. It's lightweight and very easy to use making it appropriate to many people.

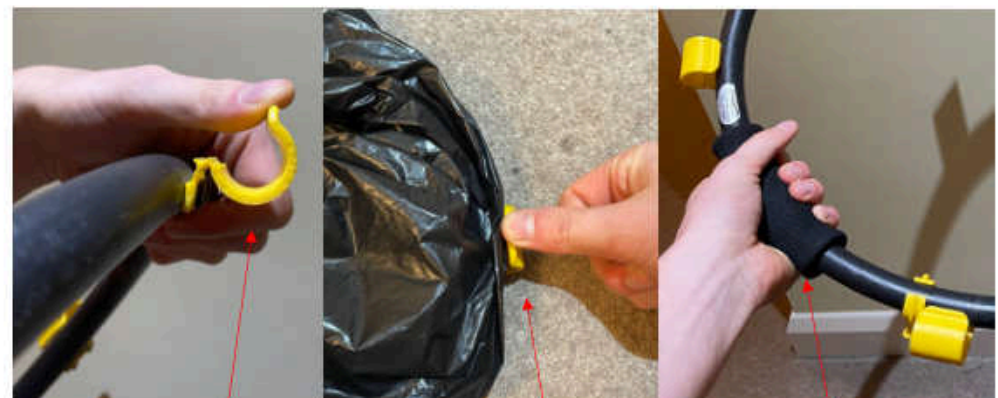
Environmental effects

The product is made from HDPE, this material is recyclable so is beneficial at reducing the effect of pollution on the environment, as well the product is a tool that is used in the process to remove litter and pollution which is damaging the environment. The only negative effect is if the HDPE is not recycled to produce the product and is instead produced for the first time, this damages the environment by releasing large amount of various greenhouse gases, which as a result contribute to global warming, severely affecting the environment. Additionally, the handle is produced from foam which is plastic based, this causes an environmental impact due to the extraction of oils used to produce it. That being said, the product is used in the service of removing litter from the environment, which themselves produce greenhouse gases and can cause harm to animals.



How to use a bin bag holder:

- Get a bin bag and the holder, with the holder push open all the 5 hooks, this can be done easily by pushing the curved end of the hook
- Place a bin bag inside of the hoop, ensuring that there is a small amount of the bag that is poking out of the holder
- As the hooks are open, push in the top end of the bag till it hits the inside wall of the hooks, then push the hooks back over the layer of bin bag till it is closed fully and fully secure, the clip will make a 'clicking' sound when closed
- Complete this for all the clips, if there is any part of the bag covering the handle, push the bag downwards so that it is secured below the handle, this will remove discomfort.
- Pick the holder upwards, and the bag will be tightly fitted and secure, this can now be used ideally for litter picking and placing the litter inside
- To remove the bag, place the bottom of the bag on the floor, so the bag won't fall and spill litter, push open the hooks open and dispose of it in a bin, then push close the clips



Curved end of the hook, makes pushing open the hook easy. Hooks have been cut so that it can bend without snapping

Push close the hooks on the end of the bin bag, the hooks will 'click', letting you know that the hook is closed

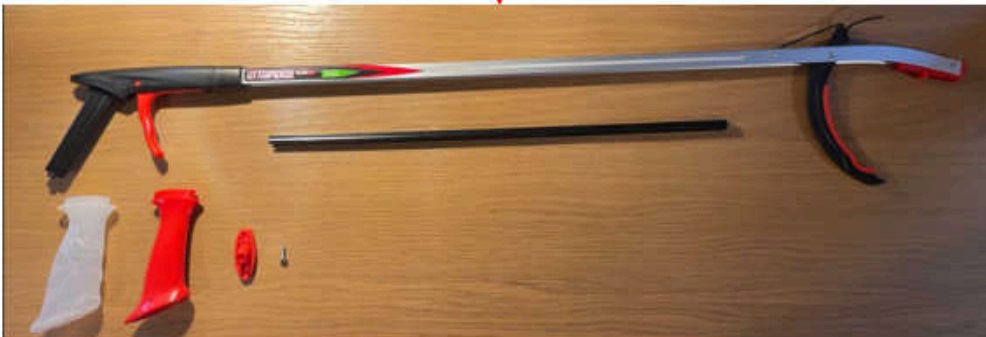
Ergonomic handle that provides comfort, shaped with a curve from end to end allowing the palm to fit comfortably

Disassembly of litter picker

The litter picker is the tool that I will use in my disassembly, being a tool that is commonly used to remove litter and pollution from the environment it is necessary that I investigate how the tool functions. Being able to fully understand that shape, size and how the tool functions will help me develop my design for a product. Although, the product is useful at removing litter, **disassembling the product will also enable me to find the disadvantages that come with the product, helping me find a solution to negate these problems and overall helping develop a product that is beneficial.**

Finding these problems will enable me to innovate a new one or create a new unique product that doesn't have these defects.

Photos of disassembly:



When disassembling the litter picker, I was able to investigate and understand how the product functions and how it has been designed so that it is appropriate for the user and that it is comfortable and easy to use.

The handle of the design is produced by having a handle covering the internal design, this is secured by a screw at the bottom of the design, there is then a gelled grip which is then layered over the handle, this provides comfort to the user as the gel morphs around the user's hand. Due to **the gel morphing around the user's hand, it allows various amounts of users to use the design with comfort.** I could incorporate this idea into my design as an easy way to develop an ergonomic grip for my future design that will be accessible to many.

The trigger of the design is additionally ergonomic, as the components of the design would have been produced by a method such as injection moulding or a similar process, the design has been shaped accurately to the shape of a user's hand. The curves on the trigger are placed so that the fingers can be placed for comfort and grip, allowing the user to pull the trigger with ease.

There is also a covering on the underside of the body of the design, this **covering is a bended sheet of aluminium which covers the pulley system and helps keep it protected, it is not connected to the design physically and is remaining in position by its shape.**

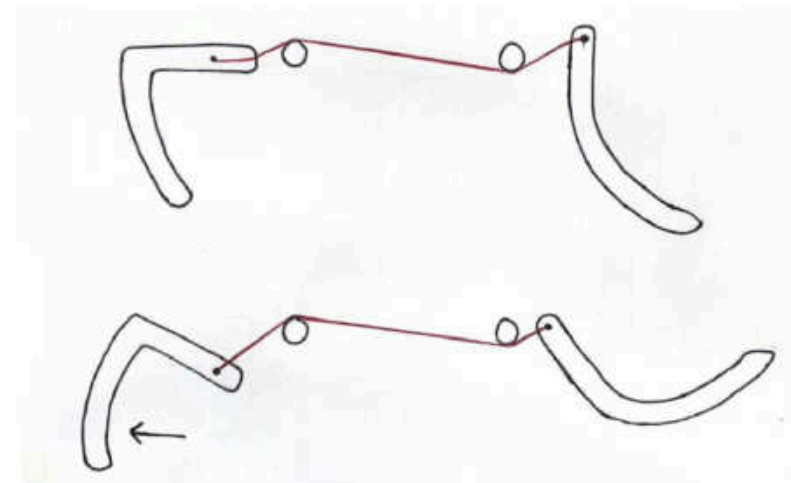
As the handle and size of the design is ergonomic it provides a comfortable and easy to use experience for the user, I should incorporate this into my design so that the product is easy to use.

The pulley system

This litter picker and most litter pickers that I have seen whilst researching this tool, contains a pulley system for the claw to open and close, investigating how this will allow me to understand how a pulley system works and could be a part in the designing of my product.

The design functions by the trigger on the handle being squeezed, a string is connected inside the handle, so when squeezed the string is pulled further from its resting position. The **string is run through a pulley located just before the silver body of the design to ensure that it remains inside the body of the design, then a pulley changes the direction of the string so that it connects to the beginning of the bottom jaw.** The squeezing force on the trigger pulls the string which pulls the beginning of the jaw closer, making the grabber close. There is also a spring attached just before the beginning of the jaw so that the grabber will remain open, and the user won't have to squeeze continuously which may strain the user's hand.

How the pulley looks without frame:



When force is applied to the handle and is squeezed, the string that is connected inside the handle is pulled, as a result the string which is connected to the beginning end of the claw will also be pulled. **The claw can rotate and due to the string being pulled closes the claw, allowing for litter to be picked up.** Behind the claw there is also a spring, this spring provides a force so that the claw will remain open when the trigger is not being pulled, putting less strain on the user and making the tool more comfortable. The pulleys simply allow for the string to be positioned in the correct place so that the pulley system can function appropriately.



Environment

The environment that my designed product will be able to be used in will decide whether my product is of a high quality. Due to my product being focused around limiting pollution and litter, which can be found in all/most environments, **the tool that I will produce will have to be able to perform at a high quality in a certain environment or a wider range.**

Sandy environment:

Another environment that I could develop my product to be used in is a sandy environment, such as a **beach or the banks of a lake or pond.** The litter found in dry conditions may often rest of the surface of the sand, however picking up the litter may be difficult due to the shifting sand and it moving when trying to be picked up, making the user must be more precise. This can be even more difficult for smaller pieces of litter such as **litter butts which require accuracy whilst it may be moving and sliding down the dunes.** Wet sand can also form in clumps which when removing litter may get caught in the object, making the object heavier and more difficult to carry and move, and may cause objects to sink slightly which then requires more force to remove. Due to the **moving environment, it may be tiring for the user to remove the litter as it uneven, which may require for a more accurate tool.** I will take these problems with this environment into consideration if designing my product for a sandy environment so that I can negate these problems, and they won't be an issue.

Grass environment:

A grassy environment can either have short or long grass, which can affect the variation of tools that can be used on it. Short grass allows for an easy pick up of litter due to there being a limited amount of grass blocking the user when removing the litter. When dry the user can easily pick the litter up, however in wetter conditions it may be more difficult due to the wetness of the grass, which may cause the litter to be caught inside the ground or behind grass, making it more difficult for the user to pick up. **Long grass will make it more difficult for the user to pick up the litter due to more grass covering the object, as well it may be more difficult to reach the object without damaging the environment or being able to be within access of reach.** In dry conditions, the litter may be caught in the grass but is often laying on top and is easier to get than in wet conditions. This is due to the wet grass being wetter, which can make the litter more difficult to pick up, as well the litter can often sink into the grass making it more often to be caught or tangled. **The wet conditions can also create mud which often makes the litter stick to the ground more making the user have to use more precision.** If I design a tool to help remove litter in this environment, I will also have to focus on the problems and adapt my design so that there are solution to these problems.

Watered environment:

Another environment that my product may have to be used in is a watered environment, this could be in the sea, lake or a pond to capture the litter. The litter can often be difficult to retrieve due to objects being constantly moving in water due to factors such as wind, this make the litter difficult for the user to obtain and often takes more precision for the user to pick up, tools also may not be long enough to reach the required litter. **As well, the user often must use a tool like a net to pick up the litter as a litter picker, which is commonly used for litter picking in general, as it doesn't have enough grip to obtain the litter, however a net is able to capture a large quantity.** The net therefore may be difficult for the user to carry if they are litter picking as well. Wet and slippery litter can be a challenge for the user to obtain and could drift with the currents of the water. Mud may make the litter stuck and difficult for the user to capture and applying more force to remove litter may cause damage to the environment. **If I design a tool to remove litter from the watered conditions, I will have to take these factors into consideration when designing my product to ensure my product doesn't face these problems or could easily limit the effect of them.**

Woodland environment:

Another environment that my product may be used in, is the woodland environment where there can be a large amount of litter carried from wind and people littering which often lays on top of the woodland floor or can get caught in bushes making it difficult to access and remove. **The uneven terrain may make this even more difficult to retrieve litter as well. The wet conditions can also cause for there to be mud, making some of the litter stick inside the ground and be difficult to remove fully.** This environment is also densely packed with wildlife and biodiversity so the dangers that litter has to the animals is high so is needed to be removed. As well, some litter may be blocked or can't be accessed due to bushes or branches blocking the user from retrieving the litter. **I will take factors into consideration when designing a product so that it is able to function well if I decide to develop a product for this environment.**

Urban/concrete environment:

The urban environment is hard and mostly flat which would allow for a wide range of products to be used on it. When it is a dry environment, it is easy to remove the litter, however when it becomes wet after rain it can become more difficult to remove this litter. **For example, when trying to remove soggy paper or cardboard, sometimes it can become stuck to the flat floor, this becomes difficult to remove the litter as there is limited angles to grip the litter, requiring more precision and force, which can often tear the litter, spreading it further.** Additionally, as the roads are flat, there can often be an accumulation of litter in a lower area due to litter being carried by wind or water, this therefore requires more intensity for someone to pick up. **If I design my tool to be able to remove litter from this environment, then I will also have to focus on these problems that occur.**

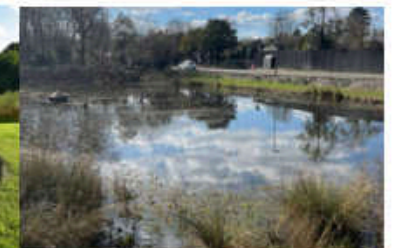
Sandy:



Grass:



Watered:



Woodland:



Urban/ concrete:



From the research I have conducted on the different types of environments and challenges that come when attempting to remove litter, I have been able to identify the problems, which I will take into consideration for when I am developing a product which will be used in one or more of these environments. **One of the main problems that I found with all environments was when the environment had wet conditions which made the process of removing litter more difficult, with it making litter stuck to the ground or trapped as water moves the litter difficult to retrieve.** With this research I will be able to adapt and develop my designs to fit the requirements of a product within certain environments for it to perform at its best and be a high-quality product that is adapted to the user's needs.

Design brief

A design brief will allow for a short summary of my current findings and the effect that they have had on the development of my design. It will allow for me to state what my research has led me to and enable to focus on the type of product I will produce that has been developed around the user.

Problem:

From the research that I was able to conduct and with **speaking with my client and having a questionnaire answered by users**, it came clear that there were issues with the current products used to remove litter from the environment that my client and other users faced when removing litter from the environment, and targeting their problems is essential for producing a product to allow for the users to have an easier session. These problems were:

- Being a difficulty to remove litter in a wide range of terrain as the change would make the litter get stuck differently and make the removal difficult
- Small objects were difficult to pick up such as cigarette butts due to their small size, making the user have to be more accurate when picking up, and would be a slow process if there were large quantities
- Tools being straining due to the user having to carry them around in various types of weather and conditions and having to carry them for long periods of time
- The strain of picking up numerous amounts of litter in a short period of time which makes the user strained as they have to clear a larger amount of litter, making the user take longer and not be able to take much of a break/rest
- The difficulty to place litter in the correct bin due to it litter often being carried around in one bag for ease, user then has to remove litter and organise it into its specific bin
- The litter carrying equipment can be easily damaged due to the litter being placed in a bag, this bag often doesn't have much support and can be torn or damaged which can make the process more difficult

From these problems that I have been able to identify, it has focused me on the problems, which I am now able to focus on creating solutions for to provide a better and easier experience for the user when removing litter.

From the problems that I have identified I can write a brief for a product which will be focused on limiting the effects of these problems.

Design brief:

Design a product that can help reduce pollution and litter in a wide range of environments and terrains so that the removal of litter can be done more effectively and be easier for the user. The product should not cause strain on the user or be difficult to use and should provide the user with a simple and easy experience so that litter can be removed more effectively.

This brief allows me to direct my product onto the needs of the user and the experience that they will have when using the product. I will have to ensure that the product is able to complete all the requirements on the design brief for it to be a good product. Not only should my product be effective in all terrains, but it should additionally not cause strain on the user. This was something that my client and those who answered the questionnaire often faced when removing litter, so it is an essential part of my production and development process to ensure that the product doesn't cause strain on the user. Creating a product that is easier for the user is additionally essential, with it being able to work in multiple types of terrains and environments, so the user can use the product diversely. Directing my product on this brief will allow for a user-centred design to ensure a developed product, with the user being cleaners, volunteers and workers who clean an area, it is imperative that the design is efficient and easy to use which will allow for the job to be performed at a high quality.

Design specification

In order for my product to be useful, I will need to stick and stay close to the requirements of a checklist, to ensure that I am not diverging too far from creating a solution to the problem identified. By sticking to the points, I will develop a product focused on the problem identified, ensuring that the product developed is useful and as well meeting the design brief, resulting in the user of the product being able to be remove litter more effectively and have an easier experience.

Function of the product:

The function of the product is to help the user remove litter and reduce the effects of pollution and the damage that litter can cause to the environment, by **making the removal process simpler and more effective, allowing for an easier experience for the user, reducing the difficulty of the process which can be stressful and difficult.**

The process is often difficult to remove all types of litter, as some pieces of litter can be difficult to retrieve such as small pieces of litter, like cigarette butts and corners of food wrapping. Evidence of this was found in my primary investigation, where I went around the Millenium Green and counted the types of litter, where the proportion of cigarette butts found was much larger than other types of litter, and as this area is often litter picked, inferring that the removal of these small pieces was difficult. This can be further proven from my questionnaire, where many users found small pieces of litter difficult to remove due to there small size and their tool, which was a litter picker, being too large to grip the desired litter. As well, this can be further proven, with my conversation with my expert and client where they found that the removal of these small pieces was very difficult and time consuming, as well as very straining as they would then resort to kneeling to remove the litter. Additionally, removing litter from a wide range of environments such as litter stuck to the ground in an urban environment and litter in a watered environment could often be difficult due to the lack of appropriate tools and grip that can be achieved with current tools to remove this litter, which would often lead to the user being unable to retrieve the litter as they were unable to grip the litter effectively. Furthermore, the tools used could often be straining when using for extended periods of time which was found in my questionnaire and interview with my client and expert, this would cause the user to feel discomfort when using for extended periods of time which would make the process of removing litter more difficult, going against the design brief where the removal of litter should be simple and effective.

My product must be functional and effective for the user so that it is able to help remove litter from various environments, be able to remove all types of litter so no size of litter is in the environment and can cause damage and negative effects and not be straining so that the user is able to remove litter from the environment without discomfort. When designing my product, if I follow these check points, the product will be able to function to the needs of the user and accommodate for the problems that I investigated, the end product will be able to function effectively and be a simple process for the user to remove litter from an environment, and to decrease the overall effects of pollution and litter.

Performance requirements of my product

My product must have elements based around simplifying the removal of litter from various environments to allow for an easier removal process. The product must be able to accommodate the needs of the user. For example, as the product must help make the removal process of litter easier from a wide range of environment, the removal of litter from a watered environment, must be accommodated in order to remove litter from a watered environment which is difficult with current tools and products, this is found in my conversation with my client and expert where they were often unable to remove litter due to not being able to grip the litter. As well, the tool must be able to help remove small types of litter such as cigarette butts and food packaging which was found often difficult to remove due to their small size and lack of grip that the user was able to get on the tool, this evidence was provided in my questionnaire for users and conversation with my client and expert, where they were unable to gain sufficient grip on the litter with the tools.

Material and product features:

As my product is to be used in the environment and help to limit the amount of litter, helping contribute to decreasing the overall effect of pollution, even just within a given area, the user is most likely to carry the design around. As the user is likely to be carrying the design around, the design would have to be supportive so that the user feels less of the weight that they are carrying, a design that is not carried, or one that is lightweight, this is important so that the product would not cause strain or discomfort to the user so that they are able to continue removing litter, and continue to decrease the amount of litter within an environment. As well, the material of the product needs to be able to deal with the wear and tear that may come with use, due to knocks and bashes against parts of the environment and potentially misuse of the product to prevent it breaking and maintain a long lifespan of the product. By making the product from a durable material, the user will not have to replace it or repair it as often, meaning an easier maintenance of the product. By aiming for the product to last decades with proper care and rigorous use, the product will cause minimum inconvenience for the user, which can be done using lightweight, durable materials.

User requirements

As my product is most likely to be held or moved by the user, the product must be ergonomic and comfortable for the user, the sizing of the product must be ideal to fit within the range of the 5th to 95th percentile of users, this is essential in order for the product to meet the requirement of the brief where the product mustn't be straining or be difficult to use by the user. As well, providing a strain-free or decreased straining product which will allow for the product to be a solution to the problem that my client/expert faced where the tools they use to remove litter could often be straining on the them. By using anthropometric data and testing the sizes of components with users, I will be able to achieve the comfortable and ergonomic size of the product to accommodate for users in the 5th to 95th percentile, meet the design brief and meet the requirement of the client. The target market for the product is for those who remove litter from environments, this could be those who are cleaners, and their jobs requires them to remove litter from an environment, such as a festival cleaner, to volunteers such as my client and expert who remove litter from a chosen environment on their own accord and their good will. Allowing for a product that can assist with the removal of litter and pollution that the users find when removing litter, will help for an easier experience, make the process of removing litter quicker and therefore allowing for the user to be more effective and remove more litter in a given time, reducing the effect that the litter and pollution will have on the environment, even if the difference is microscopic, helping remove this litter will help sustain certain locations and environments so that they are more pleasant and less environmentally damaging to animals and to contributing to climate change, which I investigated in my research.

Sustainability

In order for my product to maintain the environmental friendliness, the product should be produced from sustainable materials so that in the case that my product breaks and is unrepairable, the user is able to dispose of the product properly and with care (trusting in the fact that when the litter is disposed of in the bin, those who take the litter from the bin, recycle it and take care with the process), trusting that the product will not be sent to a landfill site, where the product will degrade and cause damage to the environment, and that the product can be recycled. Producing the product from sustainable materials will also limit the damage caused to produce them, such as energy and fuel damage to the environment, as well if the product is produced from a polymer, a sustainable product will reduce the effects of the microplastics that will be exposed into the environment.

Scale of production

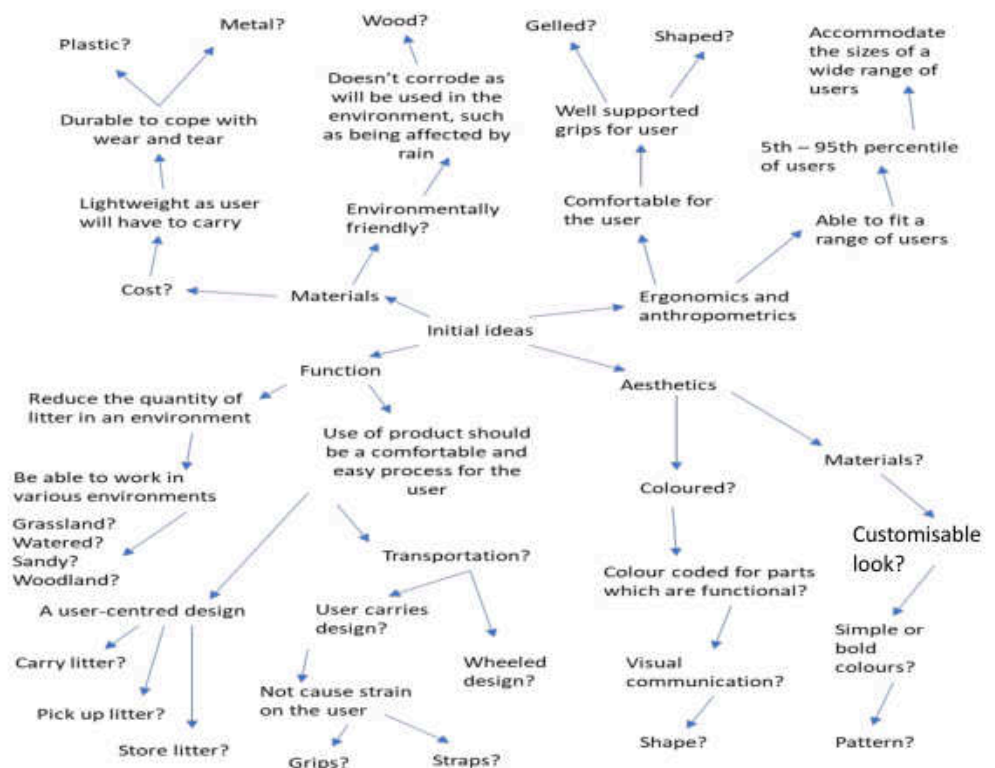
Further speaking to my client and considering other products on the market that help to remove litter, by making the product be accessible for many with a price range that is affordable and a quality of product that meets the demands of the user, the product will allow for the user to remove more litter from the environment and decrease the effect that litter and pollution has on the

environment. As well ensuring that the price of the product remains affordable, would allow for more user's to acquire the product and be able to remove litter from the environment, creating a positive cycle where more litter can be removed and reduce the effect that the litter has on the environment. As well, when developing my product, I will have to consider if the product was to be mass or batch produced and the requirements that would need to be met to facilitate this demand.

By ensuring that I follow these checkpoints on the specification, the design will meet the demands required and be a solution to the problem identified, leading to an easier experience with the removal of litter and therefore helping cause a reduction in the quantity of litter in the environment. Combining the checkpoints of the specification and the demands of the design brief, I will be able to produce a product centred to the user's problems and help with removing litter, resulting in a good and effective product.

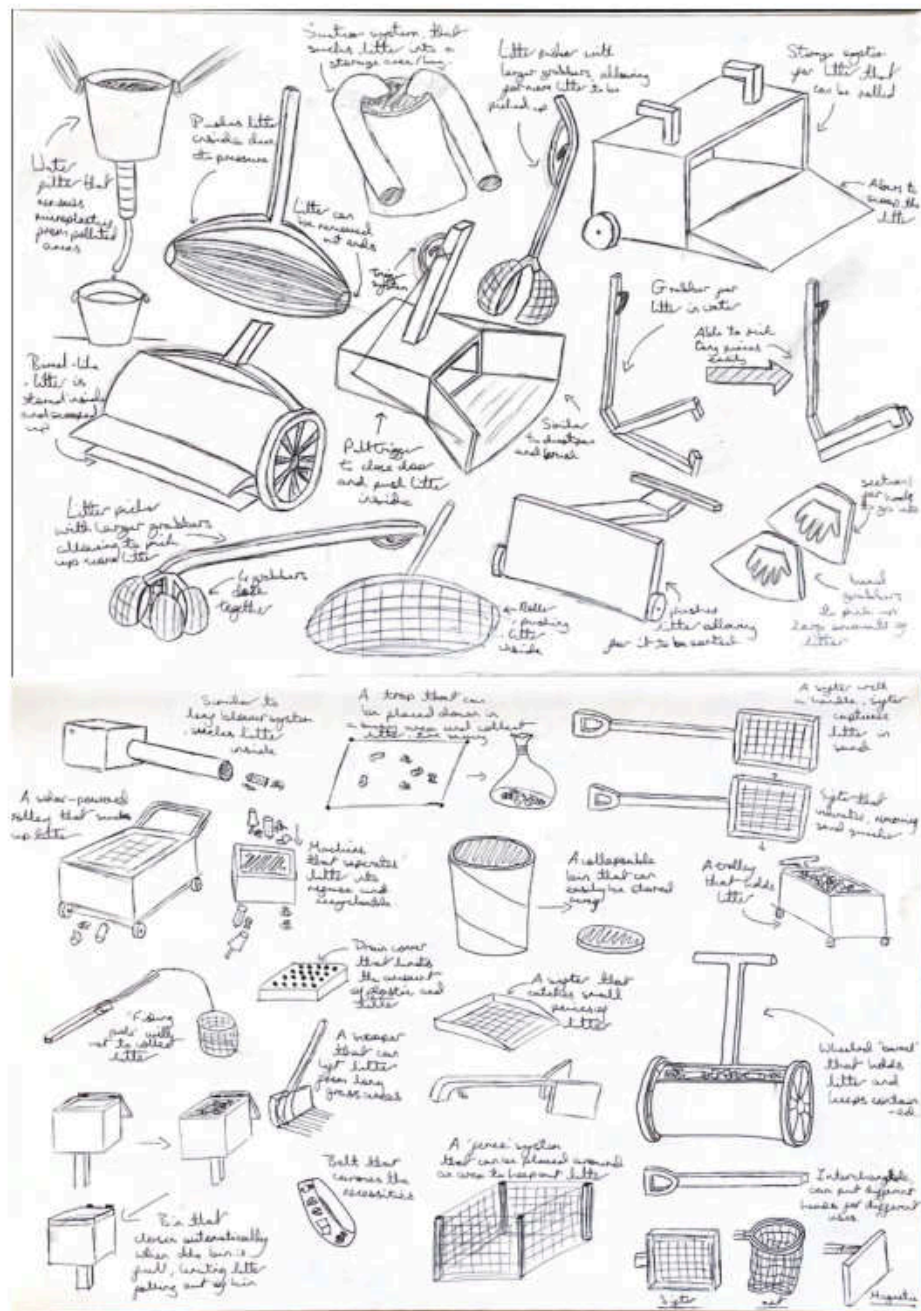
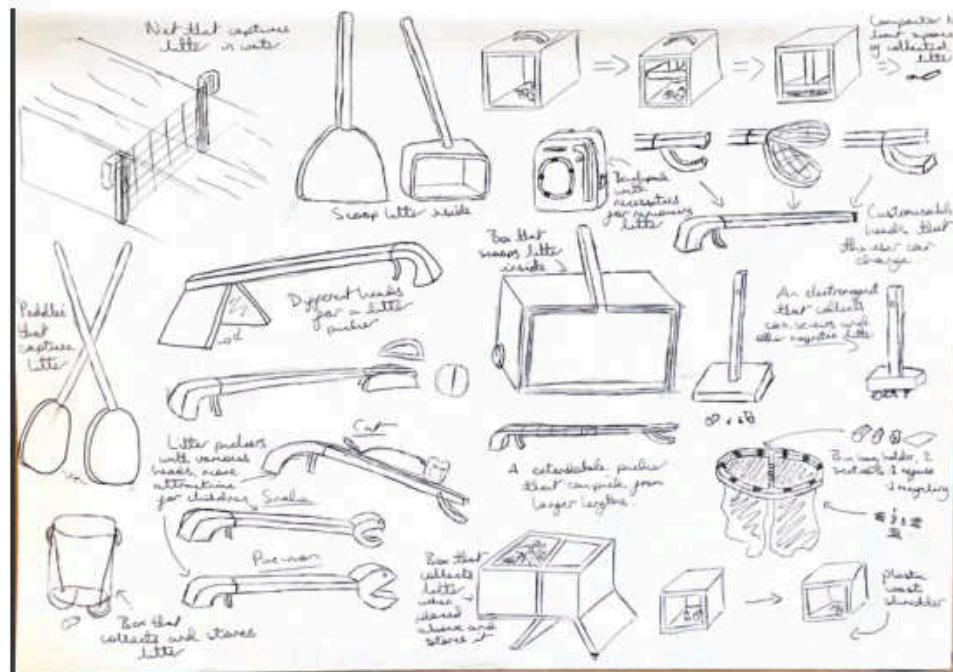
Mind map of areas to consider when developing

As I now have my specification and design brief in place, which I will follow strictly in order to for my designing of my product to not stray from the designing of a solution to the problem identified, I am now able to begin the designing process of a solution to the problem of litter and designing a product which reduces the quantity of pollution as researched. Below is a mind map that I have created clarifying the needs that my product will have to fulfil for it to be a good functional product that can meet the demands of the user, and will allow for my design ideas to follow around these chosen aspects to ensure an efficient product:



This mind map outlines the necessity of my initial ideas for my product through the materials, ergonomics and anthropometrics, function and aesthetic which if all are done correctly and with consideration with the client and user in mind will result in a good product. From these ideas, I am now able to begin the design process of my product, sticking strictly to the specification and brief.

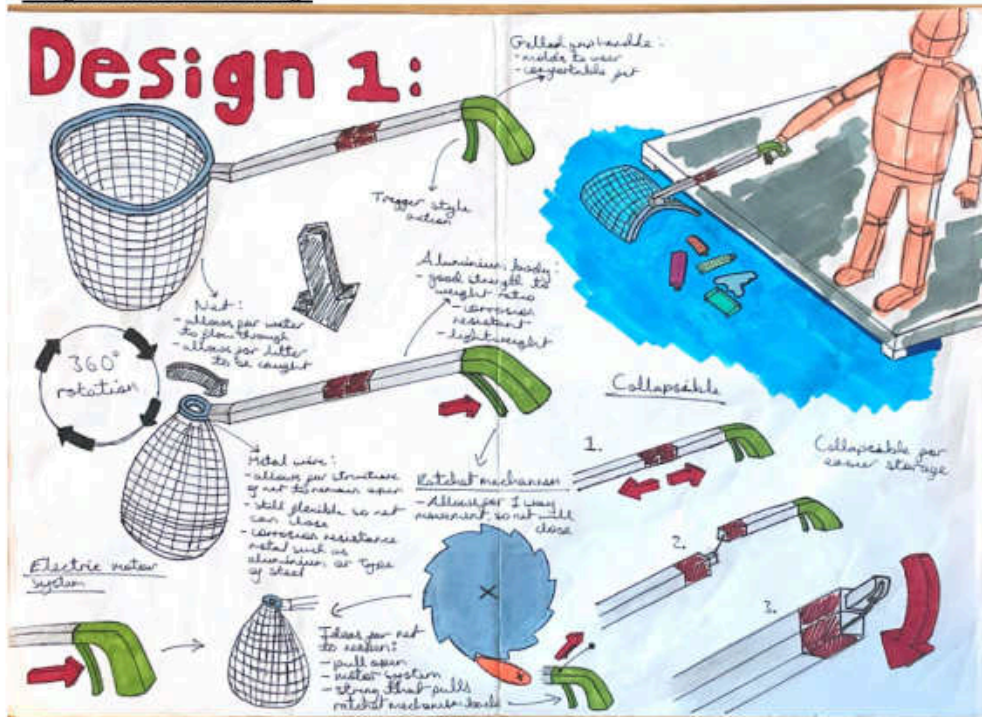
To start the development of a product that can meet the design brief and reduce pollution, I have produced initial sketches of potential solutions that I can develop and target to produce a product that makes the removal of litter an easier process and one that can be done effectively.



From the initial ideas sketches that I have produced, I will choose and develop the ones that I believe to most meet the design brief and target the problems that users have when removing litter, by targeting the products with the most problems my design will allow to provide solutions which would give the user an easier experience when removing litter, allowing for an easier removal or sorting of litter and pollution, meaning that the user can remove more litter, and keep an area of the environment less contaminated from pollution.

Further developed designs

Design 1: Closeable net design



Closeable net design:

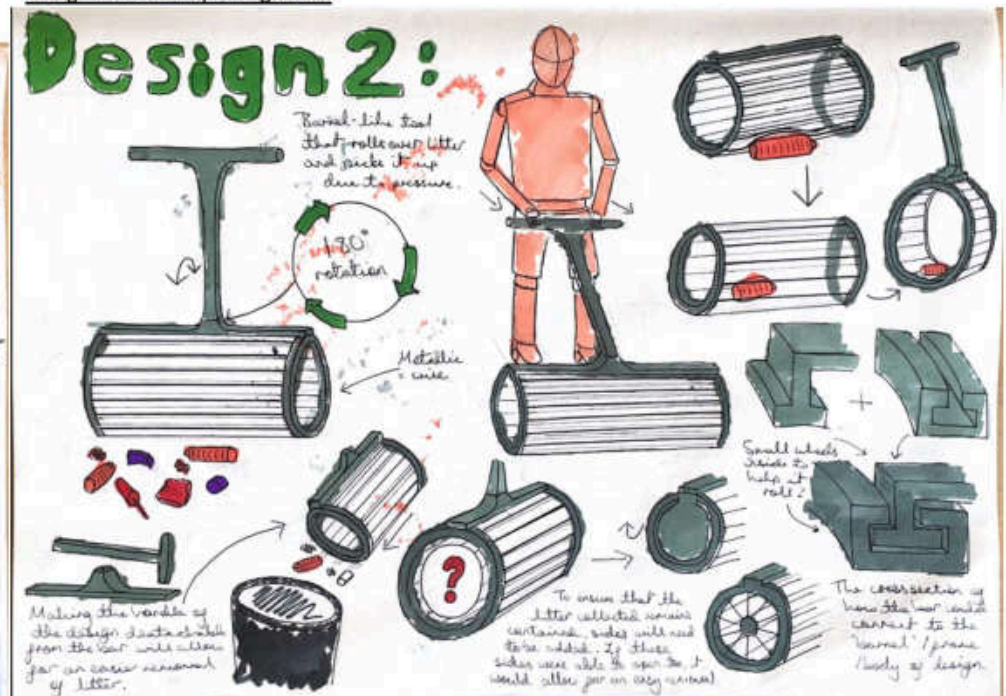
This developed design functions as a net that users use to capture litter in water, the net can close by the pull of a trigger if I use an electric motor or numerous pulls of a trigger if I use a ratchet mechanism for this design. Aswell, the design is collapsible which provides an easier and tidier storage for the user. The design is inspired by a litter picker as some features are like the design of a litter picker. **The main function of the design is to close so that litter is unable to escape after being picked up and being inside the net, allowing the user to remove litter that is inside a watered environment,** where the litter would be secure and not leave the net which would be an inconvenience

To achieve the net which can close by the pull of a trigger I could use two types of mechanisms:

- **Ratchet mechanism**: This mechanism allows for continuous pulls of the trigger to make the net tighter and closer together to remove the chance of litter escaping from the net. It consists of a round gear with teeth and a pawl that locks onto the gear, making it unable to move unless the pawl is moved, my current idea for this design is that the trigger of the design is attached to the pawl, so when the trigger is pulled it pulls the pawl too, which as a result would make the pawl move to the next tooth on the gear, the only problem that I have found with this design is that loosening the net after being tightened may be a challenge as it goes against the teeth of the ratchet mechanism.

- **Electric motor**: An electric motor would allow for an easy opening and closing of the net as it could be done with the pull of a trigger, this mechanism would allow for opening and closing to occur quicker and therefore be much more efficient with the design. With this would be that as it is electric, the design would either must be charged or have batteries inside to provide the energy that is required to open and close the net. It wouldn't be logical to have a wired design either as it is impractical due to the user being outside in an environment.

Design 2 : Litter capturing barrel



Litter capturing barrel design:

This developed design functions by the pressure exerted onto the floor and on the wires of the design, when the design has rolled over a piece of litter the wires will be moved outwards due to being slightly flexible, and the litter will then move inside the design, allowing for it to be stored inside the wired design.

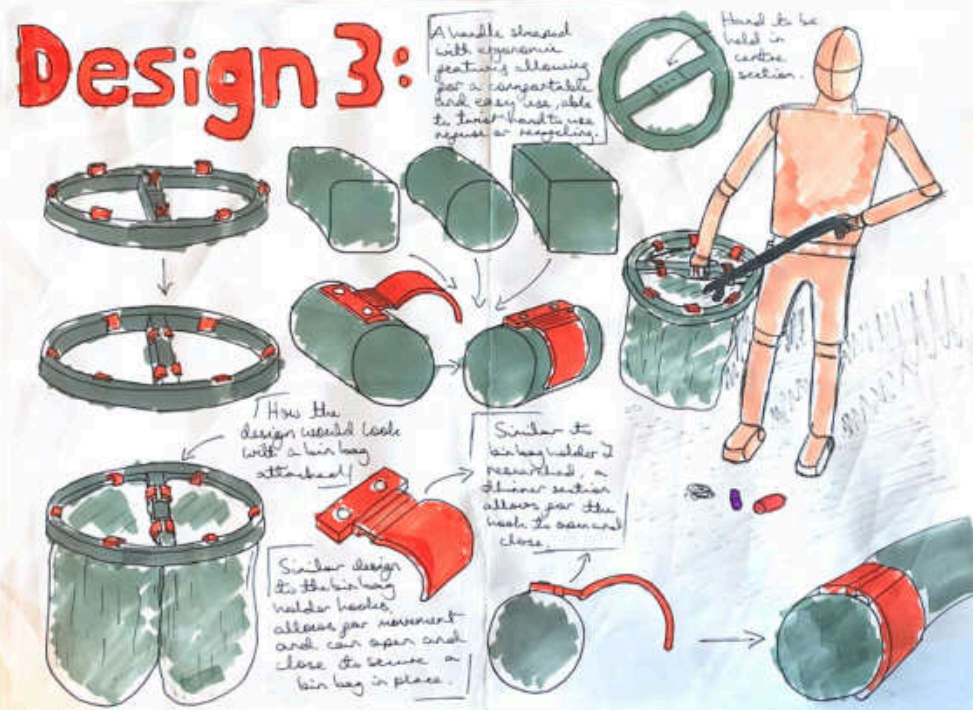
The design functions by the **force exerted onto the litter when the design rolls over, which widens the wires, allowing for the litter to be captured inside the design.**

The inspiration behind this design is a gold ball collector which allows for golf balls to be easily picked up when the design is rolled over the golf balls, the wires are slightly flexible which allows for the balls to be pushed through into the barrel, the golf balls are then stored inside the design, the gaps of the wires are smaller than the size of the golf balls in order to store the gold balls inside. I have taken the idea of this design and changed it so that instead of the barrel picking up golf balls, the barrel picks up litter.



Design allows for golf balls to be picked up, my design would be based for litter instead and pick litter up instead of golf balls

Design 3: Bin bag holder

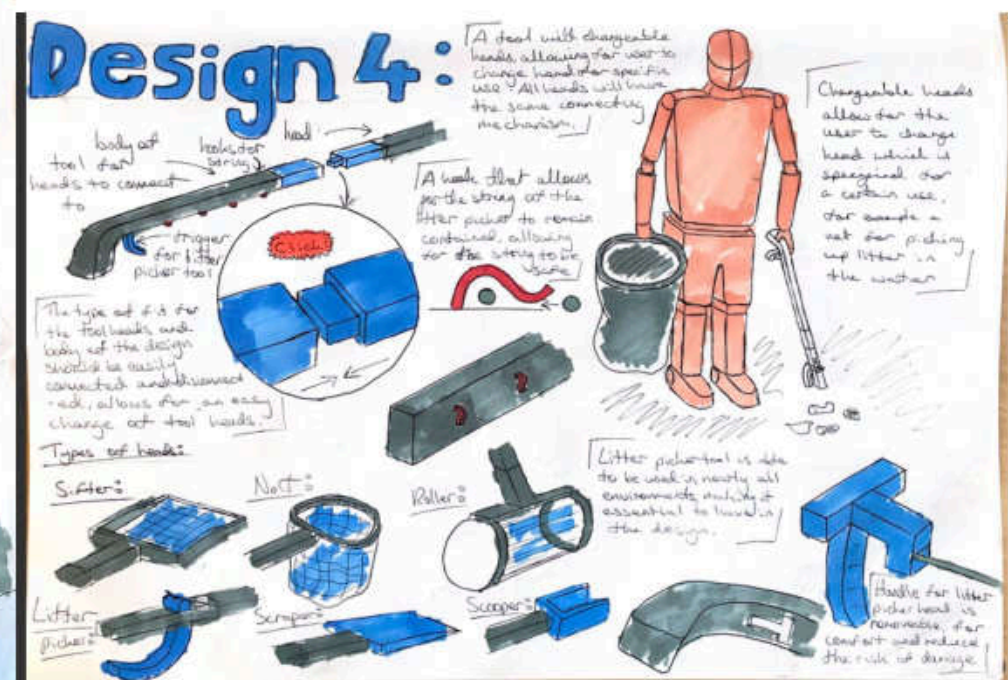


Bin bag holder design:

This developed design functions identically to a single bin bag holder however this design has an area for two bin bags, one for recycling and one for refuse. This design allows for a much more environmentally friendly removal of litter as the litter that is collected can be designated into a recycling and refuse section. With current bin bag holders in the market only allowing for one bag to be carried and held open, it means that users have to place all their litter which they find into one bag when removing litter, commonly with a litter picker. This means that the recycled litter is now contaminated with the litter which can't be recycled, meaning the user may have to be manually sort out and separate the litter which can be unpleasant. This design removes the need for this process and allows for the user to sort out the type of litter whilst removing litter from an environment, saving time and decreasing the quantity of recycled litter that could be recycled but is instead sent to landfill. The design has clips to secure the bag to the hoop, these clips will be colour coded, so the user knows what type of litter to place in each bag, green for the recyclable bin and black for the refuse bin.



Design 4 : Changeable head litter removal tool



Changeable head litter removal tool:

This developed design is a combination of many tool heads onto one, by making the heads attachable and removable the user can easily change the tool head for their required use. The litter picker head will allow for a use in many environments, the scraper can scrape stuck litter off the floor, the scooper is able to scoop up large amounts of litter, the roller is able to pick up litter on a flat environment easily, the net can capture litter in a watered environment and a sifter can let mud, sand and water pass through, each of these tool heads will allow for the user to capture and remove litter from a certain environment and location where the litter is different or difficult to capture. With there being multiple tool heads, the user will not have to carry numerous tools and instead only can carry the tool heads which can connect to the body of the design. This design will allow for the user to change the head of the design to accommodate their needs when removing litter from an environment in order to make the litter removal process for them an easy and simple job. If the removal process is easier then then the user will spend less time on removing litter and can focus on other pieces of litter, allowing for the user to remove a larger quantity of litter at an easier rate.



Decision on solution and analysis of further developed designs

With the four further- developed designs that I produced, I need to choose one of the designs to be a solution to my problem and develop even further, this design needs to meet the design brief and specification which as a result would make it appropriate for the user and my client and they would then therefore face fewer problems due to the design providing solutions to the current problems that occur for them.

To choose the correct product that can meet the requirements of the design brief, I need to be critical in my analysis of the product to ensure that I can outline the advantages of the product to compare with the other designs I have produced to ensure that I have chosen the best and correct product that meets the design brief directly. However, if there is a potential problem with one of the further developed designs then it is imperative that I highlight it, outlining the problems will not only help me decide whether the product is appropriate but also help me direct my development on my chosen solution to deal with the problems that I have outlined.

Reminder of design brief:

Design a product that can help reduce pollution and litter in a wide range of environments and terrains so that the removal of litter can be done more effectively and be easier for the user. The product should not cause strain on the user or be difficult to use and should provide the user with a simple and easy experience so that litter can be removed more effectively.

Following the intentions and requirements of the design brief, when analysing the potential solutions will allow me to focus on the product required. For example, the product may be good, but it needs to fit the requirements of the design brief in order to fix the problems faced by the users. These problems that have been faced have been found from a questionnaire and talking to a volunteer about the problems that they themselves have faced. Ensuring that my design can provide a solution to these problems would allow for an easier experience for users when they are removing litter, which would then allow for more litter to be removed from the environment as they are able to remove litter at a faster and easier rate, which would reduce the environmental impact that litter has.

This would then target my chosen area of interest which is pollution and provide a solution and would decrease the environmental effects caused such as damaging wildlife and contributing to global warming, even if the design is only able to partially help speed up the litter removal process, less litter would be in the environment and allow for there to be less of an environmental impact due to pollution and litter.

Closeable net (Design 1)

The first design is a closeable net design that can capture litter in water and then close the net client's contain litter and prevent it from escaping. The design has a trigger inspired by the litter picker tool that when pulled will close the net.

By closing the net, the litter that is captured inside the design will remain captured and will not escape the design, if the litter escapes it could cause an inconvenience for the user which is not wanted.

Advantages:

- Design allows user to retrieve litter that is in water which is often difficult to retrieve with other tools
- Can close the net which can contain the litter to prevent it from escaping
- Able to pick up larger amounts of litter due to a net which has a large opening, reducing time taken to pick up litter

- Able to pick and capture litter that would have been extremely difficult to get without the net tool, my client often had to use a litter picker to obtain the litter but was difficult to pick up due to the slippery surface of the litter, or push the litter in the water to them until it was easily accessible, and they could easily pick the litter up
- Solution to one of my client's problems being that they found it difficult to remove litter from water

Disadvantages:

- Design can only be used in water environments due to the net tool
- Unable to pick up litter that could be on the shore of the water
- May be difficult to reopen the design again with the trigger mechanism which only functions in one direction
- Doesn't meet the part of design brief where it is able to be used in a wide range of environments as is only able to be used in a watered environment and not a wide range of environments, however, is useful as litter in a watered environment can be difficult to remove due to the litter being wet and difficult to grip
- Litter may be difficult for the user to remove as the litter may get caught inside the net, this would require for the user to have to touch the net to remove the litter, which they may find unpleasant and wouldn't want to do
- May also pick up items such as twigs and these may get caught inside the design which is a hindrance for the user
- Litter will be at bottom of the net due to gravity so closing may not be needed

Litter capturing barrel (Design 2)

My second design is a portable barrel-like tool that can be rolled over litter and captures it inside due to the pressure of it being rolled over the top. The litter goes through wires which moves inside the design due to the pressure applied which occurs when rolling over.

Advantages:

- Design can easily pick up a remove litter in flat environments, such as urban or roads
- Design can pick up large amounts of litter with ease due to large size of barrel
- Able to contain litter without the litter falling out from inside the barrel due to a limited amount of force being placed on the inside of the wires of the barrel
- Able to remove litter easily due to detachable handle and sides which can open, leaving litter to fall out of the design
- Can be used on any hard surfaced environment, due to hard surface being able to resist the force applied, resulting in the litter moving inside
- Can be easily moved due to barrel shape allowing for it to be rolled
- Meets the design brief as the design can be used in most environments, allows user to pick up litter with ease due to pressure applied and simplifies the removal of litter whilst making the removal more effective

Disadvantages:

- Unable to pick up litter stuck to the ground and unable to pick up litter in 'off-road' terrain due to the wires requiring tension to be separated
- May be difficult to pick up litter such as paper, cardboard or flat cans due to the barrel rolling over the top of the litter
- May be difficult to pick up litter in a terrain which is varied in height, as design can't apply pressure easily to the design
- Unable to pick up small sized litter due to litter being smaller than the gap of the wires, so the design is unable to place pressure on the design
- Connection between handle and barrel could get dirty easily due to being rolled against the floor, could it be pressure washed?

Bin bag holder (Design 3)

My third design is a bin bag holder, however instead of being a typical bin bag holder, this has two sections with hooks, being able to hold two different bin bags, one for recycling and one for refuse. The handle for this design is in the centre so the user can easily turn or twist their hand to place the litter in the correct bin bag, typically a litter picker tool would be used as this is able to function in various scenarios. The user will capture the litter with the litter picker and release it into the chosen bag, the clips that attach the bag to the holder will be colour coded so the user can identify which bag to put what type of litter inside, green for recycling and black for refuse.

Advantages:

- Allows for more environmentally friendly removal of litter as litter can be placed into specific bin
- Reducing contamination of different types of litter, such as recyclable litter being placed in a refuse bin, which may go to landfill, which is essentially just removing litter from one location to the other however the damaging effects will continue to occur as some litter can be recycled
- Saving time of the sorting type of litter at the end of a cleaning session as litter has already been sorted into specific type
- Colour coded hooks that keep the bag attached, green for recycling, black for refuse, allowing for clear communication on where the type of litter should go
- Handle in centre of the design, allows for an easy turn of the hand in order to access correct bin bag for storage of the collected litter
- Meets the brief as the design simplifies the removal of pollution by reducing time taken to organise the collected litter into either refuse or recycling, as it is already organised into its refuse or recycling
- Meets the brief as the design reduces the effects of pollution due to more litter already being organised into its specific type, so recyclable litter can be recycled and not placed in landfill where it would degrade and release greenhouse gases and chemicals
- More litter can be obtained than a normal bin bag holder due to being able to hold two bin bags

Disadvantages:

- Design is unable to pick up litter due to being a bin bag holder however can store the litter
- The importance of the design relies on the user knowing what types of litter are for the different bins being refuse or recycling and could lead to the wrong type of litter being placed in the wrong bin
- Due to more litter being able to be stored, the weight of the design may increase which may cause the user more strain
- The bin bags may swing which may cause discomfort for the user, the user could be hit in the leg when using the product due to the swing
- Bin bags may get caught on the handle due to it being centred and placed between the hooks which could cause discomfort

Changeable head litter removal tool (Design 4)

My fourth design is a litter removal tool that has changeable heads, each head is designed for a specific use, designed around a litter picker design as it is the most versatile and effective to remove litter but adding other heads where the litter picker is unable to be as effective in, such as a net which can be used for litter in water which a litter picker can find difficult to grip.

Advantages:

- Allows for litter to be removed in a wide and varied amount of environments to remove litter effectively

- Meets the design brief as the design is able to be used in various types of environments and still function effectively as each head is designed for a specific use
- Trigger connected to the litter picker head is used in the design without attaching, allowing for easy attachment and removal
- Due to trigger being removed it allows for the trigger to not get in the way of the users hand and allows for extra comfort and freedom of hand when using
- Attachable trigger allows for the user to keep all parts of certain tool heads together to prevent the user losing items and to not be an inconvenience when the user is using other tool heads
- Litter in environments can be captured; litter can be captured using the net head, litter on the ground can be captured using the litter picker head, litter stuck to the ground can be captured using the scraper, scooper tool can pick large amounts of litter, sifter allows for litter to be picked up in sandy environments such as the bay area next to water, roller allows for a large amount of litter to be easily picked up specifically in urban environments
- Hooks on the underside of the tool allow for the string for the litter picker tool head to be contained and helps keep it safe from damage
- Shape of litter picker trigger section contains the trigger and ensures it stays in that position (remaining straight) and will not become loose or fall out, trigger is secure in place when put in the triggers position

Disadvantages:

- Due to the large amount of different types of heads, it may be difficult to carry them all efficiently at the same time, efficiency being important as it is one of the factors of the design brief
- The heads could be damaged due to them not being directly on the tool
- Transportation of tools may be a problem as there are many tool heads
- User will have to decide what type of tool head is appropriate for the type of litter that they encounter
- User may get dirty due to being in contact with the heads of the tool that are

Conclusion for my decision on solution product:

From weighing up the advantages and disadvantages, talking with my client about my four further developed designs and comparing the designs to how well the design meets the design brief and specification I have decided to choose my fourth design which is the changeable head litter removal tool to develop even further and be the product that I will take forward.

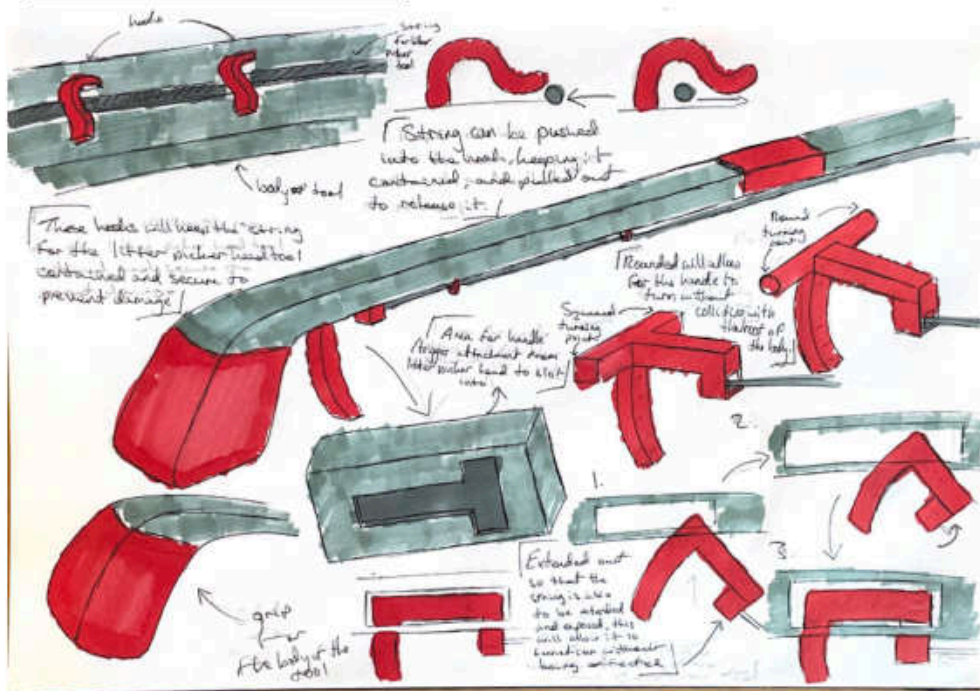
This product is able to meet all the problems that my client faced when removing litter and able to provide a solution to problems other users currently face when removing litter. Being a tool that has changeable heads allows for the user to choose the appropriate tool head for their required situation, therefore as a tool head can be specialised for a certain situation or environment it will enable for the litter removal process to be performed quicker and more efficiently.

Whilst the other developed designs were also good ideas which were able to be beneficial and remove litter effectively (apart from Design 3), this option outweighs the other designs, not only can this tool be used in more environments which fits the design brief better, but it additionally could use some of the aspects from the other designs which I could take forwards, such as the barrel-like design and a closeable net.

This design allows for the user to always be able to remove litter effectively with the tool heads being specialised for certain environments. Additionally, the design is able to meet the problems faced by my client and other users easily and meets the design brief; the design can help reduce pollution and litter, able to function in many different environments and conditions as each tool head is specialised for a certain function and I can develop a design to be less straining on the user. Throughout the development of this product I will continue to check up with my client to ensure the product meets her needs.

Developing the multi-head litter removal tool

Development of the body of the design:

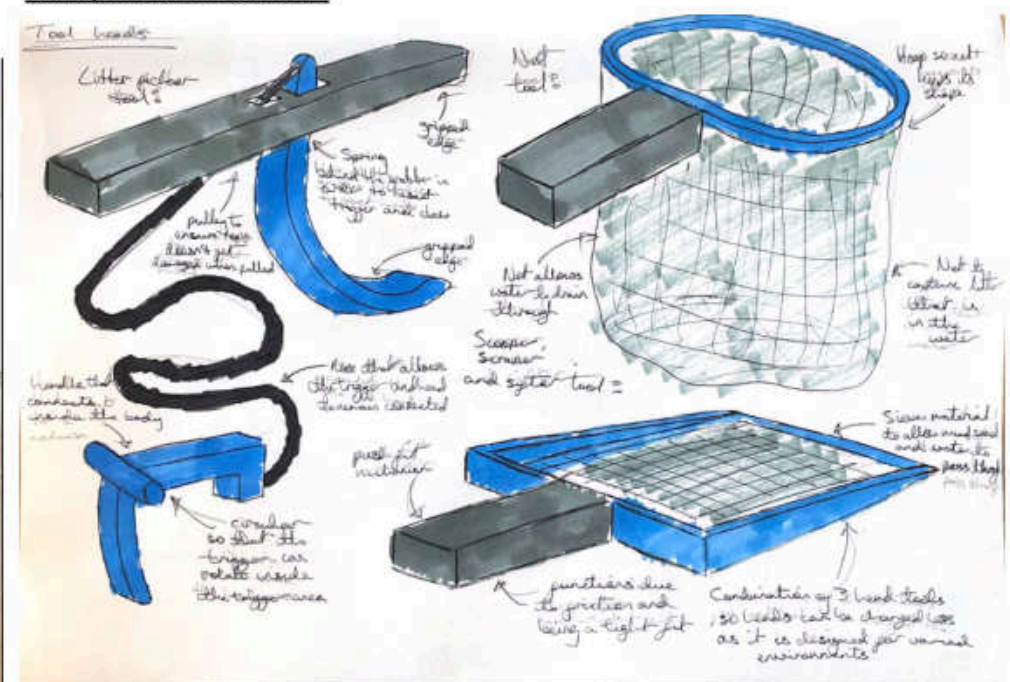


When developing my product, I initially focused on the storage area for the litter picker head tool trigger, by allowing for the trigger to be removable, attachable or detachable, this is essential for the design as otherwise the connection that is required for the trigger may have to be done via the rope which is small, as it needs to be lightweight and would be an inconvenience to the user due to the small size. By making the trigger removable the whole design will stay together, tool head will stay with the trigger which will help remove confusion for the tool head. As well, the trigger would not be in the way of the user, causing less discomfort and would provide a simpler experience for the user.

This trigger connection relies on the rotation of the trigger once it has been pulled, the trigger can be inserted into the design at an angle and pushed inside into the storage area until it is fully inside the design. The trigger can be pulled and should return back to its original resting position inside the storage area due to the spring inside at the litter picker head tool, this would allow for repeatability of the tool head and cause minimum inconvenience for the user. The rope attached to the trigger has been connected to an extension downwards in order for the rope to be outside of the storage area to cause minimum wearing to the design.

The hooks designed will be placed on the underside of the design, and they function by the rope of the litter picker head tool being inserted inside the design. Most commonly, user's will be holding the design with their strongest hand being their right hand, making the gap of the hooks facing upwards when the tool has been turned to the right will allow for an easy and simple insertion of the rope for most users. The hooks rely on the pressure and bending of the rope when inserted, causing the rope to deform in size and will return to its size once inside the design, and is unable to escape due to no force acting on the rope causing it to deform.

Development of the tool heads:



When first developing the tool heads, I was initially intending to do 6 heads so that all types of litter in all conditions would be able to be captured:

- **Litter picker tool head** – Able to use in most environments and conditions, the main tool for the design as it is appropriate in many conditions and circumstances
- **Net tool head** – Able to retrieve litter that is inside a watered system that would have been difficult to remove with another tool due to the litter being wet and the tool unable to get grip
- **Sifter tool head** – Tool that allows water, mud and sand to pass through so that only litter is removed in an environment and not the environment itself
- **Scraper tool head** – Able to scrape litter that is stuck to the ground and collect it, litter that could not be captured or would be more difficult to remove with other tools
- **Roller head tool** – Rolls over litter and picks it up due to the pressure exerted on the wires causing them to separate, allowing for the litter to be captured
- **Scooper head tool** – Able to scoop up large amounts of litter in order to make the process of removing litter easier, scoop up litter in one go instead of taking time to remove large amounts

However, the scraper, sifter and scooper head tool can be combined to make one head which can incorporate all these design ideas. This would make the process of attaching and choosing the different tool heads for a certain environment easier as the tool is able to be used for a variety of situations, allow water, mud and sand to pass through, whilst collecting large amounts of litter and being able to scrape litter which is stuck to the ground off the ground. By having a tool which can do more functions, it is more effective in its use, the user wouldn't have to think as much about what type of tool head is needed as the tool head is more effective as it can do more functions. As well, as there are less tool heads the transportation of the tool heads will be easier, this is because the user would have less tool heads to carry, reducing the weight that the user is carrying and amount of tools that the user needs to carry, these tools could be carried in a backpack, or the user could decide which ones are needed based on the environment that the user will be removing litter in.

Modelling #1

From producing various sketches on my chosen product, which helped give me ideas and solutions to make the product easier to use and more effective; to overall create a better product and an overall better experience for the user, I was able to produce a model based on these designs. This model consisted of the main body of the tool and three heads which could be attached to the body of the design, these heads being a net, a scraper/sifter/scooper and a litter picker. When producing my first model of my design, my main priority was to test if the design was able to function effectively or even function at all to what my sketches and research had planned for.

Body of the design



When modelling the body of the tool, my main priority for this model was to ensure that it was able to function. This function was to attach to all the heads, allow for the rope from the litter picker head to remain secure and to allow for the trigger of the litter picker head design to function when placed in the body of the tool. This model hasn't considered ergonomic features or anthropometric data and is purely to test if the design is able to function.

With this model it allowed me to assess the design of my product and visualise and test if it functions:

Advantages:

- The trigger of the design can function, with it being able to be placed inside the design and be removed, whilst remaining secure inside and not falling out, the trigger was able to move when pulled due to the circular turning point on the trigger allowing for the litter picker head to function
- The hooks placed on the underside of the body kept the rope from the litter picker tool to remain contained, ensuring a reduced chance that the rope would become damaged, additionally the shape of the hook allowed for the rope to be placed in and be removed easily

Disadvantages:

- When the trigger was placed inside the design, the rope end of the design would drop down past the storage area of the design due to there being no force stopping the design from turning, which as a result decreased the range that the trigger turned, which made the litter picker hook more shut, decreasing the range and size of litter that it can capture
- When attaching the tool heads, there was not maximum distance that it could be attached into, which would mean the tool heads could have kept on being pushed in to the design

When developing the body of the design further, I will take into consideration these disadvantages to ensure that the user of the design has an easy and effective use of the product.

Litter picker head

The litter picker head is the most complex part of the design, allowing for it to function and be removable will allow for more comfort for the user which is a necessity and ease to remove litter in various conditions due to a litter picker being more appropriate for capturing litter in a wide range of environments compared to the other heads which in theory can still remove litter however are more focused on specific areas of removal, and not as much wide of a range as a litter picker. This makes it a necessity to have the litter picker design embedded throughout the whole design of my product.

For this model, I have just incorporated the trigger design along with a functional hook to pick up litter, as a base for my modelling of this specific design.

Advantages:

- The rope that was connected to the litter picker head was able to slot into the hooks on the underside of the body, allowing for the rope to remain contained when the trigger was pulled which would pull the rope leaving some hanging, the hooks then also helped keep the trigger vertically in line with its storage area which would help with the handles return
- With the trigger being able to be placed inside the design, it was able to function which allowed for the hook to shut close, allowing for pressure to be placed onto pieces of litter and capturing them to be disposed of
- The push fit mechanism allowed for the head to be easily applied and removed and allowed for a secure hold of the head, keeping the head and allowing it to remain functioning

Disadvantages:

- Due to there not being a spring or any elastic tension behind the litter picker hook, the hook wasn't able to return back to its open state, this also left the trigger being left hanging out the storage area, requiring for me to push the rope end of the trigger back into the storage area, this means that the user would currently have to push the trigger back inside the design after every pull of the trigger, which is not effective
- The end of the trigger would become angled once the trigger had been pulled, making it difficult to push back inside the storage area of the design
- The push fit mechanism did not have a limit on how far the head and body could be pushed into each other as the head is smaller than the body of the design width wise, so the user would have to decide for themselves on how far to push the design, which could result in damage to the body and head

When developing the litter picker head further, I will take into consideration the disadvantages that come with this tool head and ensure that the user doesn't face these problems when using it to allow for an easy and effective use when removing litter.



A net head tool

Similar to the scraper/sifter/scooper head tool design, this net design is relatively simple in comparison to the design of the litter picker head tool. This net design would allow users to obtain litter from a watered source such as a river or pond easily and dispose off the litter, this would result in only litter being picked up and the water passing through the design, which would not be an inconvenience for the user as they do not have to worry about carrying the weight of the water. For this model, I focused on a simple design that would allow for the tool to take a net shape and resemble a net. The hoop of the design would be solid hoop so that it can retain its size.

Advantages:

- Litter would be easily captured by the user due to the net allowing the litter to get caught inside
- Water would be able to pass through the design due to it being a net which has holes in, this allows the water to pass through almost instantaneously, so when design is taken out of the water it won't take a long period of time due to a lack of holes to leave from
- Net allows for a somewhat collapsable design as the net can be easily moved due to it being made of rope
- The solid structure of the hoop would allow for the design to maximise the quantity of litter captured every time the tool is used which would then minimise effort required to obtain large amounts of litter

Disadvantage:

- Due to the hoop being a certain size it may not be able to pick up large amount of litter and may have trouble doing so
- Some litter may get caught in the net of the design, which would be inconvenient for the user as they would have to physically remove the litter from the net, which would most likely be dirty due to it being placed in water and capturing litter
- Some small pieces of litter may pass through the design, as the net would have gaps in the design which are too large to capture the litter, which may result in the user having to leave the litter which they are unable to capture
- Due to the tool head being in the water, it will get wet, so may be difficult or inconvenient for the user to carry if they want to change tool heads, such as if they carry the tool heads in a rucksack, the rucksack then may get wet if the net head tool is swapped due to the user requiring another head tool for a different use

When developing my design further, I will take into account these disadvantages to produce an improved product and one that is able to meet all the demands of the user to provide a good experience.

A scraper/sifter/scooper head tool

When developing my designs I had three designs which were all somewhat similar as the theory for the design is that it can either scrape litter which is stuck to the ground, be a sieve when picking up objects in environments with water and sand so that only the litter is picked up and the sand and water are able to pass through easily and a scooper which is able to scoop and shovel large amounts of litter, allowing for a large amount of litter captured in one movement from the user. With this design I have attempted to combine them, allowing for less head tools needed required overall, meaning more simplicity for the user as they will have to think less on what tools are required for their use, and tool heads can be used for multiple types of environments and scenarios. For this model, I focused on the scraping and scooping part of the design, and tested if this design could function and what could be done to make improvements

Advantages:

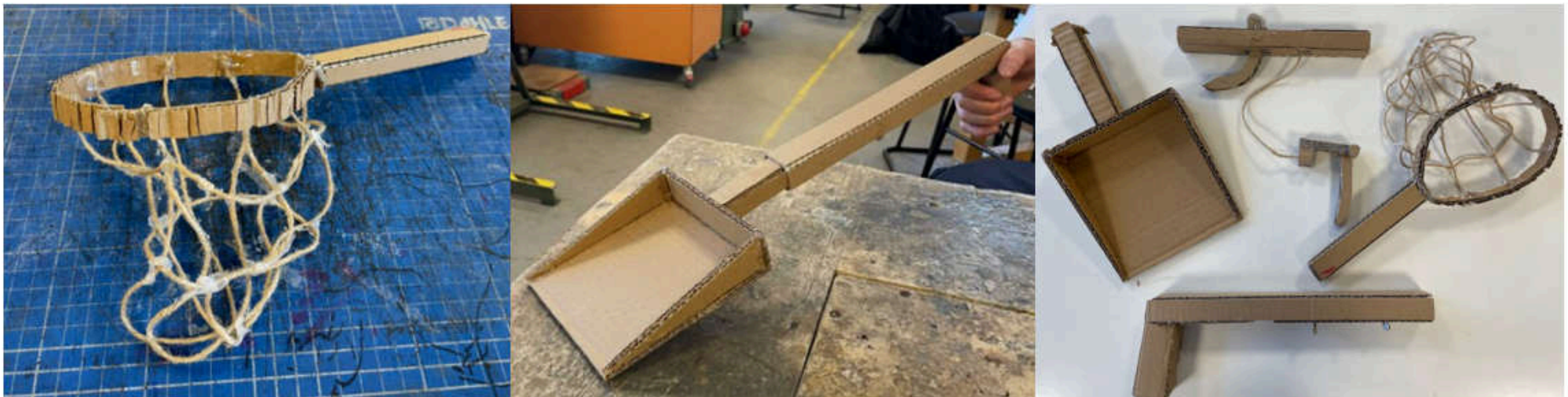
- Design would be able to scrape litter stuck to the ground off easily due to thin end which can get underneath the litter
- The design would be able to store litter due to the walls of the head which keep litter inside stored
- Litter can be easily removed by turning the scooper upside down which as a result would mean that the litter would fall out of the scooper due to gravity and can be disposed off
- A simple design which users would feel comfortable using due to it being similar to a shovel which users would have probably used before

Disadvantages:

- Design was not angled to accommodate the user, so would be extremely difficult to get under any litter to capture it, user would have to squat down or get low for them to be able to do it, which is straining and inconvenient

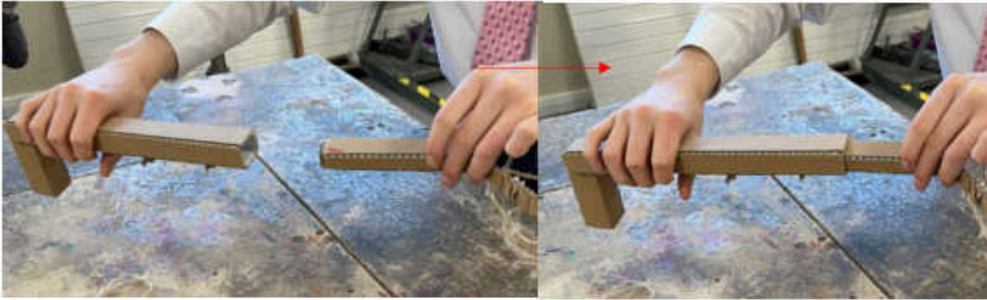
When developing this tool head further, I will take the disadvantages into account so that my product produced will not face these problems.

With all the designs produced with this model I was able to see the advantages and disadvantages with my designs, from these advantages I will be able to keep the aspects of the design that benefit the user and allow for an easier and effective use of the tool. From the disadvantages of the designs, I will be able to see the areas of my design that I need to improve to ensure that the design would not only function but function well and provide an experience for the user that is simple and effective at removing litter.



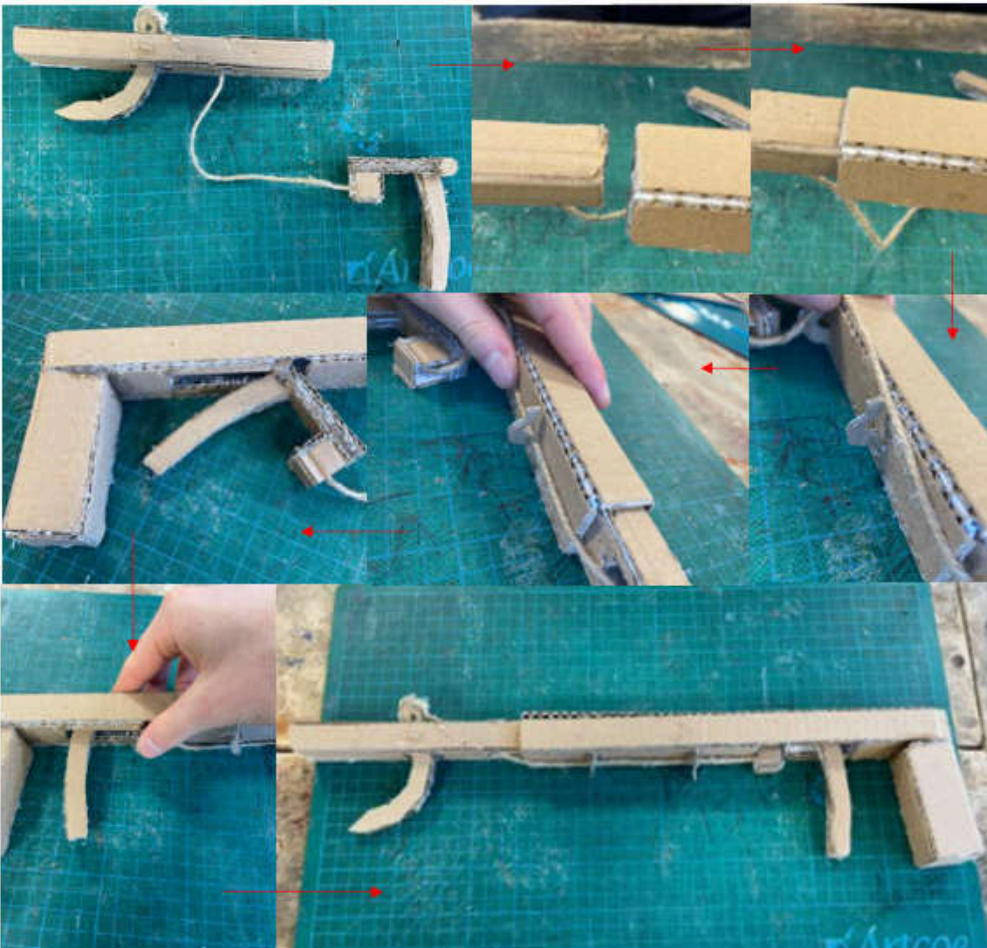
Modelling #1

How the tool heads are attached to the body:



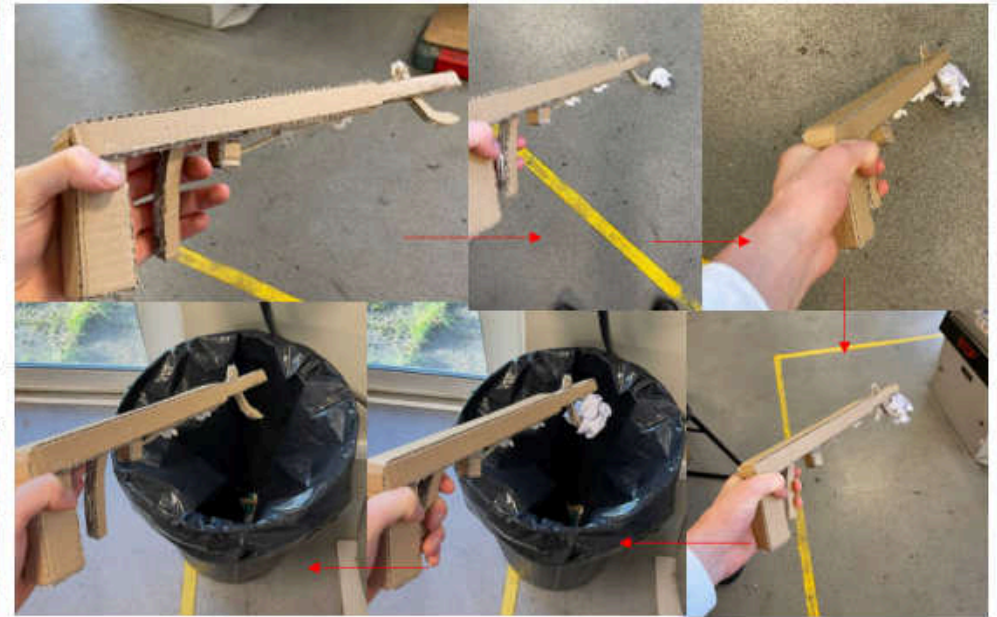
Design is a push fit mechanism, meaning that the tool head can be pushed inside the body of the design and will stay locked in due to friction. The problem found was that the design didn't have a limit of how far the tool head could be pushed into the body so the user wouldn't know when to stop.

How the litter picker head tool is attached to the body:



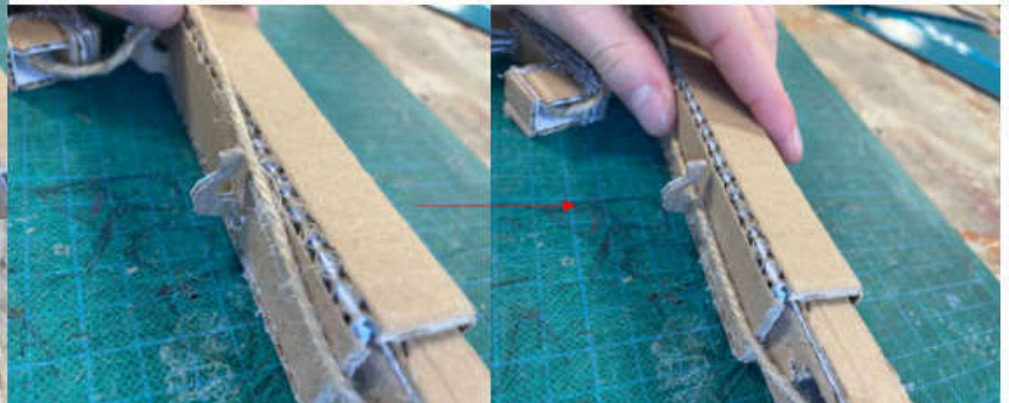
Design of the litter picker head tool can be attached by the push fit mechanism, passing the rope through the hooks to keep the rope contained, and then inserting the trigger into the storage area.

How the litter picker is used:



To use the litter picker the hook should be placed over the piece of litter, and then the trigger should be squeezed, the user can then hold the tool over the disposal location and stop squeezing, for the litter to drop due to gravity as there is no force holding it.

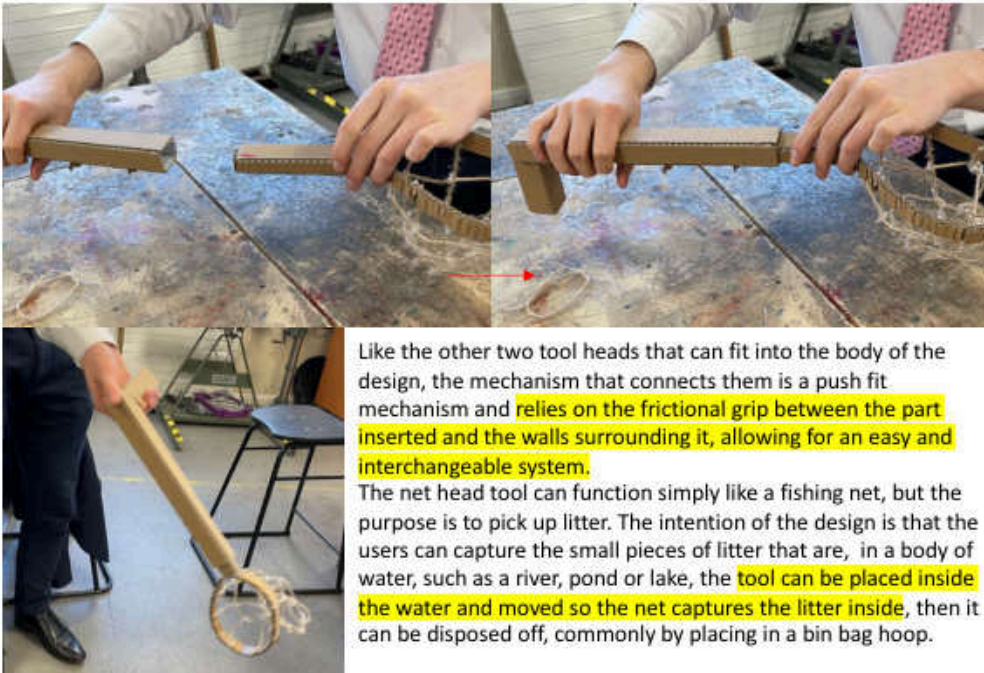
How the rope from the litter picker head tool is attached:



The rope from the litter picker head tool can be attached by pushing the rope down into the hook so that it is inside and against the body of the design. It is now unable to leave and is secure due to it requiring a pushing force to be slightly deformed and leave the hook. When pushed the rope slightly deforms and allows for the rope to be inserted or removed.

Modelling #1

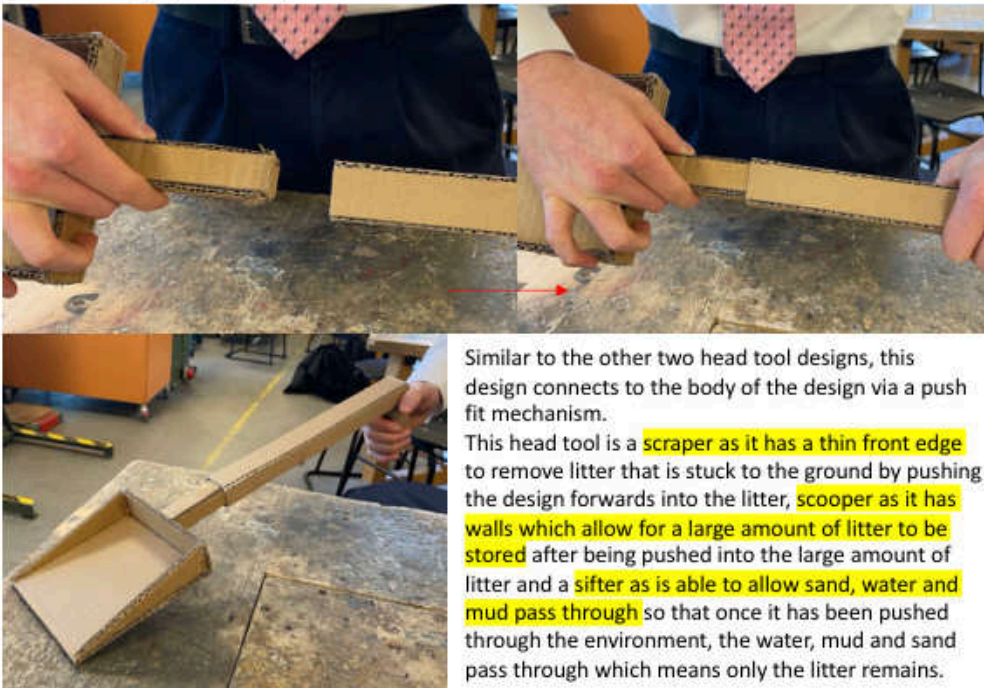
How the net head tool is attached and used:



Like the other two tool heads that can fit into the body of the design, the mechanism that connects them is a push fit mechanism and **relies on the frictional grip between the part inserted and the walls surrounding it, allowing for an easy and interchangeable system.**

The net head tool can function simply like a fishing net, but the purpose is to pick up litter. The intention of the design is that the users can capture the small pieces of litter that are, in a body of water, such as a river, pond or lake, the **tool can be placed inside the water and moved so the net captures the litter inside**, then it can be disposed off, commonly by placing in a bin bag hoop.

How the scraper/sifter/scooper head tool is attached and used:



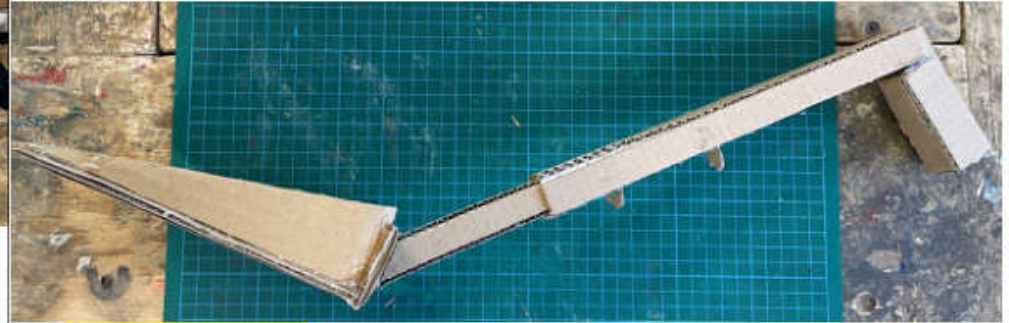
Similar to the other two head tool designs, this design connects to the body of the design via a push fit mechanism.

This head tool is a **scraper as it has a thin front edge to remove litter that is stuck to the ground by pushing the design forwards into the litter, scooper as it has walls which allow for a large amount of litter to be stored after being pushed into the large amount of litter and a sifter as is able to allow sand, water and mud pass through** so that once it has been pushed through the environment, the water, mud and sand pass through which means only the litter remains.

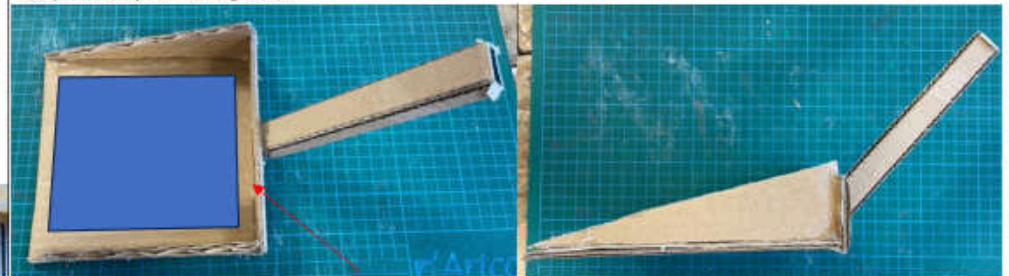
Redesign of scraper/sifter/scooper tool

The **current issue with the design of the scraper head tool is the actual head of the tool hasn't been angled** to the user when they hold the design. It currently remains flat and this would result in the user having to kneel to get a flat surface of the head so that the litter can be scraped, picked up and removed from the environment.

New design:

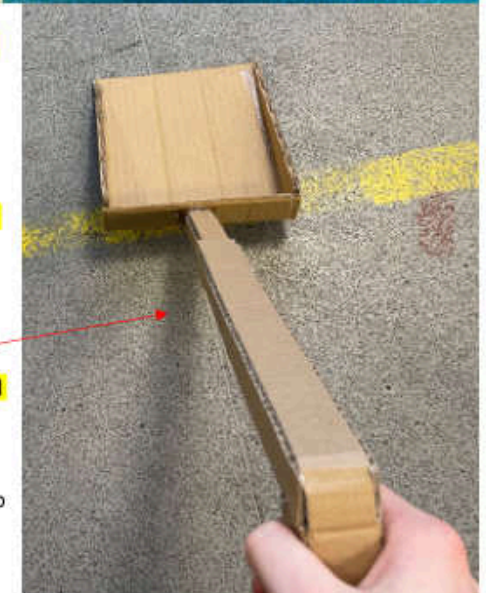


This **new design is now angled**, this is done by connecting the head of the head tool at an angle so that it will now be flat for the user when they are using the tool, allowing for them to **easily scrape litter that is stuck to the floor**, scoop it up and for any water, mud or sand to drain through so that they are only removing litter.



The area highlighted here is where I intend to place the sift so that water, sand and mud are able to pass through, this allows for the user to only capture the litter and dispose of only the litter.

As the head tool is now angled it allows for the user to now easily scrape and scoop up the litter for it to be removed from the environment. It is **now able to actually function and be effective**, whereas before the user would have to kneel down and scrape and scoop so that the design would be able to function. This would not only be very inconvenient for the user but would also cause unneeded stress and strain onto the user. **This redesign allows for the tool to be used at the users standing height**, allowing for it to be a more comfortable and provide an easier experience for the user. Additionally, the design allows for a simple use, where the user simply has to push the design into the litter so that it can collect the litter and scrape it off the floor if it is stuck.



Modelling #2

For this model, I focused on the litter picker head and the body of the design as the current design would not function whereas the design for the other two heads would now function, my main intention was to develop and improve the litter picker head to remove the problems that occurred with it, these being that the claw of the head was unable to return to its original position or even maintain it as there was no spring which meant no force to pull the claw back to its full open position. This problem then also affected the trigger which meant that it was unable to return back to its original resting position inside its storage area, this would then affect the next pull of the trigger, as the trigger would not be in its resting position so then the user would pull the trigger but would not receive a full range of motion of the claw, and may need to be pushed back into its position which would be an inconvenience for the user. As well, another problem was that when the trigger was pulled it would not return back to its resting position as the trigger wasn't aligned with the opening for it, this meant that the trigger would be at an angle, meaning a full range of the trigger wouldn't be achieved and as the trigger isn't in the correct position it could result in parts of the product becoming damaged.

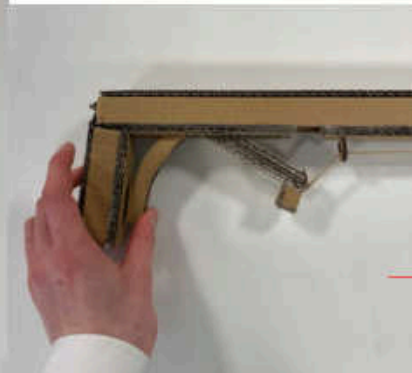


When first producing this model, I attached some key features to the design intending to meet their intention to solve a current problem:

- Such as a limit for distance that the push fit mechanism would be able to be pushed into the body of the design, this limit would allow for the user to know when they have pushed the tool head far enough into the body so that it is properly connected and will not fall off or detach when the user is using the tool
- Making the design the size that a litter picker would roughly be, this would allow me to test the features of the design better and allow me to produce a higher quality product
- Adding a hook just outside the area of the trigger storage area, this would help align the trigger so that after it was pulled it would be able to return back to its original position inside the trigger storage area and not be at an angle
- Making the storage area a tighter fit so that it would help align the trigger in order to keep the trigger straight and not at an angle
- Increasing the length of the top part of the trigger that the rope is connected to, as the length would be larger, when the trigger is pulled the distance travelled would be greater, allowing for the user to achieve a full turn of the litter picker claw with a smaller force applied to the trigger
- Adding an elastic material to the claw of the litter picker, this allowed for the litter picker claw to return back to its original position after being pulled by the trigger, which would allow for the litter picker claw to remain open and allow for an easy collection of litter as the claw can be fully around the litter

Positives about this model:

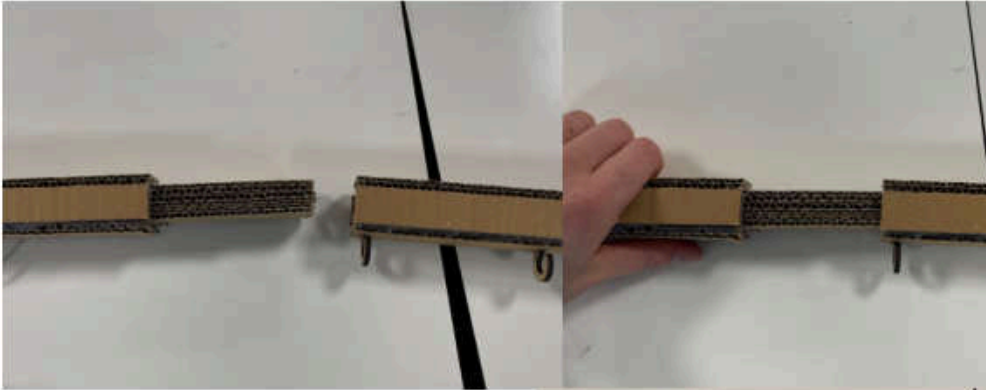
- The push fit mechanism allowed for a strong and secure grip between the tool head and the tool body, ensuring that it would remain in contact with the tool
- The elastic material allowed for the claw to return back to its original opening position as there was the elastic force pulling the claw back, ensuring that the head of the tool could function
- Hook near the opening of the tool allowed for the trigger to remain straight so it wouldn't be angled, and be closer to its resting position inside its storage area



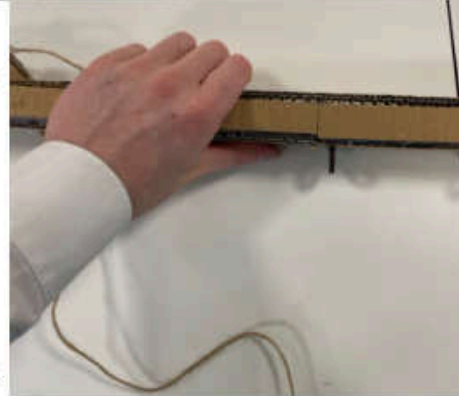
Problems that occurred with this model:

- Although the hook near the end of the storage area helped keep the trigger straight and not angled, and helped bring the trigger closer to its resting position, the trigger still not be in its resting position and would be slightly angled downwards. This would cause the trigger to slip out of its position which would decrease the tension in the rope and would therefore make the claw move less if the trigger is pulled. This would also make the trigger less effective and would be an inconvenience as the user may have to consistently push the trigger back inside the storage area of the design
- Although the push fit mechanism was secure, it would slightly slip, requiring the user to push the head tool back inside the body of the design

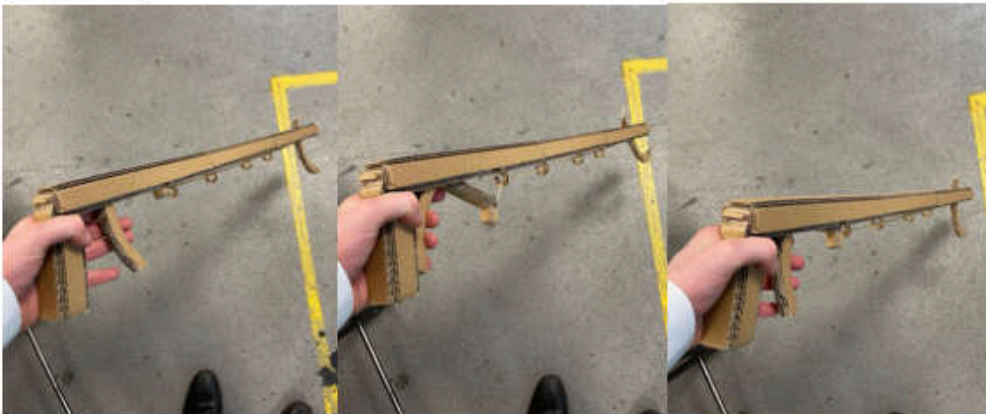
Push fit mechanism



With this push fit mechanism, the user will now know when they have connected the mechanism fully as the push fit would be fully inside the body of the tool. When the design is fully inside the fit can not be seen. The user would simply have to insert the design to connect the head and body, and to remove would have to pull the head apart from the body. The problem with this design is that the mechanism can sometimes become loose which requires the user to push the tool head back in, which can be inconvenient. The push fit as well cannot be too long as the tool heads will need to remain portable so the user can change whilst using.



Using the tool



1. Place the claw over the piece of litter that you want to pick up

2. Squeeze the trigger so that the claw captures the litter and continue squeezing the trigger

3. Place the litter over the area where you will dispose of, and stop squeezing

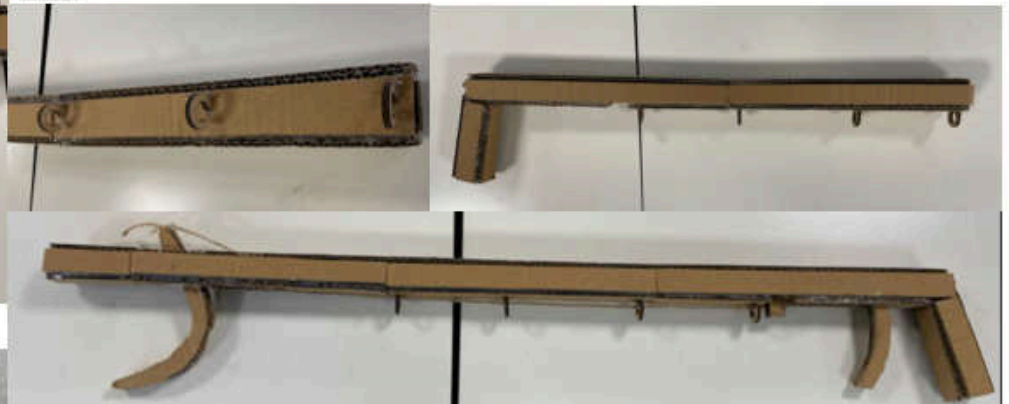
The problem with this design is that the trigger is angled downwards after it is returned after the trigger has been squeezed, this will reduce the distance that the claw will close and reduce the size of litter that can be picked up from the tool.

Trigger not returning to storage area position



The design does not return back to its original position inside the storage area, which combined with the trigger moving forwards can result in the tension in the string to decrease, which would decrease the distance the claw can travel and therefore limit the size of litter that can be picked up.

Hooks



These hooks help contain the rope so that it remains close to the body of the design, as well the hook closest to the trigger acts as a point of tension so that the claw can achieve a greater distance.

Hook closest to the trigger

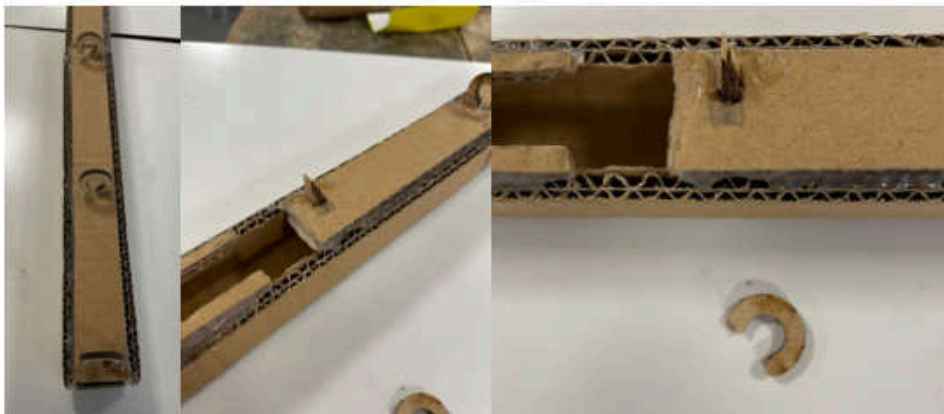


Due to there now being a hook which is close to the trigger of the design, the insert of the trigger is more difficult due to there being less of an angle and therefore a room for the trigger to enter the design of the tool, which if caught could cause damage to the design.

Before I would attempt to develop a new and improved design for the storage area of my litter picker, I attempted to resolve my problem of the trigger moving forwards and slipping out which reduced the tension in the string and resulted in decreased movement by the claw of the litter picker. To do this, I decreased the height of the storage area by adding another layer of cardboard, attempting to increase the tightness between the trigger and the walls of the storage area so that the trigger wouldn't move as much forwards and would be more secure.

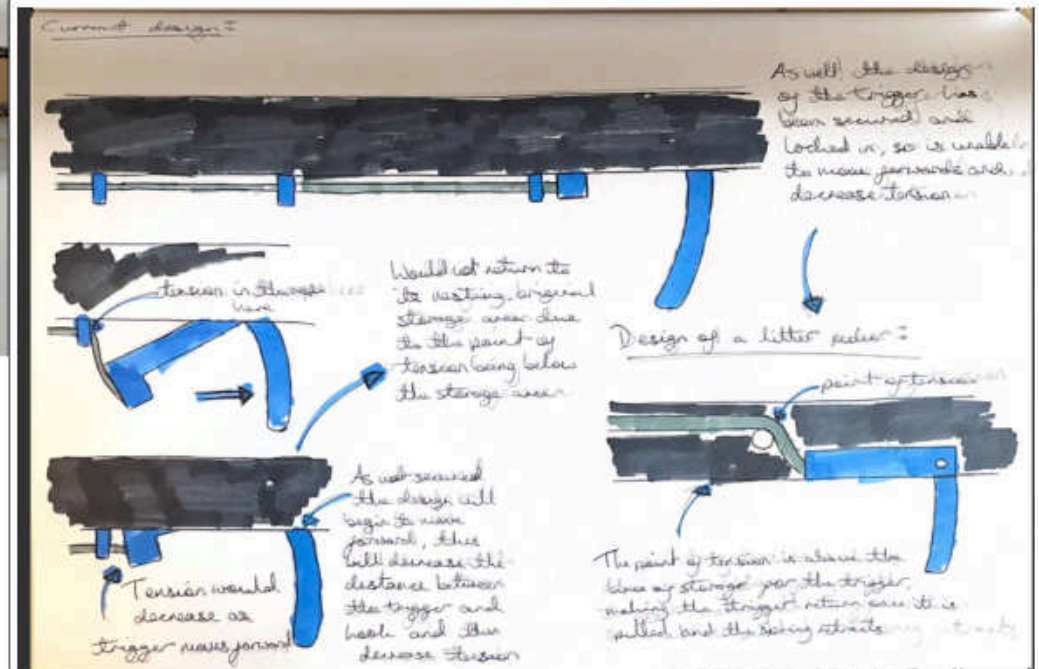


When adding this piece of cardboard to reduce the height of the storage area, it resulted in the trigger not moving forwards and keeping tension in the rope so that the claw would achieve a full range of motion. This allowed for a functional tool and the litter picker tool to function effectively, even if the trigger didn't return to its full height, it still allowed for the tool to function with only a slightly decreased range of motion with the trigger not going fully back up into the design. It would allow for the user to use the tool to remove litter from an environment as intended, however with the decreased storage area height came problems. One being that due to the turning point of the trigger being in tight contact with the storage area it would mean that the design would decrease quality as it is being rubbed and large amounts of friction would occur, and erode the design, meaning the user may have to get another trigger, which would be inconvenient for them. Another issue is due to the tightness of the design, it made the placing and removal of the trigger into or out of the storage area much more difficult, and when removing the trigger it resulted in a hook breaking, and as this hook is the point of tension for the design, and where the trigger can achieve a large amount of distance travelled, pulling the hook so that the claw would be able to close, it would make the design not work as effectively. Due to this, a new design is needed to remove these issues that occur to make a useful and functional product and one that will not cause damage.



Litter picker investigation

To improve my design to ensure it is functional, I re-analysed a litter picker and compared to my current design to see what I could change to make sure that the litter picker head could function effectively, was fully removable and that the trigger would return to its resting position to get a full range of motion. I still believe that keeping the design fully removeable would allow for a better, more simple and neater product as the tool would only have what is required for each tool head when the user is using it.



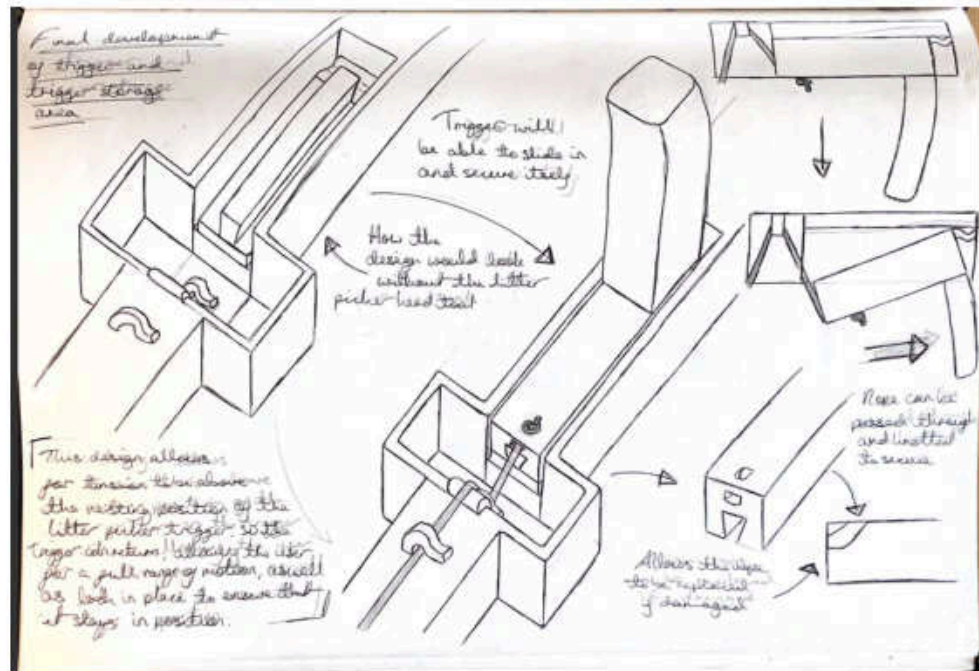
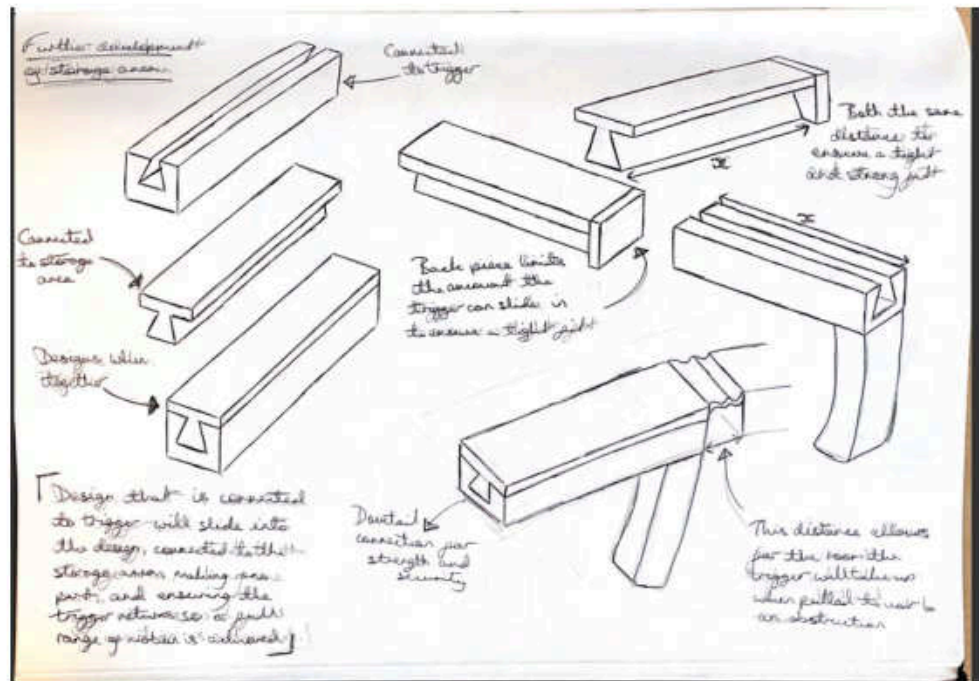
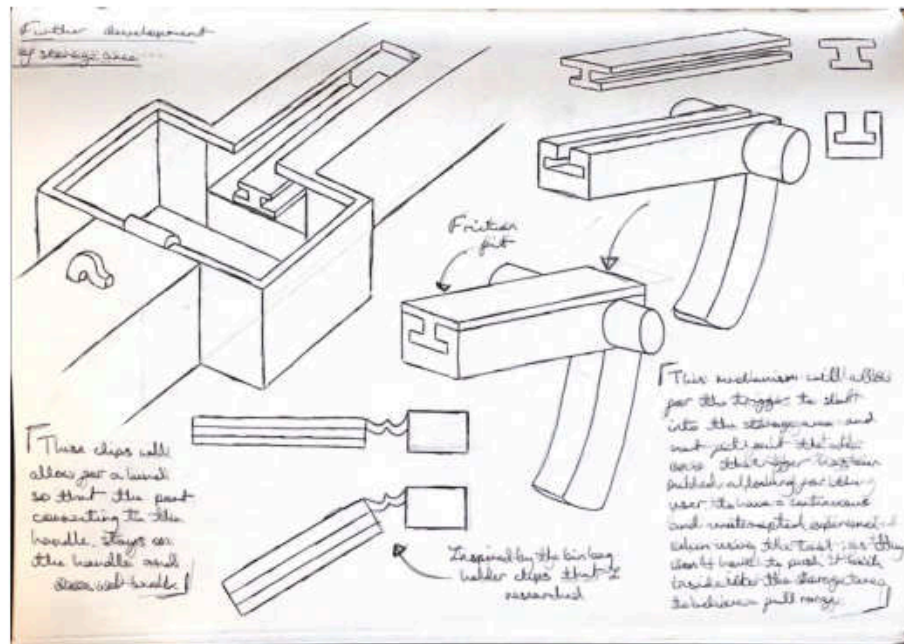
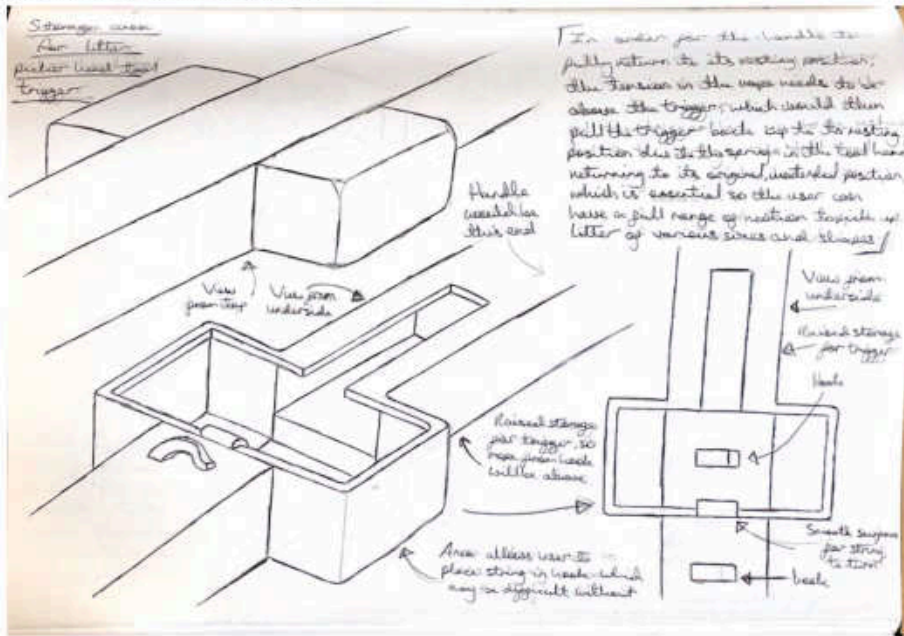
When comparing my design to the design of a litter picker, there are two differences:

- My design of the trigger has not been locked into position which would allow for the trigger to move forwards in the trigger storage area which would decrease the tension in the rope, whereas the litter picker trigger is locked into position so the tension in the rope would always be constant and the rope would remain at the same tightness. This would allow for consistency in results when the trigger is pulled.
- My design has the resting area of the trigger above the point of tension (which is at the hook closest to the storage area), this means that when the trigger has been pulled and then released, and the elastic material pulls the hook back, the trigger would also be pulled back but would not return fully back into its resting position as there is no force pushing the trigger back up into its resting position as the force would only exert it up to the resting hook, which is not in the storage area. In comparison, the tension in a litter picker is above its resting position, this means that when the elastic material pulls the claw back, the trigger would return to its resting position as the force of the spring pulling the rope would pull the trigger back up, and as the resting position is below the point of tension, the trigger would return back to its resting position. Which would allow for a full range of motion and to keep a strong tension in the rope.

In order to improve my design, I will have to add these designs from a litter picker into my design, which would allow for a functional product.

Development of litter picker trigger redesign

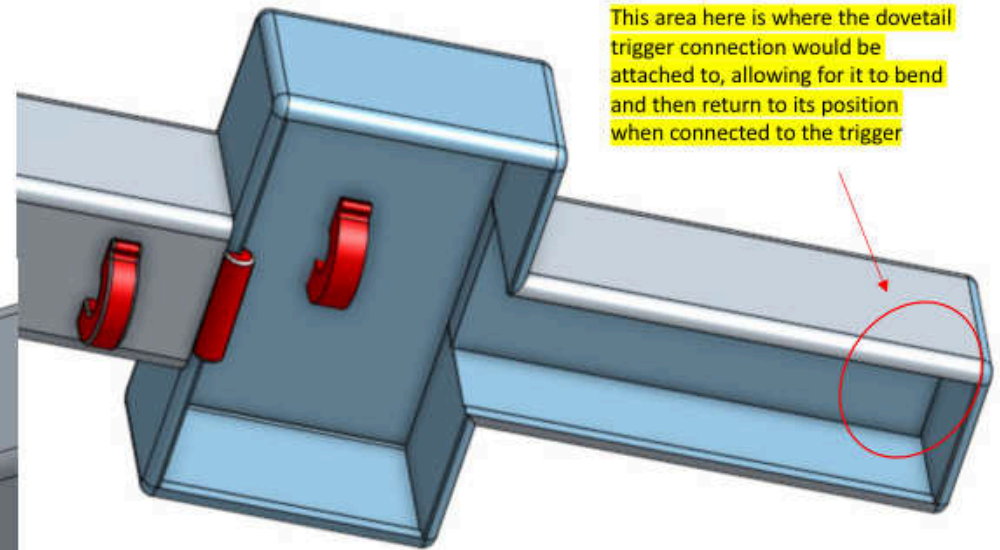
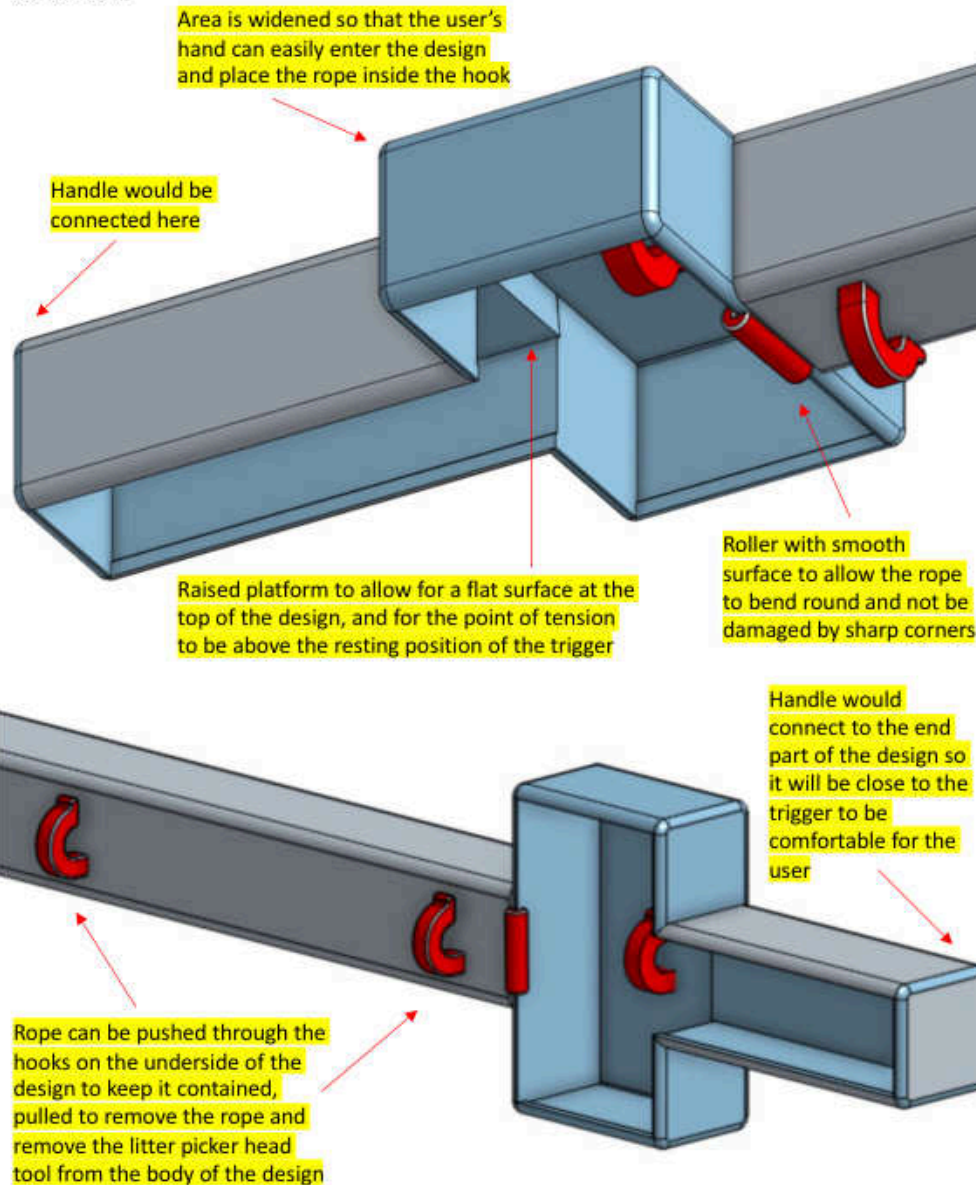
From the problems that I encountered during my modelling of my product, I will have to develop parts of the design so that they are not just functional but are effective. From the investigation of the litter picker that I analysed I will have to develop my product to ensure that these problems faced do not occur in my product.



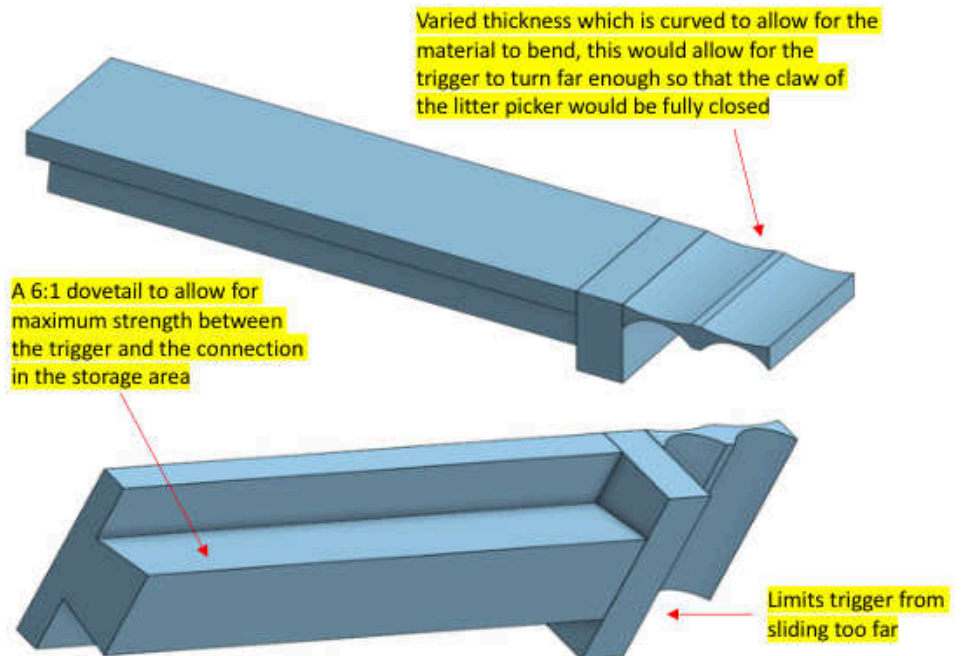
With the new design of the storage area, I will test it and develop it further if necessary to ensure that the product is not only functional but can perform effectively at its job to ensure an easy and simple experience for the user.

CAD of litter picker storage area redesign

For this model, I have removed the dovetail trigger connection mechanism, as this piece can be added on to the design and is easier to be modelled separately due to the moving and flexible connection to the body. This model shows the new design of the storage area to allow for the point of tension to be above the storage and resting area of the trigger for the litter picker head tool. I have changed the design of the body to allow for the litter picker head tool to now function effectively and well. The net head tool and sifter/scraper/scooper head tool are not affected by the redesign due to them not having to connect to the body other than the push fit mechanism connection.



Dovetail trigger connection:



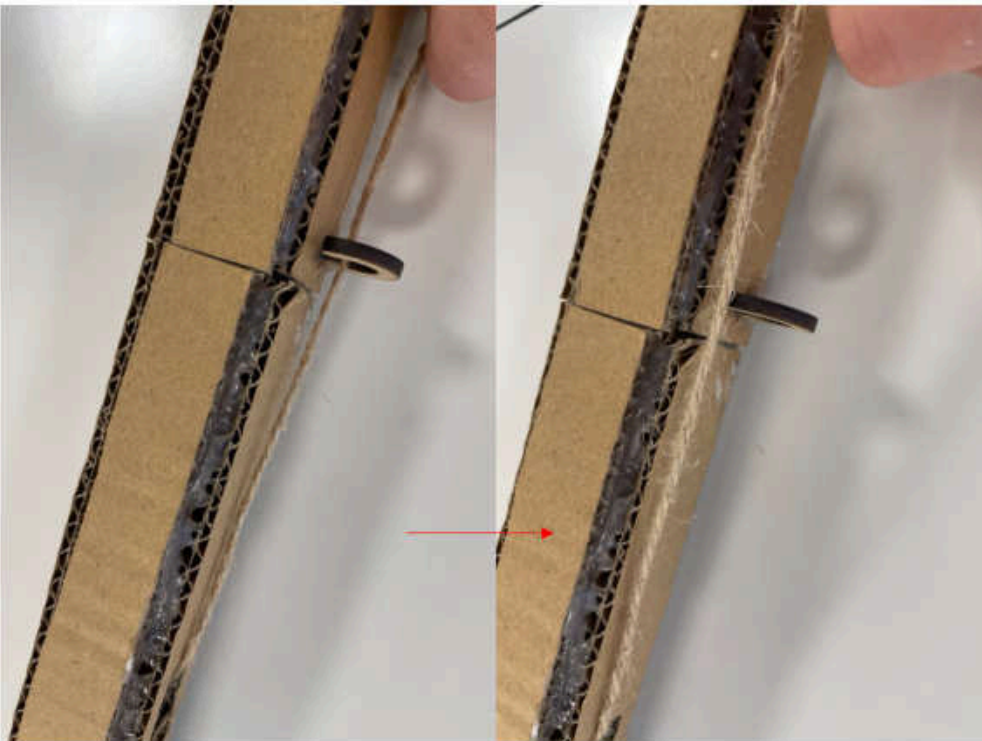
This design of the dovetail trigger connection would allow for the trigger to be pushed into the design of the connection piece in order for it to remain connected and be a tight fit so that the trigger feels like one solid piece and not two connected. The piece connected to the design which is varied in thickness would allow for the design to bend, supporting the turning of the trigger, as well the strength of the shape would allow for the connection to stay inside the storage area when it is not being used, keeping the part contained and not being an inconvenience for the user.

Hooks

A key feature for the litter picker head tool to function effectively are the hooks, these help keep the rope inside the design so that the rope is less likely to become damaged, help the set up of the litter picker head tool and creates a point of tension in the new design of the tool so that the trigger is able to return back to its original position and the claw can gain a full range of motion, the hook allows for the litter picker to function effectively and for the trigger to return back to its resting position, inside the storage area, which has been developed to be comfortable for the user.

The hooks relies on the material of the rope to bend and compress when the rope is pushed in or pulled out of the hook, This then allows for the rope to stay inside the design of the hook unless a force is applied onto the rope to either push it or pull it. Providing a secure aspect to the design that will keep the rope inside the hook when the user is using the litter picker tool. Due to there being numerous hooks, the hooks will remain on the underside of the design and be connected to it, this allows for the hooks to remain in position and reduce the set-up time for the litter picker head tool as the user will only have to thread the rope through these hooks and not have to set the hooks up.

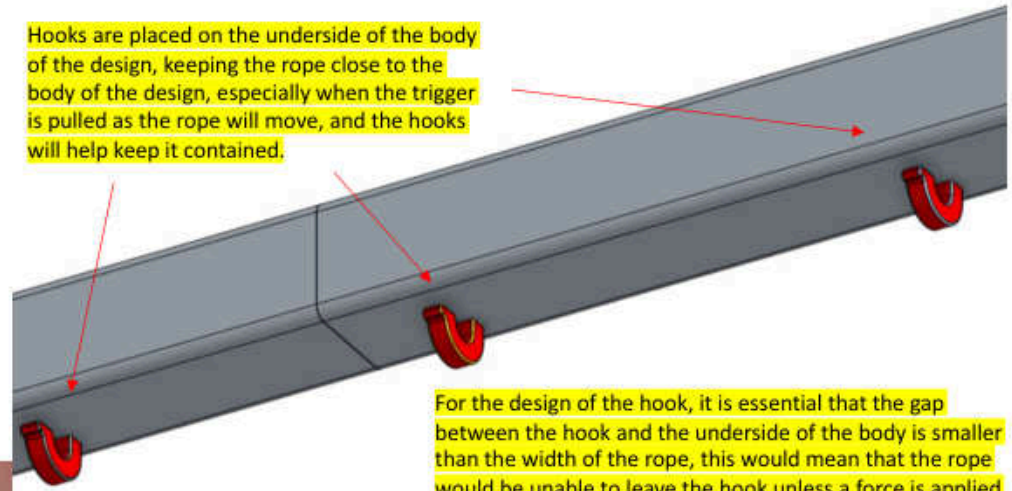
How the rope can enter and be removed from the hooks:



The two main aspects of the design is that the entry and exit of the rope to or from the hook requires for the rope material to bend or compress which allows for a secure connection, the other aspect is that when inside the hook the rope is not compressed or has forces on as this may then affect the pulling of the trigger and therefore the distance that the claw will move, which could affect the size of litter that can be captured. The hooks should just touch the surface of the design to ensure that the rope is secure, connected and able to move when the trigger is pulled, pulling the rope across the surface of the hook design.

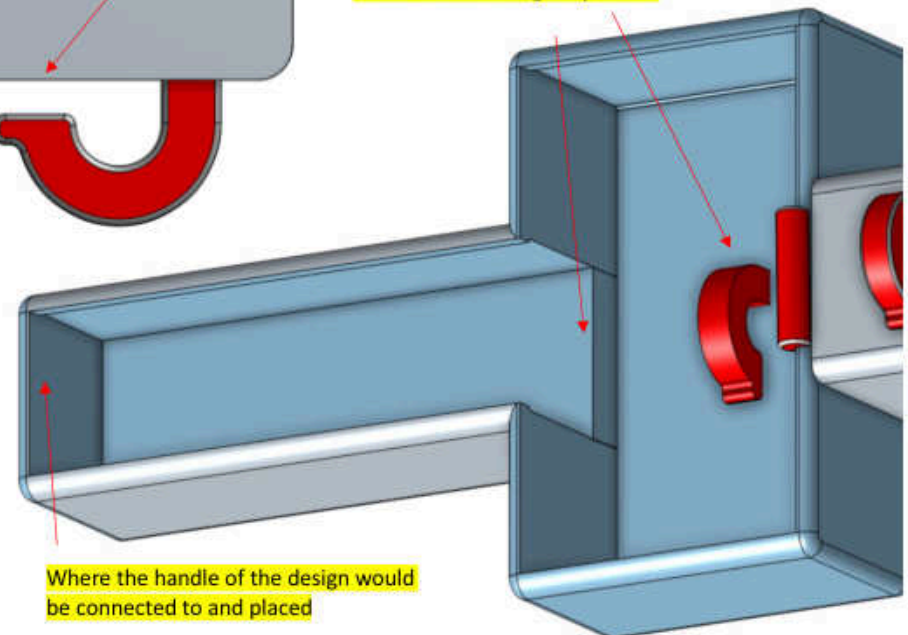
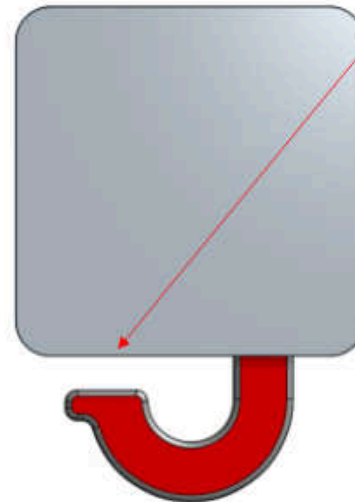
CAD of hooks:

Hooks are placed on the underside of the body of the design, keeping the rope close to the body of the design, especially when the trigger is pulled as the rope will move, and the hooks will help keep it contained.



For the design of the hook, it is essential that the gap between the hook and the underside of the body is smaller than the width of the rope, this would mean that the rope would be unable to leave the hook unless a force is applied from the user, this would help keep the rope contained. Gap needs to be smaller so that the rope must compress which it is unable to do if a force is not applied to it.

It is essential that the hook that is located closest to the trigger, in the widened area, has the tallest point inside the hook above the starting point of the trigger, this would allow for the trigger to return to its original position once pulled, and the spring returning the claw back to its original position



Where the handle of the design would be connected to and placed

Research of tool handles

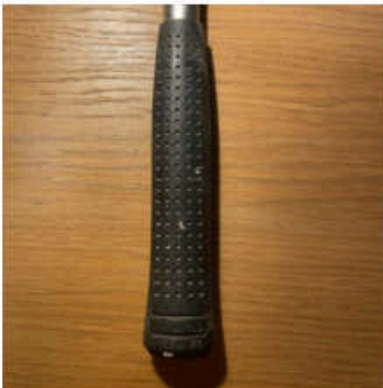
An important feature of my design is to make it ergonomic and comfortable for user so that when they are using the tool, it does not cause strain or discomfort on the user. Allowing for the user to use for the tool for an extended period without being uncomfortable. By analysing features that are in current designs of handles, I could incorporate these into my design and make it more comfortable and ergonomic for the user that they will be provided with a better experience.

Design of litter picker handle:



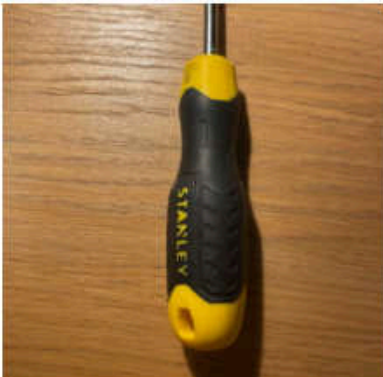
The design of this litter picker has many features that enable it to be ergonomic and comfortable for the user when holding. It is covered in a gelled grip enabling for the handle to be moulded and shaped to fit the user's hand providing comfort. There are slight indentations which have been made to accommodate the fingers and when they rest against the back of the handle by the user's hand. The handle has a large base at the bottom of the design, allowing for the hand to rest on this base, providing support to the user's hand. As well, there is a curved section at the top of the design allowing the base of the thumb to wrap around this part easily, allowing for further comfort when holding the design in this trigger grip.

Design of hammer handle:



This design of a handle for a hammer is simply shaped yet is able to meet the ergonomic needs that a user would have. The design is an extremely simple shape that has a small groove across the edges of the design in order for the base of the thumb to fit comfortably when holding the handle. As well there is a wider part of the handle near the top of the design that allows the palm of the user's hand to fit comfortably when holding the design. With the change in widths of the handle either side, the user's hand can comfortably hold the design with a large amount of comfort. Another feature is the dots that are on the sides of the design, these allow for an improved grip as the dots increase friction and therefore reduce the chances of the design slipping out of the user's hand.

Design of screwdriver handle:



This design of a screwdriver is also very simply shaped, with very few features and a shape that allows for the hand of the user to wrap around it is able to be comfortable for the user. With this design the width is largest at the bottom as it fits the palm of the user's hand around it, allowing for control of the screwdriver. As well this thinner section at the top of the design is so the users that can rest in this position, and fingers can be held on this part with a large control of the design. Additionally, there are gripped features of the design, this allows for friction between the user's hand and the tool so that they can gain grip of the design, meaning that the design is less likely to slip out of the users hand.

From analysing these handles, the biggest feature of the designs in order to make a comfortable design that stuck out to me was a design of a simple shape. A simple shape of the handles allowed for the hand of the user to wrap around the design, such as differences in width of the handle and grooves to accommodate for parts of the user's hand. When developing my product, these features are essential so that the handle design is comfortable, and the user has a good experience.

Ergonomics and anthropometrics

Removing litter can often be a time-consuming process so it is essential that the handle for my tool is ergonomic and comfortable for the user, which if done will allow for a better experience for the user and allow for them to be more effective with the removal process. In order for the handle to be ergonomic it will need to be shaped to an appropriate size. The size needs to fit the users in a 5th to 95th percentile, doing this would remove the need to accommodate outliers who have hands that are very large or small, allowing for the handle to function effectively for 90% of the population and being widely usable.

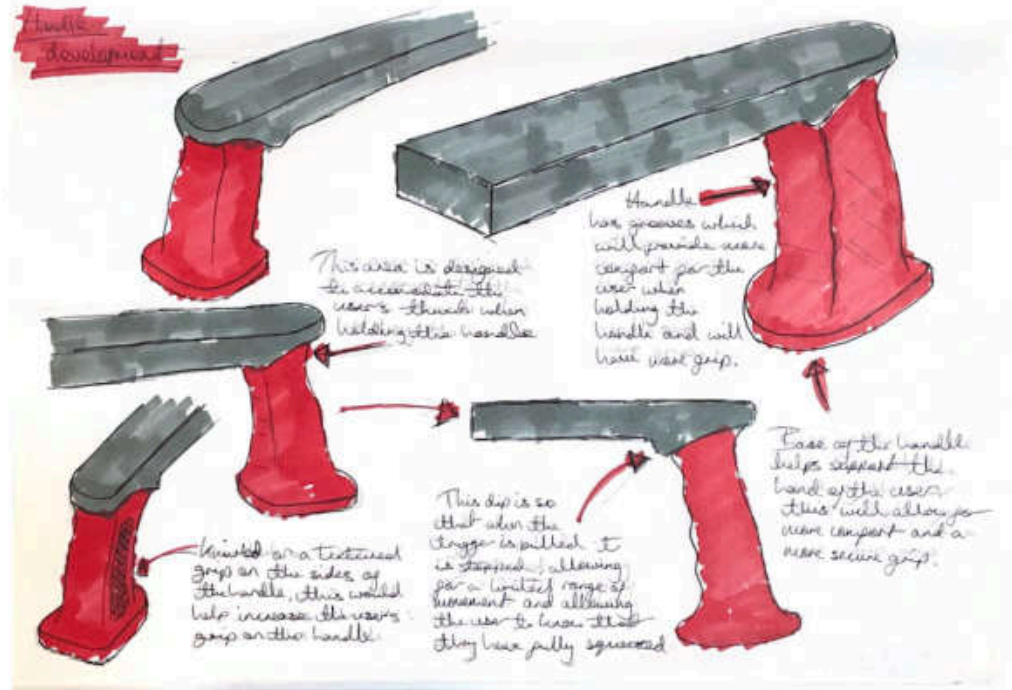
Data that I will use and consider when developing my handle to ensure that the size of the handle is appropriate to 90% of the population:

- Wrist centre of grip length
- Hand breadth
- Hand length
- Hand circumference
- Thumb breadth

The link for this data can be found at the back page (links for information), the bullet points are titles of each page's anthropometric data and I will take these into consideration when designing and developing the sizes and dimensions for the product

Handle development

By taking the features of the other handles that I investigated, I began designing the design for my handle with some key features that I may incorporate into my design in order to make it appropriate and comfortable for the user and reducing the strain when they are using it.



Handle development

Handle modelling

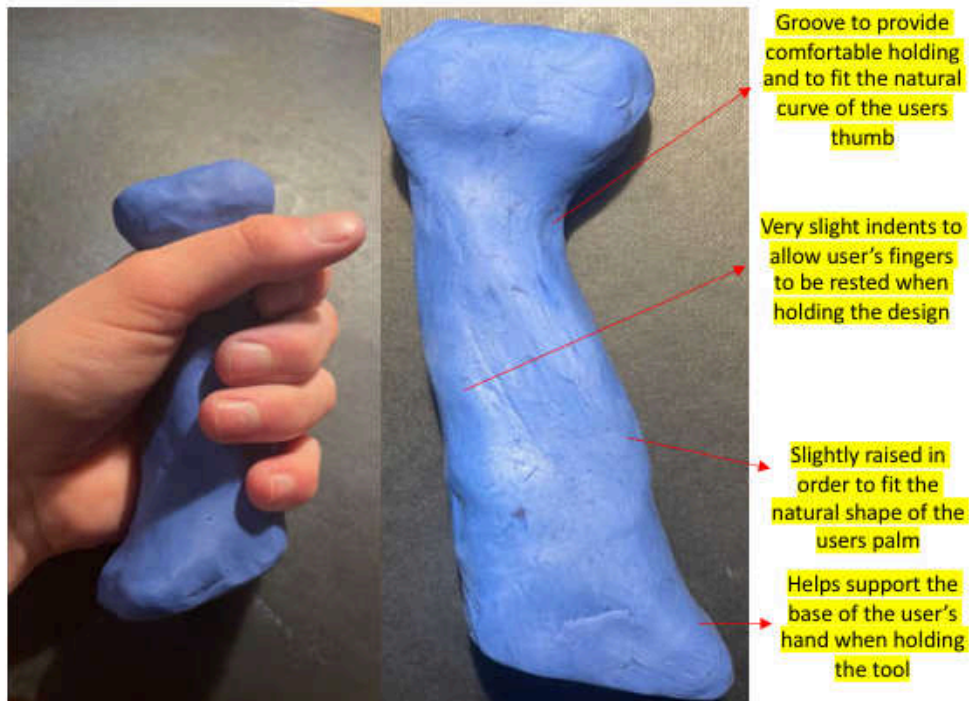
To make the handle of the design comfortable for the user, I developed designs and elements that I wanted to add to ensure it was comfortable fit for the user. When analysing the other handles from different tools it was necessary that I took some features into the design of my handle which were:

- A simple and comfortable design
- Able to fit the 5th to 95th percentile of users comfortably
- Be slightly angled to better fit the natural curve of the user's hand
- Provide support to the user
- Shape of the handle should allow for the thumb to grip it comfortably

In order to show and test these features, I produced a model of a handle to see what needed to be developed or changed.

Model #1

For this model I developed a design that would be able to fit the 5th to 95th percentile of users to allow for ease of use for a majority of users as the design would be ergonomic. As developed in my ergonomic and anthropometric data page, I took the anthropometric data which helped me consider the sizing's and dimensions required to allow and comfortable and ergonomic fit to produce a model which I developed and applied the wanted features to ensure a comfortable fit.



When producing this model of my ergonomic handle, I attempted to add the key features of my design that I wanted, such as those labelled. In order to make a comfortable handle, I ensured that there weren't too many small features and mostly only simple and needed features. It would ensure that the design can accommodate a large majority of users and not certain user's handle grip. This model allowed for a comfortable grip that would suit a users hand, however there was a lack of support for the user on the base of the tool, as well the model was not tall enough to support the larger end of users and too thin, which made the fingers overlap when holding the tool, which was uncomfortable.

Model #2

For this model, I tried to add more support for the user's hand when they are holding the handle, to do this I added a larger section at the bottom of the design as this would provide support to the user's fingers. As well, I increased the height and width to match the orthographic data and become more comfortable when the user is holding the handle.



User's hand would rest on the side of the design when holding the handle, this part will provide a supporting force to their hand, making their grip more comfortable

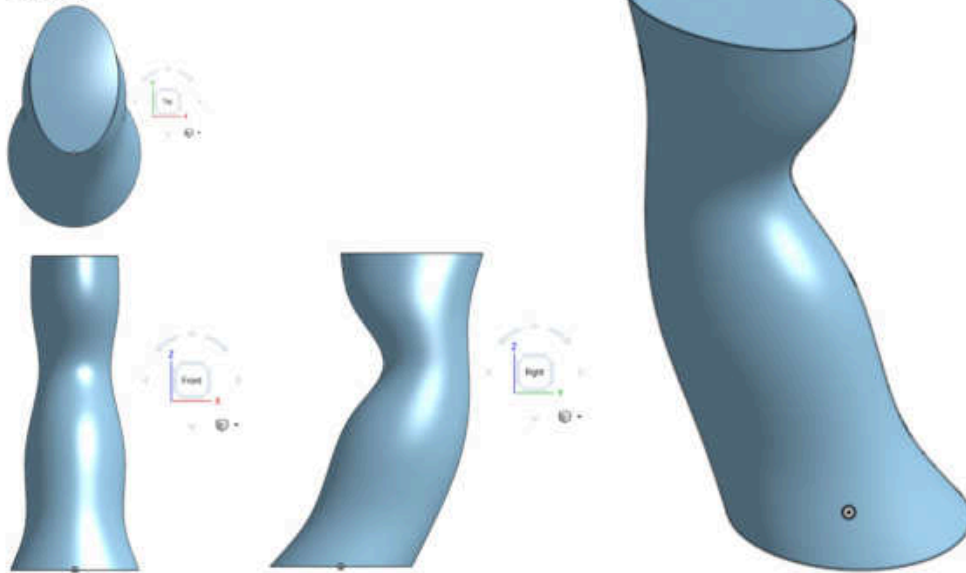
A larger base on the bottom of the design allows for the user's hand to be focused into the position of the handle as their hand is unable to leave the handle from moving down so will keep the hand locked into a position where it can use the tool effectively. Aswell, the larger base is extended all around the bottom of the design, this allows for the user's fingers to rest on the design when holding the handle, this will provide a larger support force on the user's hand, allowing for the grip of the user to be less which would overall allow for a more relaxed and comfortable experience for the user. There would be a decrease in strain on the user's hand which would allow for a more comfortable experience when holding the handle.

This handle shows the key features of the handle that I want for my design, as well as simple features which allow for the design to be comfortable for a wide range of users.



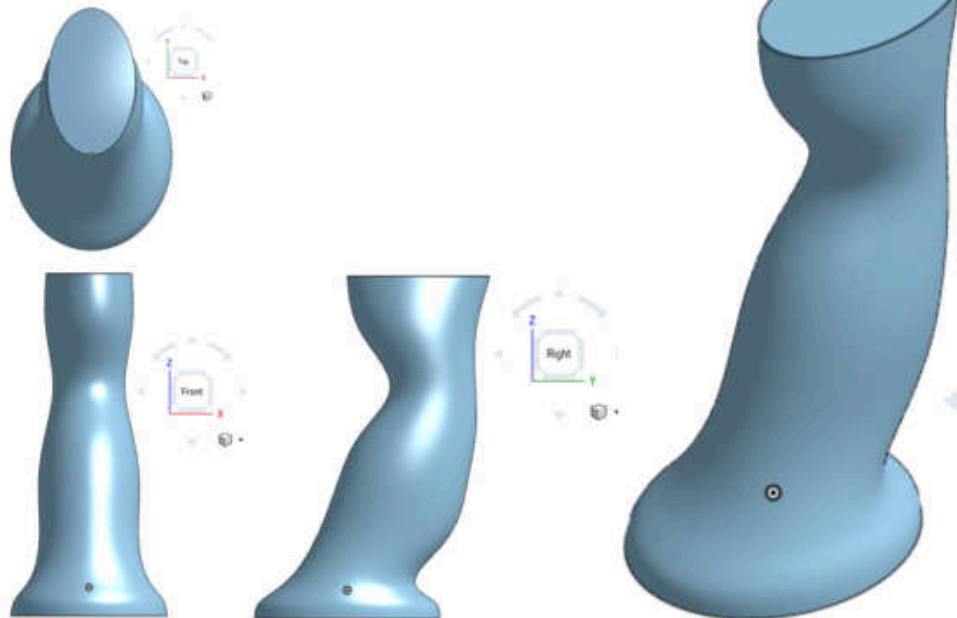
CAD of handle

CAD #1



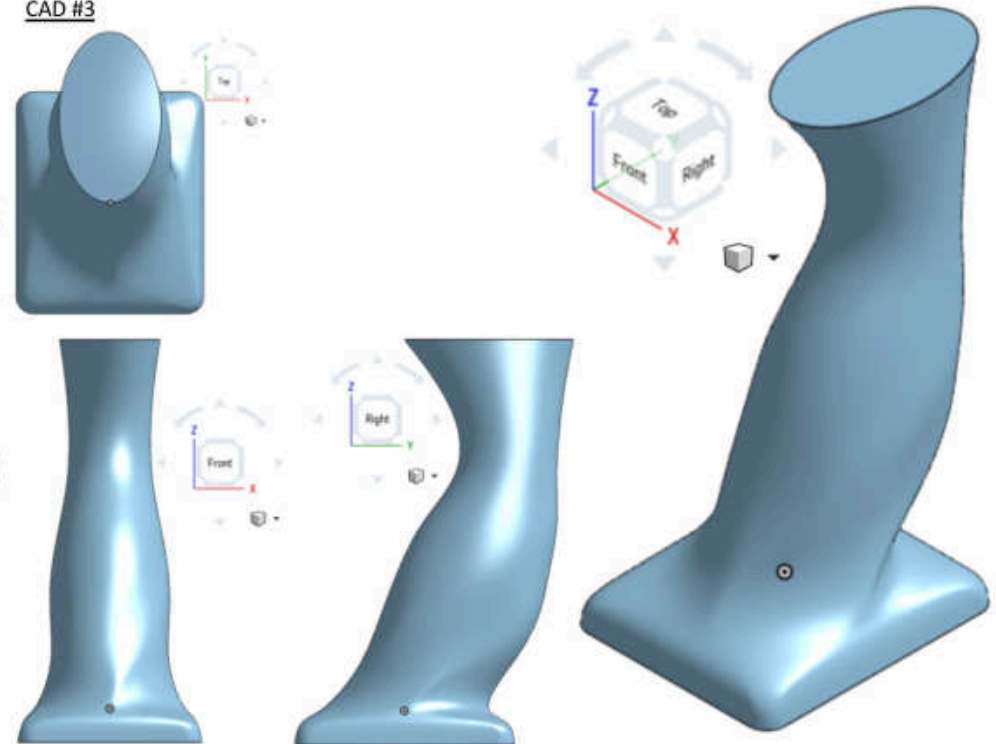
To transfer my modelling clay model onto CAD, I measured individual sections of the physical handle, by measuring the length and width at every key point turning point or shape that the handle had. This allowed me to transfer my key features accurately onto CAD so that the design would be correct. For this model, I focused on achieving the key features of the design, such as the groove for the thumb and the widened centre for the palm of the hand.

CAD #2



For this model, I attached a base at the bottom of the design from my first model. For this base, I made the shape of it an eclipse to see if a change in shape of the base may change the grip of the design. The bottom piece would be round and allow for the user's hand to be wrapped around the design and use this base as support for when they are holding the handle. Not only would this provide support for the user, but it would somewhat keep the hand of the user trapped between the top and bottom of the design so that they would not let go of the handle, meaning that the only way the user could remove their grip from the handle would be to remove the grip of their fingers from the back of the design, allowing for a better security when holding the design as their hand would be less likely to slip out of the handle. The issue that would happen with this is that the hands of the user would not be an even width all around, meaning more of a larger base would be needed at the front side of the base due to the base of the hand being thicker than the fingers of the design, meaning the base wouldn't provide as much support on the user's hand if the front part of the base was larger.

CAD #3



With this model, it is a direct copy of my modelling clay model design as I have measured from the individual sections of changes in size and shape, this allowed me to achieve a copy of my design that is ergonomic and would be comfortable for the user when they are holding. The base of this design accommodates for the change in width of the user's hand, for the front of the base to allow for the base of the user's hand which is the thickest part of a hand., as well as a thinner length of the far end of the base to allow the user's little finger to rest comfortably. This ergonomic handle would allow for support to be given to the hand of the user so that the design is more comfortable. The key features of the design being the base which provides support to the bottom of the user's hand, the widened area which is around halfway up the design to allow for the palm of the user to rest comfortably when holding the design and the thinner section at the top of the handle which allows for the thumb of the design to wrap around the design and allow for a comfortable grip for the user. These features that I have added allow for my handle to be comfortable and ergonomic for the user.

Further modelling

To further improve the accuracy of my CAD design, I produced another model that was made from modelling foam. I did this as when measuring the handle of the design, due to the modelling clay being a compliant material, the measurements may have been inaccurate which would lead to an inaccurate design, as well producing a new stronger handle which would allow for me to add the design into another physical model.

To produce this foam model, I placed the modelling clay model against a piece of modelling foam and drew around the shape of the handle so the outline of my handle would now be on the foam. This was then cut on a bandsaw to remove excess material and leave me with a very rough shape of my ergonomic handle. I then ran a pen across either side of each corner of my design, so that I would be able to shape the handle to be ergonomic, by removing the excess material between the lines. This would then create two more edges on the corner where I would repeat this process so that the shape would become more circular and meet the needs of an ergonomic grip.

With repeating the stage of sanding down the corners of the design, the handle started to form the ergonomic shape of the clay model. With a basic shape now achieved, I used a file and removed areas of the handle that needed material to be removed. Using a file, I removed larger amounts of material to make parts of the design more defined and clearer so that they would be more comfortable, such as making the base of the handle more of the base that I intended. As well, as ensuring all the corners and edges of the design were round so that the design would be ergonomic and comfortable for the user. Additionally, checking with others of varied hand sizes to hear their input on the design and changing parts of the design to meet ergonomic needs as parts may have not been comfortable for them, and not just changing the design to my ergonomic needs which may result in a handle which has been made specifically for me and not for a wide range of users between the 5th and 95th percentile of hand sizes. When testing from others, they used both hands to see if the comfort was different and if anything needed to be changed on the design. Making the design symmetrical so that it can comfortably fit both left-handed users and right-handed users.



With this new model, I will be able to use it on more developed models to show my design clearly and test the function of the tool, whilst having an ergonomic grip which I could use later for CAD.

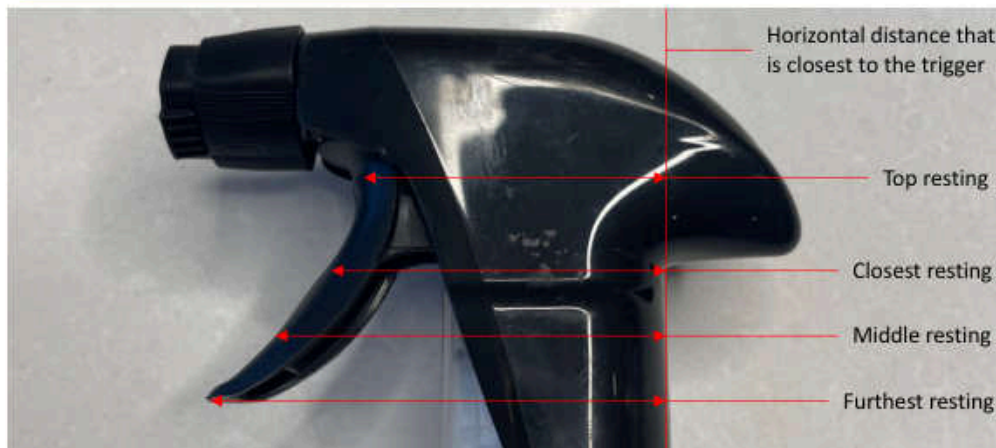
Trigger investigation

An essential feature of the design is the trigger, having varied the design of the product around the trigger, and it being removable, it is essential that the design is comfortable and functional for the user. Being the correct distance from the handle so the user's hand is not strained, being the correct thickness so the handle is sturdy and having the right shape to fit the hand of the user accurately, so it is comfortable. Another essential feature is the trigger being removable to allow for the head tool of the litter picker to be separate, which would help prevent damage to the rope as it is connected to the trigger and can be easily damaged, and the trigger may could be in the way of the user which could be an inconvenience to them. As well, keeping the trigger in this section would mean another type of connection, which could increase the complexity of the product which would make it more difficult the user to use.

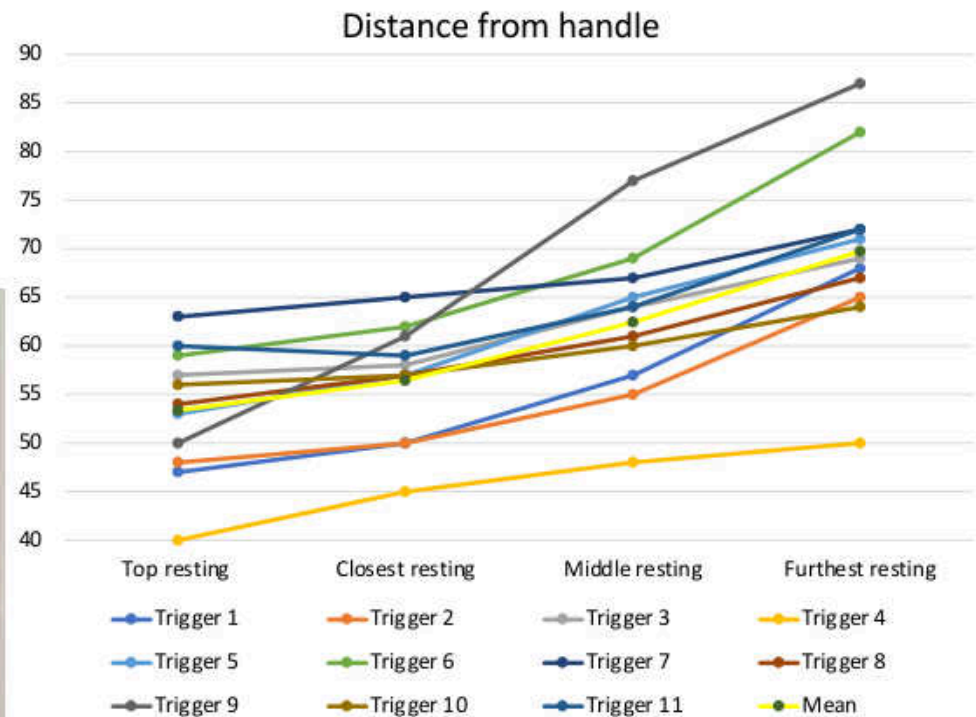
In order to find out more information and develop my trigger, I investigated the shapes and sizes of other various trigger mechanisms which would help with the development and help me to produce an ergonomic and functional trigger.



By measuring the distances on the key points of the triggers, I would be able to develop the shape and size of my trigger and how far the trigger should be from certain points. The key points were the top resting, closest resting, middle resting and furthest resting.

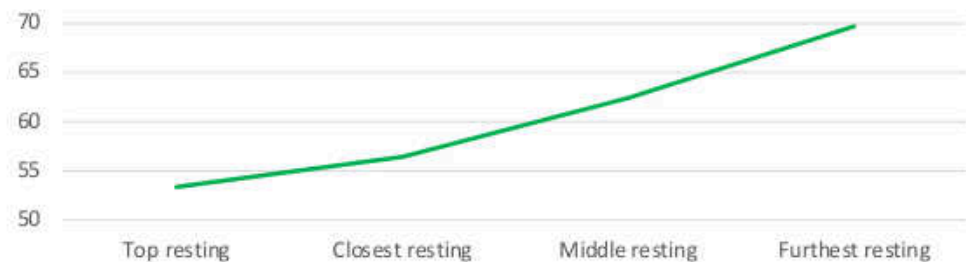


By measuring all 11 triggers at their key points from the horizontal distance of the part of the handle which is closest to the trigger I was able to get a consistent set of results that showed the shape of the trigger and would not be affected by the shape of the handle. By getting this rough shape of a trigger, I would be able to adapt it and to add it to my design. As my design already incorporates a dovetail connection for the trigger, finding the sizes of triggers will allow me to identify how far the trigger should be from my handle and the shape of my handle.

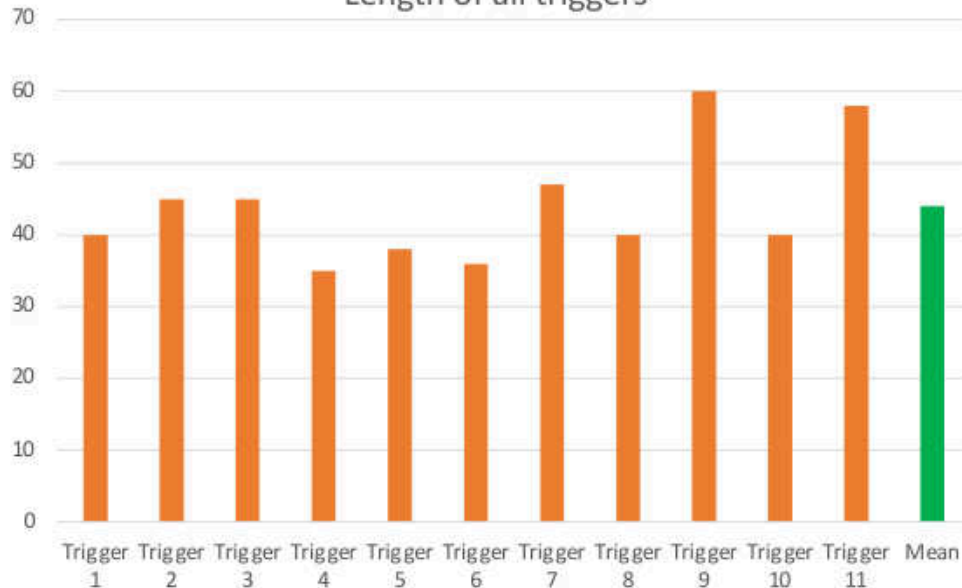


By plotting these key points and distances and getting a mean value for each of the key points, I can get a mean distance from the horizontal distance that is closest to the trigger (section of the handle where the thumb rests and is the closest part of the handle to the trigger).

This is the shape of the handle and the distance from the horizontal part of the handle that is closest to the trigger that I will use for my design, as it is the mean of all the handles that I investigated it will be ergonomic for a user as it incorporates some of the ergonomic distances and sizes from the triggers. This is the rough shape of the handle that I will have to develop further, and I will test and develop if any of the key point distances need changing too.

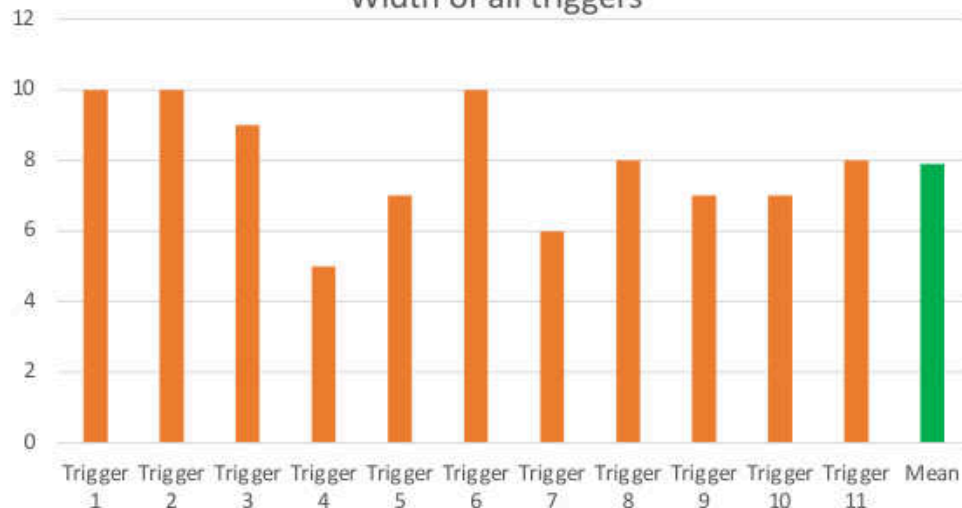


Length of all triggers



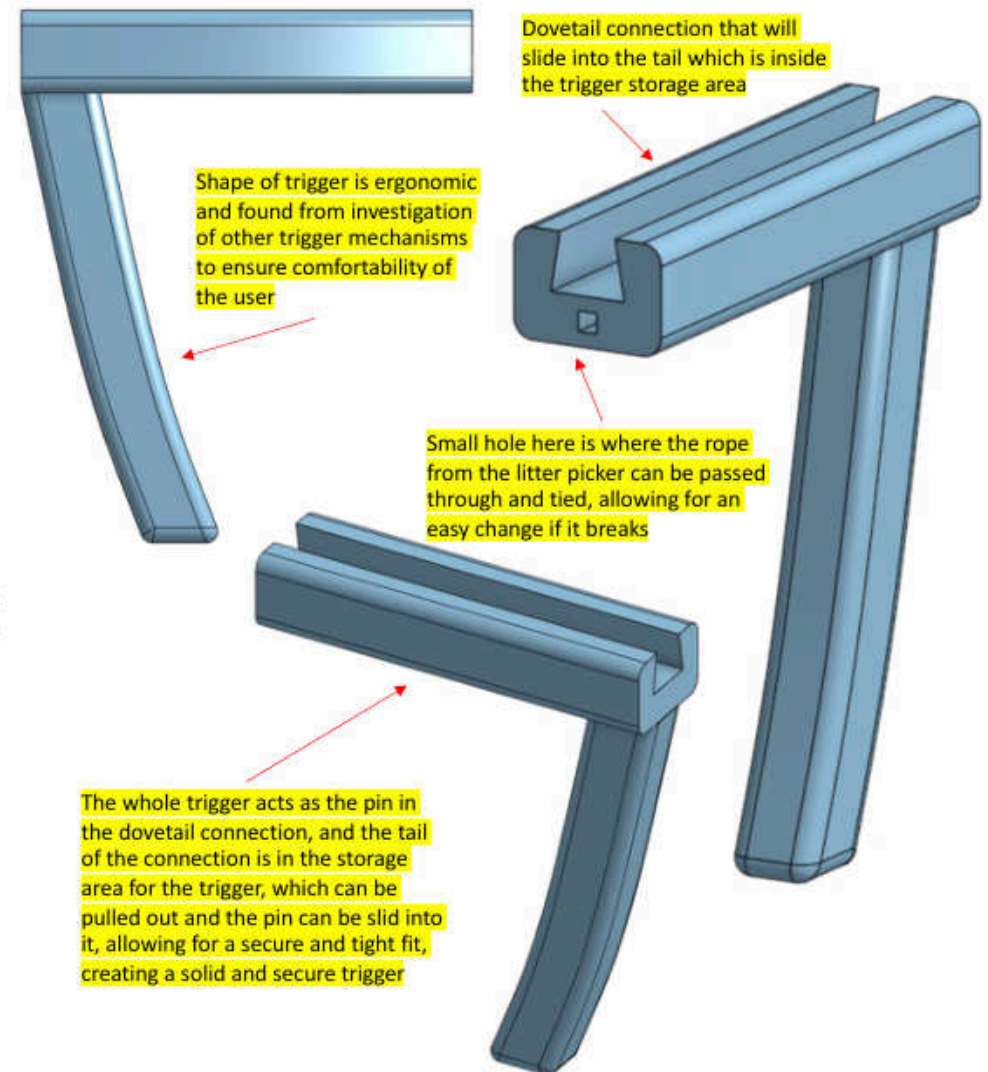
By measuring all the lengths of the triggers, I was able to get the mean value of length for all the triggers investigated, with the average length being 44mm in length. However, when holding the triggers when holding the handle, the most comfortable types of triggers were the longer ones, specifically trigger 9 and 11, this is due to there being a larger area where the fingers of the user can rest on and pull the trigger from., allowing for the user to accommodate for their comfortability and meet their demands as they can then move their fingers to find a comfortable position. Allowing for the user to easily meet their ergonomic demands, due to this when developing the shape and size of my trigger, the size I will choose will be larger and around the size of trigger 9 and 11 which allow for a more ergonomic fit.

Width of all triggers



By measuring the widths of all the triggers, I was able to find a mean value for the widths, with the average being just under 8mm in width. Which is similar to trigger 11, which is the litter picker and one I should have similar results to as the products are very similar. The benefits of the litter picker is that the width is a good width so that the user's hand can easily wrap around the trigger, and easily pull the trigger to the them, as well as being wide enough to retain the strength that is needed to accommodate for the pull of the trigger from the suer.

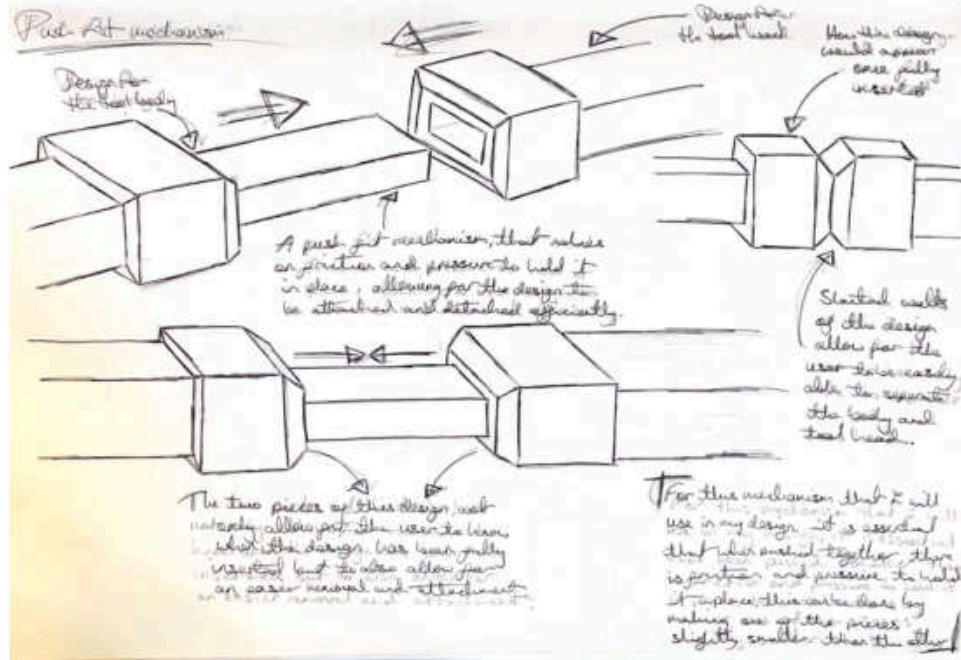
Combination of results:



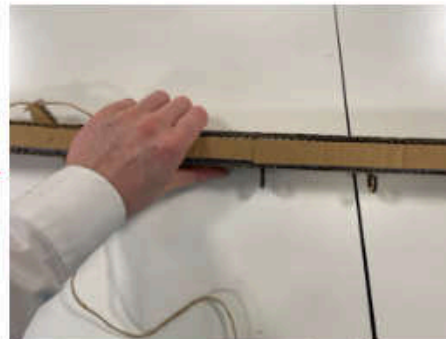
Combining the results of my investigation, I have produced a trigger that is ergonomic and will fit the user comfortably, allowing for an easy use. This trigger will be connected to a dovetail connection for it to slide into position, which will allow for the trigger to turn and change angle easily, allowing for the claw of the litter picker to move, close and capture litter which the user can then easily dispose off.

Connecting mechanism

The current design of the connecting mechanism between the head tool and the body:

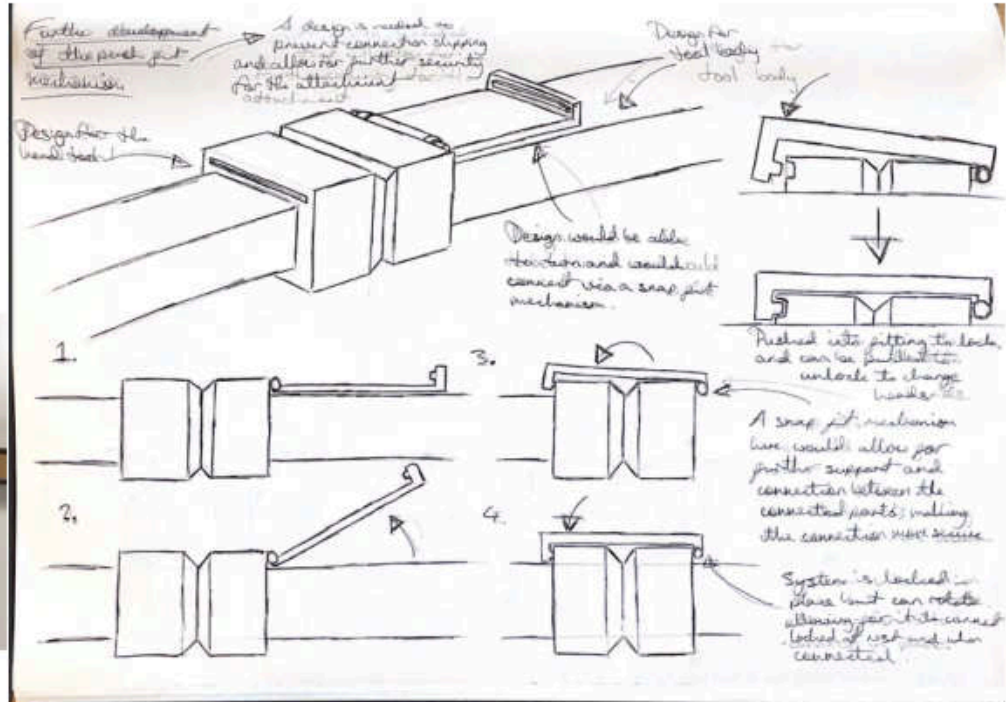
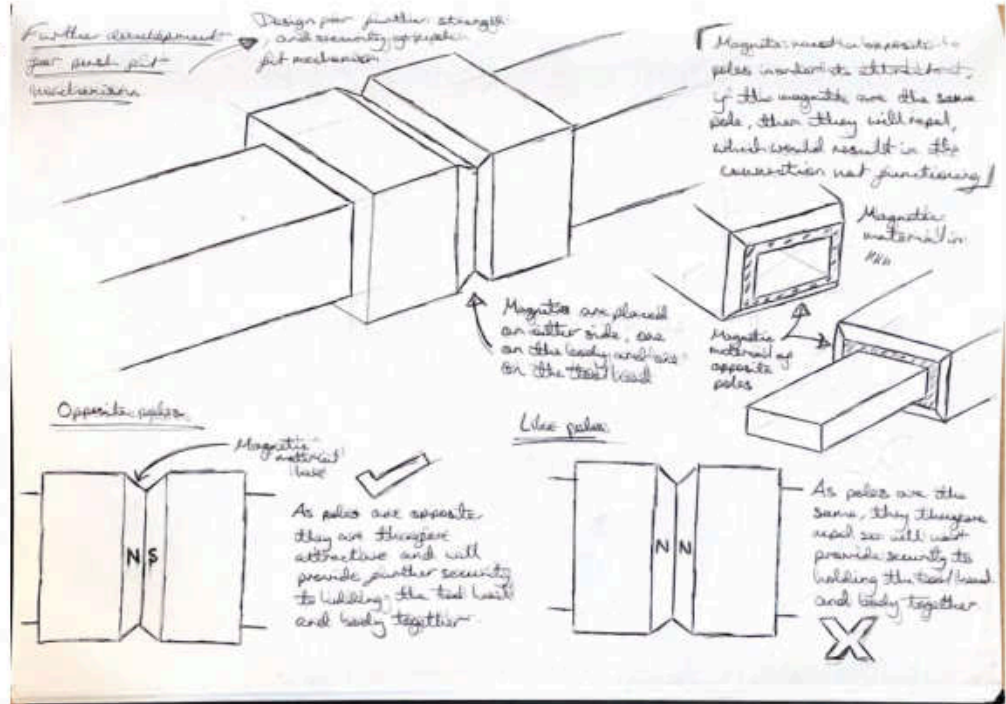


As found from my modelling, another mechanism is needed to secure the head tool and the body of the design, due to sometimes the push fit mechanism slipping out, which is inconvenient for the user as they would have to push the head tool back into the body of the design. A solution to this would be to either make the push fit mechanism longer so that there is a larger surface area contact, meaning more grip and a more secure connection, however this is just not practical as then the head tools would be a larger size, making it inconvenient to carry the heads of the tools. Another option could be to make the push fit tighter; however, this can cause numerous problems such as excessive insertion force which may damage the material, as there is an increased friction the lifespan of the product would decrease, and the high pressure could cause deformation of the body of the design.

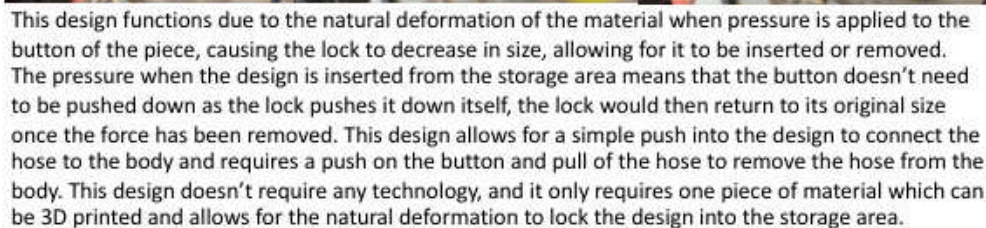
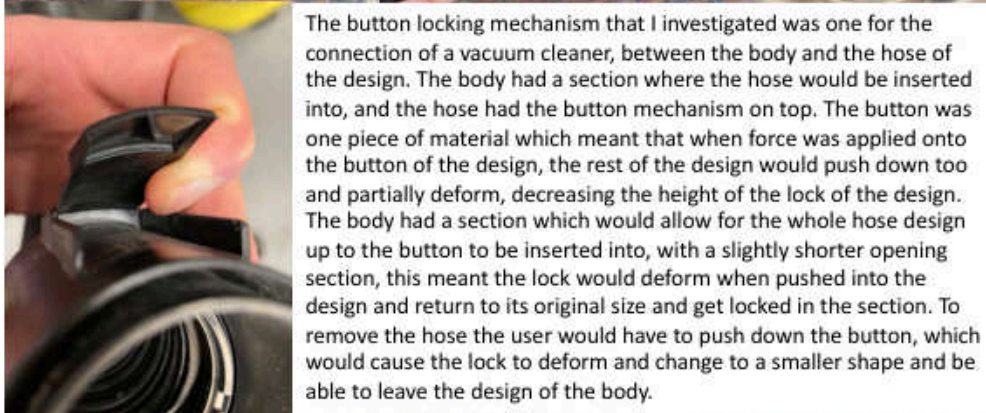


Therefore, a new design is needed that can increase the security of the connection and would not affect the push fit mechanism.

Developments of solutions to secure the connecting mechanism:



Button locking mechanism investigation:



Final design of connection mechanism:

One whole component / piece of material will work.

Control design

Allow the split to maintain shape and will resist and hold.

Control design is only permitted near the bottom and to allow for the material to hold under pressure.

Loading mechanism

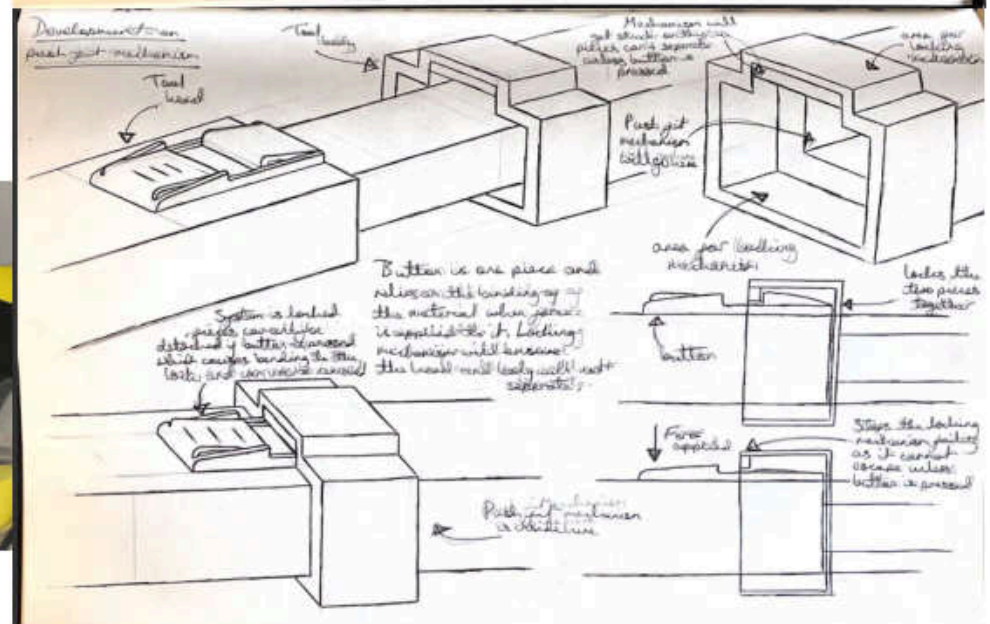
push-pit mechanism

Design is able to be pushed in and will hold in as material returns to original shape, bottom steel pulled to remove with mechanism.

Design allows part of the material to be released, design will return to shape so that the user can gain a greater view when attaching and detaching to improve accuracy.

Barriers will set up to the bottom of the mechanism.

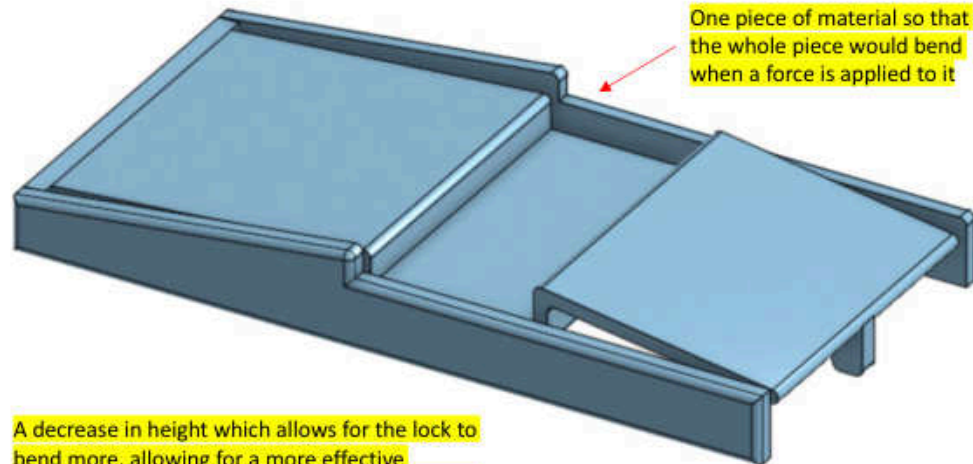
The design incorporates both a push-pit mechanism and a locking mechanism to ensure the strength and security of the attachment between the handle and the body of the device. With push strength, the design will become secure and push better when using the tool.



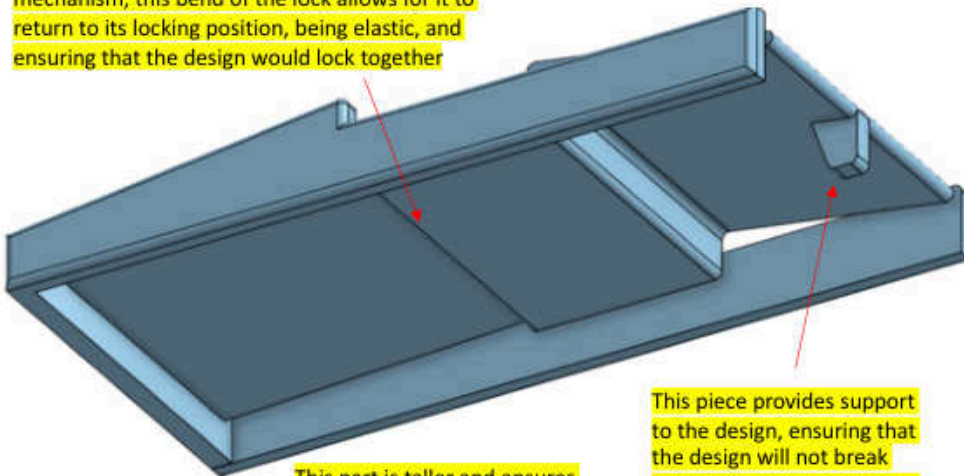
When comparing the designs that could be used to further secure the connection between the head tools and body, it seems most appropriate to use the button locking mechanism due to the effortless insertion and easy removal, as well as providing a very secure and strong connection which would remove the current problem of the design slipping out. The design will remain together unless the user pushes the button and pulls the head tool, due to the deformation of the one piece of material.

CAD of connecting mechanism

This button locking mechanism would rely on the movement and bending of the material when it has been pressed down at the button to move down the lock, allowing for it to be inserted into an area and when the force is removed, will return to its original size and be locked into position, connecting the two pieces; the body and a head tool. Allowing for a secure connection that relies on the bending of the piece to lock it into place. The insertion is simply a push as the lock would be pushed down, the removal would be a push of the button and then a pull.

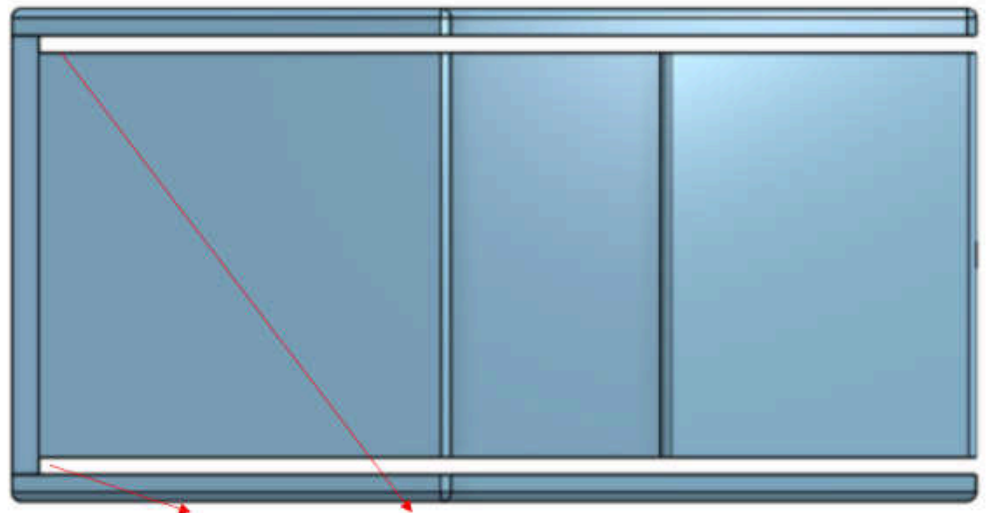
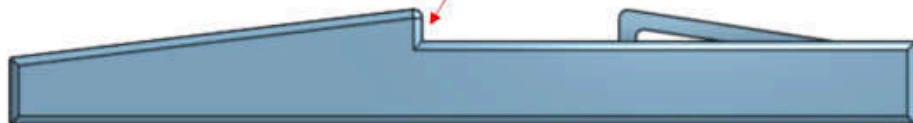


A decrease in height which allows for the lock to bend more, allowing for a more effective mechanism, this bend of the lock allows for it to return to its locking position, being elastic, and ensuring that the design would lock together



This part is taller and ensures that the button mechanism can't enter the locking system

This piece provides support to the design, ensuring that the design will not break when parts of it are pushed downwards and are under pressure and are bending

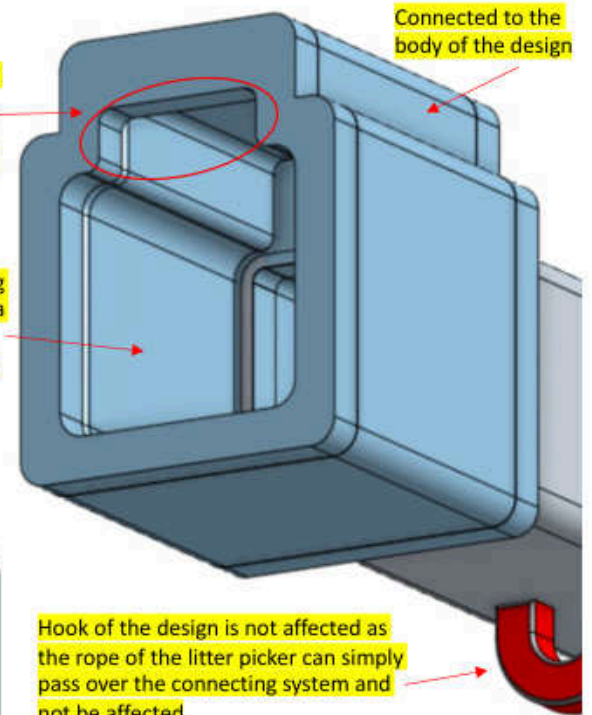


This is the only part of connection between the walls of the design and the main design (the button and the lock), this being the only part allows for the piece to be more flexible and allow for the design to bend, ensuring that the lock can move downwards and lock into place

Locking system:

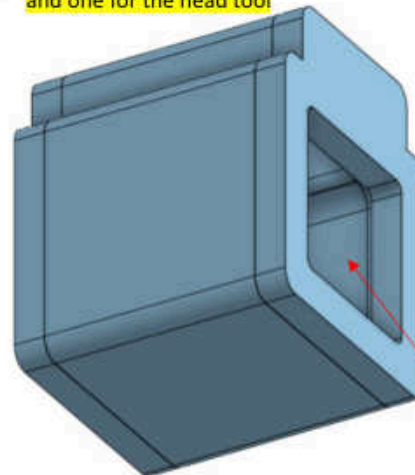
Button locking mechanism is on top of the head tool, and there is a smaller push fit in front, the button locking is pushed into here, top part has a shorter opening allowing for the system to lock into place after it has been pushed through and stops bending

Whole piece of the head tool up to the button is engulfed by the system, allowing for it to be locked together and ensuring a secure fit, in some way there are then 2 push fit mechanisms, one for the push fit and one for the head tool



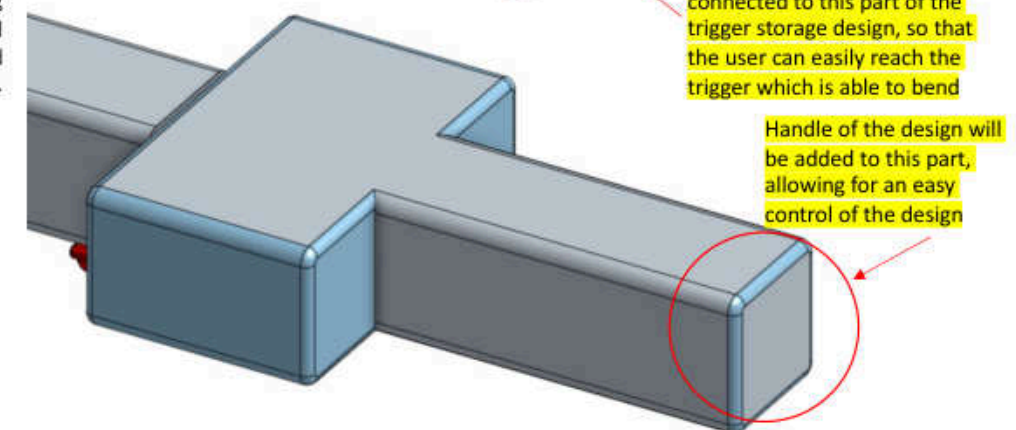
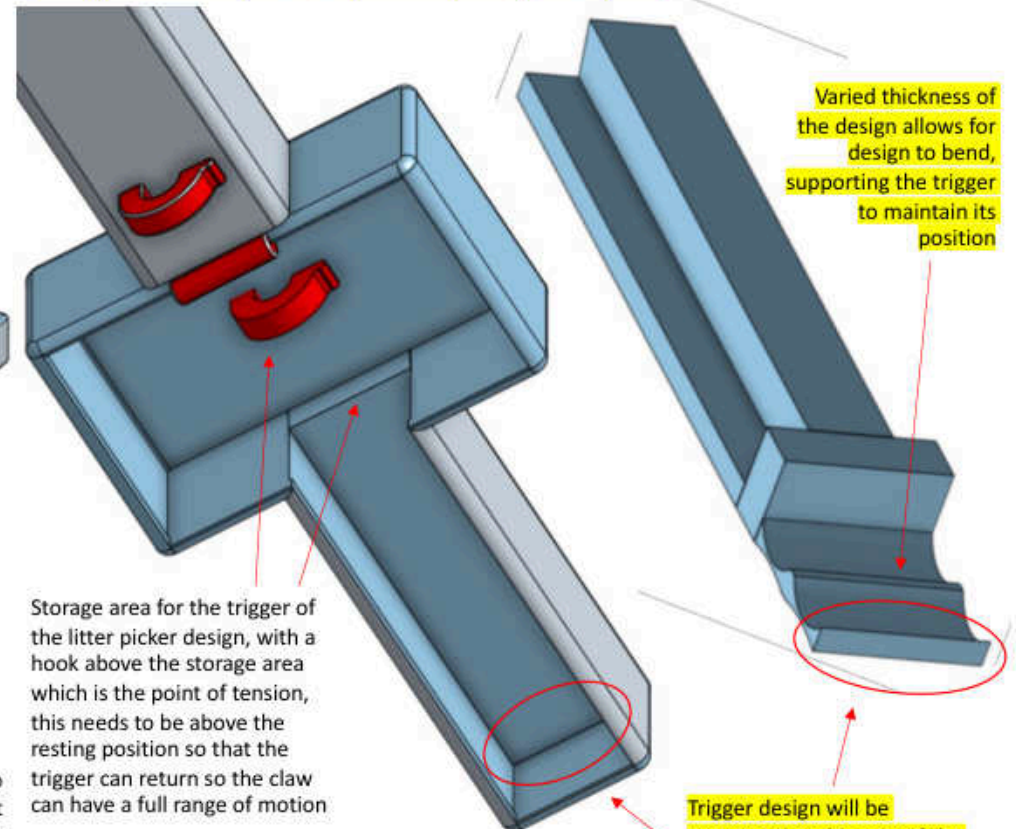
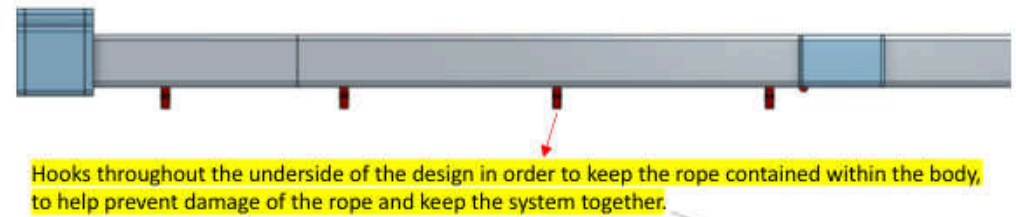
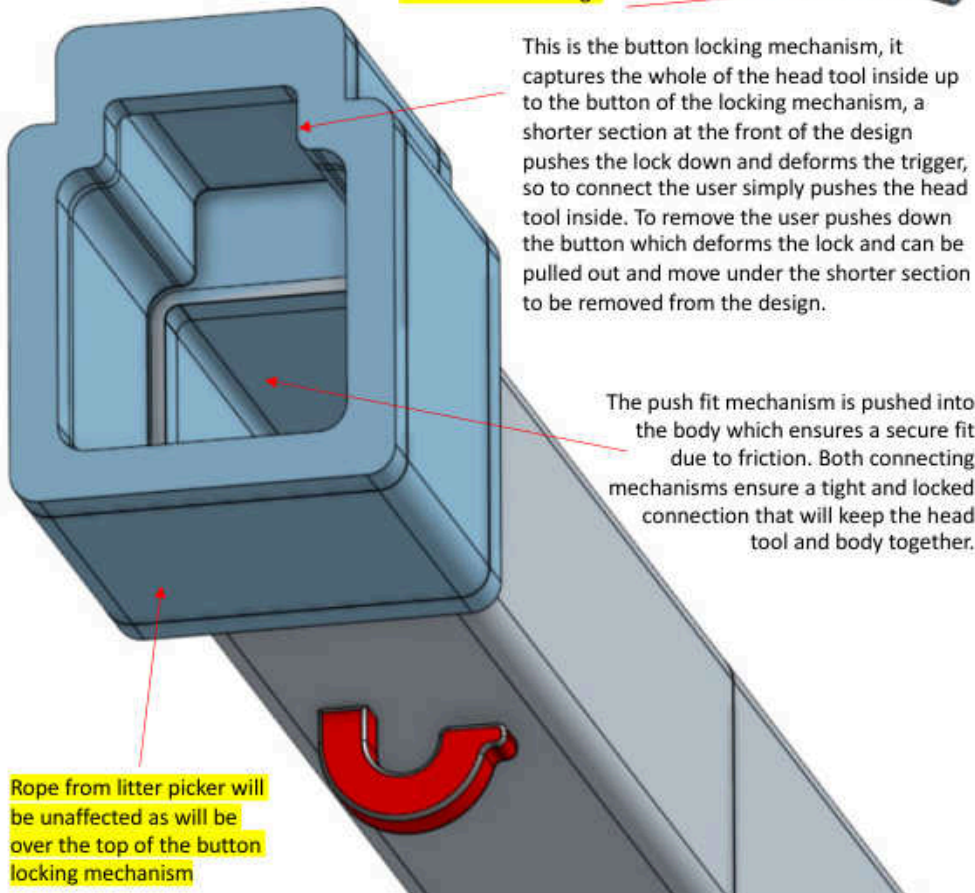
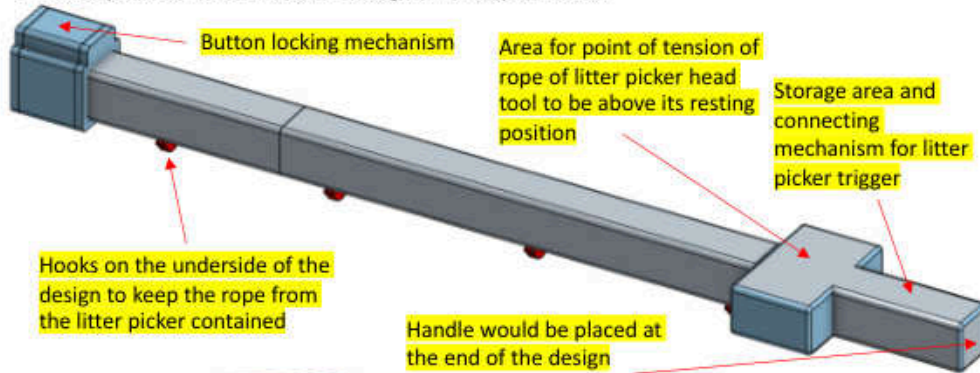
Hook of the design is not affected as the rope of the litter picker can simply pass over the connecting system and not be affected

This part is where the design is connected to the body, part of the body can be inserted here and secured to ensure that the design is solid and will remain connected and together



CAD of body of tool

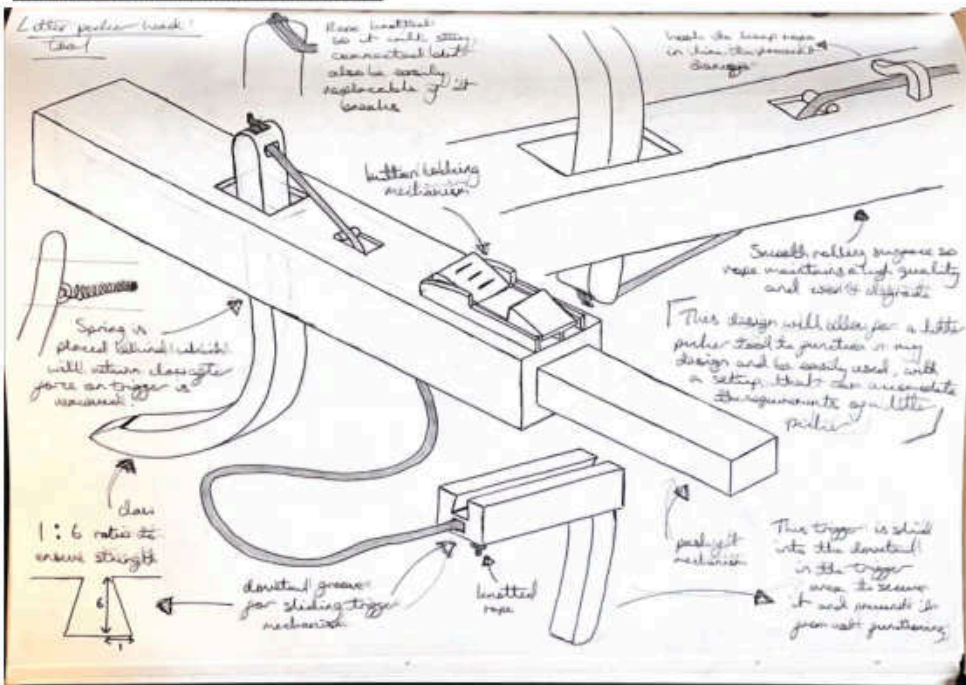
The body is the main part of the design, all the tool heads can connect to the body of the design so that it becomes one piece and one tool. Adding different heads to the body allows for different tools to be used on the same design, which is beneficial for the user who can save storage space as they don't have to carry numerous types of tools to remove litter, only the heads and one body. This allows for the user to be prepared to pick and remove litter from many environments easily and effectively as each of the tools are designed for a specific use.



Litter picker head tool

With my development of the design of the body complete, I can now finalise the development of the tool heads which would be connected to the body of the design, to make a product that has changeable heads for certain scenarios when removing litter.

Development of litter picker head tool:



Parts of a litter picker that I need to add to ensure that it is functional



- Adding a spring to the back of the claw is essential for this tool to function, when the trigger is pulled and the claw is turned, the spring is extended and when the force on the trigger is removed, the spring returns to its original position, this brings the claw back and the trigger too. The spring essentially allows for the whole system to be recycled and start again so that the user can consistently use the tool
- Another essential feature is the rope being connected to the top of the claw, the shorter the distance from the top of the claw where the rope is connected, the shorter the distance the claw must be pulled so that the claw can fully close, however the shorter the distance applied, the larger the force required to move the claw is required

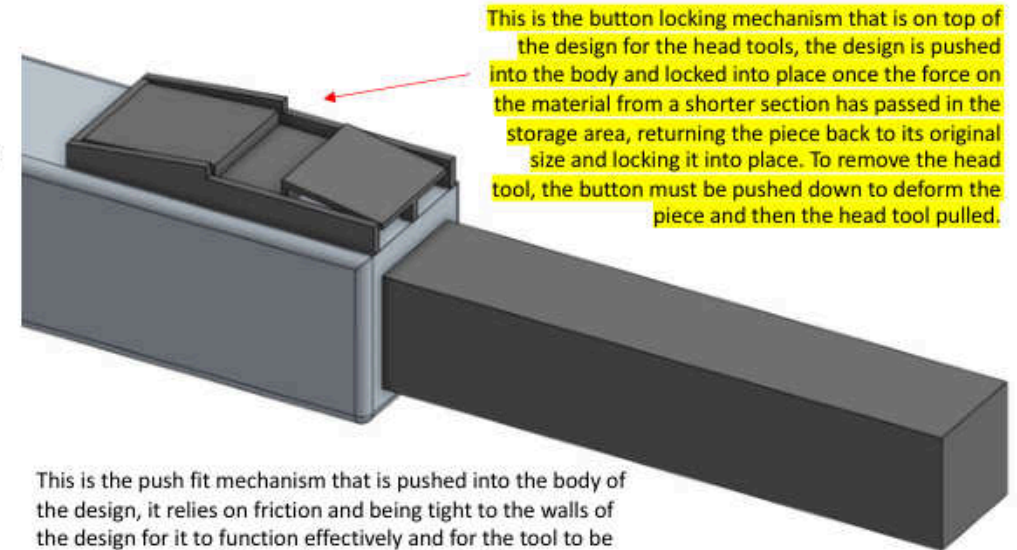
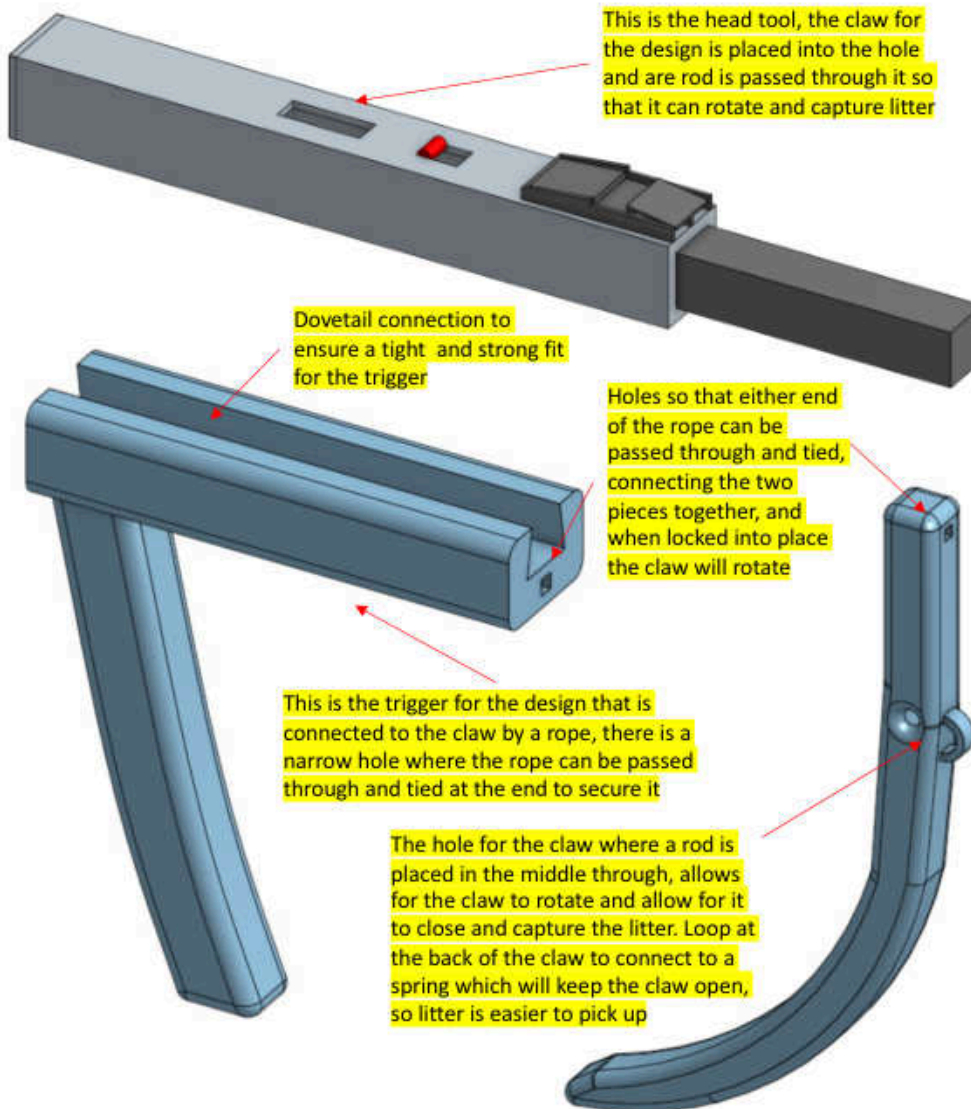
Claw investigation



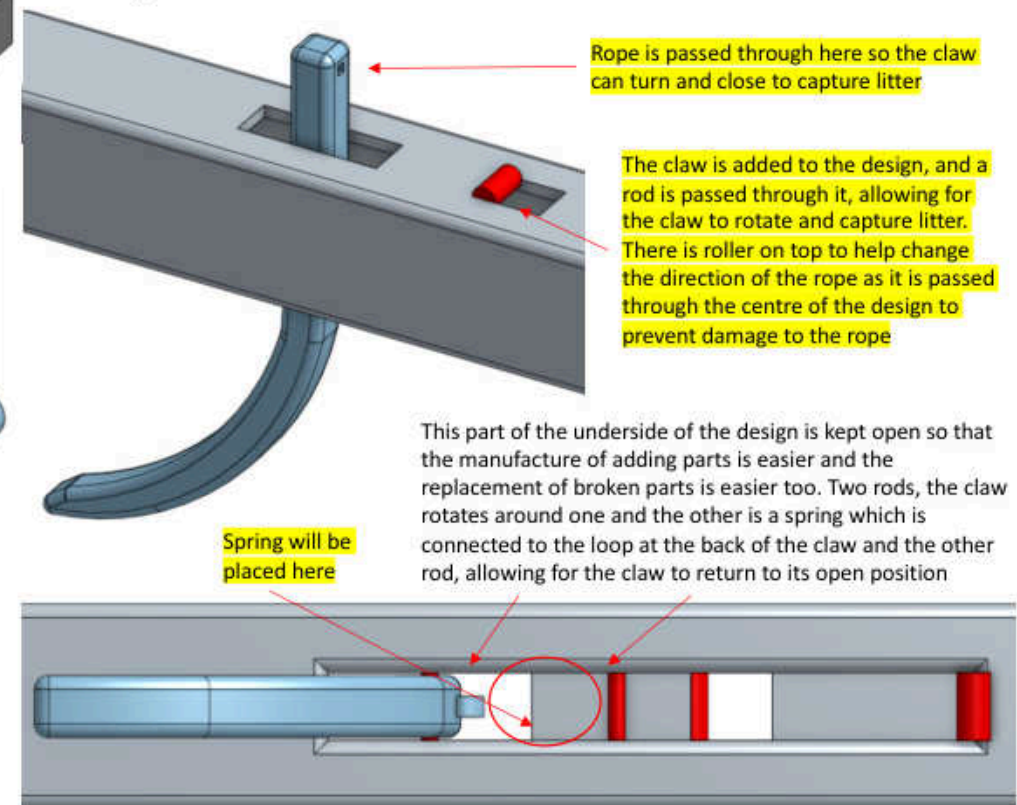
When the trigger is pulled, it moves around a circular piece that it can rotate around, and moves a distance of x , the distance of x is the same as the distance that the trigger has been pulled from. The smaller the distance means a larger force required which may be uncomfortable for the user, but a longer distance would mean a further distance the trigger has to be pulled, which may be uncomfortable or not possible. The design won't be able to move further when the claw is fully closed. The distance that the claw should move should not be short or long but, in the middle, so that the user does not have to use a large amount of force to close the claw of the design, and that the user doesn't have to pull the trigger too much which could be uncomfortable.

CAD of litter picker tool head

For the litter picker tool there are 3 parts that will need to be combined in order to make a functional and efficient product, this is the head of the tool, the claw and the trigger. The claw will be connected to the head around a rod so that it can rotate and the claw is able to close, a spring will be added to the back of this claw for it to close once the trigger has stopped being pulled which is connected to another rod. At the top of the claw there is a section where a rope can be passed through and tied, the other end of the rope can then be passed through the hole of the trigger and tied, connecting all the parts. The head of the tool can then be pushed into the body, rope passed through the hooks and the trigger slid into the dovetail connecting mechanism to connect the trigger, once this has all been done the product is now functional and the litter picker can now be used.

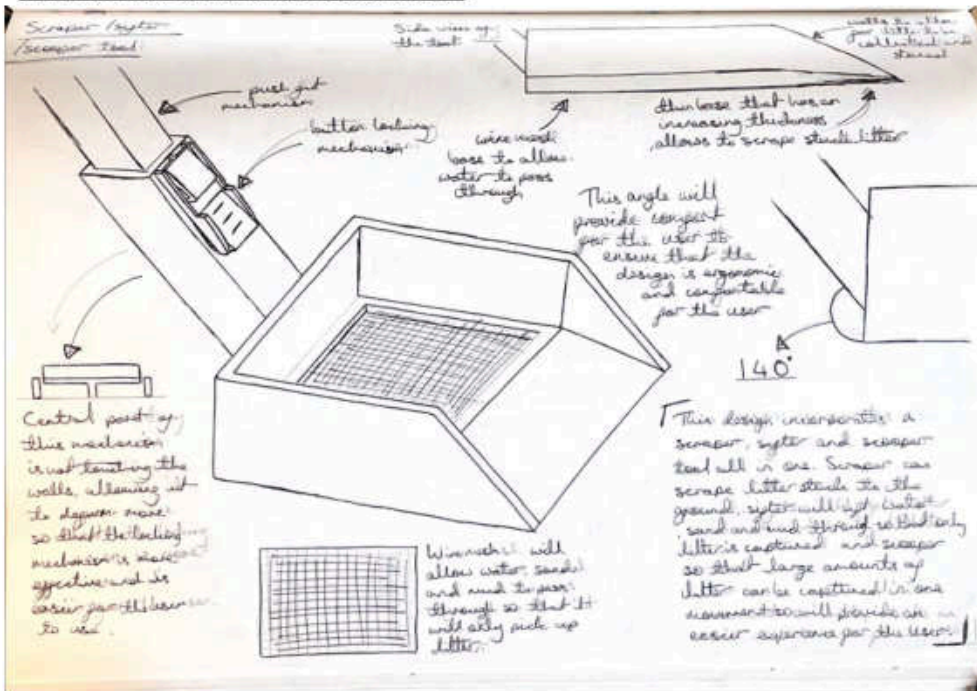


This is the push fit mechanism that is pushed into the body of the design, it relies on friction and being tight to the walls of the design for it to function effectively and for the tool to be strongly connected. The push fit and button locking mechanism have both been added to ensure a tight and secure connection so that the head tools and body won't separate when using which could be an inconvenience for the user.



Scrapper head tool

Development of scraper/sifter/scooper design:



This head tool is a combination of three initial head tool ideas that have been combined into one, combining a scraper that can scrape litter which is stuck to the floor, a scooper that is able to scoop up large amounts of litter and a sieve which is able to let sand, water and mud pass through so that the user only captures litter and not items that belong to the environment. With the combination of these three different head tools, it not only allows the tool to be more versatile but allows for an easier experience for the user, as they will have to carry less head tools and the tool that they use can be used in more scenarios, this would overall lead to an easier and less confusing use of the tool, as the user can spend less time deciding what type of tools they will have to use.

Key features of the design:

In order for the scraper head tool to be functional and effective, some key features are required to ensure that the tool can perform well and remove litter efficiently from the environment:

- Sieve material needs to be have large enough gaps that water, mud and sand can pass through, as well as gaps that allow small stones to pass through other wise the user may have to remove them which would be an inconvenience, but the gaps also need to be small enough so that the small pieces of litter that are captured can be stored
- Needs to have a thin width so that the design is able to get under the litter which is stuck and lift it from the ground so that the litter is no longer stuck, and the design is able to be pushed under the litter and captures it inside the design. Making the design thin enough at the front is essential so litter will roll over the front and be captured by the design, if the front is too tall or wide then the design would not scrape the litter effectively or well at all, and would instead push the litter and be moved over it
- The design must be angled to accommodate for the angle that the user will be holding the design at to the ground, ensuring that the angle is appropriate for the user which would allow for an easy experience when the user pushes the design under the litter stuck on the ground

Sieve size investigation:

In order for the design to be appropriate and being convenient for the user, the size of the sieve should allow for small stones, water, sand and mud to pass through but be able to capture small litter. The most common type of litter by count according to 'Keep Britain Tidy' are cigarettes and is also one of the smallest types of litter that can be found. By developing the size of the sieve around the size of cigarettes it would allow for the design to be accommodated to litter in the environment. The size of cigarette butts can vary due to the type of cigarette and how much was smoked; according to study on 'how close to the tipping do consumers actually smoke?', the mean butt length from 2000 cigarettes was 33.2mm.

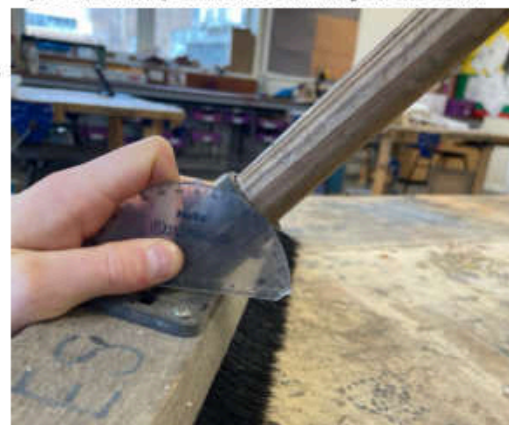
Data on cigarette butts from Research Gate:

Code	FTC 'tar' (mg)	Nominal length (mm)	Nom. circ. * (mm)	Filter length (mm)	Tipping length (mm)	Menthol flavor	Subject count	Cigarette count
A	1	100	25	30	36	Yes	50	641
B	1	100	25	30	36	No	49	662
C	1	85	25	27	32	No	49	658
D	3	100	25	31	35	No	48	619
E	5	100	25	31	35	No	49	680
F	5	100	25	30	36	No	44	687
G	6	100	23	31.5	36	Yes	48	682
H	6	100	23	31.5	37	Yes	49	582
I	9	100	17	27	32	No	36	266
J	9	100	25	30	36	No	49	657
K	9	85	25	27	31	Yes	49	552
L	10	85	25	27	32	No	50	716
M	11	85	25	27	32	Yes	49	717
N	11	100	25	31.5	36	No	48	538
O	14	100	25	30	36	No	47	600
P	16	85	25	21	26	Yes	47	651
Q	18	85	25	21	25	Yes	45	620

By dividing the circumference of the cigarettes by pi, the diameter of the cigarettes ranges from 5.41mm to 7.96mm, with the mean diameter being 7.73mm. The width of the actual cigarettes would be mostly larger at around half of the circumference of the cigarette, this is due to the butt being squashed and being a flat object. For the sieve to allow for the capture of all types of cigarettes, the size should allow for all the cigarettes to be obtained, with the smallest being 5.41mm, the sieve should have holes slightly smaller to allow for the smallest of butts to be obtained, yet allow sand, water, mud and some small stones to pass through.

Angle of the tool head:

The angle that the head is connected to the body is essential so that the scraper is parallel with the ground and the user is easily able to get underneath the stuck litter so that the litter removal process is simple and not straining on the user.

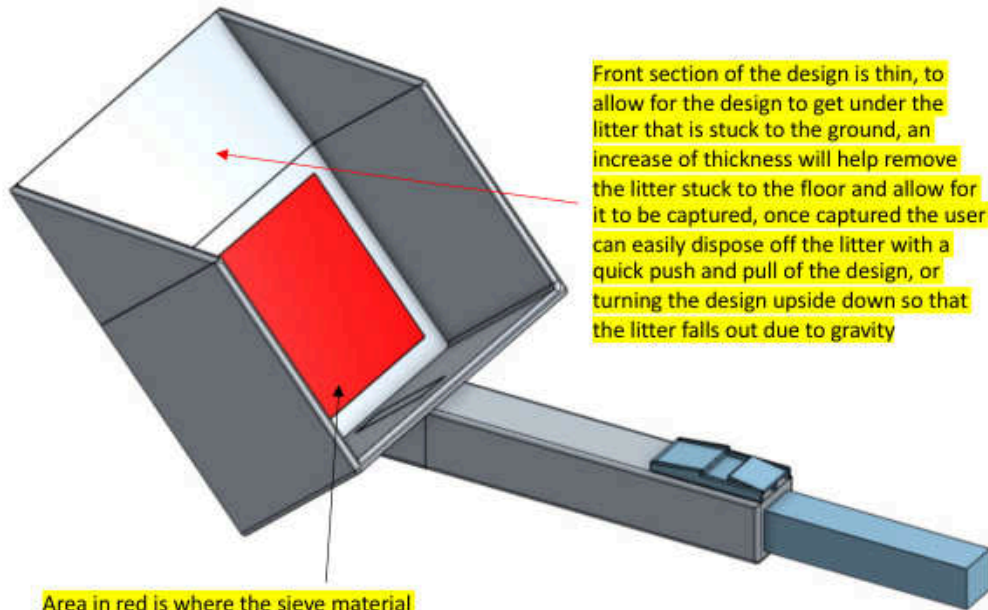


A product with a similar angle to this design would be a broom, the angle connecting the handle to the bristles is the angle that is required for m design as it allows for an ergonomic fit when the user is pushing the design forwards in order to scrape the litter of the ground and obtain it so that the litter can be disposed off.

From the angle measured, the design has an angle of 140 degrees, this angle allows for the user to be in a comfortable position whilst standing and pushing the design and is an angle that I should use for the design to ensure that it is comfortable and appropriate for the user.

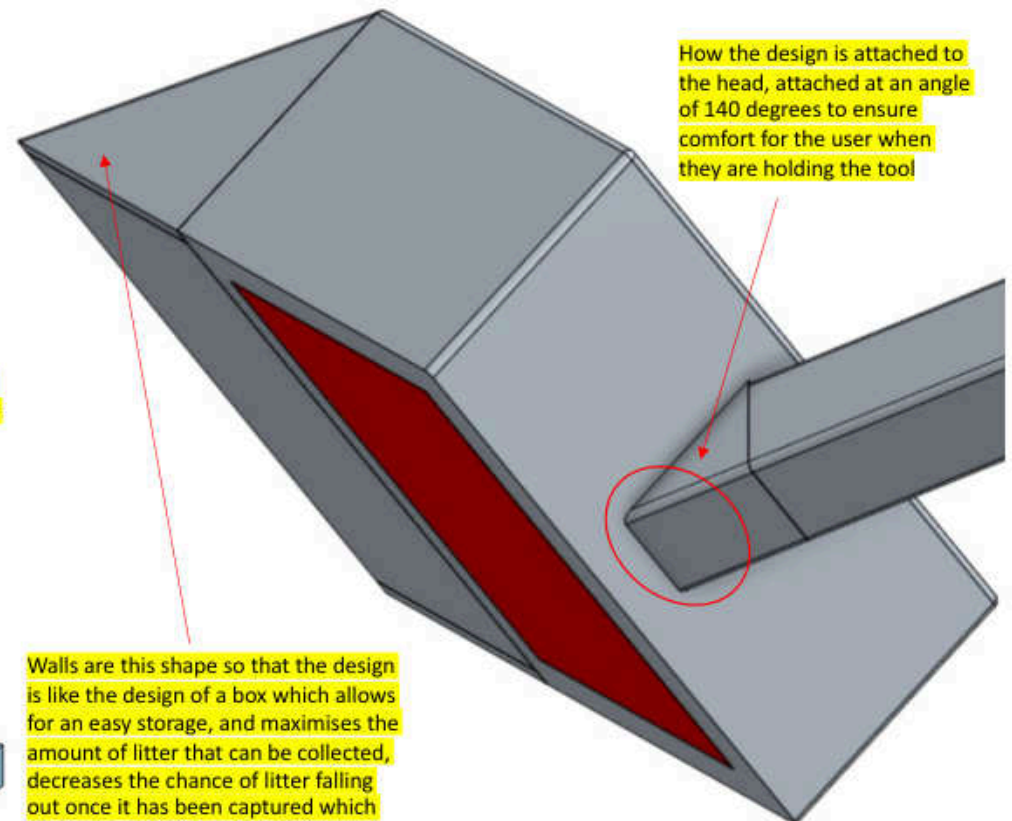
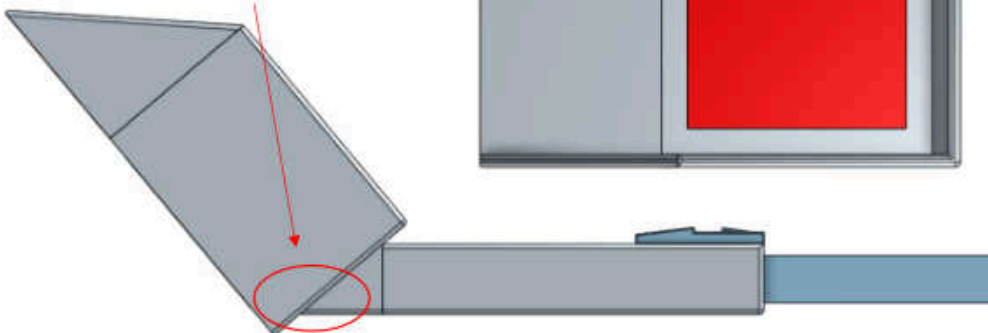
CAD of scraper head tool

The scraper tool is one of the head tools which can be connected to the body of the design. The intention for the design is to scoop up large amounts of litter such as large amounts of cigarette butts and scrape litter that is stuck on the floor such as cigarette butts or packaging. It allows for litter to be picked up that would have been difficult to do so with a litter picker due to the pieces of litter being too small or that litter picker is unable to get to an angle to obtain the litter.



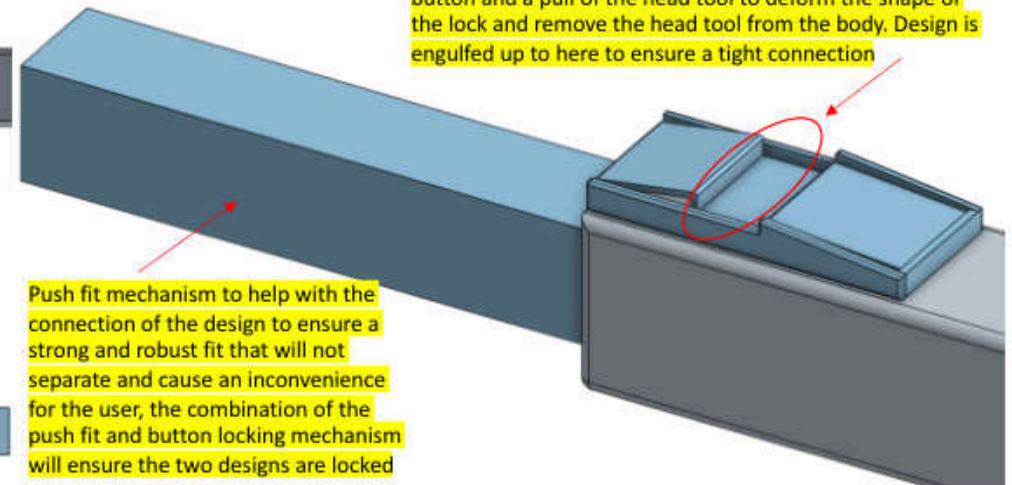
Area in red is where the sieve material will be, as investigated the gaps will be less than 5.41mm to ensure that water, sand, mud and small stones are able to pass through, allowing for an easy removal of litter for the user

Angled at 140 degrees to provide a comfortable use when using, meeting the natural position of the user so that the head will remain parallel, and is able to scrape litter off the ground



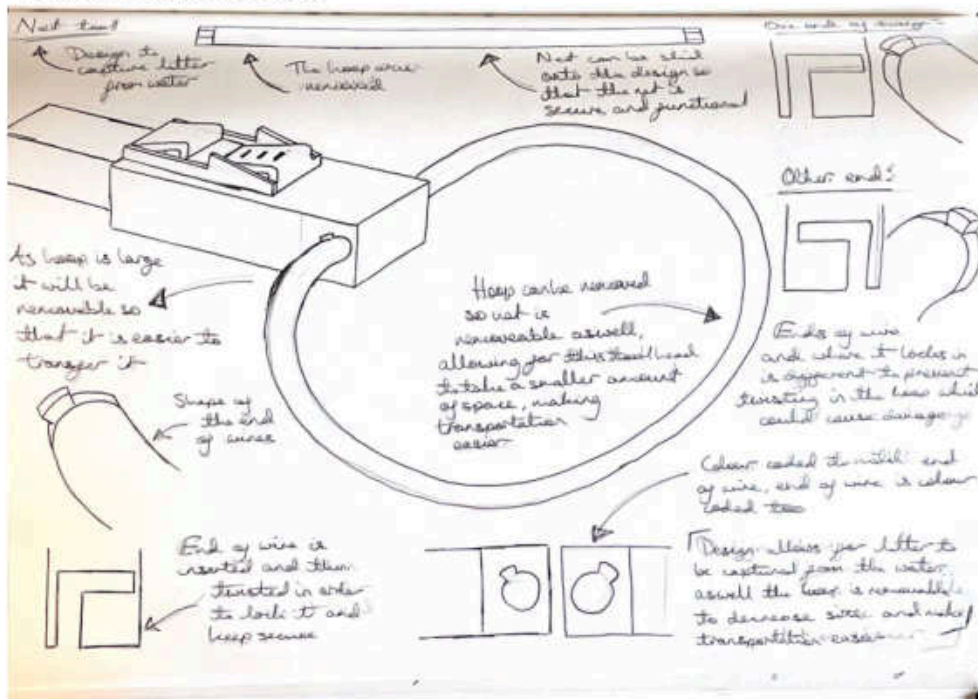
How the design is attached to the head, attached at an angle of 140 degrees to ensure comfort for the user when they are holding the tool

Button locking mechanism which will allow for the design to be locked when connected to the design, can be pushed into the body and will automatically lock due to the material not returning to its original position, and a push of the button and a pull of the head tool to deform the shape of the lock and remove the head tool from the body. Design is engulfed up to here to ensure a tight connection



Net head tool

Development of net head tool:



This head tool is a net and is intended to be used to collect litter from a watered system such as lake, river or pond where otherwise the litter is extremely difficult to remove due to the litter being wet and there being a lack of grip that the tool can gain in order to capture the litter. By using a net, it removes the need to grip the litter as the net can go around the litter and engulf it and then capture it, the user can then pull the design out of the water where the excess water can drain off, the user can then dispose of the captured litter by turning the design upside down, allowing for the captured litter to fall due to the force of gravity on the litter.

Problems with a net design:

Commonly nets such as fishing nets are very large in order to maximise the area captured in one use of the tool, however this would cause transportation problems as the tool heads are meant to be carried, however the net head tool will be unable to carry easily if it is too large. Developing a design that would allow for the hoop of the head to be collapsible or removable would allow for easy transportation of the head tool.



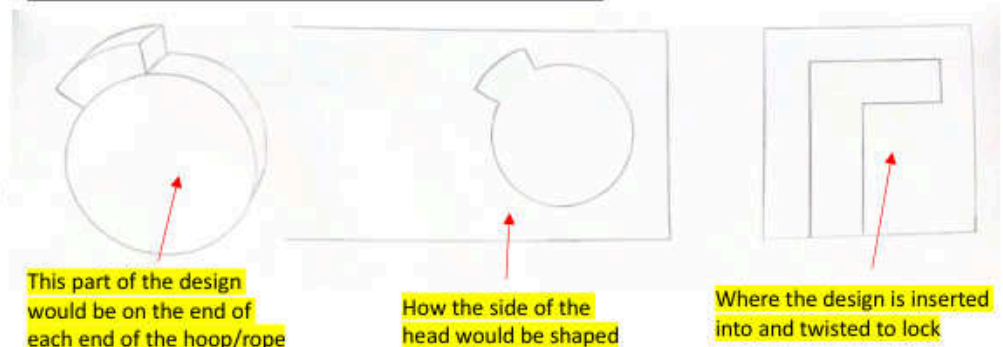
As shown in the development of my design, making the hoop of the net tool removable would save space and allow for the net of the design to be easily replaced if it got damaged which can be likely as there may be sharp objects in the watered environment.

The idea behind the redesign of the hoop is to allow for the net and hoop to be removed, which allow for the user to easily replace the design if it was to be damaged. The design works by having the hoop which would be made of a material such as a rope, as this is able to keep a circular shape and bend, allowing for the hoop shape to be achieved. The hoop/rope would be passed through a net, and the ends of the rope would be connected to the head of the design. This would allow for a design that can be easily transported which is an essential feature for the design of my litter removal tool.



By having a net similar to these two, with a loop section at the top, a rope would be able to pass through and create the net shape so that the design can capture litter. The design can then be connected to the head of the design to secure it.

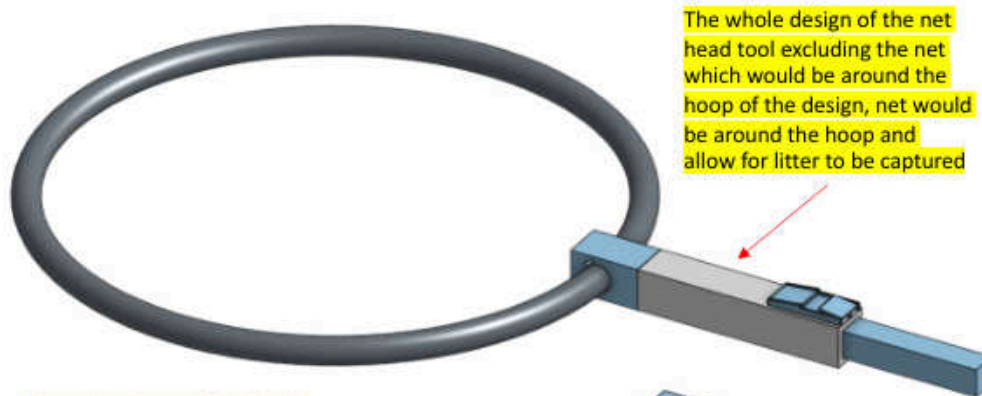
Connecting mechanism for the hoop to the head of the tool:



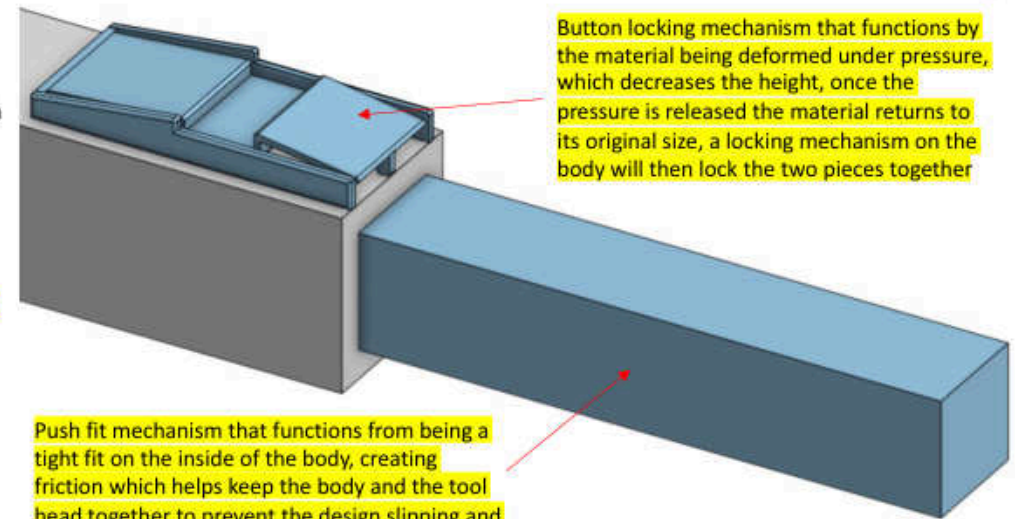
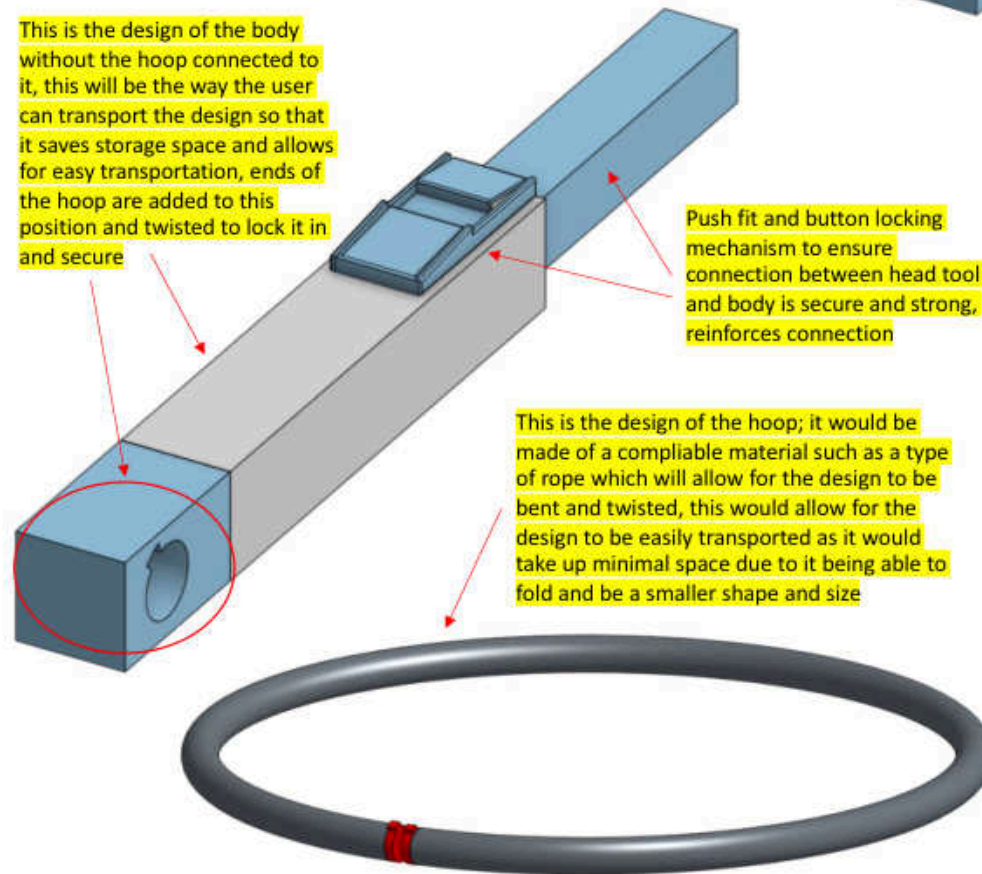
This connection mechanism functions by there being two of the parts at each end of the hoop/rope, both ends of the rope would then be inserted into the hole in the side of the design, and twisted which would secure the design together. By twisting the hoop, the design locks as it is now unable to be pulled out from the hole, securing the hoop, and can then be twisted and pulled again to release the mechanism. This allows for the hoop of the design to be removed so the net can be changed if it is damaged and the storage space for the tool is decreased, allowing for an easier transportation of the tool head.

CAD of net head tool

The net tool is one of the three head tools for my design of a multi-head litter removal tool, this head allows the user to remove litter that is in a watered environment such as a river, lake or pond. The user can attach the head tool and can capture the litter in the net, they can then turn the design upside down so that the litter will fall out into an area for litter storage, such as a bin bag.

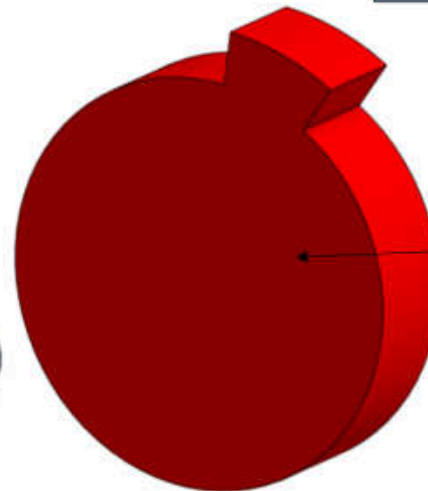
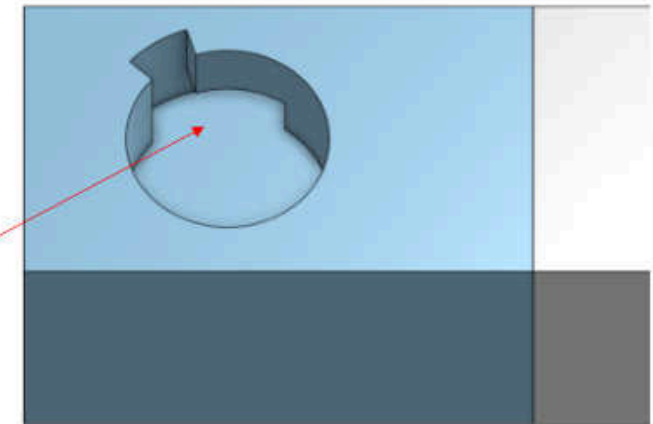


This is the design of the body without the hoop connected to it, this will be the way the user can transport the design so that it saves storage space and allows for easy transportation, ends of the hoop are added to this position and twisted to lock it in and secure



Push fit mechanism that functions from being a tight fit on the inside of the body, creating friction which helps keep the body and the tool head together to prevent the design slipping and causing an inconvenience

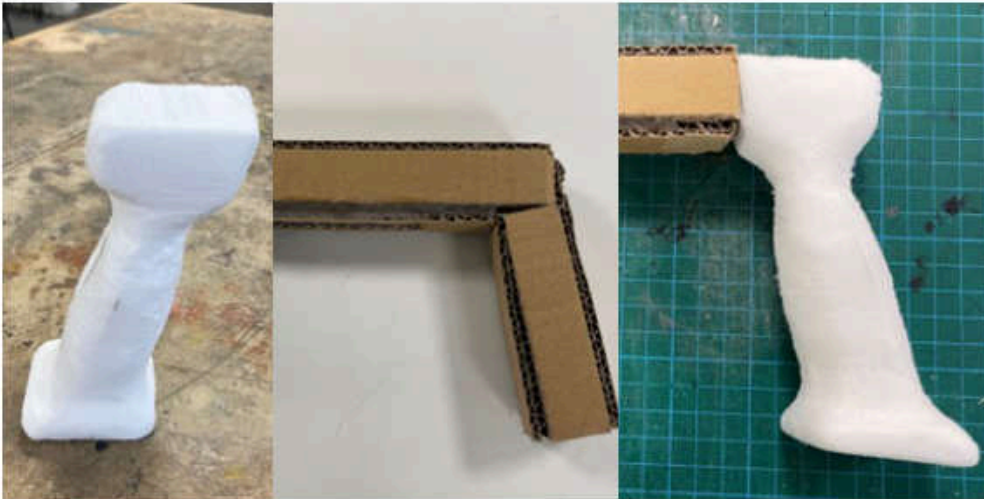
Design that is connected to the tool head and helps lock the hoop to the tool head, front part of the design allows for the hoop design to be inserted, the second part allows for the hoop to be turned so that the hoop can lock into position, design needs to be tight otherwise the hoop may slip out



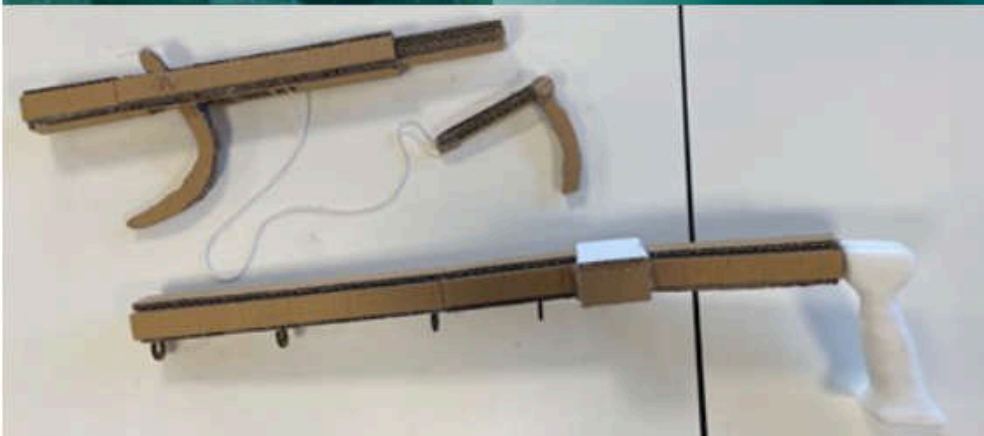
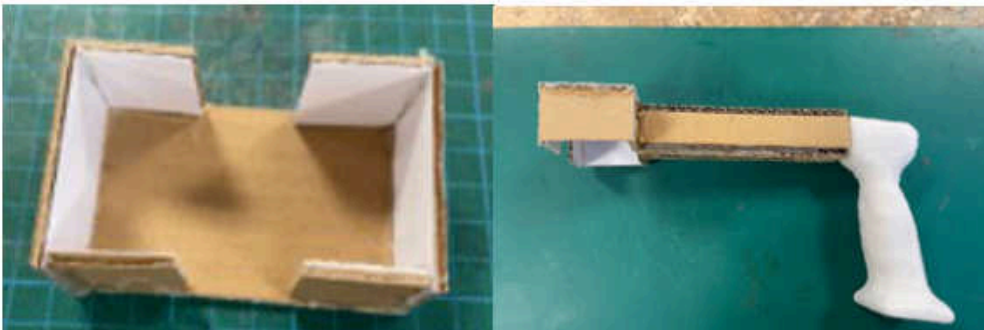
This part is connected to both ends of the hoop/rope, it allows for the hoop design to lock into place, the design should be pushed into the hole of the design and once it can not be moved further, the design should be twisted in the direction that is allowed to move so that the design will be locked in, to ensure that the design is tight and secure, the thickness of the design needs to be close to the thickness of the hole so that there is somewhat of a push fit mechanism and friction will keep the hoop locked into position so that it will not fall out

Model of final product

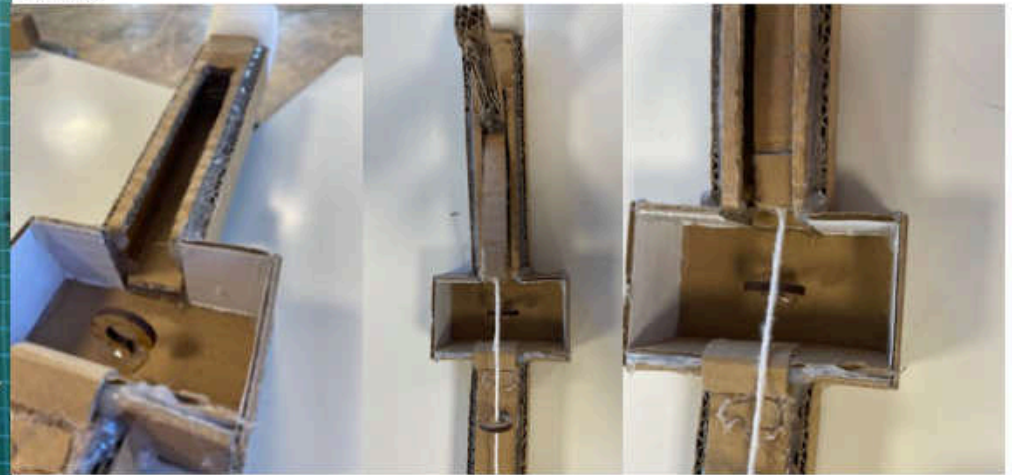
With there now being redesigns of my product which will allowed for it to be functional, I produced a model where I was able to see these new designs and test them to ensure that they effective and allowed the product to function.



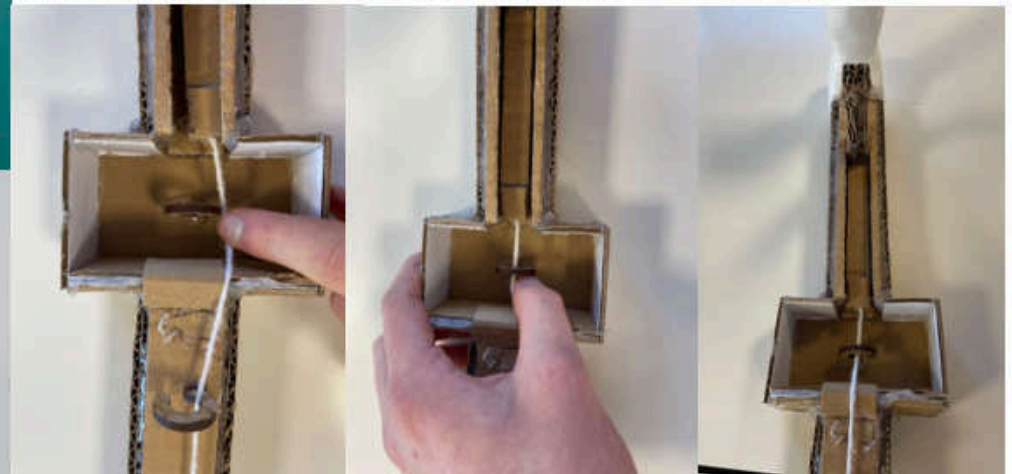
By removing the current handle of my design and adding the foamed model, I was able to add the ergonomic handle to my design which would allow for a closer look at the final product of my design.



Adding the new storage area, which is a widened area would allow for the user to easily attach the string to the hook inside which is above the storage area, so the tension is above the storage area allowing for the trigger to return to the storage area, so that the hook will remain as open as it can so that the collection of litter can be as simple as possible for the user. Aswell, I added back a hook which was broken due to the tightness of the trigger in the last model which made it difficult to remove.

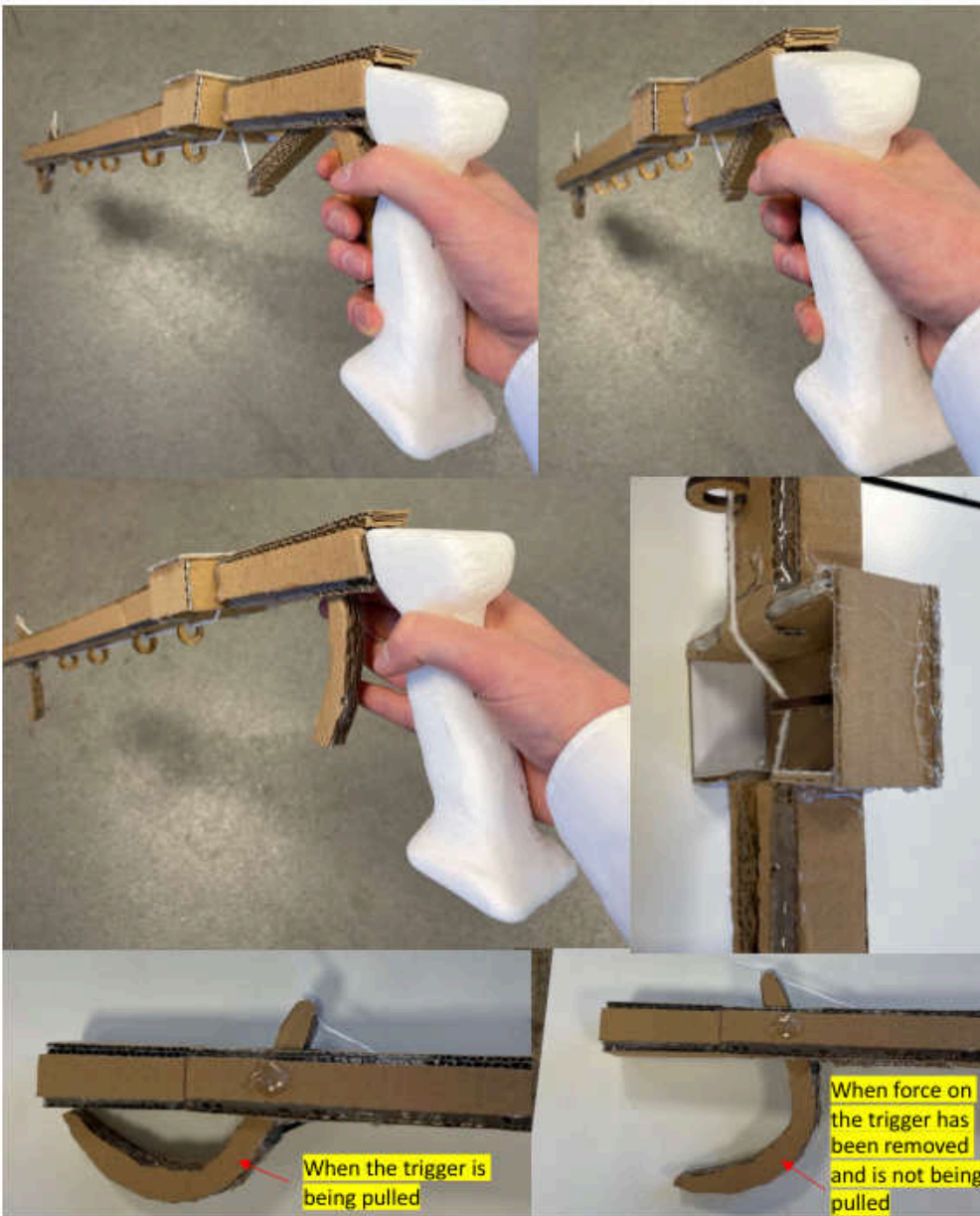


By now adding a hook into the area in front of the trigger storage area, it would now allow for the trigger to return to its original position, allowing for the claw to remain as wide open as possible. Due to the materials being weak for this model and not strong as the materials that will be used for the final product, I was unable to add the dovetail design for the trigger. Instead, I used the trigger from the last model to test if the trigger would be able to return to its resting position. The connection for the dovetail design would be similar to the one modelled, the tail would be connected inside the storage area and the pins (the trigger of the design) would slide into the tail which connects it together, making one piece and allowing for the trigger to be removable.



By pushing the trigger back into the storage area, pulling the rope to the right side of the hook and pushing the rope inside the hook of the design, the design would now be able to function properly, with the trigger being able to return to its original position which would allow for the claw of the litter picker to return to its original position and remain as wide open as possible.

For this model, I didn't add the locking mechanism that would be on the head of the tool due to it being intricate and precise which is unable to do with cardboard and requires to be 3D printed. This mechanism will be tested in the future of the design process.



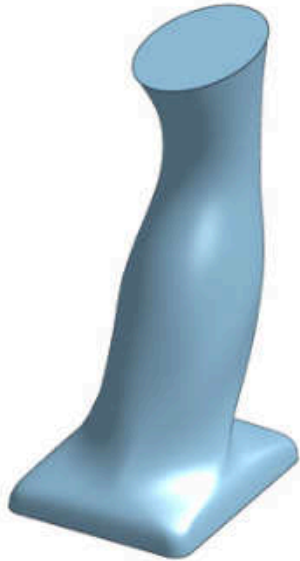
With this final design of the litter picker head tool and body, the design of the tool is functional due to the hook of the design being above the resting position of the trigger, this would then cause trigger to return to its resting position as the point of tension is above the storage area and is located at the hook, in front of the storage area. With the help of the spring that would be located behind the claw of the litter picker, when the claw is closed and the trigger is pulled the spring would be extended, and when the trigger has stopped being pulled, the spring would return to its original shape, making the claw open, pulling the trigger back up to its original position. With the design also keeping tension in the rope so that when the trigger is pulled, the rope would not be weak or have extra length, keeping it tight and functional.



This model allowed for me to test the features of my design to ensure that the litter picker tool and the body of the design would be able to work together, due to the litter picker head being the only tool that requires a trigger to be attached to the body. The net head tool and the scraper/sifter/scooper head tool only require a connection to the body and can function well. This model being an accurate size at 840 mm allowed for an accurate model with features that could be tested. Testing these features will allow me to incorporate them into my design so that the product is effective. The real size model allowed me to test how the features would act in a product this size, such as the trigger mechanism and being able to test if it would return back to its original position.

Requirements for each component

Handle of the tool



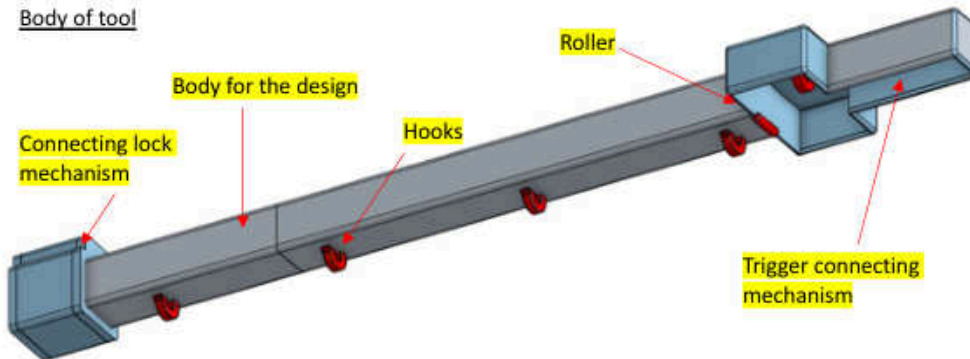
The handle of the design would have to be produced in one shape which would allow for the complex shape of the handle to be produced. For this unique and complex shape, the best process to produce it would either be 3D printing or injection moulding, this is because of the complex and ergonomic, strength and durability and surface finishing, all of which will allow for a comfortable and ergonomic handle that is suitable for users in the 5th to 95th percentile. Both allow for the intricate shape of the handle design but are different depending on the type of production.

Injection moulding for this design would be best for mass production process due to the high consistency in quality that it provides. 3D printing would be best for rapid prototyping and testing the shape of the handle when creating a prototype.

For my prototype, creating the ergonomic handle would be best with 3D printing as I am able to achieve the complex shape of the handle at a low cost and be able to achieve strength within the handle, that allows for the product to be tested and evaluated as well as being able to be produced from a variety of materials.

As shown in the shape of the modelling foam model and the connection to the body in my modelling, having a taller section above the ergonomic area the handle would allow the strength of the handle required to be achieved to ensure the design will be supportive.

Body of tool



Components of the design:

Connecting lock mechanism - this mechanism has a complex shape which would require detail and precision with the change in thickness and holes inside the design, it would therefore be appropriate for the design to 3D printed for my prototype, which would allow me to achieve the complex shape of the design and ensure that it would fit tightly into the design and size of the body

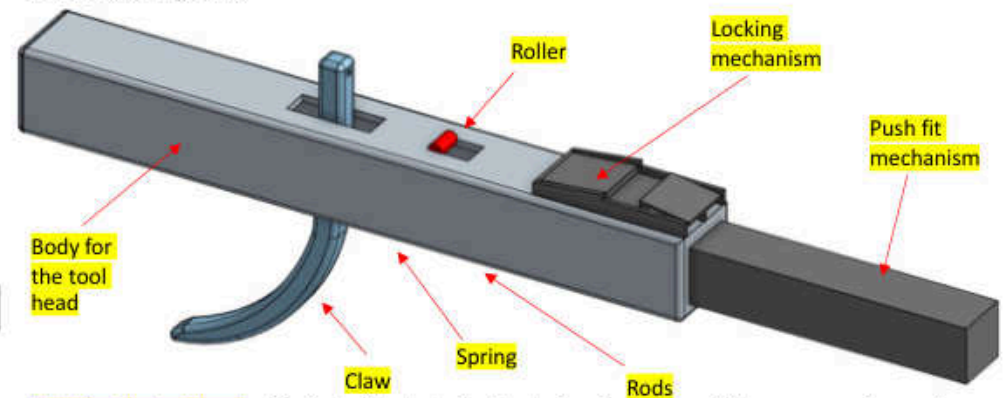
Body for the design - For the body of the design, a few requirements must be met to ensure a functional and effective product, the product needs to have high corrosion resistance due to it being out the environment, lightweight as the user would have to carry the design which if the design is heavy will cause strain and not meet the design brief and long lasting to ensure that the design will function for an extended period of time and not break or have to be replaced. This would ideally be made from a metal, as this is stronger than a plastic and can meet the strength requirements, similar to the litter picker I investigated, a metal tube would be ideal to meet the demands of the product.

Hooks - The hooks would have to be water resistant due to the top end of the tool potentially entering the water due to having a net tool; to keep the high quality of the component as well as the aesthetic of the design, all the hooks should be produced from the same material. As well the unique shape that allows for the rope to be contained and the consistency for the repeatability needing for the hooks can be done by 3D printing for the design, which would allow for consistent sized hooks which would allow for a good and aesthetic product.

Roller - The roller is required so that the rope of the design is less likely to be damaged and as well to help direct the rope so that the litter picker tool head can function effectively. Like the hooks, the design can be 3D printed which would allow for the cylinder shape to be produced, as well a polymer can be used for this which can be corrosion resistant so that the quality doesn't degrade.

Trigger connecting mechanism - The trigger connecting mechanism is a separate piece that can be added to the inside of the body, this material needs to be flexible to be able to accommodate for the turning of the trigger and to keep the trigger connected to the design. Due to the precision of the turning piece, the design should be 3D printed so that the accuracy of the design can be met and the material which would be a polymer would be suitable as it would be able to bend when pressure is applied to it

Litter picker head tool



Body for the tool head - Similar to the body for the design, it must meet the same requirements, and should therefore be the same material potentially a metal tube as this would allow for easy input of the push fit mechanism as well as the claw, spring and rods being simple and easy to connect and allowing for the rope to easily pass through

Claw - Due to the claw needing a precise sizing, it would be best for it to be 3D printed to allow for the holes and hook at the back of the design to be connected and allow for a strong connection

Spring - The spring connected at the back of the claw should be from a corrosive resistant material, to allow for a long-lasting piece that doesn't require to be changed or replaced and provides the least inconvenience for the user

Rods - The rods of the design require to be accurately sized so that it can easily fit in the underside of the design, and not angled as this would affect the turning of the claw, to achieve this the rods should be 3D printed, as this would allow for strength to be achieved and the accurate sizing. As well as being from a polymer so that the design can be corrosion resistant

Roller - Similar to the roller in the body, it should be 3D printed for the curved shape

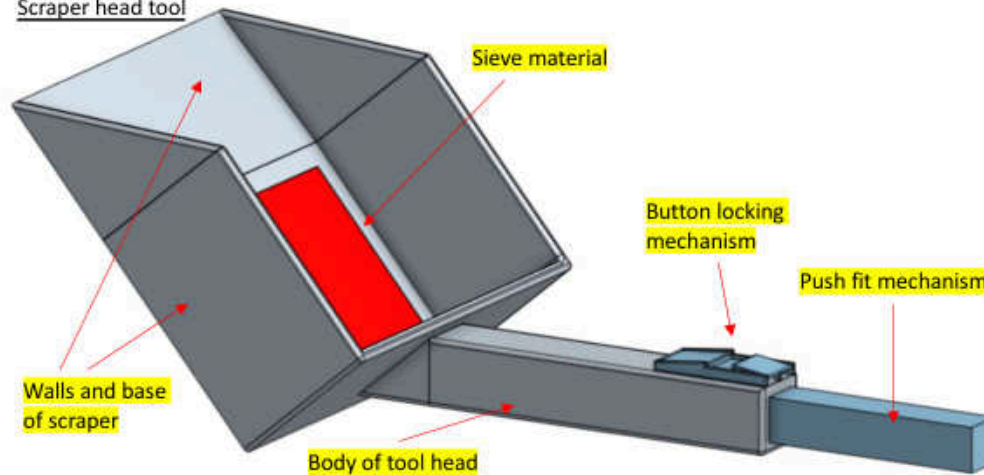
Locking mechanism - The locking mechanism has to be made from one piece of material that has a very complex shape, this complex shape could be achieved by a process like 3D printing which would be more appropriate as there are three locking mechanisms required for each tool, which would take time to 3D print, however for my prototype 3D printing would be ideal to get the small features, bend that allows for the system to lock and achieve the complex structure

Push fit - The push fit mechanism relies on friction which can be achieved by most materials, the design could be 3D printed or a metal tube, allowing for the shape of the push fit to be produced.

Trigger- Due to the ergonomic shape of the design, 3D printing would be most appropriate so that the smooth and ergonomic trigger is able to be produced so that it is comfortable for the user

Rope- The rope that is connected to the claw and trigger must be corrosion resistant to allow for the long lifespan of the design, as well as being able to deal with the wear and tear that will occur when using

Scraper head tool



Body of the tool head – Similar to the body of the design, the material for this should be the same as the body to allow for strength throughout the design, allowing for a clean aesthetic too

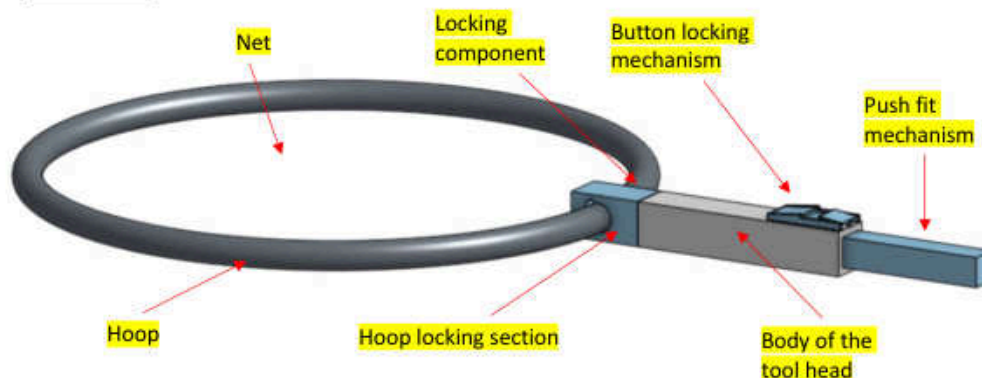
Push fit mechanism- Like the litter picker head tool this push fit mechanism can be 3D printed, or a tubing could be used, this would allow for a tight fit that relies on friction, the only requirement is that the fit must be tight for the head tool and body so that it seems like the design is just one piece.

Button locking mechanism- As discussed, this part should be produced from a polymer as it would allow for the elastic properties required, the process for this piece could be injection moulded or 3D printed, however it is more appropriate to be 3D printed for my prototype

Walls and base of scraper- This part of the design must be corrosion resistant and hard wearing to ensure a long lifespan of the design, and to prevent damage that could occur when scraping the design against the floor

Sieve material- As developed the sieve material is most appropriate if it has gaps under 5.41mm to allow for the smallest cigarette butts to be picked up yet small stones, water, sand and mud to pass through. This part of the design must be chemical and corrosion resistant to ensure that the quality of the design doesn't degrade so that litter is able to be collected

Net head tool



Body of the tool head- Like the other tool heads, keeping this part of the design the same material as the body would allow for strength and support of the litter captured and a clean aesthetic

Button locking mechanism – Like the other tool heads, allowing for this part to be produced from a polymer would allow for the elastic feature to be achieved, which can be done by 3D printing

Push fit mechanism – Like the other head tools, this part can be 3D printed or produced from another head or tubing to allow for a tight fit that relies on friction and keep the design together

Hoop locking mechanism- Due to this part having a complex shape, it would be ideal to have this part 3D printed or put together in numerous layers so that the shape and thickness for the lock can be achieved, however it is essential that it is chemical and corrosion resistant, as the design will be placed in water which would degrade the quality of the design

Locking component- The locking component too must be chemical and corrosion resistant as this would allow for a long lifespan and for the quality to not degrade, the design could be layered with pieces, or 3D printed so that the complex shape can be easily achieved and would be appropriate for my prototype

Hoop- As required for my design and for it to take as little storage space as possible, making this shape from some form of rope would allow for the bend that is required to allow litter to easily be placed, as well as being rot and fray resistant so that the lifespan of the piece is as long lasting

Net – Just like the rope for the hoop, it will get wet as this part of the design will be placed into the water, so it is essential that the design is resistant to rot and being fray resistant, as well as having a hole for where the rope/hoop can be passed through, allowing for the design to take up less storage space and be easier to carry

To allow for the accuracy required and correct material, 3D printing is a necessary option to produce parts of my design for my prototype, if my design was to go into mass production a method such as injection moulding would be appropriate due to the speed and quality at which the designs can be produced, however for my prototype 3D printing is the best choice to produce these components.

- Handle
- Connecting lock mechanism
- Hooks
- Rollers
- Claw
- Trigger connecting mechanism
- Trigger
- Button locking mechanism
- Hoop locking component
- Hoop locking section

Similar to the litter picker that I investigated it would be most appropriate for the body of the design and the head tools to be produced from metal tubing, this would allow for the most strength to support the tool heads and weight of the litter that the design would be able to pick up. Whilst also being able to deal with the wear and tear that the tool would have to deal with. With the shape of the design being able to cut and changed to allow for the design to be made.

The other parts of the design are the hoop/rope, rope for litter picker, net, spring and sieve material, these would be most appropriate to be pre-made and bought in as well as the metal tubing for the body, due it being more cost effective, having a high-quality assurance and not having the machines to develop these components.

By producing my product with a mixed approach (producing some components and buying in some components), I can produce a product that is optimal on cost, quality and efficiency to ensure the highest quality prototype or product for the user. By 3D printing, I am to rapidly prototype the components and change them, if necessary, at lower costs and at faster and more efficient rate. By outsourcing, I can have the high-quality design with components produced from manufacturers that have the machinery and expertise to provide the high-quality piece that would have little to no defects. By combining both methods, I will be able to achieve a high quality and professional design that allows a representation of what my product would be like, as well as being realistic if this product was to be batch or mass produced in order to save money and ensure a high quality of product that the user will have to ensure an easy and simple experience when removing litter.

Material research

Finding the most appropriate material for the design will allow for the tool to be used easily and perform and be efficient, allowing for an easy experience for the user.

Body and body of head tools

The body of the design and body for the head tools is required to provide strength to the tool so that litter can be picked up easily, lightweight as the user will be carrying the design around for long periods which would cause strain if heavy, able to deal with wear and tear and to have corrosion resistance as the tool will be used in the environment and will get wet due to one of the tool heads being meant for the water. The best choice would be a metal tubing which would allow for pieces to be slotted inside of the design, as well as keeping the tool lightweight and giving strength. Whilst researching there were two clear options to choose from:

Aluminium tubing:

- 2.7g/cm³ weight ratio allowing for a lightweight option that can be easily carried
- A high strength to weight ratio at 1 to 8 allowing for litter to be easily carried and stored
- An excellent corrosive resistance due to it naturally forming a thin protective layer of aluminium oxide on the outer layer, this acts as a barrier for further corrosion and increases resistance
- Easily machinable and bent, allowing for a compliant material so manufacture is easier

Stainless steel tubing:

- 8.0g/cm³ weight ratio, being a heavier option than aluminium and being more difficult to carry
- A lower strength to weight ratio due to high weight, but very high strength at 1 to 16, allowing for all litter to be easily picked and carried with very little stress on the tool
- Excellent corrosion resistance due to it containing little chromium, which reacts to the oxygen and forms an extremely strong protective layer, this acts as a barrier and prevents further corrosion and increases resistance to rust due to it containing iron
- More difficult to work with than aluminium due to the very high strength, but can still be worked with

When comparing both properties of aluminium and stainless steel, aluminium is the ideal choice for my tool due to the lightweight properties and high strength that it provides, whilst also being corrosion resistant so the tool will stay a high quality throughout its lifespan. The weight to strength ratio allows for it to be more appropriate for my product as it can be carried around easily and the user will not be as strained, the strength ratio compared to steel is lower but is made up due to the light weight of the material, allowing for the user to carry the tool for extended periods of time and having high strength to pick and carry most types of litter easily.

Spring for litter picker head tool

The spring is connected to the back of the claw inside the design of the litter picker, allowing for the claw to return back to its original position once the trigger has been pulled, due to this spring being consistently used it is essential that it has a long lifespan, as well as being long enough so that the elastic limit is not reached as otherwise the spring would have to be replaced as it will not return to its original position. When researching the materials that could be used for the spring there was one option that was superior to the others:

Stainless steel spring:

- Strength so that the claw can withstand the stretching and not break or reach the elastic limit
- A very high corrosion resistance so the quality of the spring will be easy to maintain, allowing for it to perform at a high quality, a low maintenance is required for this spring
- A very high durability so the spring can be repeatedly compressed and extended without cracking or breaking, able to prevent fatigue cracks so the quality of the spring will not degrade

Using a stainless-steel spring in my prototype would allow for the repeatability of the litter picker tool head to be achieved so the claw and trigger can return back to their positions.

3D printing components

By making all the 3D printed components from the same material, it allows for a simplified design process that reduces cost as buying in bulk can be done for the material, ensures consistent results for the design and reduces manufacturing time as the material for the 3D printing or injection moulding would not need to be changed or reheated allowing for a continuous production. If this product was to be produced for batch or mass production, it is likely the components would be injection moulded due to the time saving aspects of this process, and as there are many components that are requiring this process, 3D printing would take too long and just wouldn't be optimal for the design process. Injection moulding would allow for the accuracy of the component to be achieved as well as being a faster process to meet production demands. The demands for these pieces are strength and being able to withstand the forces that will be placed upon the tool when it is used. When researching the materials for my product, there were two options to choose from that would be appropriate for the design of my product:

ABS:

- Good strength so is impact resistant and will not break when used
- Can handle forces being applied that would bend the material without breaking, would be able to be the button locking mechanism easily
- High chemical resistance to acids, alkalis and various solvents so the quality of the component will not degrade over time and will stay a high quality, increasing lifespan of the product, allowing for a better experience for the user
- Considered easy to injection mould, allowing for the design to be easily achieved and consistent
- Considered difficult to 3D print due to the tendency to warp, shrink and require precise temperature control which would affect the quality of the finished component

PLA:

- Moderate strength, can break under stress and is weak and brittle in some situations
- Low flexibility meaning the button connection component may be difficult to produce and design may need to be changed to meet requirements
- A low chemical resistance due to it being produced from cornstarch and sugarcane, does not decompose effectively in water, allowing for it to be appropriate for a prototype
- Considered easy to injection mould due to its low melting point, allowing for the design to be easily achieved and consistent
- Considered very easy to 3D print due to it having minimal warping and low melting point, allowing for the design of the components to be easily achieved

When comparing the characteristics of ABS or PLA for my design, PLA would be more appropriate for my prototype that I will produce due to it being cheaper and easier to 3D print, allowing for the components to be produced easily, the low chemical resistance and decomposing of the material would take time to happen and I would be able to test the features to see if they function effectively before the quality decreases, allowing for it to be appropriate for my prototype. Whereas ABS would be more appropriate for a final product, that is being batch or mass produced and done in the polymer process of injection moulding, due to ABS having a higher strength than PLA so the components are less likely to break or snap, having a very good chemical resistance so the quality would not degrade and being easy injection moulded which would allow for a quicker production of the components.

For my prototype, the components will be produced by 3D printing and would be made from PLA as it is ideal for modelling and prototyping whereas for the final product, if the design was to be batch or mass produced it would be injection moulded, the material would be ABS due to the ease of injection moulding and strength that it would provide to the components, allowing for a longer lasting product that is less likely to break and degrade in quality.

Sieve material

As investigated, the sieve material must have a distance smaller than 5.41mm x 5.41mm to allow for the most common and one of the smallest types of litter being cigarette butts to not fall through and remain stored. Due to the mesh allowing water through it, the mesh must be corrosion resistant in order for it to maintain its quality and not degrade. When researching the materials for this mesh, one material was superior to the others:

Stainless steel mesh:

- Corrosion and rust resistance so the component will remain a high quality and not degrade
- Strength in order for a litter to be supported so the mesh will not be damaged
- Can be easily cleaned due to having a smooth surface, so blockages between the wires of the mesh can be easily removed, allowing for better water flow

Using a stainless steel mesh will allow for support for the litter captured and low maintenance.

Rope for the tool heads

The tool heads that require rope to function are the litter picker tool head where the rope is connected between the claw and the trigger and the net tool head where there is a net made from a rope and a rope passed through the top section of the net so that the net can be connected to the body of the tool head. The rope must be able to resist moisture so that mould and rot is prevented in order to keep the high quality of the net, as well the rope shouldn't fray as this would degrade the quality of the rope. When researching options for the rope there were three options to choose from:

Polypropylene rope:

- Doesn't absorb water or moisture so will resist rot and mould, keeping the high quality of the product
- Will float when placed in water, so will not absorb the water, allowing for the weight of the rope to remain light, causing minimal discomfort for the user and decreasing strain. Being able to float allow for the rope to remain visible too so the user can clearly see where the rope is when capturing litter
- Lightweight so the net will not put strain on the user
- Cheap so the rope can be easily replaced by the user if the rope was to be damaged

Polyester rope:

- A strong and durable rope, allowing for the litter to be easily carried without causing damage
- Doesn't absorb water, so will resist rot and mould, keeping the high quality of the product
- Sinks in water due to having a greater density than water, meaning that the net will not be as visible when using the tool, making the removal of litter more difficult for the user
- Not that expensive so the user is able to easily replace the rope if it is damaged and decreases in quality which affects the performance of the tool

Nylon rope:

- A very strong rope, stronger than polypropylene and polyester, allowing for any kind of litter to be easily picked up and removed from a watered environment
- Resistant to rot and mould allowing for the high quality to be maintained and not degrade so that the part does not have to be replaced
- Nylon absorbs water which would increase the weight of the rope, and therefore increase the weight of the tool which could put more strain on the user and be more uncomfortable
- Nylon is also more expensive than polypropylene and polyester so the rope may be more difficult for the user to replace

When comparing the characteristic of each of the materials, a polypropylene rope is most ideal due to it having enough strength to support the weight of the litter captured, being able to float when placed in water so the user can easily see and target litter, being lightweight and remaining lightweight as it is water resistant so will not absorb water and increase the weight of the tool and resist rot and mould so the quality will remain high and not degrade.

Materials that will be used for my product

For my prototype I will use the following materials:

- Aluminium for the body of the design and the tool heads due to it being lightweight and strong
- Aluminium for the rods, providing strength and allowing the claw to rotate
- Polypropylene rope for the parts that use rope as it is water resistant and lightweight
- Stainless steel spring to allow for repeatability and the quality to not degrade
- Stainless steel mesh to allow for flow of water through the mesh and low maintenance
- PLA for the 3D printed components of the design due to it being easy to 3D print and model the components of the design, however if the product was to be moved into batch or mass production these components would be injection moulded and from ABS due to the high strength

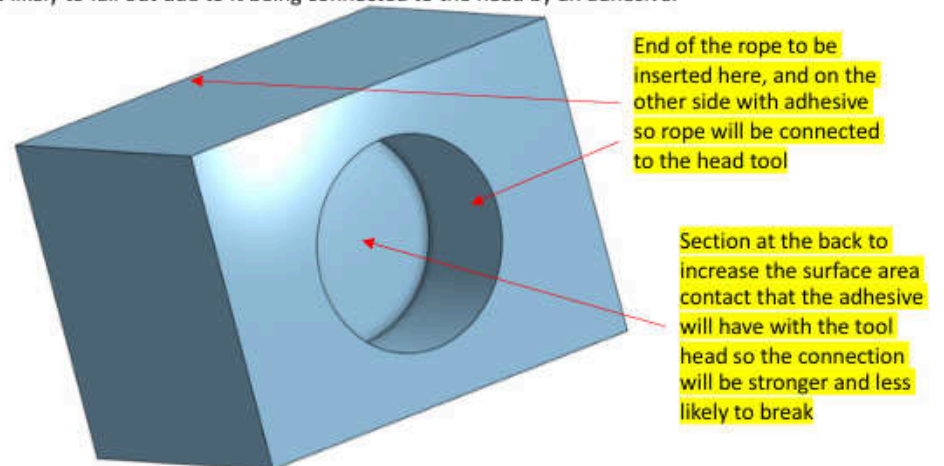
By using these materials for my design, I will be able to achieve a professional and clean looking product that is able to function effectively due to the appropriate materials being used

Development for problem of net design

A key feature for the redesign of the net head tool was to reduce the space required for storage so that transportation with the tool was easy, initially I solved this by making the head removable and being produced from rope that had a lock mechanism that would be pushed into the head and twisted. With this design problems could occur, such as the rope slipping out which could result in the head falling into the water, the tension on the locking pieces would be extremely high for there small size if a heavy piece is captured, as they are supporting, and locking the rope into place and the force is applied on the locking mechanism which could easily break.

For an improved design, so there is little tension where the locks are placed to prevent breakage of the tool head, the end of the rope could be placed into the end of the design and an adhesive could be applied in order to connect the two pieces. This design would still be able to meet the need to reduce storage space as the hoop will be produced from a polypropylene rope, which is a compliant material and able to bend easily, this would then allow for easy storage as the design would be able to fold and bend, reducing the storage space used.

This re-design would make the replacement of the hoop or net more difficult as the ends would be glued to the head and are unable to access, the rope could be replaced if the end of the rope is removed by a force and the correct tools are used, which would be an inconvenience for the user however this design would allow for a much stronger and more reliable design as the rope will be less likely to fall out due to it being connected to the head by an adhesive.



This piece can be 3D printed for my prototype and would be injection moulded if the product was to be batch or mass produced to meet demands. The design allows for a much stronger connection between the hoop and head which is less likely to break and can still be easily stored due to the hoop of the tool being from polypropylene rope which can be bent to store.

Final design before testing

My final design for the product before testing is a changeable head litter removal tool that has heads that can be connected, disconnected and swapped in to help the user remove litter from an environment as the tool heads are specialised to meet the demands in a range of environments. The tool heads can be connected by pushing the tool head into the body of the design, and to be removed the button must be pushed down and held and then pulled out, allowing for a simple connection and removal so the change of heads is efficient. Once the litter has been captured the litter can be placed inside a bin bag and removed off appropriately. The tools heads are:

- **a litter picker head tool** which will be the most common tool head used due to it being able to be used in nearly all environments and functions by squeezing the trigger which closes the claw onto the body of the tool head which captures the litter. This is the only design which requires a trigger to be attached to the body of the design or any setup to ensure the design can function; the tool head is pushed into the head of the design, the rope connected to the trigger is pushed through the hooks on the underside of the body, the trigger is then pushed into the dovetail connecting mechanism, the loose rope near the trigger is then pushed into the hook, the litter picker is now able to be used to remove litter from an environment. To remove the tool head, undo the steps and squeeze the button and pull on the connecting mechanism
- **a net head tool** which allows for litter in a watered system such as a river, lake or pond can be captured by passing the net around the litter, which can often be difficult to remove with a tool such as a litter picker due to the litter being wet and difficult to grip
- **a scraper head tool** that functions by the user pushing the design under the litter, allowing for large amounts of litter to be captured at one time, saving time. The design is most effective for smaller objects such as cigarette butts which are often very difficult to remove with a tool such as a litter picker due to the small size of the cigarette butts, this tool can be scooped under the butt and captured. The design also has a wire mesh in the centre which allows for sand, water, small stones and litter to pass through so the only item that is captured is the litter

Handle



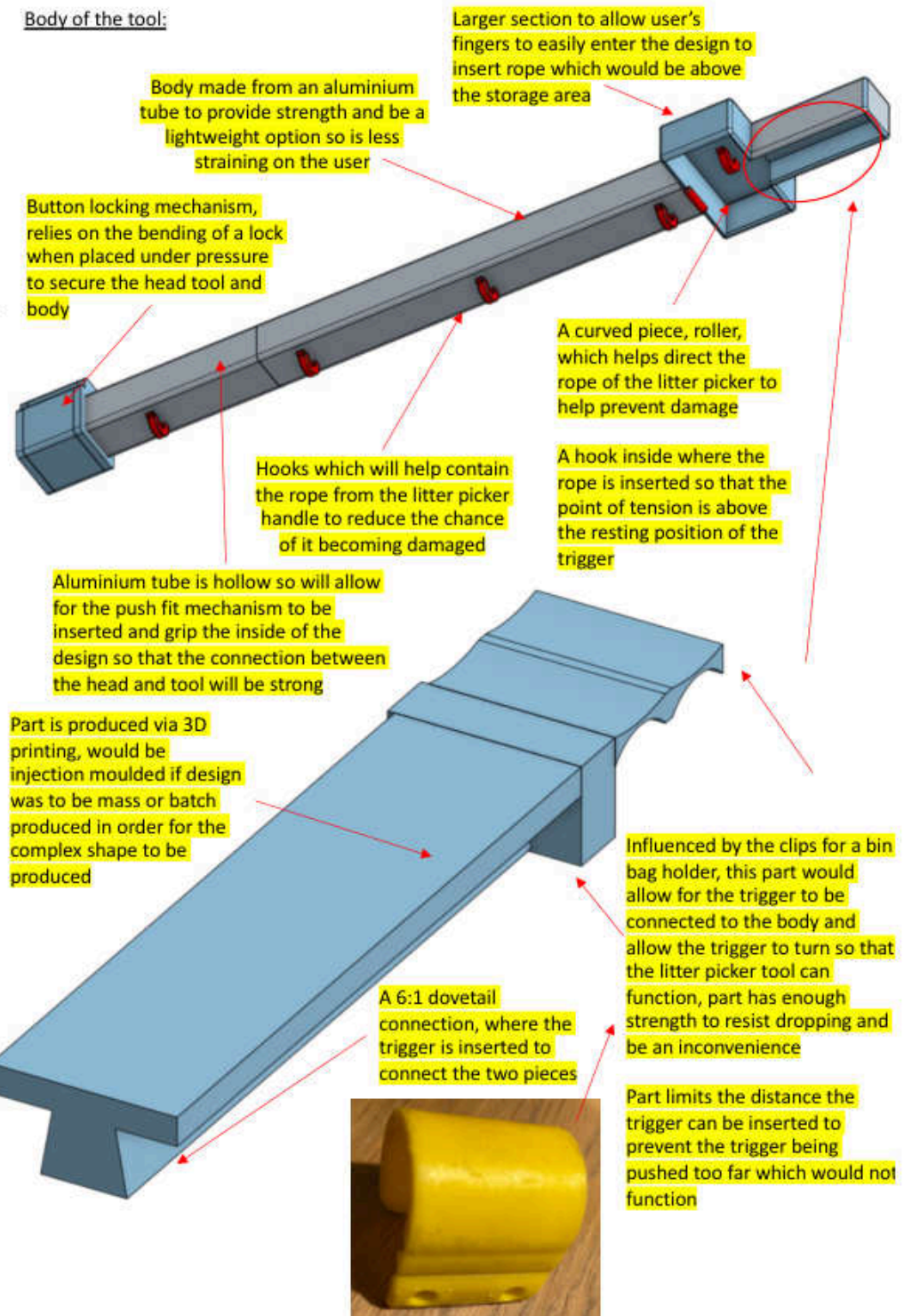
This part of handle will have a hole in so that the body made from an aluminium tube can be inserted, allowing for a tight fit and a secure connection

Ergonomic handle, allows for a comfortable grip on the tool to reduce strain when the user is holding it

Handle to be 3D printed and be produced from PLA for the prototype, if design was to be batch or mass produced the handle would be injection moulded so that the production demands would be met

Modelled handle to be cut into key sections, measured and placed onto CAD so the design can be transferred to 3D print

Body of the tool:



Larger section to allow user's fingers to easily enter the design to insert rope which would be above the storage area

Body made from an aluminium tube to provide strength and be a lightweight option so is less straining on the user

Button locking mechanism, relies on the bending of a lock when placed under pressure to secure the head tool and body

A curved piece, roller, which helps direct the rope of the litter picker to help prevent damage

Hooks which will help contain the rope from the litter picker handle to reduce the chance of it becoming damaged

A hook inside where the rope is inserted so that the point of tension is above the resting position of the trigger

Aluminium tube is hollow so will allow for the push fit mechanism to be inserted and grip the inside of the design so that the connection between the head and tool will be strong

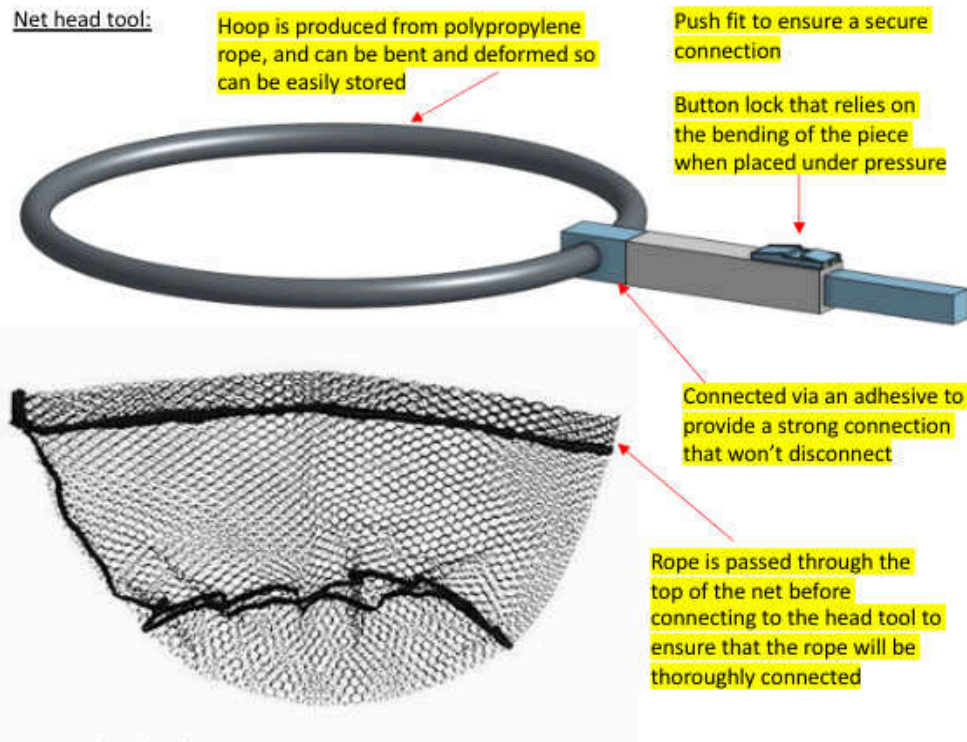
Part is produced via 3D printing, would be injection moulded if design was to be mass or batch produced in order for the complex shape to be produced

Influenced by the clips for a bin bag holder, this part would allow for the trigger to be connected to the body and allow the trigger to turn so that the litter picker tool can function, part has enough strength to resist dropping and be an inconvenience

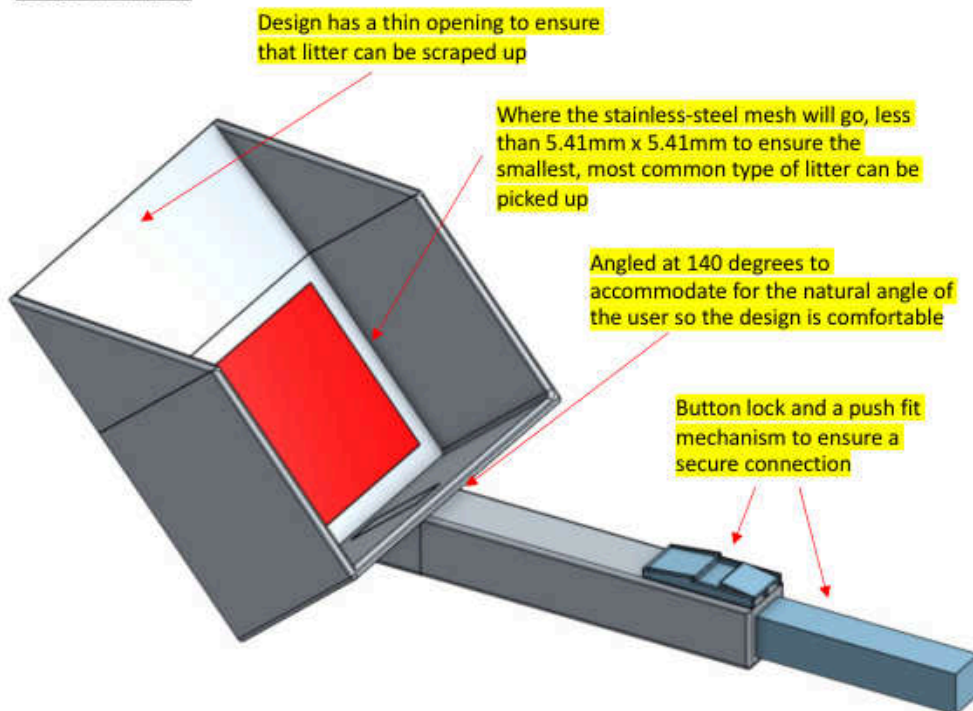
A 6:1 dovetail connection, where the trigger is inserted to connect the two pieces

Part limits the distance the trigger can be inserted to prevent the trigger being pushed too far which would not function

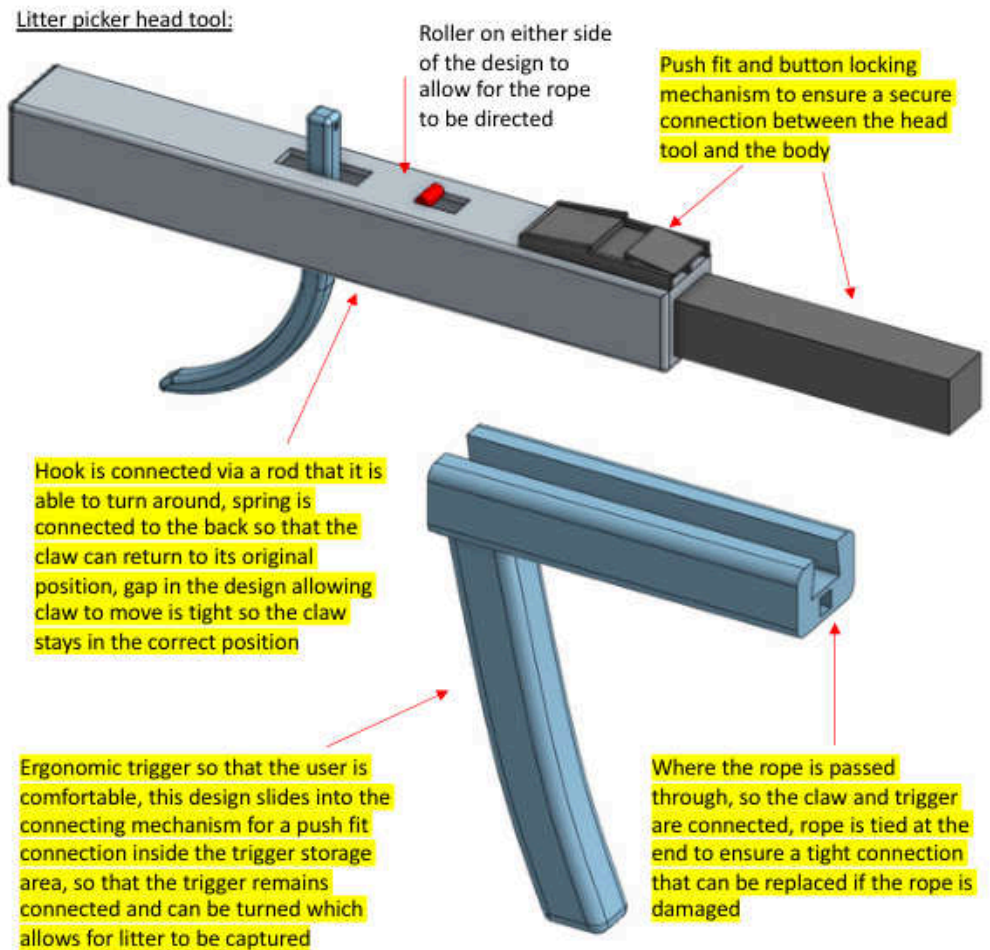
Net head tool:



Scraper head tool:



Litter picker head tool:



Client check-up:

By checking up with my client I am able to show her my design intentions for my product, I then asked her how she thinks this would affect her removal of litter and if anything needed changing. My client said that this product would help with the removal of litter due to the heads being developed for certain scenarios

Design brief:

Design a product that can help reduce pollution and litter in a wide range of environments and terrains so that the removal of litter can be done more effectively and be easier for the user. The product should not cause strain on the user or be difficult to use and should provide the user with a simple and easy experience so that litter can be removed more effectively.

This final design fully meets the requirements of the design brief, as it is able to reduce the litter within a range of environments by being able to collect it and remove it as it can be placed inside a bin bag and be removed appropriately by tools that have been specialised for certain conditions and environments allowing for a more effective removal of litter. As well, the product has ergonomic parts which have been designed to meet a wide range of users and reduce the strain on the user.

Industrial manufacture

If my design was going to be mass/batch produced, the manufacturing process would be different to the process of producing my prototype. This is due to many factors such as:

- **Cost** – If the product was to be mass/batch produced, it would have a higher funding, allowing for an availability on different tools, processes, equipment and skilled manufacturers. This would allow for the product to be produced at a higher quality and at a faster rate, allowing for a better product and for a lower cost per unit produced compared to my prototype
- **Processes** – Automated machinery is likely to be used, allowing for fast repeatability of products with high accuracy as it is produced from machinery, compared to my prototype which would take longer due to development and not being as efficient as the machinery
- **Time** – As a batch/mass production is fast once it has been set up due to machinery, allowing for a more efficient production, whereas with my prototype the design will take time to produce due to testing and manual labour taking place, however the batch/mass production may be difficult to change if an alteration to the design has been made
- **Skill** – As the skill of the machines will be higher and funding for the product would be higher, better tools, manufacturing processes and workers (who can be trained better), the overall product and consistency of the product would be high, whereas for my prototype would be a lower quality as I have less training on the processes, and fewer equipment to achieve the high accuracy product that would be achieved in mass/batch production

Components of my design

For my planned design so far, the design and components may change when testing the product as the component doesn't work effectively or would cause a difficulty in producing the prototype. For my design there is mostly two types of components, the 3D printed components and the bodies of the tool heads and main body of the design.

3D printed components for my prototype:

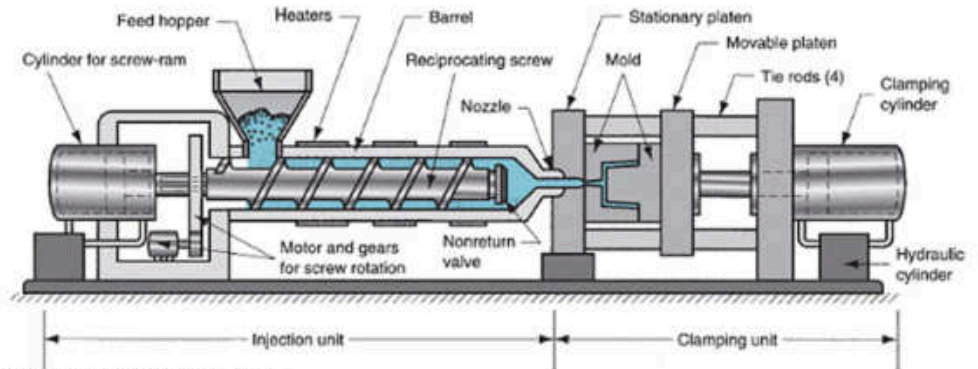
- | | |
|---------------------------------------|--|
| - Handle | - Trigger connecting mechanism |
| - Button for lock mechanism | - Trigger |
| - Connecting piece for lock mechanism | - Hoop locking section for net head tool |
| - Hooks | - Claw for litter picker head tool |
| - Rollers | |

For my prototype, I will be 3D printing the components for my design due to their complex and unique shape which would be difficult to produce otherwise, I am producing them from PLA due to it being easy to print, relatively cheap and being environmentally friendly as it is produced from renewable sources such as cornstarch. However, PLA is not a good shock absorber and may break if used incorrectly which would result in broken components and the design not functioning effectively.

Therefore, if my design was to be mass or batch produced, it would be more appropriate for the design to be produced from ABS. By the process of injection moulding. ABS would allow for the components to function effectively as it has shock absorbing properties, decreasing the chance of the components breaking or deforming, as well as it having enough flexibility so that the button for the lock mechanism is able to function effectively.

Injection moulding would allow for the repeatability of the components to be produced, and the short time taken to produce these components would allow for a continuous production to meet the greater demands of mass production. ABS is a thermoplastic so is suitable to be injection moulded, as well it can be pigmented, reducing the time taken to finish and make the components aesthetic as they can come out of the injection moulding machine with an aesthetic and glossy finish. The complex shape of the designs can be achieved due to the polymer being heated and being pushed into all gaps inside the design, this would therefore result in all aspects of the design

being achieved and produced, allowing for the components to be accurately produced. The complex shape of the trigger connecting mechanism (the thin section inspired by the hooks from the bin bag hoop) and the unique shape of the button for the lock mechanism can be easily achieved. This would then result in simple manufacture of the design, as the pieces are self-finishing, and can then be taken to connect to the body of the design. In conclusion, an injection moulding process for the components of the design if it were to be in the process of batch/mass production is the most appropriate process to produce the components.

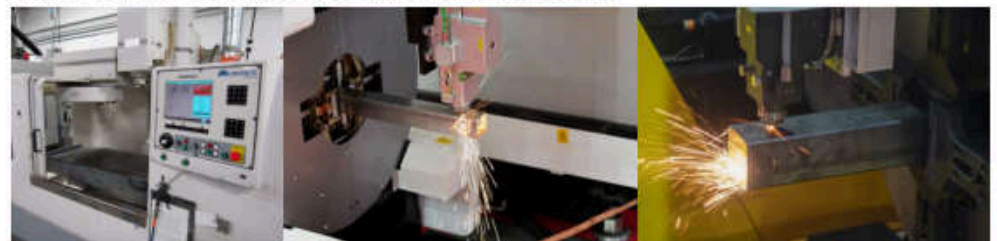


Body and body for tool heads

For my planned design so far, the tubing for the body and for the litter picker tool head are required to have holes cut into where components will be placed on the inside of the design, for my prototype it is most appropriate to use a milling machine to cut the holes due to the high accuracy that can be achieved when cutting a hole in the design. However, if my prototype were to be moved to batch/mass production then there are a few options that could be chosen:

- **Stamping with progressive die** – Able to punch rectangular holes into the tube in a single movement, can be done through an automatic press system so less effort work is required, able to be very fast and is cost effective for large batches. The punch can sometimes cause damage to walls which are too thin, and can be difficult for walls that are too thick
- **CNC milling** – A high precision cut that allows for repeatability when cutting as the machine will stay in line, and the X-axis of the machine can be moved to cut the design. The only problem with this is that the edges will have a circular profile which will need to be adjusted for it to be square
- **Laser cutting** – Able to cut rectangular holes inside the tubing, can handle different sizes and holes very easily, able to complete complex hole shapes for high precision work. The only problem for this method is that it is the slowest of the options, as well the set up cost is very expensive. However, numerous pieces can be cut at the same time, allowing for the head tool and body to be cut at the same time, allowing for a faster production

When analysing the different types of processes that I could use, laser cutting or CNC milling makes the most sense as the cuts for the holes would be very accurate and precise, allowing for a repeatable process of cutting holes into the design. As well, the laser cutter could also cut the length of the tubing, decreasing the time taken to produce the product and tools required as the laser is able to easily and precisely cut the required parts of the tubing.



Rods and scraper head tool

The rods that will be inside the litter picker head tool and will allow for the litter picker claw to rotate easily around the design, as well as supporting the spring that will be connected to the claw, which will allow for the claw and whole system of the litter picker head tool to reset and allow for repetitive movement. The spring will be connected to one of the rods by one of the ends being hooked around the rods. As well, the scraper head tool would have to be produced from an aluminium sheet, then bent and welded in order for the complex shape to be produced. If my design were to be batch/mass produced, the rods would be produced from aluminium, and would be welded to the inside frame of the tubing, this would allow for the claw and spring to be secure and have maximum strength. Welding would also allow for the scraper to achieve its unique shape yet still obtain its strength and rigid qualities required for the tool to be effective. As well as having a smooth finish which would provide an improved aesthetic. The types of welding that could be used for the rods and scraper head tool:

- **TIG Welding** – Allows for precise control over-heat input which is essential as the tubing could burn through if the temperature is too high, able to achieve clean welds which would allow for a clean and simple aesthetic. However, it can generally be a slow process and takes time which could cause disruption for a batch/mass production line, as well it requires high skill in order for the connection to be clean, aesthetic and strong
- **MIG Welding** – Allows for relatively fast welding to occur, meeting the demands of a batch/mass production line and is often simple to learn so skill required for the tool is less and has a continuous wire feed meaning a simple experience. However, this welding provides less control which could result in burn through occurring, leading to a damaged and unsellable product and may provide a product which isn't as aesthetic
- **Laser welding** – Allowing for extremely precise welding, meaning for an aesthetic and clean finish that is pleasing, as well is extremely fast allowing for it to meet the demands of a batch/mass production. However, it may be difficult to connect the rods to the inside frame of the tool head body, as well as being very expensive

When analysing the different types of welding methods that I could use for rods and scraper head tool if my design were to be batch/mass produced, **laser welding** seems to be the most efficient method to choose from due to its high precision and high speeds, allowing for the tools to be produced at a fast rate and meet the demands. However, **TIG welding** would also be another appropriate process that I could use due to the high precision too, as well the laser may not be able to access the angles required for the welding as easily as with TIG welding, so that the strength and connection of the pieces is solid and secure. Tests may be done to test which manufacturing process is the most appropriate to ensure a high-quality design of the product.



Additionally, **injection moulded components** such as **surface finishing techniques** like **texturing for the handle** are key so that the can maintain large amounts of grip, as well for the aluminium body a process such as anodising would allow for an improved and more protected product. Furthermore, the body of the design and tool heads could be manufactured by processes such as seamless tube manufacturing and welded tube manufacturing which would allow for in-house production, resulting in a cheaper manufacturing for the tool as the tubing wouldn't have to be bought in, reducing the costs and reliability of others if the product was to be mass/batch produced.

Material research – Adhesives

In order to connect the components and pieces together into my design I will have to use an adhesive, this will make the component remain connected to the tubing and is the most appropriate method to connect the complex components to the tubing as their shape and size will not be changed or damaged. If I used another joining method such as screws, the components would get damaged and not be able to function effectively as they are too small and sensitive, making it an inappropriate joining method and one that mustn't be used in order for the high quality and professional design to be maintained. The adhesive that I will use must be able to connect the desired components to the main body, the tubing, with a strong and long-lasting connection, this would then allow for the lifespan of the product to be improved and if the components are connected appropriately will then result in a good and effective product. Not only does the adhesive need to be strong, but it must be water-proof as parts of my design will be used in the water as well as being able to easily connect PLA or ABS components to aluminium tubing.

When researching different adhesives I could use for my product, there were a few options that I could choose from:

- **Epoxy resin** – Provides a very strong and durable bond that will allow for the components to remain durable and well-bonded to the tubing. Can connect PLA or ABS to aluminium tubing effectively which would result in the components being able to function, as well as being water-proof so the quality of the connection will be difficult to degrade over time, helping keep a high-quality design. A few problems that can occur with this adhesive is the slow curing time to properly connect the pieces and it not being very UV resistant which can result in discolouring and may degrade slightly
- **Polyurethane adhesive** – Provides a strong and flexible bond for the PLA or ABS to the aluminium tubing so the components will remain connected and functional. This adhesive is water-proof so the high quality connection will be maintained, it is also very good at connecting different materials together which would allow for the PLA or ABS components to maintain a strong bond. A few problems with this adhesive is that it expands when it cures which may move some components out of the desired place making the product not as accurate as it has been wished to be, as well it can often be messy and can sometimes require clamping which may be difficult to do for the smaller components of my design such as the button connecting mechanism
- **Cyanoacrylate adhesive (super glue)** – Very fast curing, able to be applied to small areas and intricate designs, as well as being waterproof when used with a primer. However, the bond of the adhesive is often brittle and can break under stress or vibration which would lead to a damaged product, as well for the adhesive to be strong and connect to aluminium the surface would have to be primed so that the connection can be secure



When comparing the different adhesives that I could use to connect the components of my design which would allow for it to be functional, all the adhesives researched would be appropriate however **epoxy resin** seems to be the most appropriate due to it having the strongest bond out of the adhesives which would allow for the design to function effectively for a long lifespan of the product, as well the lack of prep required for this adhesive will allow for it to be easily and accurately applied even with a lack of skill.

Manufacturing specification

As I have now completed most of the development of my product (further development may be necessary when testing the components and how the tool functions to ensure that the product functions with maximum efficiency) I am able to write a manufacturing specification, outlining the check points and key parts of how the design will function and perform.

Function of the product:

The product that I am going to produce is a multi head litter removal tool, which is heavily inspired by a litter picker, however with this tool the heads are able to be changed to accommodate for the different scenarios that the user may face in order for litter to be removed from environment effectively. The different heads that are offered is a litter picker tool head, a net tool head and a scraper tool head.

I have chosen a litter picker tool head to be one of the heads due to it being an appropriate tool for most circumstances when removing litter, which is why all of the users from the questionnaire and my client/expert chose to use a litter picker, as it can be used in various scenarios. I chose to do the net tool head to allow for litter inside a watered environment to be retrieved, allowing for the user to easily retrieve litter, minimising the litter that cannot be captured by the user which results in a decreased environmental impact, this head targets the problems faced by my client/expert where they found it difficult to remove litter from a watered environment with a litter picker due to the lack of grip that could be achieved with the picker. The last head was a scraper head, this allows for litter that is stuck to the ground to be scraped up due to its angled and thin size which allows for the tool to get underneath the stuck litter and remove it, the tool also has walls, allowing for large amounts of litter to be stored in one movement and decreasing the time taken for the litter removal process, additionally there is a 5mm wire mesh gap which will allow for even the smallest types of litter such as cigarette butts to be stored, this was decided in my cigarette size investigation during my development of the head tool, yet still allowing water, mud, small stones and sand to pass through. This means that this tool could be used in a sanded environment such as a beach, to capture litter that is stuck withing the sand as the tool can be pushed into the sand and is able to pass through, allowing for only the litter to be removed and not the sand with it.

With the combination of these three tools heads, the user can capture nearly all litter when removing litter from a range of environments successfully. This will allow for the product to be functional and for the user to be effective at the removal of litter within an environment, as the tools heads can be easily changed due to the button locking mechanism and the heads are designed for specific instances and circumstances for the removal of litter. In order for the product to be effective and meet the demands of the design brief and specification, the tool must be able to remove litter from the environment. From my clients needs, the different tool heads focus on the problems that they faced and allow for the client to have reduced negative impacts when removing litter from an environment.

Performance requirements of my product

With the different heads that the user is able to choose from, litter is able to be retrieved from numerous different scenarios and environments to ensure that the litter can be effectively removed. For example, the scraper tool is able to directly focus on the problem that my client/expert faced when removing litter, where they found it difficult to remove cigarette butts due to their small size. However, this tool should be able to get underneath the cigarette butts and retrieve them so that they are not left in the environment, causing further environmental damage and problems. As well, the net tool head focuses on the problem of my client/expert where they found it difficult to remove litter from within a watered environment, as the net is able to engulf and capture the litter, allowing for it to be removed. The performance of the tool allows for changeable heads to focus on certain scenarios faced when removing litter to ensure that it can be done effectively. From my designing

and development of the product the product has been made to be effective by the user of lightweight durable materials, and simple to follow yet reliable mechanisms that will allow for the tool to be used effectively and with ease.

Materials and product features

The main body of the design is produced from aluminium tubing, resulting in a lightweight product that will not degrade in quality, helping to sustain the high quality of the product and resisting being damaged as it has hard and tough qualities, allowing for it to resist scratches and dents to the design. The complex components are produced from PLA (ABS for the real product) due to the unique shape being able to be achieved when designing my prototype, this allows for the components to be effectively sized and produced in order for the prototype to have a high quality. These materials will allow for the complex design to be produced and result in an effective product and was decided in my materials research page where I discussed various materials which could be used for the design.

User requirements

The product allows for an ergonomic and comfortable fit when the user is holding the design due to numerous testing and development of the handle in my development pages, this helps to reduce the strain on the user, increasing the comfortability of the product, which may make the process of removing litter easier, resulting in litter being removed more efficiently and reducing the time taken. The handle and sizing of the tool is able to accommodate users in the 5th to the 95th percentile of users, resulting in 90% of those who use the tool to find it comfortable, focusing on the main user of the product. The product meets the needs of the target market, as the problems found out in my questionnaire and client interview, have solutions which are accommodated by the different tool heads. By manufacturing the handle on a 3D printer, I am able to produce a high quality and accurate handle that will allow for the prototype produced to be an accurate version if the final product, this would also allow for the disadvantages of the product to be identified too.

Sustainability

As my prototype will be produced from recyclable materials, which are aluminium and PLA (ABS if the product was in batch/mass production), the product can be recycled and disposed off appropriately if the product has broken or become damaged, limiting the environmental effect caused by the product. Furthermore, the product helps sustain the high quality of the environment by removing litter from the environment, decreasing the environmental effects that it will cause which is essential as it is limiting the effect of pollution which is the key point of my design. With the resources such as aluminium being finite, being able to recycle the tool helps sustain the quantity of aluminium, decreasing the amount that has to be extracted and helping to decrease the environmental effects that will be caused because of this.

Scale of production

By using processes such as a milling machine and a pillar drill, I am able to accurately produce the holes in the tubing wanted which will allow for the holes wanted in the tubing to be accurately produced. With the tool being produced as a prototype the initial price of the tool would be expensive, however as the scale of production increases, such as beginning batch or mass production the price of the tool will decrease and allow for the product to become more affordable. As well more advanced machinery could be used, decreasing the labour costs and overall price of the product, making it more accessible to users. Using processes such as welding will decrease the material required, as well as various scales of production where components can be mass produced, allowing for cheaper production and therefore an overall cheaper product.

With these key points in place about the final product, the design of my product should be able to meet the demands required by this design specification and the design brief to allow for a good product. Certain components and parts of the design may change if necessary, in order for the design to function effectively throughout testing the components and mechanisms of the design.

Manufacturing diary – Testing of components

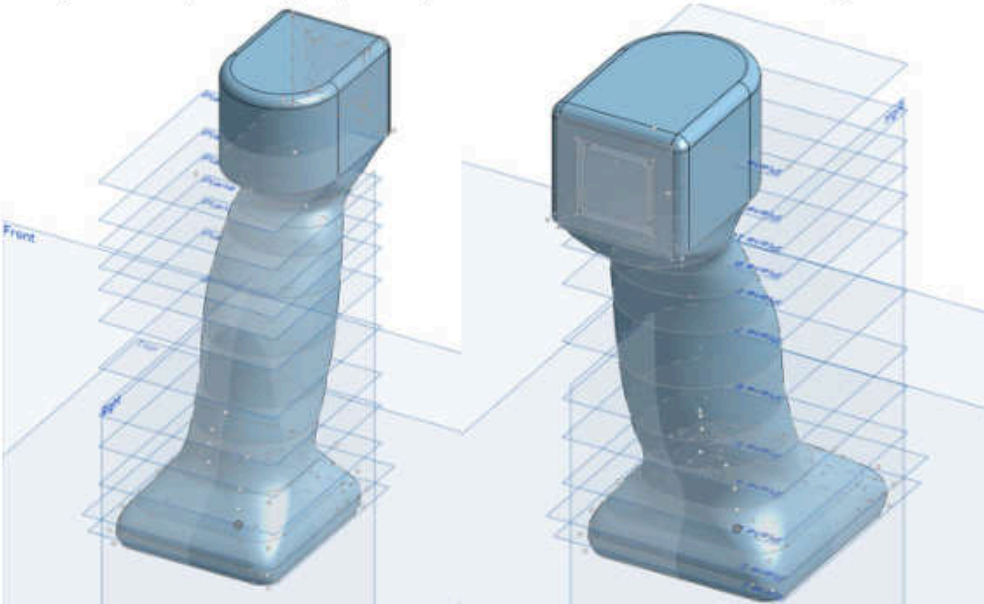
Production on 3D printed components

Many of the components in my design must be 3D printed in order to achieve their unique shape and size, this can be achieved by transferring my design from CAD to the 3D printer, where it will be printed. Due to the parts being 3D printed; I am able to rapidly prototype them, which will allow for me to test the parts of the design to ensure that they function effectively, and if they do not, I am able to rapidly prototype a new redesigned piece which I can test to see if it functions in my design. Testing the pieces first is essential in order for the prototype to function accordingly.

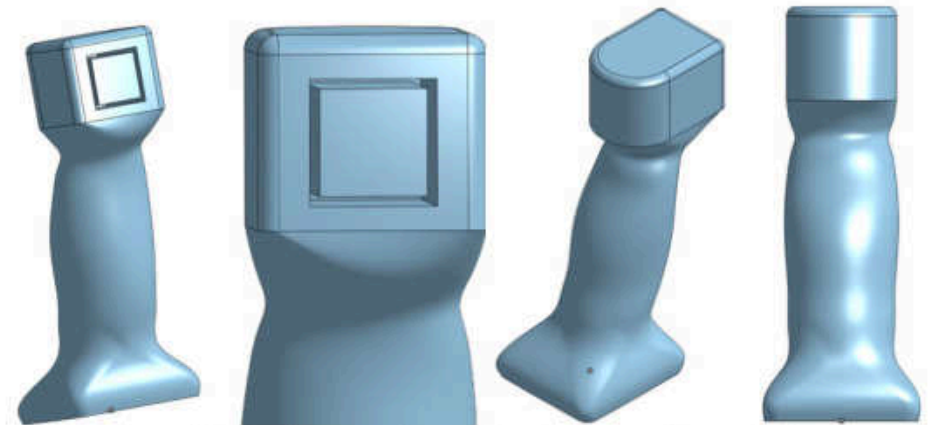
Production of handle:



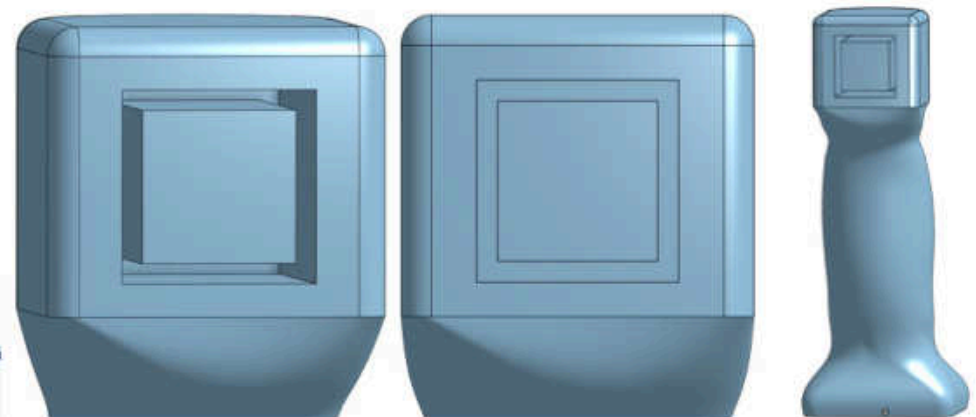
Having the foam ergonomic model that I produced in my development, I measured and traced this onto a piece of paper which would help guide the sizing of the cut parts to be transferred onto CAD. The lines drawn onto the handle were the key points that had a change in shape or sizing, cutting these parts would allow for the shape of the parts to be measured and to be accurately transferred. Using a bandsaw, the handle was cut at the marked stages and were numbered so that the sizes of the handle would be transferred across in order and not in a wrong order which would result in a disfigured handle, numbering the parts helped remove confusion of the manufacturing.



By individually measuring each top face of the cut parts, I was able to produce the handle onto CAD, as well by measuring the distance between the tops of the faces on the paper which had the traced handle on, I was able to ensure that the key points of the handle were an accurate distance apart as the distance between the key points would have been decreased due to the bandsaw removing material, so that the planes for each face where an accurate separation like the foam model, allowing for an accurate transference of the ergonomic model.



The ergonomic shape of the handle has now been produced and accurately transferred across onto CAD, there is a square section at the top of the handle which will allow for the aluminium tube to be slotted into and secured tightly so that the handle and the body do not feel loose and are tightly connected. However, for the hollow section I did not add a tolerance, which is essential so that the aluminium tube can be placed inside the design and not be forced inside which could result in damage. As the square section will be narrow, I will be unable to sand it to decrease the size, by adding an offset of 0.3mm either side, 0.6mm in total to the square section, the aluminium tube would have a snug fit inside the handle and allow for a secure and tight connection.



With there now being a tolerance of 0.3mm either side of the square, 0.6mm in total, the aluminium tubing will be able to fit into the design with a snug fit, allowing for the design to be added and not damage the handle. The CAD of the handle is now complete and can be sent to be 3D printed and will be produced from PLA for my prototype. The square section where the aluminium tube will be inserted into will allow for the two components to be connected, by adding an adhesive or epoxy, the two will remain connected and ensure a tight and secure connection, with the centre piece providing additional surface area connection so that the connection is tight and secure.

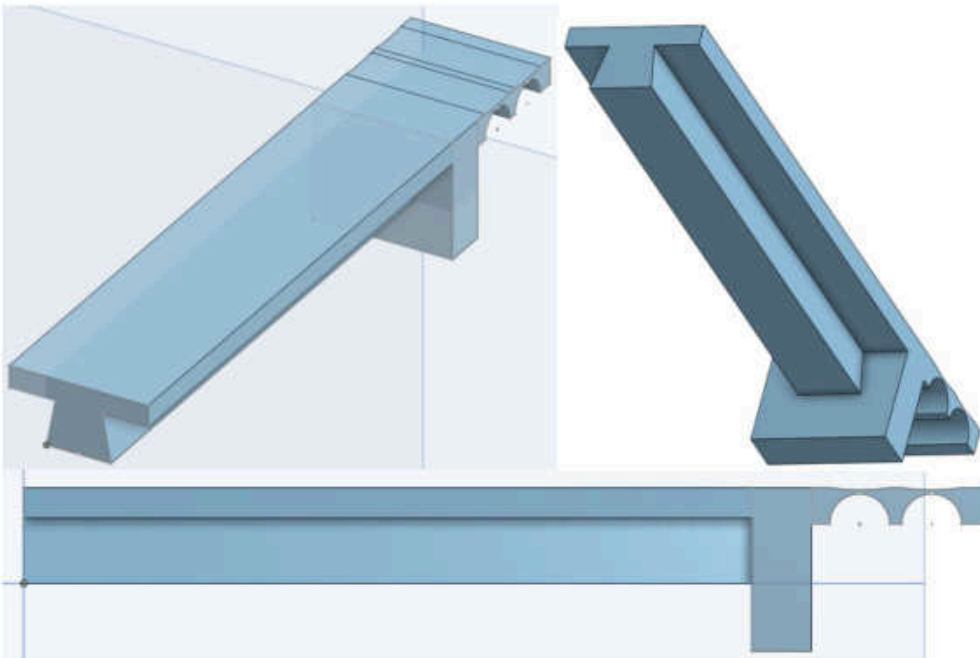
Production of trigger mechanism:

The sizing of the tube I am using is 19.5mm x 19.5mm, and has a thickness of 1.5mm, having produced the shape and size of the trigger in my development stages, the shape has been achieved but will need to be adjusted to meet the size requirements to fit inside the aluminium tubing so that the design is able to function effectively and efficiently, as well as having a 0.3mm tolerance so that the 3D printed design will be able to move and be functional.



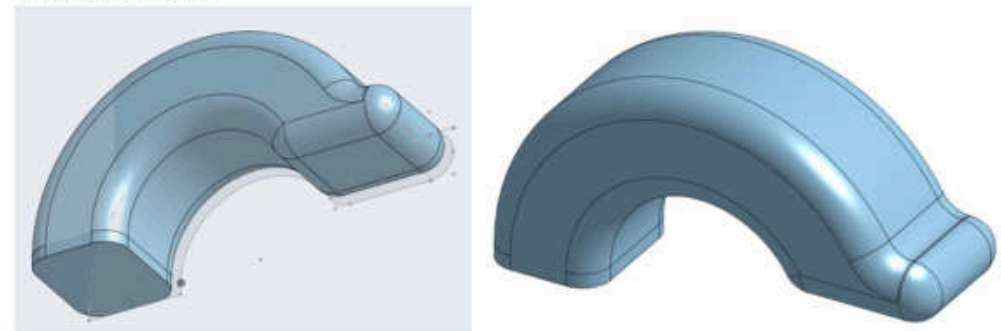
Having already produced the trigger in my development, I only had to adjust the sizing of the trigger so that it would be able to fit inside the aluminium tubing, I decreased the sizing further to allow for the trigger to be able to turn inside the tubing and not touch the sides where it may degrade over time and affect the user's experience.

To allow for the trigger to be connected, I produced the dovetail connection mechanism, which would allow for the trigger to be connected to the body of the design, by taking the sizes of the trigger, I produced the connection with 0.6mm tolerances, so that the design would easily connect.



This design is one that I will test to ensure that it is durable and functional and may have to be reprinted to get the sizing of the thin, bending mechanism correct, to ensure that the quality doesn't degrade and is able to function effectively.

Production of hooks:

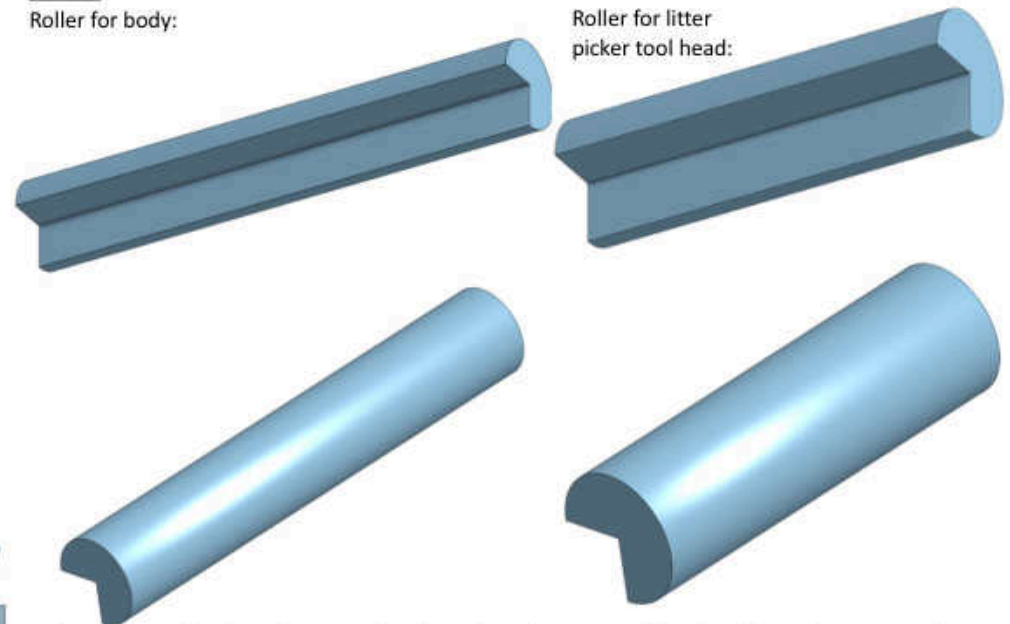


The hooks are used to help keep the rope contained inside the design so that the rope will not leave the system and become damaged. By adding curved to the design, the rope and hooks are less likely to become damaged and need to be replaced. By ensuring that the height where the rope is inserted is less than the height of the rope, it will mean the rope would have to disfigure to be pushed into or out the design, securing it inside the hook. As I am using a rope of 2mm, I have made the height 1.8mm to allow for a slight change in thickness for the hook, but not too much that the user would have to apply large forces which could damage the design.

Rollers:

Roller for body:

Roller for litter
picker tool head:



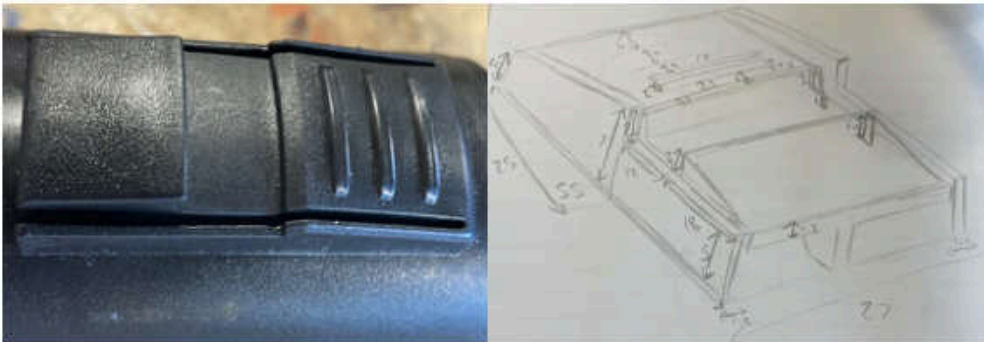
Both of these rollers have been sized with a radius of 1.5mm so that they are able to be positioned around the thickness of the aluminium body. These components are ones that I will test as due to the very small size of the roller, the 3D printer may not be able to produce the design effectively, and it may result in a poor-quality component which would need to be reprinted. The only difference between these rollers is the length as the size required for the body is larger than for the tool head.

Production of button lock connecting mechanism:

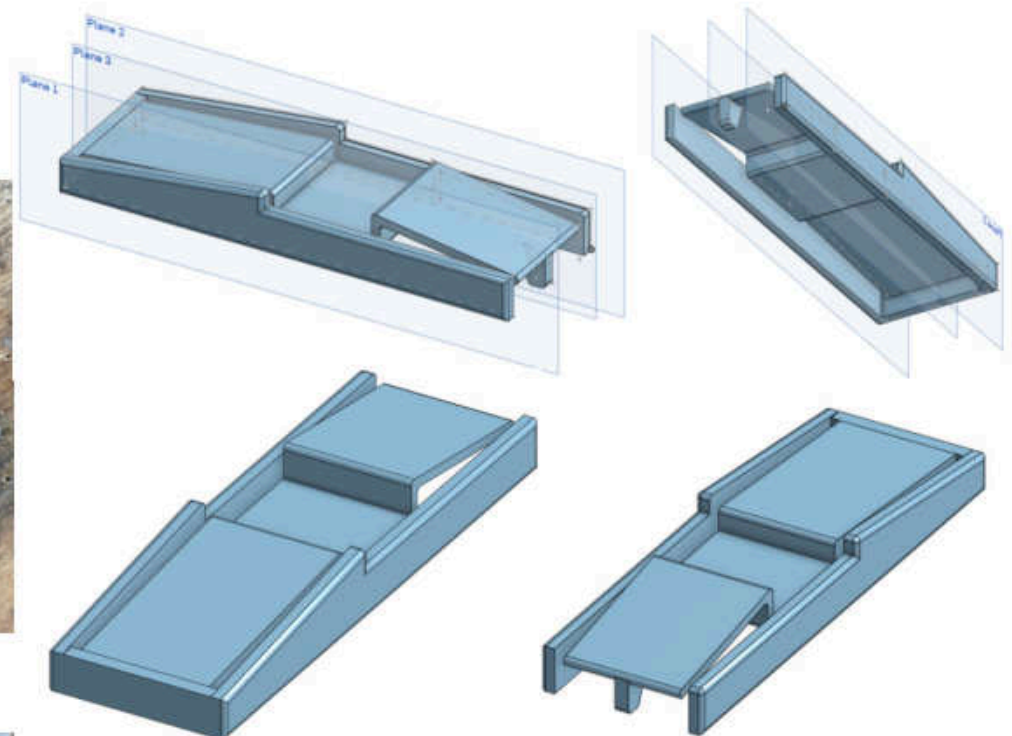
The button lock connecting mechanism is inspired from the connecting between the body and hose of a vacuum cleaner, with the design incorporated into my design, a tight connection between the tool head and body can be achieved. The size of the body I am using is 19.5mm x 19.5mm so the design will have to be adjusted to fit the shape of the body.



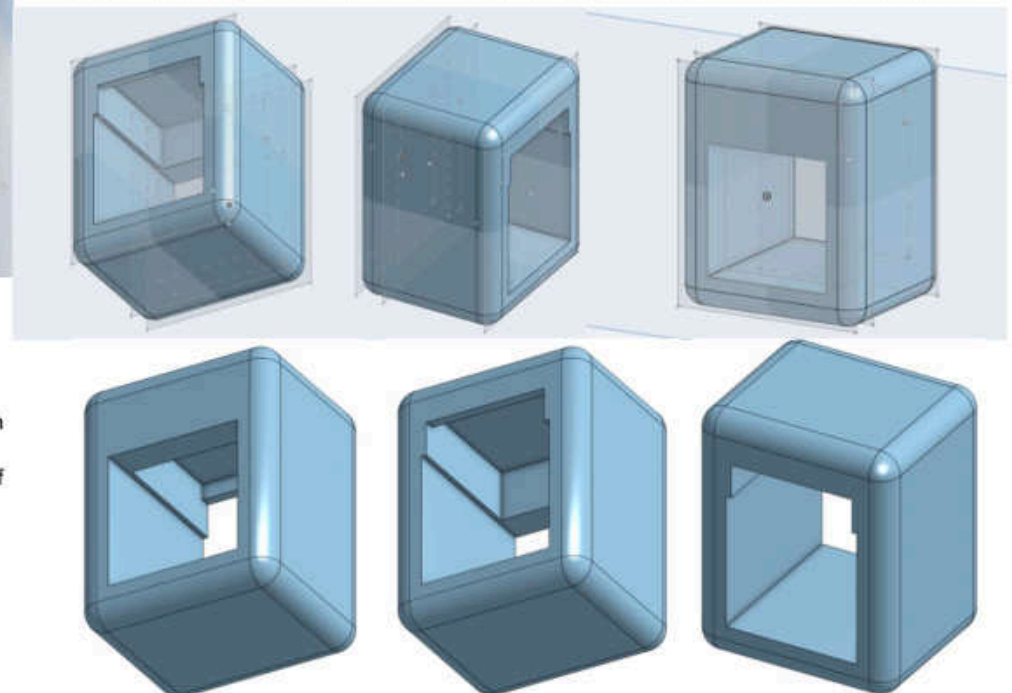
In order to scale the button lock down so that it can fit onto my design, I measured the height of the shorter section, which was 2.3mm, but pressing clay into the design so there would be an imprint of the height which I could measure, allowing for me to work my design around this.



By measuring the size and dimensions of the button lock, I can scale it down to fit into my design, as well as measuring it will allow for me to easily transfer my scaled version of the button lock into CAD, where I can then transfer it to be 3D printed. When developing my scaled version, I only changed the horizontal dimensions to fit into the 19.5mm x 19.5mm aluminium tubing, so that the design would be able to achieve the bending when pressure has been applied, by keeping the same vertical dimensions, the bend of my button lock would be similar to the one investigated, allowing for an easy design that can be easily transferred. If the height was to be reduced, it would reduce the distance that the button would bend under pressure and would affect the strength of the system as there would be less material that is behind the shorter piece and locks the design into place. Keeping the same vertical distance allows for a reliable component as the piece has been designed and placed onto a product, meaning the sizing has been tested to ensure it functions; by using these vertical measurements I will be more likely to achieve the design investigated. I will have to test this design once it has been 3D printed, to ensure that it can function, and may need to change various aspects to ensure the design is effective.



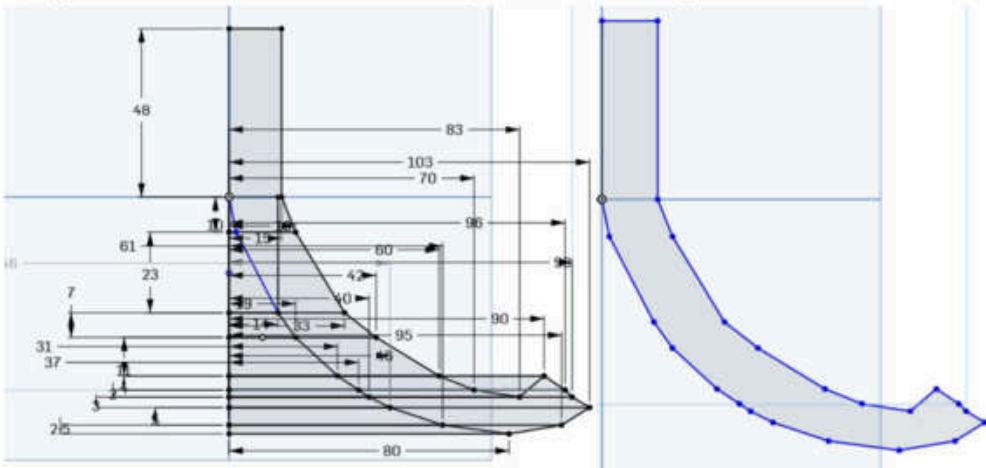
To test this mechanism, I need to produce the component that locks the button into place, as investigated the shorter section at the front should be 2.3mm to allow for the button lock to bend.



Production of claw for litter picker head tool

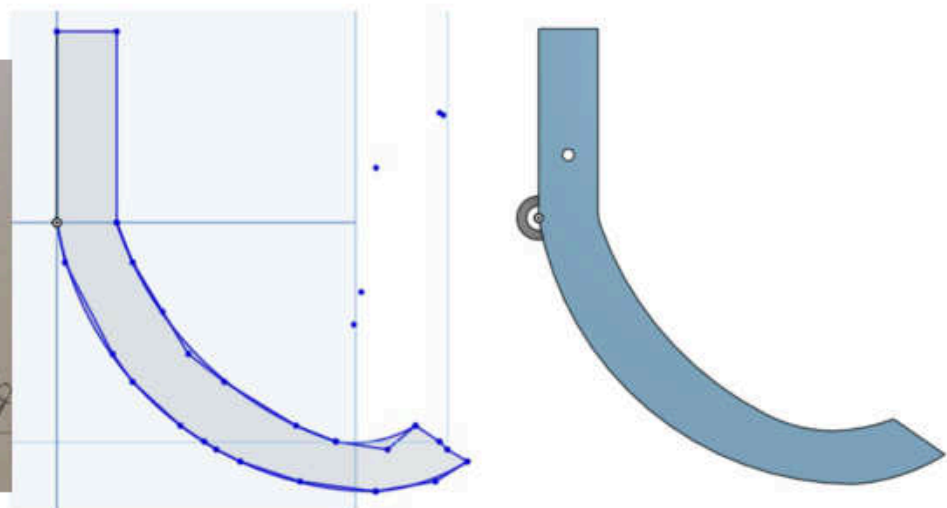


By tracing the claw of a litter picker, and drawing a vertical line, I was able to find the dimensions and shape of the claw that the claw had. I then placed a 2L drinks bottle between the claw and the body, as this is one of the thickest types of litter that isn't compliant when the lid is screwed on, so the claw must be larger than the size for it to be able to pick it up easily. I adjusted the sizing in order to meet the size of the litter and pick it up easily. By measuring the distance from the vertical line to key parts of the claw, I was able to find the shape of the claw, allowing me to transfer it into CAD.

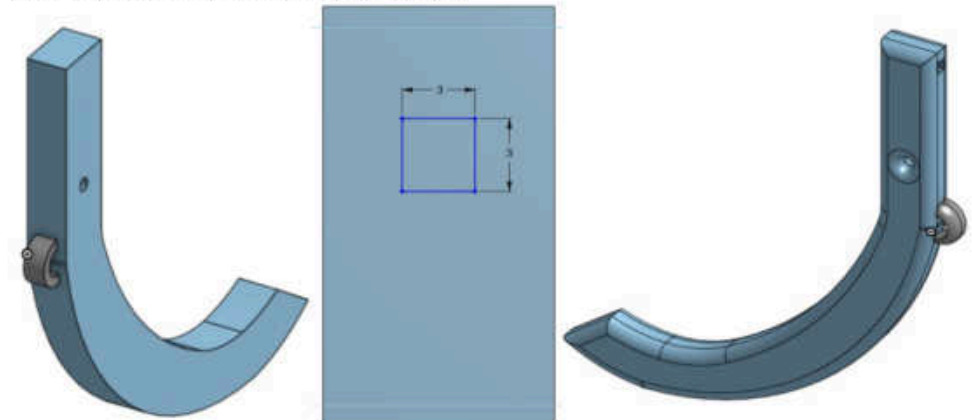


When transferring the design into CAD, I used the vertical line that I drew as the vertical plane and measured the distances between the vertical plane and edges of the claw at the key points of the claw, where the claw is curved or has a change in shape. Allowing me to accurately transfer the key points of the designed claw that can pick up the larger sizes of litter onto CAD.

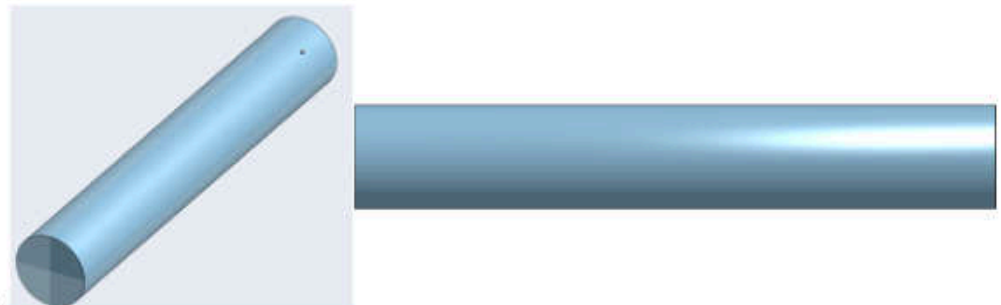
As I only measured the key points the design had very sharp and straight sides, and not the curved edge that would easily wrap around the litter and capture it, to solve this I smoothed the sides out, so that the design would achieve the curve that it required. At the back end of the design, I added the hook where the spring is attached too and allows for the claw to return back to its original and open position. As well, I added a hole at the centre of the vertical section of the top of the hook, this hole is where a rod will be inserted into and whole allow for the claw to rotate when the trigger is pulled, and allow for the litter to be captured



By using arcs and rounded edges I was able to achieve the required shape for the litter picker claw which would allow for it to function effectively.

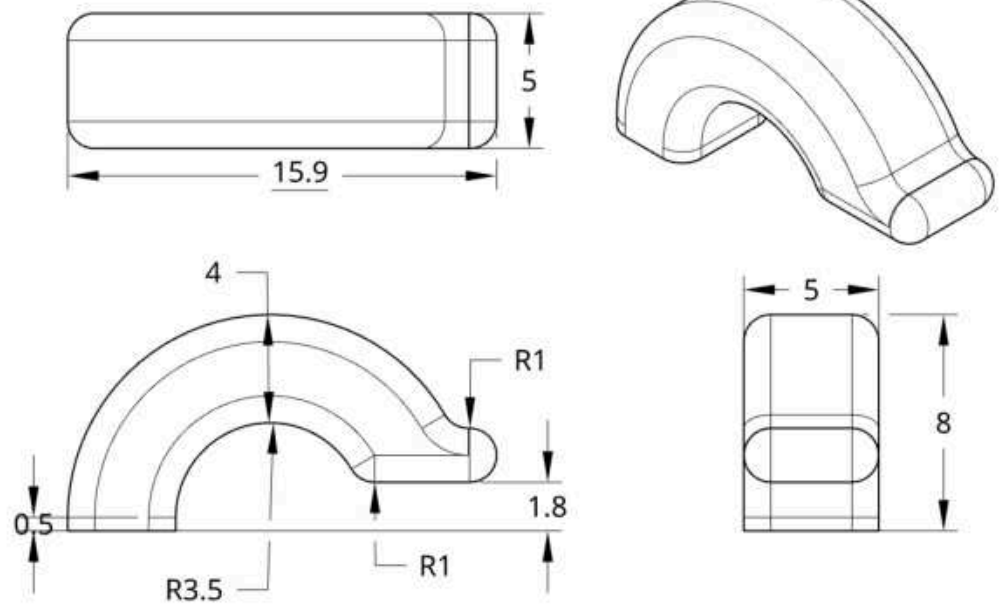
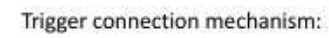


I thickened the design so that the 3D shape of the claw is achieved, and then created a 3mm x 3mm horizontal hole through the top end of the design, which would allow for the 2mm thick rope to pass through and be tied at the end, allowing for the trigger and claw to be connected. I then rounded off all edges of the design so that it would be more resistant and less likely to break.

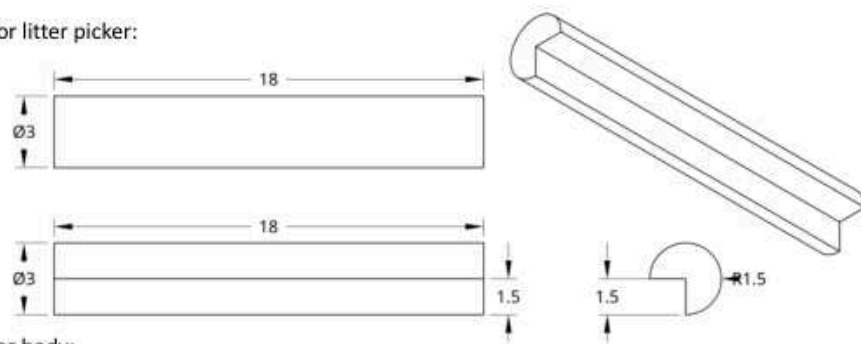


I then created the rods which will allow for the claw and spring to connect, so that the tool is able to function. I will then send these components to be 3D printed so they can be tested and added used.

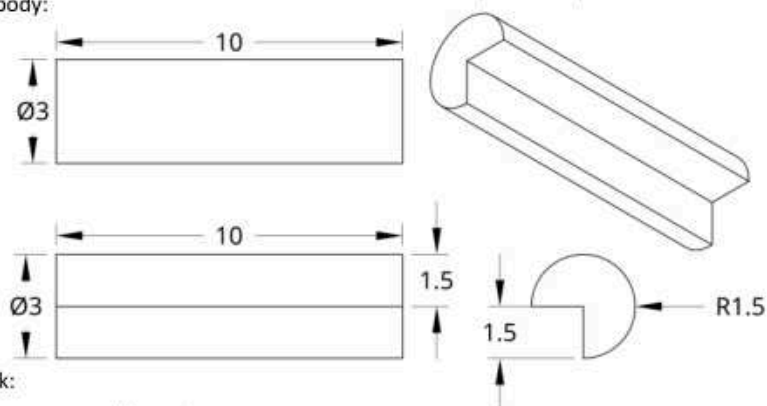
Handle:



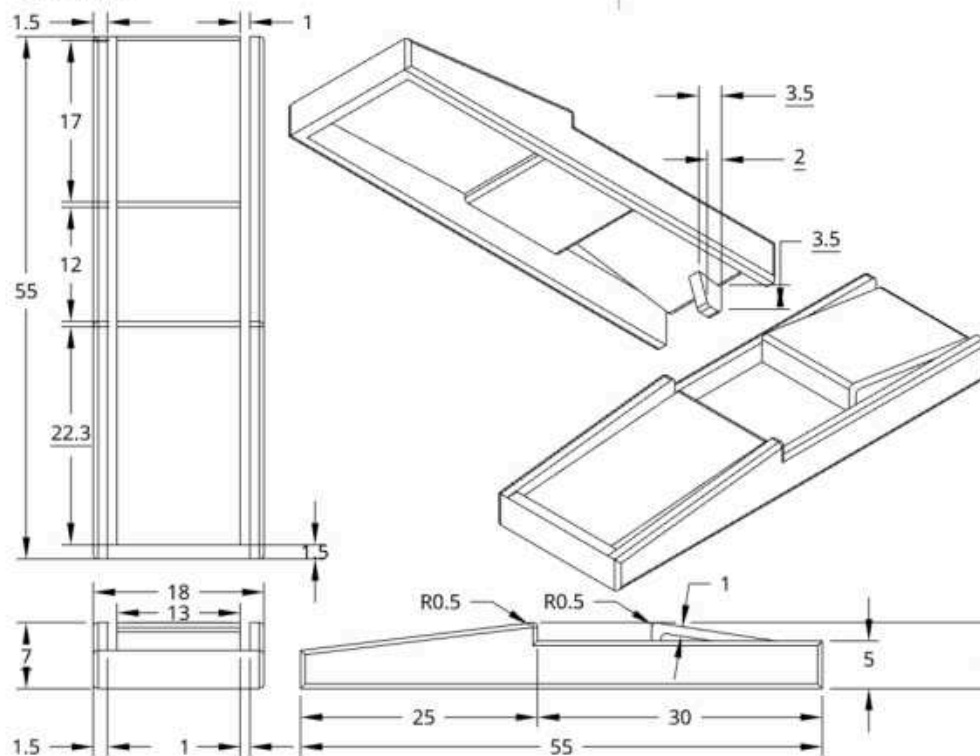
Roller for litter picker:



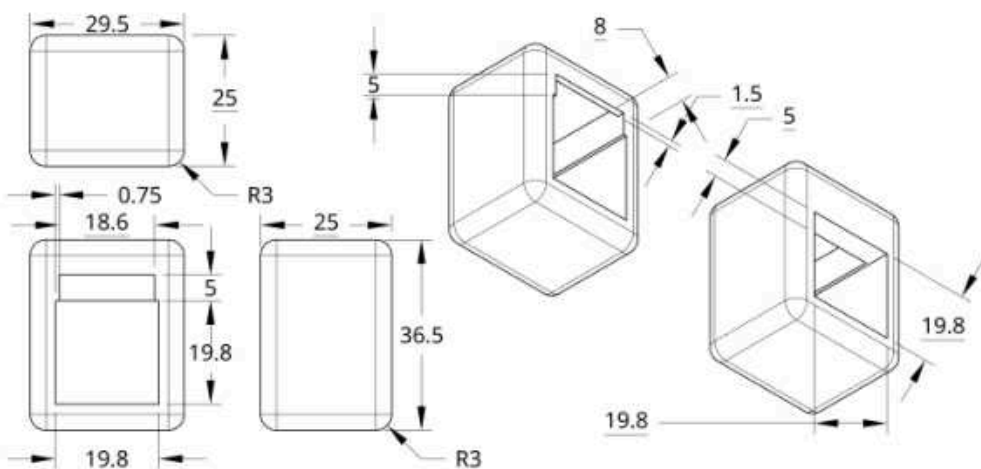
Roller for body:



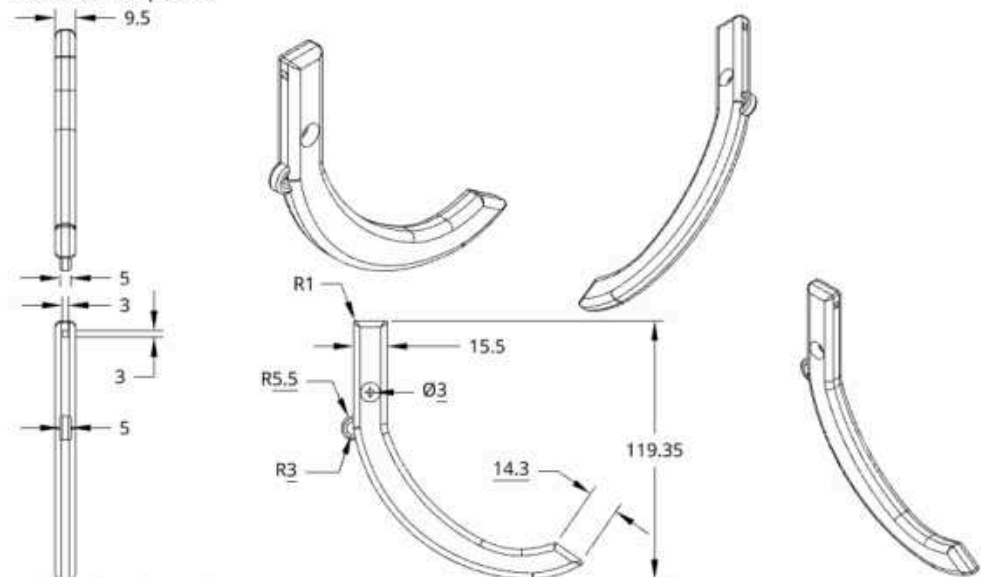
Button lock:



Button locking component:

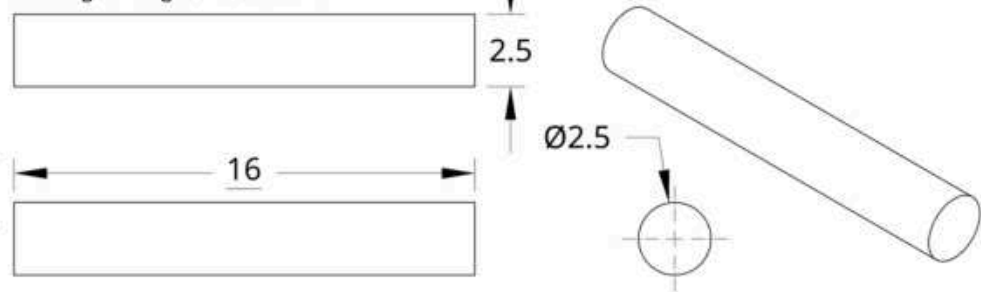


Claw for litter picker:



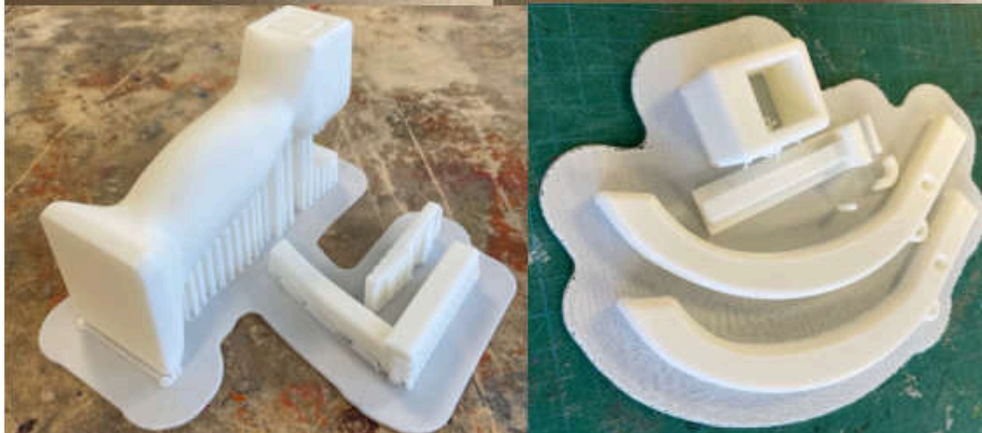
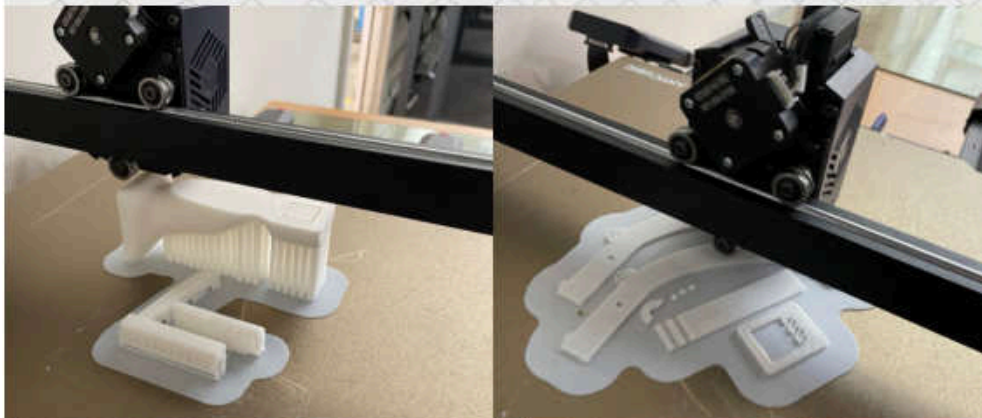
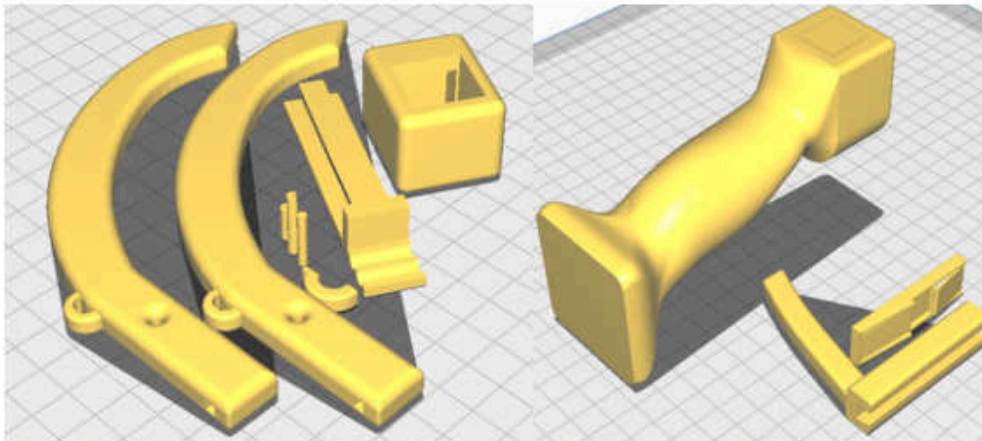
Cylinder for claw and spring:

No tolerance has been given, as component can be shortened if it is unable to fit inside the tubing, allowing for a tight and secure fit.



Printing of components

By sending the CAD of the components I developed to the 3D printer, I was able to have a 3D versions of the parts printed, allowing for me to add them to my prototype. By producing one of each of the components (apart from the claw as this was accidentally printed twice), I would be able to test the component to see if it is able to function, if the test is successful, I would re-print the design, and if the test failed, I would develop a new version with the problems fixed.



Failed print:



Removal of rafts:



(Photo taken from print that occurred further on in development)

In order for the complex shape of the components to be achieved, linear supports were added when the design was being printed, as well as a flat surface called a raft underneath the components to provide support and stabilization to the components whilst printing. When removing the components, the raft could be easily removed by hand, but the linear supports were more difficult to remove due to them being connected to the design, to remove the supports I used a jeweller's screwdriver. When using the screwdriver, I placed it between two of the supports and pushed forwards and backwards in a slow process to ensure that only the supports were removed, and that the component wasn't damaged.



Preparation of testing for printed components

As I have printed one of each component to reduce wastage of material, I can now test each part to ensure that it is able to function effectively, which would allow for a functional and good product. However, due to removing the linear supports the area where the supports were would have remnants of the supports still on and would be uneven, to then make the components even and smooth so the components would not be uncomfortable for the user, I used 120 grit and then 240 grit sandpaper to remove the remnants and not 80 grit as this would have been too coarse and may would have caused a rougher surface which would have been uncomfortable for the user. The remnants of the linear supports were most common on the handle due to it being the larger piece and having the most linear supports to the component.

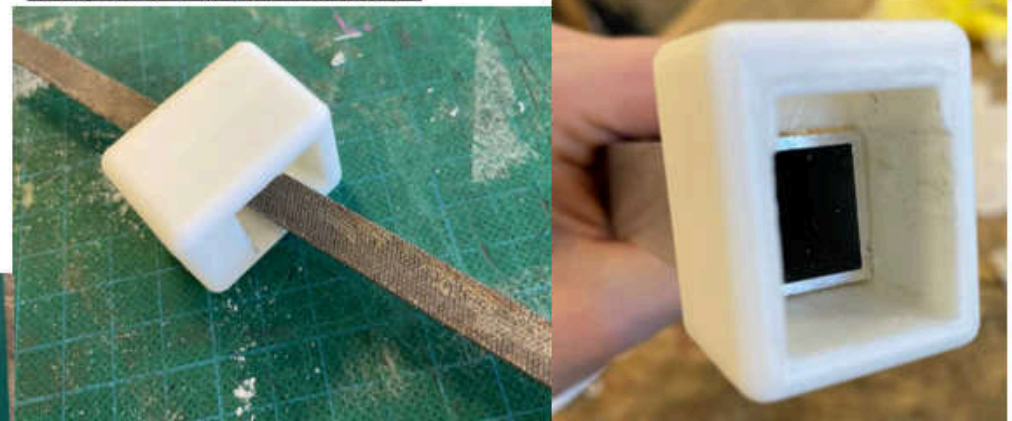


Testing of handle:

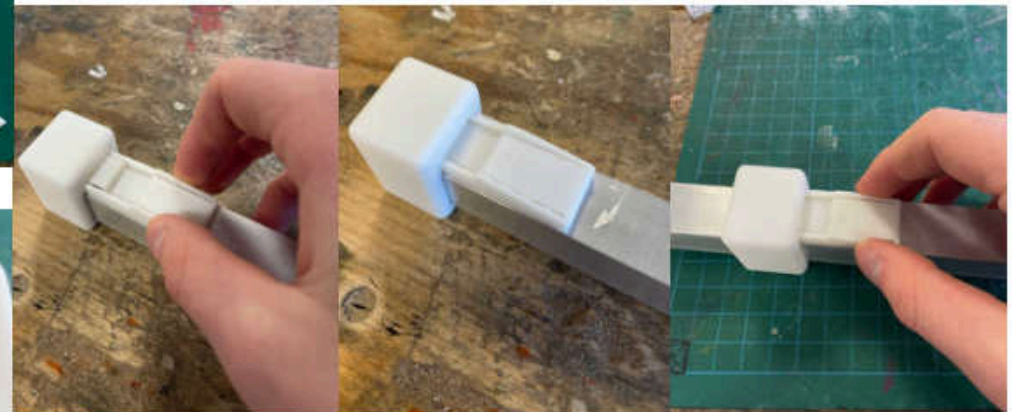


To test if the handle had the correct slot for the tubing to be pushed into, I pushed the tubing inside the part of the handle where the tubing is connected and it was a snug fit that allowed for the parts to be connected without an adhesive, combining this handle with an adhesive would allow for a secure connection to be achieved between the handle and tubing so that the connection would not be flimsy and weak. As well, when asking volunteers to hold the handle, they all thought that it was a comfortable fit, meaning the handle met the ergonomic needs of the user.

Testing of the button locking mechanism:



Due to the hole where the tubing will be passed through for the connection of this mechanism being too small for the tubing, I used a file remove excess material on the component, by consistently checking if the tubing can pass through, the connection between the body and connecting component would be secure and tight so that the head tool and body will not separate.

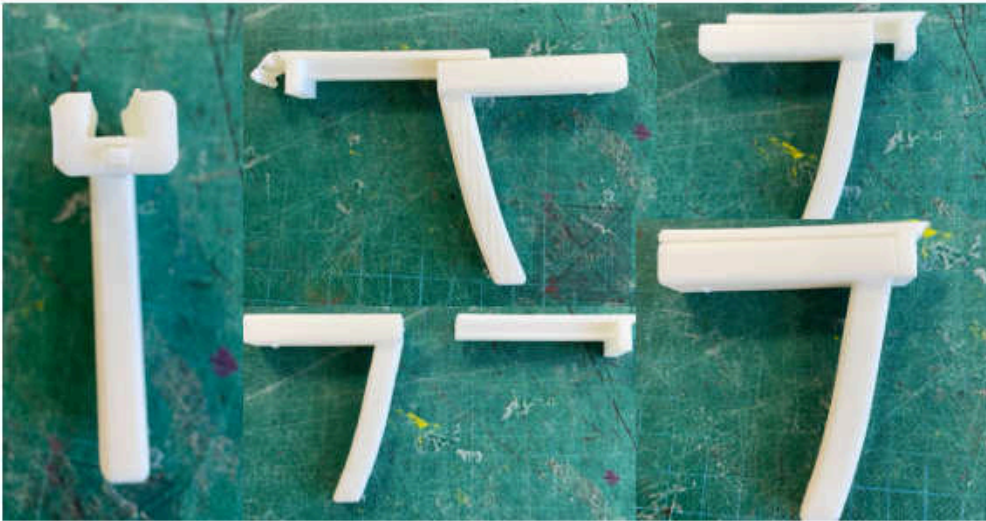


By placing the tubing through the connecting component and sliding the button lock on top of the design into the locking mechanism, I was able to test the connection between the button lock and locking mechanism as this would act how my design would function. The connection between these two parts was secure and would allow for the body to be connected to the tool head, the locking mechanism was able to slide on the top of the design and remain connected, when these two parts are connected to there body, the connection between the body and tool would therefore be secure.



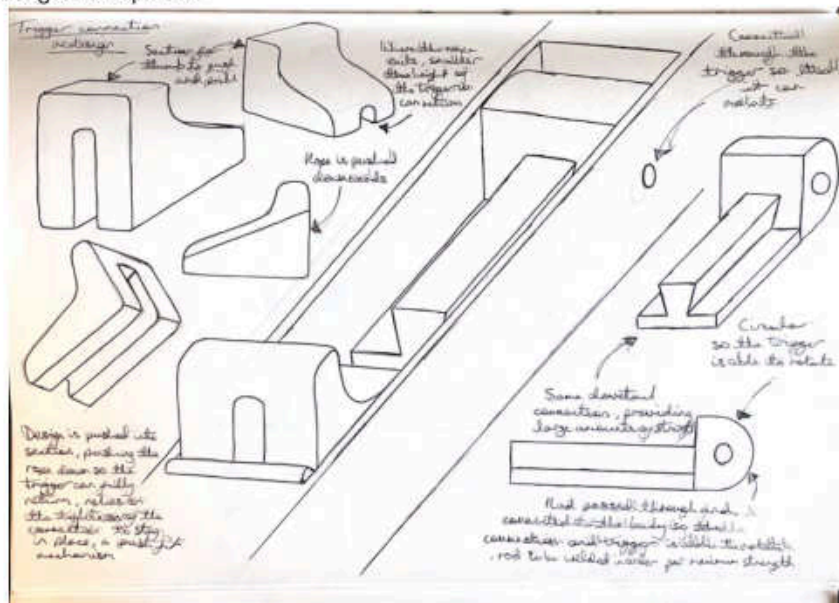
When sliding the design into the locking mechanism, the height of the lock would decrease due to the pressure applied from the shorter section at the front and would return to its original position and lock into place securing the two pieces. The button lock could then be removed if the button was pressed down, decreasing the height of the lock, and if pulled would not be connected to the head.

Testing trigger component



Due to the tight-fitting connection between the trigger and trigger connection required, I sanded the sides of the tail and pin of the connection and checked often to see if the design would have a snug fit together yet also be easy to push together. As I produced the tail with less of a tolerance, sanding the tail down afterwards allowed for me to control the size and tightness of the connection so that the two pieces would be well connected and be secure together, and allow for the desired connection mechanism that I had planned. The problem that I encountered with this mechanism from the 3D printed component, was that the thickness of the turning part on the trigger connecting component was too thin and the 3D printer was unable to effectively print the design, resulting in a weak design that could be easily broken and would result in the tool not functioning effectively for the user.

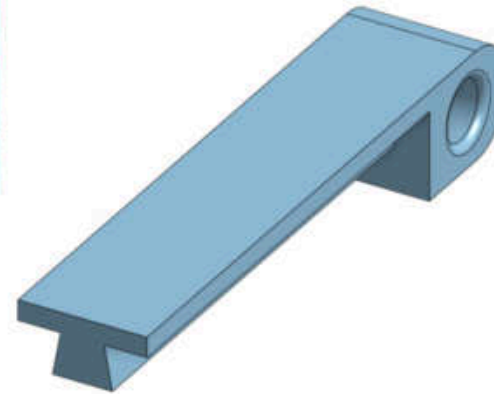
New design development:



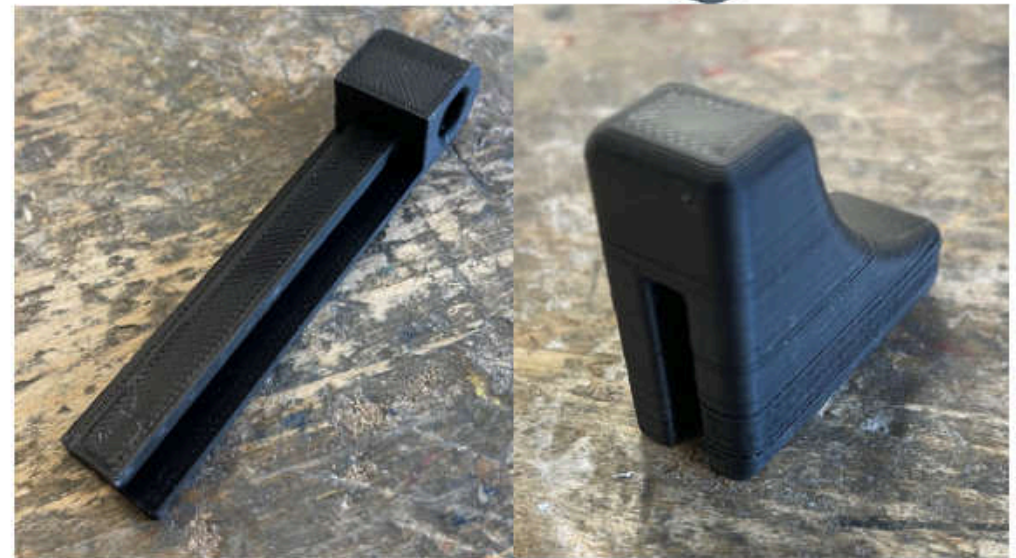
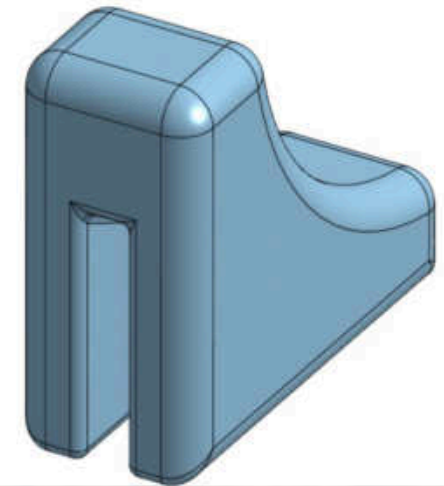
This new trigger connection will be connected to the inside of the design, with a rod passed through the hole and connected to the sides of the body, allowing for a connection which is much more secure and allows for the rotation required to be achieved. The dovetail connection remains the same due to the simple yet strong connection that it provides.

This new component to move the rope down functions by having a slanted piece in the centre of the design, where the rope follows and is pushed downwards. The opening at the bottom of the design, which is closest to the trigger has a height lower than the trigger so that the trigger is able to return back to its original position. The fit of this piece relies on friction and the component being tight to the body of the design

CAD of new trigger connection:



CAD of component to move rope down:



By using these two new redesigned pieces I will be able to test the design to see if it is able to function effectively, I will test this by using a test piece of a section similar to the hole where these components will be inserted into. Allowing to test if the redesign is able to function and if not, another redesign is necessary to ensure that the design can function effectively.

Development of rods

The rods that are planned are intended to allow for the trigger connection, claw and spring that is hooked on the rod to be connected so that they are able to rotate when pressure has been applied and perform their function effectively. The rods printed were too small and therefore wouldn't be able to support the weight and force that would be applied, leading to them breaking and the product not performing effectively. The idea for the final product is for the rods to be produced from aluminium and be welded to the frame and body of the design, allowing for a secure and strong connection. However, for the manufacture of my prototype this is not available, and I will instead have to develop or come up with a new mechanism to produce the function of the product. If the design is successful, it may also be able to replace the need for welding for the final product. A rod replacement that I will test are inter screws, allowing for the trigger connection, claw and spring to be attached to which would allow for it to function effectively. I would have to alter the design however it should allow for the design to function effectively. I will then test this piece with the testing of the newly developed trigger connections too.

Testing of components connected to tubing

On a test piece of aluminium tube measured at 19.5mm x 19.5mm, I used a milling machine with a 16mm drill bit to cut a hole through one face of the tubing, doing this allowed me to test and prepare the angle and sizing required so that when I cut a hole on my prototype piece it is neat and tidy to ensure a correct mechanism. I then filed the edges so the test piece would be similar to the final prototype that has been planned so that the testing would be realistic, this hole allowed for the designed piece to be achieved so that when I test this piece it can be accurately done. I as well then used a 12mm drill bit on the other side so that I would be able to test the mechanism of the claw and spring connection. I then marked out where the holes would be placed for the inter screws, then used the pillar drill with an 8mm drill bit to cut through, allowing the inter screws to be placed though. I then sized down the screws with a file so they were able to fit. I then added the screws through the claw (I had to increase the size of the hole with a drill due to it being printed too small), and the connection.



I then tested the claw, and it was able to move when a force was applied to it, yet the inter screw would remain strong and connected when fully screwed on. With this I am then able to continue with this type of connection, where the inter screws can be placed through the body and components in order to connect the component to the body.

The screws would also have an increased tightness the more they were screwed in against the body, which would be able to provide a solution to a potential problem of the litter picker trigger connection mechanism falling down and out of the body, which may cause irritation and disturb the user's experience, as they would have to consistently push it back up when not using the litter picker tool head. This connection would allow for the trigger connection to work effectively as well, resulting in the litter picker head tool design being able to function and perform well.

Testing of rollers

The point of the rollers are to help direct the flow of the rope, these rollers would have a smooth and round edge and result in little damage therefore occurring to the rope, allowing for longevity of the product, however when further analysing the need for this part it is not that necessary. This is because if I sand and smoothen the edges of the tubing down so that they are soft and round, then the tubing is able to function the same as the roller. I therefore will not carry on with the production of this component as it is able to be done by the tubing already.

Testing of hooks

The intention of the hooks is to help direct the flow of the rope and keep the rope contained so that it is less likely to be damaged when using. However, upon further analysis of this component, it does not seem that necessary, this is due to the component being on the outside of the design already meaning that it could be easily damaged and broken if knocked. As well as not providing much use when the litter picker tool is not being squeezed as it will not be extended out by a larger amount due to the mechanism that allows for the point of tension to be above the resting area of the trigger.

Testing of component that pushes the rope down



By placing the component that pushes the rope down into my test piece, which was able to stay within the design due to friction, the rope would be able to be passed through and the point of tension would then rest above the resting area, allowing the trigger to return back to its original position. The friction that will take place with the tension will be able to resist from moving and allow for the litter picker head to function effectively.

Layer above the trigger

Upon further analysis the layer that would be placed above the trigger is not necessary as does not help the rope or trigger return back to its original shape, as well the design is now locked into place and rotates by the end of the trigger which is circular, meaning that the trigger will not move up and the space will not be required.

Testing of the net tool head



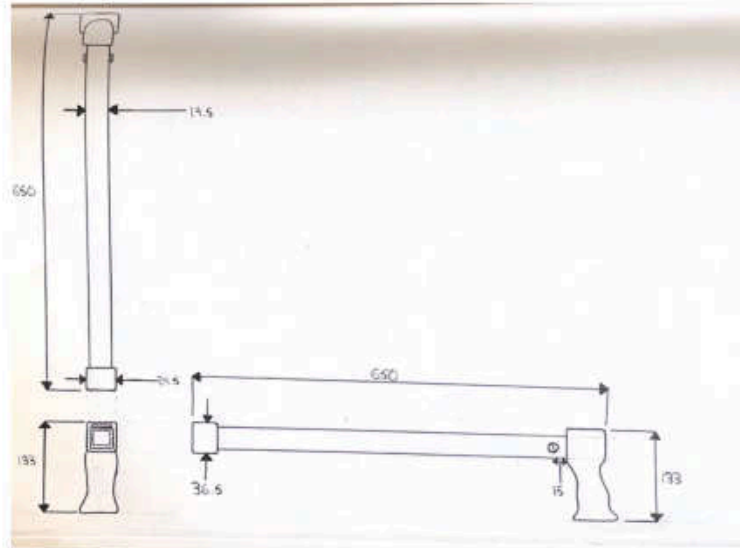
I cut holes into the sides and top of a spare aluminium tubing, passed a rope through a net and then tied the rope to the tubing. I was then able to test the shape and design of this head tool which would be able to function. The tool head would be able to be placed into water and remain open to remove any litter that is within a watered environment. By tying the rope to the body by knots which can be undone, the rope is replaceable if needed for example if the rope frays and becomes weak.

Orthographic of design

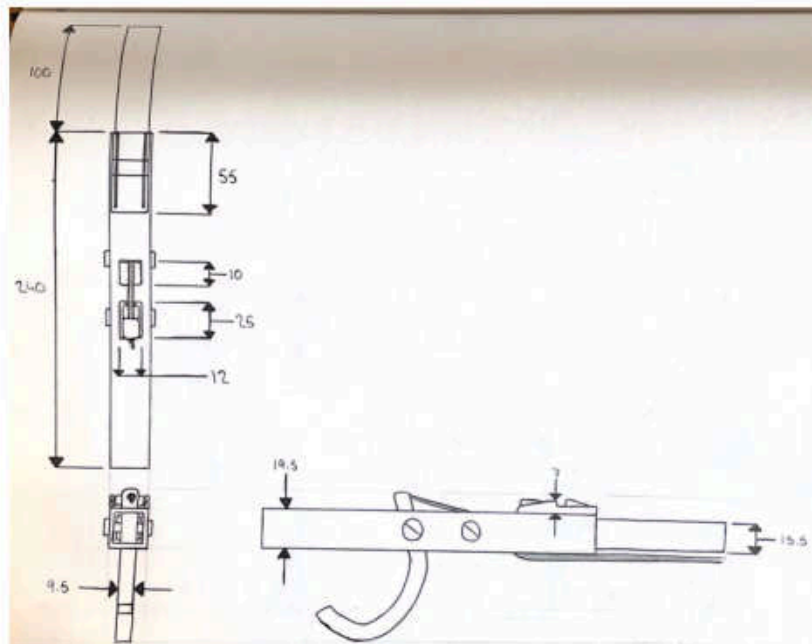
As I have completed most of my development and testing of the product, I am able to move onto the manufacturing stage of my product, the development of my product will continue through further testing to allow for a better and developed product.

This orthographic incorporates the 3D printed components inside the design and those that passed the testing stage, as well as new developed components inside the design.

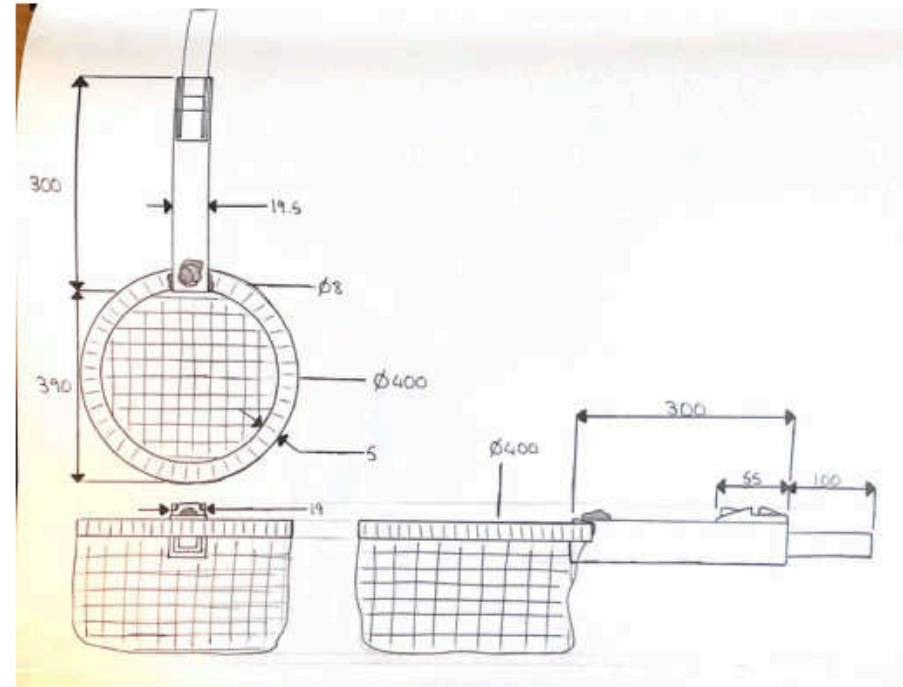
Body of the design:



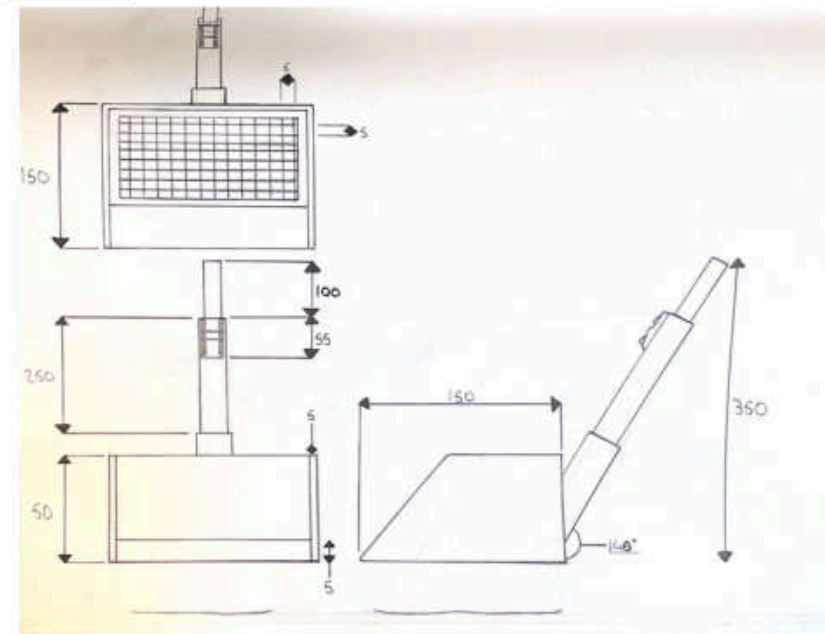
Litter picker tool head:



Net tool head:



Scraper tool head:



Here is the orthographic drawings and the measurements and dimensions that I will follow when manufacturing my product.

Cutting list and costings

Component	Location of component	Quantity	Material	Length (mm)	Width (mm)	Thickness (mm)	Cost (£)	Process to achieve material for manufacture
Body of the tool (A)	Tool body	1	Aluminium tubing	60	19.5 x 19.5	1.5	£8.10 for 1m, £4.86	Store bought
Handle (B)	Tool body	1	PLA	70	35	133	n/a	3D printing
Trigger connection (C)	Tool body	1	PLA	74	16	7.9	n/a	3D printing
Inter screws (D)	Tool body, litter picker tool head	3	PVC for plastic and zinc plated steel for screws	42mm	8mm	8mm	£2.29 for 10, £0.69 for 3, £0.23 for 1	Store bought
Component to push rope down (E)	Tool body	1	PLA	40	16.5	35	n/a	3D printing
Hooks (F)	Tool body	1	PLA	5	15.9	4	n/a	3D printing
Button lock (G)	Tool body	1	PLA	25	29.5	36.5	n/a	3D printing
Body for litter picker (H)	Litter picker tool head	1	Aluminium tubing	24	19.5 x 19.5	1.5	£1.94	Store bought
Push fit mechanism (I)	Litter picker tool head, scraper tool head, net tool head	3	Aluminium tubing	140	15.5	1.5	£6.50 for 1m, £3.12 for 3, £1.04 for 1	Store bought
Spring (J)	Litter picker tool head	1	Stainless steel	20.5	5	1	210 springs for £8.99, £0.04 for 1	Store bought
Claw (K)	Litter picker tool head	1	PLA	105	9.5	119.35	n/a	3D printing
Rope (L)	Litter picker tool head	1	Polypropylene	70	2	2	£5.99 for 10m, £0.42	Store bought
Trigger (M)	Litter picker tool head	1	PLA	60	16	71	n/a	3D printing
Button locking mechanism (N)	Litter picker tool head, scraper tool head, net tool head	3	PLA	55	18	7	n/a	3D printing
Body for scraper head tool (O)	Scraper head tool	1	Aluminium tubing	350	19.5 x 19.5	1.5	£8.10 for 1m, £2.84	Store bought
Scraper tool head (P)	Scraper head tool	1	PLA	150	150	5	n/a	3D printing
Wire mesh (Q)	Scraper head tool	1	Stainless steel	81	116	0.6	£6.99 for 3m x 0.2m, £0.11	Store bought
Body for net head tool (R)	Net tool head	1	Aluminium tubing	300	19.5 x 19.5	1.5	£2.43	Store bought
Rope for net hoop (S)	Net tool head	1	Polypropylene	150	8	8	n/a	Spare in workshop
Net (T)	Net tool head	1	Polypropylene	400	400	2mm	£10.10	Store bought

Attached above is the cutting list and costings for all the components that I will use to produce my prototype, other components and materials may have to be used to produce the product, however these will be added further on within the development of my product. Most of the complex components will be 3D printed in order for the shape to be produced, and it being cheaper due to the specific shape being costly if 3D printed elsewhere. By purchasing the aluminium tubing in 1m lengths, I am able to test the manufacturing process on excess material in order to maintain a high

quality of design, for example setting the milling machine in line correctly on my tubing to ensure the cut it straight and accurate. The 3D printed components are difficult to price, the cost of a 1kg black spool of PLA is priced at £15 and would vary for the production due to energy costs and other factors. However, the combination of all these components should not exceed 0.5kg, so £7.50 is an appropriate price tag for the components, however once the final components have been printed, they can be weighed and the cost of the manufacturing of these pieces can be calculated. The cost of the product without the 3D printed components is £24.61.

Manufacturing table

This table outlines the processes, steps and tools required to produce my product accurately, other steps may have to be taken when manufacturing due the process not going as planned. The quality control fails show the steps taken if the original procedure fails. This table is not the final process as the process may slightly alter when I am producing the product.

Process	Component	Material	Tools/process	Quality control	Quality control fails	Health and safety	Estimation of time taken
Printing of 3D printing components	Handle (B), Trigger connection (C), (E), Hooks (F), Button lock (G), Claw (K), Trigger (M), Button locking mechanism (N), Scraper tool head (P)	PLA	CAD, 3D printing	Ensure that dimensions, sizes, shapes and measurements on CAD are accurate, transfer to 3D printing, continue to check design to prevent failed print, to prevent material wastage	If print fails, reprint if necessary or alter design on CAD and then reprint, only print one component at a time to reduce time taken to print and material used if printing fails	Have good ventilation, do not touch printer whilst printing as it may fall, do not touch printer as it will be hot due to melting the PLA	Time will vary for each component based on size, density and speed of printer - 20 hours
Removal of raft and smoothing of 3D printing components	Handle (B), Trigger connection (C), (E), Hooks (F), Button lock (G), Claw (K), Trigger (M), Button locking mechanism (N), Scraper tool head (P)	PLA	Jewellery screwdriver, file, 120 grit sandpaper	Slowly remove rafts by pushing the rafts against themselves by using a jewellery screwdriver to get in the gaps of the rafts, allowing for the rafts to break, use a file and sandpaper to remove excess left from rafts	If component breaks when removing rafts, reprint or redesign component so that it will function and maintain its correct shape	Keep hands away from the rafts as they can be sharp once broken, use file to smoothen small sharp edges, don't touch broken raft edges as they will cut due to being sharp	1.5 hours
Sanding of 3D printed components	Handle (B), Trigger connection (C), (E), Hooks (F), Button lock (G), Claw (K), Trigger (M), Button locking mechanism (N), Scraper tool head (P)	PLA	120 grit sandpaper, 240 grit sandpaper	Sand down so all sides of the design are flat and smooth, Remove minimal waste, start from 120 grit and move to 240 grit for a smoother finish	Minimise removal so excessive amounts are not removed, reprint if necessary, or sand surrounding areas to smoothen area out	Don't touch eyes when sanding as particles may cause redness.	1 hour
Testing of 3D printed components	Handle (B), Trigger connection (C), (E), Hooks (F), Button lock (G), Claw (K), Trigger (M), Button locking mechanism (N), Scraper tool head (P)	PLA, Spare aluminium tubing	Milling machine, pillar drill, pencil, ruler, tri-square	Test each component to ensure it functions appropriately, develop mechanisms if necessary or required, perform appropriate tests for each component to ensure it functions effectively, essential as mechanism may not function effectively and may need to be developed	If testing of the component fails, develop a new mechanism where the component can function effectively	Wear safety glasses and apron when using machinery, keep vices tight and components secure, keep high focus an hands away from cutters and drills.	4 hours
Measuring and sizing aluminium tubing to prepare for cutting	Body of the tool (A), body of litter picker head (H), body of net head (R), body of scraper head (O)	Aluminium tubing	Pencil, metal ruler, tri-square	Ensure that the dimensions measure are accurate and straight by using a tri-square and a ruler	Redraw out measurements, measure distances from point to ensure constant and straight	N/A	10 min

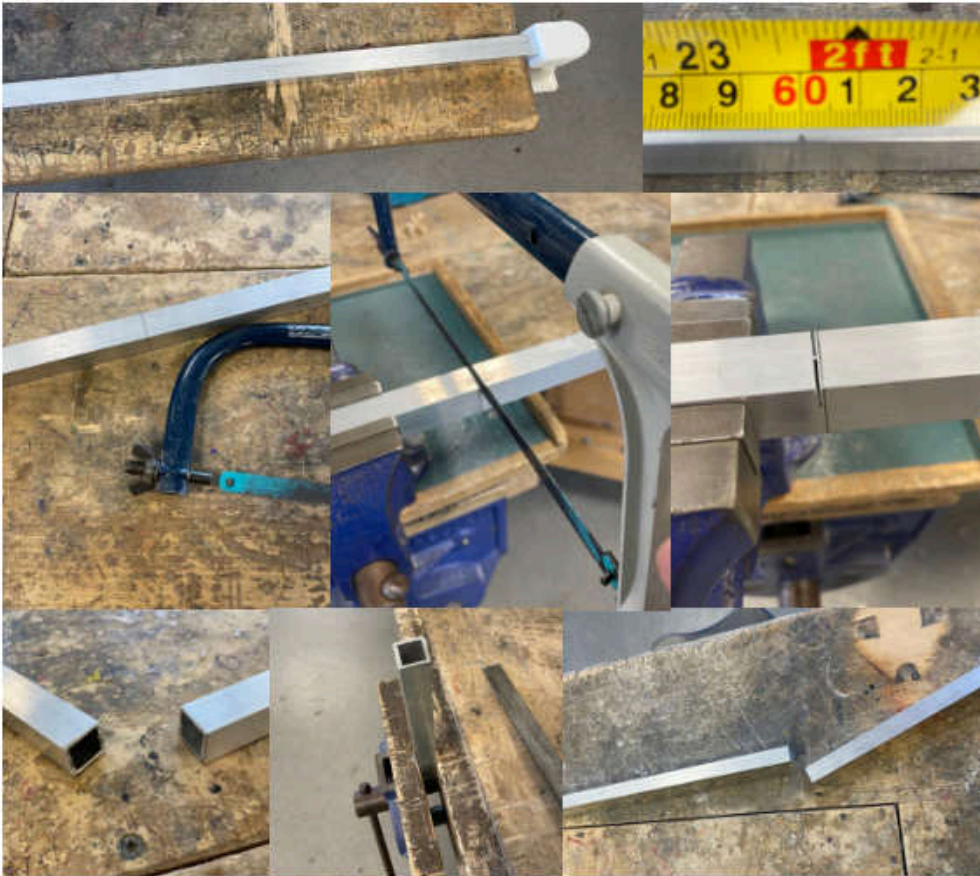
Process	Component	Material	Tools/process	Quality control	Quality control fails	Health and safety	Estimation of time taken
Cutting the aluminium tubing	Body of the tool (A), body of litter picker head (H), body of net head (R), body of scraper head (O)	Aluminium tubing	Vice, hacksaw	Ensure tubing is tightly secure in the vice, hold vice for support, keep head above cutting so that you can't see either side of the blade to ensure the cut is straight	If this fails, and tubing is cut at an angle, dimension another line and use a flat file and file to the line	Ensure vice is secure, when cutting ensure thumb is not out so if blade slips will not cut thumb, controlled strokes of the tool, keep away from blade	20 min
Measuring hole in body for trigger	Body of the tool (A)	Aluminium tubing	Pencil, metal ruler, tri-square	Use piece of the same height to balance tools on when measuring to ensure accurate and straight lines	Remeasure, use tri-square to ensure lines are straight and accurate	N/A	10 min
Cutting out hole in body for the trigger	Body of the tool (A)	Aluminium tubing	Milling machine, flat file, square file	Cut spare piece of tubing to align and straighten the machine, if necessary, move the machine horizontally slowly to ensure a clean and accurate cut	If cut is not clean, use a flat or square file to clean up and flatten the edges	Wear goggles and apron, secure tubing in vice, follow machine safety procedures, keep hands away from the cutter	40 min
Measuring holes for litter picker	Body of the litter picker tool head (H)	Aluminium tubing	Pencil, metal ruler, tri-square	Use piece of the same height to balance tools whilst measuring to ensure accurate, straight lines	Remeasure lines, using a tri-square to ensure the lines are straight	N/A	10 min
Cutting holes inside litter picker head	Body of the litter picker tool head (H)	Aluminium tubing	Milling machine, flat file, square file	Cut spare piece of tubing so that real piece will be have a straight cut, move machine horizontally slowly for a clean cut	If cut is not clean and flush, use a flat file or square file to smoothen and flatten the edges	Wear appropriate PPE, goggles and apron, keep hand away from cutter, move slowly and with care	1 hour
Measuring out holes for body and litter picker where screws will be placed through	Body of the tool (A), Body for the litter picker tool head (H)	Aluminium tubing	Pencil, metal ruler, tri-square	Place components such as trigger connection and claw above area where screws will be passed through and mark	Remark with dimensions and positions of the components	N/A	20 mins
Cutting of holes in body and litter picker head	Body of the tool (A), Body for litter picker tool head (H)	Aluminium tubing	Pillar drill	Mark out area where holes is placed, secure tubing in a vice, bring drill down to ensure in line, drill down holding the vice tight to prevent slipping	Hold the vice tightly to prevent slipping, if material damaged, try to fix but if not repeat on a new tubing	Hold the vice tightly when using, keep hands away from drill, wear safety glasses and apron	10 mins
Sizing of inter screws	Inter screws (D)	PVC for plastic and zinc plated steel screws	Hacksaw, junior hacksaw, square file	Use a junior hacksaw to cut plastic and hacksaw to cut screw to the size of the width of the body so it can fit, compare above and place inside holes, use a file to smoothen out	Slowly remove material by measuring and comparing to the hole size. If removed too much, use another screw or plastic piece	Ensure vice is secure, keep thumb away in case saw slips, controlled strokes, keep away from blade	40 mins
Placing rope inside claw and trigger	Claw (K), Trigger(M), Rope (L)	PLA, polypropylene rope	Drill	Pass rope through designed holes in the components and tie together to test	Drill through the holes to create a larger gap if necessary	Holding the component tightly when drilling, keeping hands away	10 mins

Process	Component	Material	Tools/process	Quality control	Quality control fails	Health and safety	Estimation of time taken
Measuring of push fit mechanism	Push fit mechanism (I)	Aluminium tubing	Pencil, metal ruler, tri-square	Measure out length, support ruler by using a spare piece of the same height	If fails, repeat process to ensure correct markings, check with tri-square	N/A	10 mins
Cutting of push fit mechanism	Push fit mechanism (I)	Aluminium tubing	Hacksaw, vice, square file	Ensure tubing is secure, cut with controlled strokes, head above so cut is straight, file to make smooth	If cut is angled, use file to square off the tubing to make the cut accurate	Ensure vice is secure, controlled strokes of saw, keep away from blade, keep thumb away	20 mins
Marking out holes for net head tool	Body of the net head tool (R)	Aluminium tubing	Pencil, metal ruler, tri-square	Measure out where holes must be drilled, use a piece of the same height as balance	If marking is incorrect, redo and ensure markings are straight with tri-square	N/A	10 mins
Cutting holes inside net head tool	Body of the net head tool (R)	Aluminium tubing	Pillar drill, milling machine, square file	Using machinery, remove wanted areas, ensuring tubing in vice is secure, work on test piece first in order to angle correctly	If done incorrectly use a file to neaten up the design, if this doesn't work then mark out and cut another piece, if not neat then file down holes	Ensure vices are secure, keep hands away from blades and cutters, wear safety goggles and apron	20 mins
Putting together rope head tool	Body of the net head tool (R), Rope for net hoop (S), Net (T)	Aluminium tubing, polypropylene	N/A	Pass rope through net, through holes and tie a knot to secure to head, then cut rope, ensure its secure	Tie a tighter knot if needed, ensure knot is larger than gap so is unable to leave and is secure	N/A	20 mins
Painting the 3D printed components	Handle (B), Trigger connection (C), (E), Hooks (F), Button lock (G), Claw (K), Trigger (M), Button locking mechanism (N), Scraper tool head (P)	PLA, spray paint	Finishes	Evenly apply paint on components, evenly distributed, let paint dry before touching to maintain aesthetic	Apply another coating to enhance and distribute the coating if needed, test components after first coat to ensure they still function, if not remove paint or reprint	Wear a mask and goggles, have a good ventilation system	30 mins + time to wait for components to dry
Gluing and combining components of the design together	All components, A-T	PLA, Aluminium tubing, Epoxy adhesive, Super glue	Gluing and combining components	Slowly glue components onto the design, ensure accuracy by taking time and test components work once glued on	If component doesn't function, carefully remove it, and redesign the component so it will function	Mask, gloves, a good ventilation system, prevent particles entering body system	2 hours
Polishing and finishing	All components, A-T	Aluminium tubing, polypropylene, PLA	Needle file, sandpaper	Slowly finish and smoothen edges to remove scratches	Remove scratches for aesthetic finish, increase grit of paper	N/A	1 hour

Manufacturing of product

As I have now completed and tested the mechanisms of my design, with some mechanisms and components of the design having to be changed in order for the design to function effectively and the product to be useful, I can produce my final product. By using some of the 3D printed components from the testing of my product, I am able to save material and time required in order to produce my product, as well as having a high reliability of the components as they have already been tested to ensure that they can function. By doing this I am able to produce a functional product that I am able to test with removing litter from an environment.

Manufacture of body:



For the first stage of my manufacturing process, I inserted the 19.5mm x 19.5mm tubing inside the handle and using a tape measure measured from the back of the top of the handle across the tubing till 600mm, where I marked a line with a pencil. Using a tri-square, I drew the pencil line across all the sides of the tube, ensuring that the tri-square was flat against the tubing so that the line would be accurate and straight. By locking the tubing into place and using a hacksaw I cut through the tubing in order to have two separate pieces. I cut to the side of the tubing for the body, so that I would not decrease the length and would be able to control the length easier when filing afterwards. I then filed the edges that I cut for the body so that the side would be flat and smooth to prevent damage to the user, and for a smooth finish. When doing this, I consistently checked the surface of the edges to ensure a high quality and limit mistakes such as over filing on one side, which would result in the end piece becoming wonky and slanted.



In order to create the hole where the trigger would be located, I drew a line across the edge of the top of the handle when it has been connected, from here I then drew the dimensions for the hole and added 'X' to show where to remove and cut a hole into. By measuring the distance from the handle to where the trigger would be placed, I ensure that the trigger would be able to be placed at an appropriate distance so that it is at the comfortable and ergonomic distance that I planned.



With the area to remove marked out, I used a scrap piece of the same sizing to set the position for the milling machine to cut, so that the cut on the real piece would be accurate and precise; by cutting and practising with the spare piece, I was able to position the machine correctly for the piece for the prototype. When cutting the prototype, I used a 15mm drill bit and cut from the centre so that I would not move the machine the wrong way and remove excessive amounts of material, I let the machine cut across for the most part of the removal, but until the edges where I took control and slowly moved the handwheel for the X-axis to ensure a close and accurate cut of the desired area. I then used a file to smoothen out the edges of the design to prevent damage to the user.

Manufacture of litter picker head tool and body:

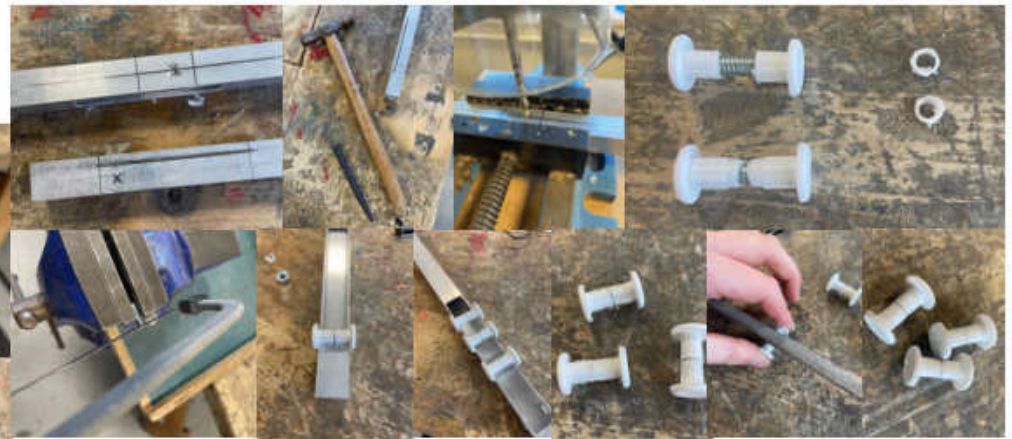
As I now have the hole and size for the frame for the body, I will now develop the litter picker head tool and the body together, as the litter picker head tool's trigger will have to be used in the body of the design, developing them together ensures accuracy and that the components will function.



With the other piece that I cut when cutting the tubing for the body, as I used this piece for testing, I cut off both the ends with a hacksaw, so that I would have the size for the head tool so that the piece would be ergonomic. When using the hacksaw, I ensured that my head was directly above the saw so that I would cut a straight line downwards. I then filed the edges until smooth and straight.



I then marked out the area where the litter picker claw would be placed, to ensure accuracy I used a piece of the same height so that when measuring, the lines would be flat and accurate. I then used the milling machine with a 12mm drill bit to cut the selected area, using the 12mm would allow for the claw to move easily once it has been pulled yet also allow me to not cut off too much so that the design wouldn't be as effective, and that I could then file and control the amount taken off easier with a file. I then filed down the sides until the edges were smooth and straight.



I then measured out the area where the trigger and claw would rest in the design, and cut with an 8mm drill bit so that the inter screws would have a snug fit inside and allow the components to function. I then cut and sized down the inter screws so that they would be appropriate for my design by using a hacksaw for the screw, and a file to shave down the plastic coating. Comparing the size of the screws above the area where the trigger would be located allowed me to accurately size the screws to ensure it would be a tight fit. I then repeated this process to produce 3 screws for my design, for the trigger, claw and spring. Then filed the inside circumference of the screws so that the design would be able to freely and easily rotate when pressure has been applied.



I then threaded the 2mm rope through the hole in the claw and secured it by tying a knot, I then cut the rope to a shorter size to allow for an easier production process but cut it long enough that there was room to accommodate for mistakes that would be made. I then removed the rope as it would get in the way of manufacturing. I then placed the claw, connected with the screw into the body of the tool head and measured the distance for the spring screw to be placed through, which the spring would be connected to, as well as adding a hole for the rope to pass through the design, and extended the bottom cut to accommodate for the extended distance. I then used a milling machine to cut the holes, starting from the centre to ensure accuracy, and then filed the edges down to prevent damage to the user. I then used the pillar drill with an 8mm drill bit and cut a hole through the design, I then placed the screws through the holes to ensure they connect and if the size needed to be changed to allow for a strong connection, but didn't as they fit well in the design.



When testing the design, the problem found was that the force exerted on the piece that pushes the rope down so that the point of tension is above the trigger, so the trigger can fully return, was too large for the component to stay fixed in the design by friction, causing it to pop or fly out. A new mechanism is need or a redesign so that the component can stay in the design.



With the holes that I cut into the specific areas for the inter screws to be passed through, I passed the screws through one side of the body, with the component in its position in between the tubing, the screws would go through the component, I then screwed on the cap of the other side of the screw design, allowing for the component to be connected to the body. As the holes of my 3D printed component where tool small, as my plan for the connection changed within my testing section, I used a pillar drill to drill an 8mm hole in the components, which would allow for a snug fit and the inter screws to fit though the design. With the claw and trigger in place by the inter screws, as well as the inter screw by itself which would be for a spring, I moved the roped through the designated hole, and over the spring screw. I then attached a spring to the claw, pushed the claw as far back as possible to minimise the distance, and then attached the spring to the inter screw by wrapping it around. In order to do this, I had to unwind the spring slightly in order to extend the distance that the spring would be able to wrap around. The spring was then connected to the claw and the body and would result in the design being able to reset after the trigger had been pulled. I then threaded the rope into the trigger so that it was connected and tied a knot to ensure that the connection would be tight. The next step for this is to test the design so that it can function effectively, and any problems can be identified before finalising.



So that a force would be applied to the component that is wedged between to lower the height of the rope, I placed a spare cut piece of the tubing that will be used for the push fit mechanism, which I cut with a hacksaw, between the walls, so that it would resist the back end of the tubing coming and result in the wedge popping or flying out. I then tested this by pulling the trigger with the attachment together, and the tool was able to function, and the wedge did not pop out. I then cut this piece even smaller so that it would be able to snugly fit inside the design, and not cause any issues or problems, as well as to maintain a clean and aesthetic design. With the piece cut smaller, I then tested again to ensure it would function which it did. Then I cut the piece in half with a hacksaw so that the litter picker rope would be able to pass through when inserting the trigger and filed down the ends and edges to ensure a smooth and clean design. I then used super glue to attach the components (would weld the components if I had the facility to do so or product was mass or batch manufactured), I then tested the product which was able to function effectively. This meant that the next stage for this product was to paint the components and glue the design.

Manufacture of net tool head:

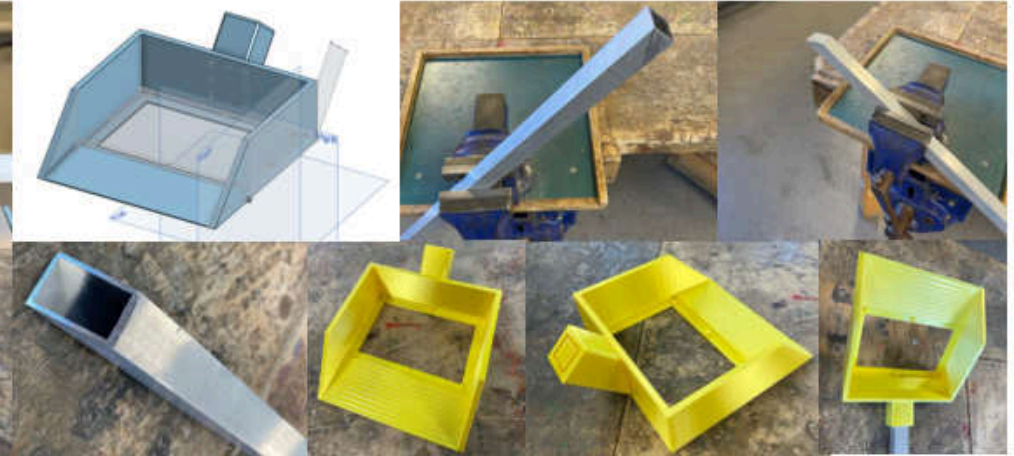


I measured the distance of the aluminium tube for the body of the net tool, marked out with a pencil and tri-square to ensure straight marking, then used a hacksaw to cut the tubing into the correct sizing, and filed the edges. By marking out the holes where I wanted the rope for the hoop to be passed through, and the area where the rope would meet at the top and be tied, I used a pillar drill to cut through the holes on the side and then a milling machine to cut the oval shaped hole at the top. Using an 8.5mm drill bit for the pillar drill allowed for a snug fit which would help keep the rope secure. As tested with the hole at the top, I created an oval shape using a 12mm drill bit on the milling machine, Starting from the centre to prevent turning the machine the wrong way and removing unnecessary material. I then filed the edges to ensure that there would be no damage to the rope when it would be passed through and tied.



With the tubing prepared for the rope, I passed through the rope, which was still connected to the rope reel which would allow me to use as much rope as necessary, through the net. I then knotted the rope that was just outside of the net so that the net would maintain its full and open and not fold on itself. Then passed the rope through the holes on either side and up to the opening at the top, where I tied the rope with together, locking the rope to the body. I then tested the tool and found that the net would sometimes slip down, to ensure this wouldn't happen when using I used the spare 2mm black rope used for the litter picker and tied the rope around the end of the net and onto the knot made against the sides of the body to secure it. To then ensure the rope will not fray, I wrapped the ends with masking tape, cut the rope up to the tape with scissors, and used a lighter to burn the ends. I did this on all of the knots tied with loose ends to ensure they wouldn't fray. The next step for this tool head is to paint and glue the components onto it for an aesthetic finish.

Manufacture of scraper tool head:



By using CAD, I produced the shape and size wanted for the scraper head tool and followed the dimensions and measurements required to meet the size needed. I then sent the file to be 3D printed, while this was 3D printing, I cut the tubing to the appropriate angle of 50 degrees by using a hacksaw. As I had to cut the tubing at an angle, I secured the tubing tightly and placed it close to the vice to prevent the tubing wobbling when cutting and cut with slow and controlled cuts, allowing for a secure and safe cut of the tubing. I then cut the other end straight down, with a 1mm allowance for both cuts next to the marked lines. I then filed down the edges until I met the measured lines, and the edges were smooth and straight.



With the scraper head tool printed, I sanded the corners and edges down to make them round and less likely to break with an impact, this was done by using a file then 120 grit and 240 grit sandpaper. With the hole for the mesh produced in the scraper, I measured out the area where the mesh would go, a 140mm x 95 mm area, and cut the mesh with wire cutters to fit this shape. I then placed the cut mesh into the area to ensure it fit. I then used 2D design to produce a the shape that would sandwich the outer side of the mesh, this was a shape of 140mm x 95mm with a side width of 10mm and a thickness of 3mm so that the design would be flat. I then sent the design to the laser cutter to be cut on acrylic. With the design cut, I sanded down the ends with 120 grit and 240 grit sandpaper and placed the acrylic on top of the mesh in the scraper. As well as testing the tubing to go inside the design which it did. The next step for this tool is to paint and glue the components together for an aesthetic finish and finished product.

Priming and painting the components:



To have an improved aesthetic for the design I sanded down all of the components with a 240 grit and then 400 grit sandpaper, I then wiped the components with a damp cloth and then a dry cloth to remove any further particles on them. I then placed all the components on top of a bin liner which was weighted down by stones, and sprayed a plastic primer on top of the pieces. I then waited for the layer to dry, and smoothed out any imperfections with 400 grit sandpaper and a microfiber cloth. I then applied another layer and repeated the smoothing process, and then flipped the components and applied two more coats, sanding and smoothing in between so that all the layers of the primer are evenly spread. To ensure safety, I primed outside and with a face mask to allow for good ventilation and safety. I then waited for the components to dry, once the components dried, I used 400 grit sandpaper and then 600 grit sandpaper to ensure a smooth finish and layer which is ready to be spray painted. When sanding I ensured I did not go through the primer layer too far, if I did I would stop or move to sandpaper with a higher grit and use gently.



As the components have been primed and sanded so that they are smooth and have little to no imperfections, I can now spray paint the components. I wiped the components with a wet cloth and then with a dried cloth to remove any particles that could cause imperfections. I then placed the components onto either the bin liner for red or black paint.. I began spray painting the components, waiting for the first layer to dry, removing any imperfections with a microfiber cloth. Then repeated this again on a second layer, waited for the component to dry, and then flipped the component so I was able to paint the other side. Here I then spray painted, waited for it to dry, removed any imperfections with a cloth and then applied another coating. I then waited for the components to

dry by waiting 24 hours. I then removed any imperfections such as drippings of paint with 600 grit sandpaper, being controlled and slow with the sanding to ensure I only took off the appropriate quantity and not an excessive amount. The components are now ready to be glued.

Cutting the push fit mechanism:



With the smaller 16mm x 16mm aluminium tubing, I cut this piece into three, 140mm lengths pieces with a hacksaw, this would mean that 40mm would get connected inside the tool head and the 100mm would be the push fit mechanism that would help align and keep the design straight. I then filed the edges to ensure they were smooth and would not cause damage to the user.

Gluing and finishing the design:

With the components all ready and tested I began the stage of gluing the components together and to the design so that the product is able to function. I glued all the components apart from the pieces that are using the inter screws as these can be locked into place already when screwed shut, meaning that the epoxy resin is not necessary, as well being able to tighten the screws further is a key part for the trigger so that it is able to remain stored inside the body when the litter picker tool head is not being used so it will not cause an inconvenience for the user.



I glued the pieces with epoxy resin, I wore gloves, an apron and was in a well-ventilated room for safety, I followed the instructions on the epoxy resin packaging, mixing for 20 seconds and applying quickly, and gluing one component at a time so that the epoxy stays a high quality. As well as using tissue to remove excess, I then let this dry for 24 hours, and once it dried used sandpaper and wet and dry paper to remove dried excess and neaten up the design to make it more professional.

Overview of manufacture process

With the manufacture of my product complete, I am ready to evaluate the product, but first by writing an overview of the manufacturing stage I am able to explain the processes that took place and the key stages. This page essentially sums up the manufacturing diary, and allows for the next stage to be the evaluation of the product where the prototype can be evaluated to find the pros and cons.

Body of the design:



This is the body of the product, the tool heads can be attached by inserting the push fit mechanism into here (where the arrow is pointing), where the connection mechanism will lock the design in place. To remove the head tool the user must push down the button and pull the head.

Litter picker tool head:



This is the litter picker head tool and will be the most common choice of head tool due to it being the most versatile and able to be used in the most environments. The trigger and wedge must be placed into the body in order for this head tool to be able to function effectively.

Net tool head:



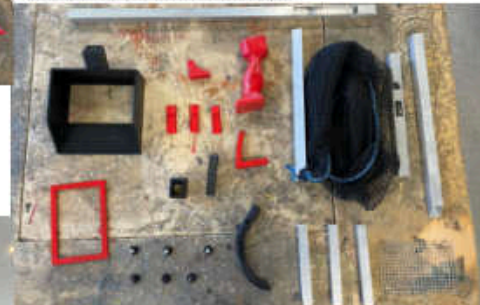
This is the net head tool and allows for litter that is within a watered environment to be removed, the hoop of the net is produced from a rope so that the head can be easily stored and allow for easy and simple transportation of the product.

Scraper tool head:



This is the scraper head tool and is used to pick up small pieces of litter such as cigarette butts, as well as be able to remove larger amounts of litter at one time and remove and scrape litter that is stuck to the ground off of it which is often difficult to do with other tools.

All components and pieces:



When manufacturing the design, I used the components from my testing due to it being time and effort saving as I would not have to re-print the designs again, as well as it was cost effective as there would not be extra components produced, and the components had already been tested so they would be able to function and using them would then result in an efficient design. However, when spray painting the components, there were added layers on the components which I had to remove in order for the component to function effectively.

Key parts of the manufacturing process:

- Cutting the body to length
- Milling and drilling holes into the body
- Cutting tool heads to length
- Milling and cutting holes into the heads for the tool head to function
- Filing and sanding down heads and body so edges are smooth
- Further sanding of components and testing to ensure the final design would work
- Attaching components to tool heads and body and testing them to ensure they can function
- Priming the components with a plastic primer so that paint can stick to the parts
- Applying red spray paint to the components that will be physically used by the user and applying black spray paint to the components that will not be physically used by the user
- Applying an epoxy resin to the components to secure them to the design, using a tissue to remove excess
- Waiting for the components to be secure and once epoxy has fully dried, using sandpaper and wet and dry paper to sand down and smoothen the edges and remove excess epoxy, making the design look professional and cleaned up

When manufacturing the prototype, I made some changes to the design in order for it to perform more effectively, this being that there is now a part in the trigger storage area which prevents the back end of the wedge (the component that pushed the rope down) from moving upwards due to the tension in the rope, this part does not affect the design but only means that the wedge has to now be inserted at an angle and placed underneath the parts to function effectively. The other was that I sandwiched the mesh of the design with a piece of acrylic for the scraper head tool, which would allow for the connection of the mesh to the tool head to be much more secure and stable which would result in the tool having a longer lifespan, although not physically used by the user, I painted this component red so that any litter that was on the edges could be easily seen. I painted other components black as these would be touching the ground, the black helps disguise marks and will help to keep a good looking, professional and aesthetic design.

Photographic sequences of how the design functions

This page shows a photographic sequence of how some of the mechanisms are able to function and allow for a functional product. This page also shows how some of the mechanisms function in order to accommodate for the user's needs and allow for a user-centred product that can help remove litter from a range of environments.

How the tool heads are connected to the body:



Video:



The tool heads are connected to the body by pushing them into the hole of the body tubing, this causes the pieces to connect as the button will automatically return to its shape without pressure

How the tool heads are removed from the body:



Video:

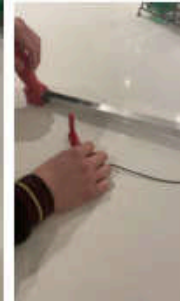


The tool heads can be removed from the body by pushing down the button and pulling out the tool head, this happens as when pressure is applied to the button, the front deforms and can be removed

How the trigger is connected to the design for the litter picker tool head:



Video:

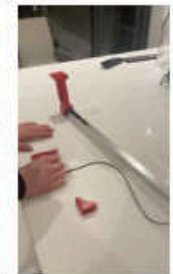


The trigger can be attached to the body by inserting the trigger into the trigger connection mechanism, the trigger then stays connected due to the strong dovetail connection

How the wedge (component that pushes the rope down) is attached:



Video of trigger and wedge:



The wedge must be placed at an angle and so that the rope can pass through it, the wedge should then be forced downwards so that it touches the bottom to be fully secured

How the litter picker tool head is used:



The litter picker head tool is used by placing the claw and body of the head above the piece of litter, then pulling the trigger which rotates the claw and locks/captures the piece of litter.

How the net tool head is used:



The net head tool is used by placing the net into the watered environment and moving the net underneath or around the litter and then bringing the tool out so the litter is captured in the net.

How the scraper head tool is used:



The scraper head tool is used by forcing the tool head underneath the piece of litter, this will allow for the litter to be scraped off the floor and captured and then disposed off appropriately.

Evaluation plan

As I have now manufactured my prototype, I am now able to evaluate the product to see if it is effective at removing litter from numerous environments without causing strain to the user, is able to meet the design brief and specification. In order for my evaluation to be thorough, I will need to find the advantages and disadvantages of my prototype so that if my product were to be taken further into development, I would have an improved design and be an overall better product. In order, to accurately find this, I will need to produce a plan and follow it so that I am able to cover all aspects of the evaluation thoroughly. By planning my evaluation, I am able to find the problems and issues associated with the product, so that if my product was to be developed even further, then the issues would be resolved, which would mean for an even better and more developed product.

Testing of my product withing a range of environments:

In order for me to assess the quality of the product and find out the advantages and disadvantages of the product I will need to test it. For this testing I will have to go out myself and remove litter from a range of environments, this can be done in the Millenium Green, which is the area where I have previously investigated with my litter count and removed litter from to test the problems with a litter picker, as well is the same area where my client/expert removes litter from. By going back to the same location, I am able to test the product on the same environment, being able to compare my experience and how the product has improved the process or made it more difficult. By doing this it will help me identify the positives and negatives of the product, which if I was to develop the product even further would allow for a better and more appropriate product that is based around the user's needs and requirements. Once I have tested the design, I will list the advantages and disadvantages of the product that I found and am able to then take forward the disadvantages and identify them clearly, so that a solution would be able to be developed and allow for an easier experience.

However, if I only test the product then the advantages and disadvantages found may not be as honest or true, which may limit the extent of the advantages and disadvantages found, in order for my product to 'appear better', resulting in a product which is not as effective. This may damage the results found and not be a true representation of my product, which would then have a flaw, and by not identifying this flaw the product will remain in the state and not have a solution, so that if the product was to be taken further then then the product would not be as appropriate at removing litter. By using my client/expert I am able to decrease the bias of the test, allowing for more reliable and accurate results, which will have valid and honest advantages and disadvantages of the product. Although my advantages and disadvantages may be honest, negligence when testing it may occur, leading to less accurate results, hindering the quality of the product if it were to be taken further into production. When comparing my advantages and disadvantages with my client's/expert's, the pros and cons that we have both found separately will help identify the true advantages and disadvantages of the product, the disadvantages both found are the ones that should then be developed in order for the product to be more appropriate and well-suited for the user to remove litter. With this, this is not disregarding the other disadvantages found, it is just prioritising the most common and problematic disadvantages with the product, so that it can be developed and become better suited at removing litter if the design was to be taken further. By using results from testing the product in an environment by removing litter from both me and my client/expert, I can improve the reliability of the testing so that the product can be better suited to the problems faced, then be developed further so that the product can then be improved.

By testing the product at the Millenium Green I am able to test the product in a range of environments to see how it is able to function, such as a grass, woodland, urban and watered environment, I am able to see how the product would truly test and function, which would allow me to list more appropriate pros and cons of the product to develop further so that the product will be more appropriate and more suited to the user's needs.

Questionnaire:

By producing a questionnaire and having potential users of the product answer it, I am able to assess the quality and product of how useful it may be to users and their process of removing litter. I am able to find further insight into the product that I may not be able to see or have disregarded due to unconscious bias of the product. This would allow me to help plan for the future development as those who answer may provide advantages and disadvantages and ways the product could be improved, as well as helping new iterations and designs of the product. They would be able to answer how useful the product would be and if it would be necessary for their product. With this questionnaire, I will be able to ensure consistency with the questions to remove unconscious bias, be efficient and time-saving to find the responses from users, and being able to provide a simple and easy response method which is cost-effective and anonymous to reduce bias. Aswell the data provided from this questionnaire will allow for quantifiable results, which would allow for me to see patterns and the problems most identified with the product, allowing for me to develop the product even further so that it can be improved and be the best that it can.

Comparing to the design brief and specification:

By comparing my final product and produced prototype to the initial design brief and specification, I am able to determine how appropriate the product actually is to the problems identified within my research and how appropriate the product is to be a solution. This process will take place after the testing has occurred in order to see how the product is able to perform and how it is able to meet the design brief and specification. If my product is able function effectively but not directly meet the brief or specification, then the product would not be bad, it would just not be as suited to the problem identified. By taking in consideration results from users in the questionnaire and my client/expert, I am able to take their responses and link it to how well the design brief and specification have met the user's needs with a smaller bias, allowing for more appropriate results.

Advantages and disadvantages of the design:

With the testing of the product from me and my client complete, as well as a questionnaire and comparing the design to how well I met and followed the design brief and specification throughout the design process, I am able to list the key advantages and disadvantages of the product. This should list what works well with the product, what doesn't work well with the product and how I have concluded that these pros and cons, they should be backed up with proof and results taken from the testing process where I will remove litter from a range of environments.

Modifications of the design

As I have now found the final advantages and disadvantages of the product, I can now suggest the further modifications that would take place if the design was to be developed further. These modifications should be able to target the problems identified so that the product would be improved and be more suitable for the user so that the removal process of litter is easier an can be done more effectively, allowing for an easier and simple experience. By doing this I am able to target the problems faced with solutions so that if the design was to be developed further then it would be successful and have fewer issues, resulting in a simpler experience.

Commercial evaluation

With the modifications that took place in my design process, the industrial manufacturing processes for the design may have changed, so it is essential that the product is appropriate to meet the requirements of the design. I will need to evaluate if any of the processes will need changing so that it is most appropriate to meet the demands of a batch or mass production manufacturing process.

Conclusion

This page should be a summary of my findings within my research and development, it should outline the problem identified and how the product that I have produced is able to meet the problem and be a solution, I should compare it to my design brief and specification to see how well I followed them in order to target the problem found and be a solution.

Self-testing and client-testing of my prototype:

As I have now manufactured my product, I am able to test it to see how the product is able to perform in its target environment. By testing the product, I am able to find the advantages and disadvantages of the product, which will allow me to develop modifications so that the product is able to perform better and be able to remove litter more effectively. If I am able to find any problems that occur with the design when using it, I will then be able to suggest solutions for this problem so that if the design process of my product were to continue, then these problems would have potential solutions that would allow for the product to function effectively.

Self-testing and testing of the product with client:

For my first stage of testing, I will go around the Millenium Green and use the product to remove litter from the environment. The other stage is for my client/expert to remove litter which will help remove bias of my prototype, to allow for more clarified results and allow me to accurately target the problems found if I were to continue my design process.

In order to save time when testing I went around the Millenium Green with my client and we both used the tools, handing the product to each other when there was litter, and we would take the product in turns so the removal of litter would be spread out and the testing would be even. In order to improve the accuracy of the testing section of my product, I let my client use the product and for them to test it around the Millenium Green; by using my client I am decreasing the bias as if only I tested the product, I may be unconsciously bias and not target the problems directly or I may miss problems. By using my client this then minimises the risk of bias results happening. By going out to remove litter with my client I am also able to instruct them on how the product is able to function so that they are able to setup and use the design appropriately.

I met with my client and gave them the product which we will share when removing the litter, I also gave them my rucksack which contained the three tool heads; litter picker, net and scraper which was evenly carried throughout the litter removal process so that the conditions of a real litter removal would be felt; the weight that would actually have to be carried by the user for this product.

I talked them through the process of attachment and how the design functions,. I additionally gave them a bin bag which was connected to a bin bag hoop to allow for easy storage of the litter collected, this litter could then be disposed of afterwards. Also, I told my client to think of the advantages and disadvantages of the tool heads and the product in general so that I would be able to get her feedback on the product which would help with further development and would lead to a better and improved product.

Initially we went around with the litter picker tool head attached to the design as this is the head which is most appropriate for a range of environments, I stored the net tool head and the scraper tool head in my rucksack which I carried on my back, as well I stored the collected litter in a bin bag which was easily carried and able for litter to be placed in as it was able to remain open. When walking around with the litter picker tool head, we were thinking and testing the tool by removing litter in order to find the advantages and disadvantages of the product. The litter picker tool head is used to remove most types of litter and be the most common tool head to use. As the scraper tool head was too large it was poking out the top of my bag in order to be carried.

By using the product within the certain environment and conditions that the tool heads have been designed for, I was able to test the efficiency of being able to remove litter within the environment; by doing this test I am able to find out any problems that would occur for the user when removing litter from the environment that the tool is designed for, this will allow me to then focus on new problems that require solutions if I were to continue the design process of my product.

What was brought when testing the product to remove litter:



Testing the litter picker tool head:



Videos of removal:



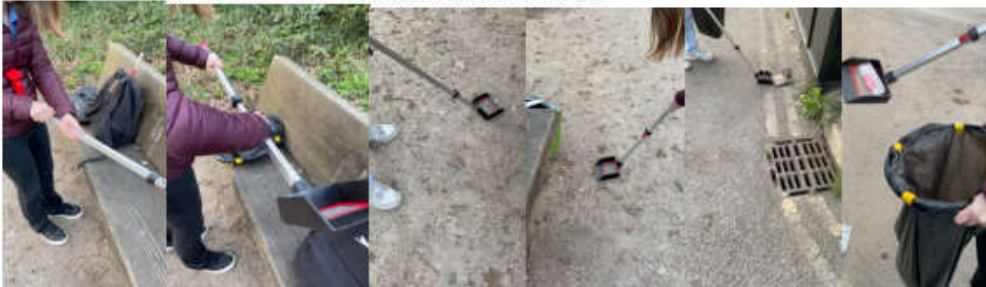
Once we had felt that we had tested the litter picker tool head sufficiently and enough to find out the advantages and disadvantages of the product, my client changed the tool head by pushing down the button of the litter picker tool head and pulling out, then putting the litter picker tool head into my rucksack, taking the scraper tool head out and pushing it into the body of the design. This tool head is designed so that small pieces of litter can be collected and removed as well as larger quantities too. I helped instruct the removal and attachment which she was able to understand quickly. For the small pieces of litter, we went to a bench where it is common to find cigarette butts to test if this product is able to efficiently remove them, as the woodland is behind a hospital and doctors and nurses come and smoke here on their break and therefore drop their cigarettes on the floor.

Testing the scraper head tool:



Video of changing tool heads:

Videos of litter removal:

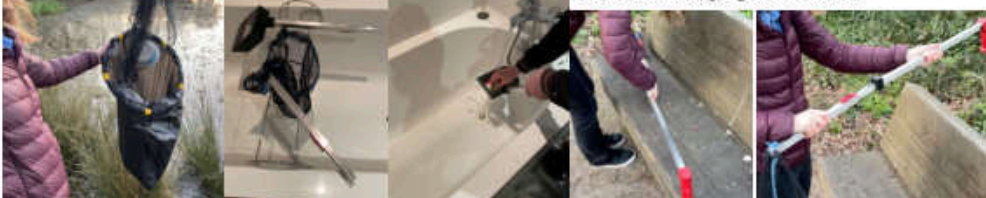


Once we had felt that we had tested the scraper tool head sufficiently, and we found the advantages and disadvantages of this tool head, my client changed the scraper tool head to the net tool head, which was done by pushing down the button of the scraper head tool and pulling, then pushing the net head tool into the design. To test this head, we went to the pond in the Millenium Green and looked to find any litter that I would be able to remove.

Testing the net head tool:



Video of changing tool heads:



As we have both used the tool and all the tool heads, we have found the advantages and disadvantages associated with the product from when testing, will then compare these with my client to ensure the findings found are not bias and are as accurate as possible.

Advantages of the product I found:

- The litter picker and scraper heads functioned effectively and allowed for an easy litter removal
- The scraper was able to get underneath the litter that was stuck to the ground and allow for an easy removal of litter, most stones were able to pass through, but all butts were caught in mesh
- Litter picker was able to pick up cigarette butts very easily which was difficult when testing with a litter picker, allowing for more use of the one head tool, minimising changing the head

Disadvantages:

- Net head tool didn't function that effectively due to the rope floating and being too light, as well as not being able to maintain its shape and overlapping on itself so would not collect litter
- If the scraper is not moved into litter at the correct angle or pushed with enough force, then the litter would often be pushed and not picked up, so would have to push litter against a wall
- The button connection mechanism was able to lock the heads together due to friction but not click into place so wasn't fully connected as button wasn't fully inside due to the area of the connecting component being the same size as the lock meaning the fit was tight and difficult

Once my client had felt that they had sufficiently tested all the tool heads and the product, I asked for her feedback on the product from when she used it and how her overall experience was with removing litter from a range of environments.

Advantages of the product my client found:

- Small litter such as cigarette butts were a lot easier to pick up than with a litter picker that she usually used, allowing for an easy and simple removal
- The scraper worked well and was able to clear up large amounts of cigarette butts quickly
- The scraper allows most stones and miscellaneous to pass through but not the cigarette butts
- The connection of the tool head to the body was very secure and the set up of the tools was very easy once you knew what to do
- Allowed for the overall experience of removing litter to be easier and improved
- Helped meet most of the problems that she faced when I interviewed her so removal was easy

Disadvantages of the product my client found:

- The net tool didn't function effectively as it was unable to go underneath the water, as well the net would lay over itself and not function properly because of this
- The net head tool would be wet after being in the water and would then cause the bag to be wet
- Scraper head tool was sometimes difficult to use if the angle of removal was incorrect

With my client and I having now listed the positives and negatives of the product, I then asked her what modifications or improvements she would suggest in order for the product to be improved so that it would be able to remove litter more effectively if I was to continue with my development.

Suggested improvements and modifications from user:

- Create the hoop from a heavier material so that the hoop is able to go underneath the litter so that it can be captured
- Have a waterproof bag or cover that goes over the top of the net head tool once it has been used so that water will not leak into the rucksack which the user is carrying
- Referring back to the problems had when I interviewed my client, she thinks that the Design 3 I did, the bin bag holder design which was able to carry two different bin bags, one for recycling and one for refuse, would fit perfectly with this design as she would then not have to sort and organise the litter after it has been removed from the environment

With me and my client having now both tested the product, I will develop ideas on modifications and improvements to provide a solution to the problems that we both or individually found. By doing this I am able to target the problems that the user would find if this prototype would be the final version of the product and released to the user. This results in the product being the best that it can be and will be able to accommodate for the user's needs for a user-centred product. When thinking of modifications and improvements I will heavily take into consideration the findings that my client had in order to minimise the bias and expand the development process so that all the problems faced when removing litter with my product can be removed.

Results of tool found from testing

When testing the product with my client, we both used all the tool heads connected to the product to remove litter from the environment. We both used the product with each tool head until we felt that we had sufficiently tested the tool heads to find out the advantages and disadvantages of the product which will now allow me to suggest iterations of the next design if the development of the design was to be continued.

When testing the product with my client by removing litter, the advantages that were found of the product were:

- The litter picker head tool functioned effectively and was able to easily pick up the litter that me or my client attempted to remove, this allowed for an easy and simple removal of litter that would result in time being saved when removing litter from the environment
- The scraper head tool was able to remove small litter such as cigarette butts effectively and let small stones pass through, this meant that only the litter was removed from the environment, and the stones, water and sand would not have to be carried in the bin bag and therefore reduce the weight that the user would have to carry around
- The scraper head tool was able to remove and pick up flat pieces of litter and litter that was stuck to the ground effectively. The scraper would sometimes push the litter forwards however the design would remain effective and useful as pushing the litter against a curb, legs of a bench or a wall resulted in the litter being captured, the technique to use the tool effectively would therefore result in the easy and simple removal of litter
- The connection between the tool heads and the body of the design was very secure and allowed for the heads to remain connected to the design, as the button mechanism wasn't able to fit fully in the design friction between the two parts ensured that the design would remain connected
- The tool heads were easy and simple to connect and remove and allow for a secure connection, my client also said that once you know the method to attach the litter picker tool head to the body it can be done very quickly and easily
- My client found that the handle was very ergonomic and comfortable and allowed for the hand to be supported so that strain wouldn't be as much as an issue with this product. As well my client found that the height of the tools allowed for an easy and simple use as the tools were able to reach the litter at a comfortable distance
- If certain components of the tool broke or came apart such as the net, net hoop and rope for the litter picker then it would be an easy and simple job to repair and reattach the components back into place so that the head tool is able to function
- The range of tool heads allowed for all litter found on land to be easily and effectively removed so that the effect that the litter would have on the environment would be limited. All various types of litter such as bottles, cans and cigarette butts could be easily removed as well in various conditions such as the litter being underneath or behind a bush, and litter in all environments such as cigarette butts next to a bench on grass, litter in a woodland area or litter that is stuck to the floor in an urban environment
- The net head tool and litter picker head tool were easily stored inside the rucksack which allowed for a simple and easy transportation of the head tools, the rope for the litter picker to connect the trigger to the claw could also be wrapped around the body of the head tool which resulted in the tool head being able to stay together effectively, and the net could be wrapped around the body of the head tool too so that the head tool can be stored effectively and not take up excessive amounts of space.
- Heads could be easily washed and cleaned afterwards to maintain a high quality and performance

When testing the product with my client by removing litter, the disadvantages that were found of the product were:

- As the locking mechanism area was slightly too small and narrow in length, the button lock couldn't fully lock in and instead allowed the tool heads to connect to the body by friction as the design was tight, the tool heads could still connect and the lock would lock into place slightly,

however, the connecting lock mechanism must be larger so that the button can return back into place once it has been deformed under pressure as currently the area where the button is pushed into is too tight.

- The net head tool didn't function that efficiently as the rope which is made from polypropylene would float due to it having a lower density than water, this meant that the hoop was difficult to move underneath the water in order to engulf the litter and capture it effectively
- The net of the net head tool was also not that effective at being able to capture the litter due to the material of the net being polypropylene which floats and does not sink so the design was unable to wrap around and engulf the litter effectively for it to be removed from the environment
- The scraper head tool would sometimes push the litter found forwards instead of the litter being captured in the design, this may be due to the user having a tight or uneven angle with the horizontal plane with the tool when trying to get underneath the litter so that it would be scraped up and removed from the environment. This then meant that the user would have to use the tool faster or with more power which could be tiring.
- When testing the scraper head tool, the litter found may have had to be pushed against a wall, legs of a bench or a curb in order for the litter to be effectively captured and removed. If a wall, legs of a bench or a curb are unavailable for the user to access then the litter may be tiring or difficult to remove from the environment as more power and speed would be required
- The litter picker head tool required the wedge component to be connected and forced into the design in order for it to function, if the wedge was not in the design then the tool would not be able to function at all and would result in the litter picker head tool not functioning as the point of tension of the rope would be below the resting position of the trigger, creating a slack rope which will not pull on the claw as much when pulled and result in the claw closing and capturing the litter that the user finds in the environment. The user would have to remember to take this wedge component as it is not connected to the design and is an external piece, so if the user forgets this piece, then the litter picker head tool will not be able to function as effectively

As I have been able to list the advantages and disadvantages of my product from when me and my client went to test the product, I have been able to identify the problems with the design, the problems of the design:

- Problem 1 - Locking mechanism was unable to fully lock in due to the connective component being slightly too small, but the components still held together due to friction
- Problem 2 - The net head tool was unable to be pushed into the water due to having a hoop and net produced from polypropylene which floats so couldn't engulf and capture the litter effectively
- Problem 3 - The scraper head tool would require a specific technique or angle to be able to function effectively, which the user would have to know to get underneath and remove the litter instead of pushing it forwards. Forcing the scraper underneath can also take more power and energy and is more straining
- Problem 4 - The scraper head tool may sometimes need to be pushed up against a surface for the litter to be captured and removed from the environment and this surface may not be available
- Problem 5 - The wedge for the litter picker is separate and would need to be taken if the user would want to use the litter picker head tool, a small and separate piece which could easily be forgotten or lost
- Problem 6 - Mesh for the scraper head tool is not removable so if the mesh breaks will be difficult to replace

As I have been able to test the product and hear from user's feedback, I now know the problems associated with my design. I am now able to suggest potential solutions to these problems so if the development of this product was continued then the next iteration may have these problems removed. Which would help meet the design brief and specification better and would result in an improved product. As I won't continue the development of my product, making suggestions to fix and eliminate the problems found would help align the direction of the development that would act as a starting point if the development of my product was to be continued.

Questionnaire of final product

Due to the manufacture of my prototype being complete, I now need to get opinions of the product from the user so that I am able to find the problems and create solutions to them so that my product is able to perform effectively. As well as finding the problems from this questionnaire, they user's response will also help with further iterations and modifications that the user would like to see to allow for their litter removal process to be easy and without any issues or problems. For example, this could be the user suggesting a new tool head or a further development of a tool head so that the transportation is easy and removal of litter from a wider range of environments is available.

In order to maintain consistent results, I used the same users for the questionnaire that I used in my original questionnaire during my development stage, allowing for the same users to answer the questionnaire allows for consistency within the group as the user's behaviour and opinions will stay the same, a reduced bias and a high engagement to respond as they have already answered one questionnaire before.

Before giving the users the questions, I should a picture of my product and described the function of the product and each of the tool heads so they would be able to answer the questionnaire based on the product that I produced. As well, by letting the user write their answer and not have a multiple-choice question, I can expand the answers available to allow for as much feedback on the product as possible so that I will have more areas of improvement and modification for the product.



Results from the questionnaire:

What are your opinions on the final product?

I think it is great and very well-made

A good product and would mean that litter is easier to pick up in different environments

It was very easy to use and was effective in picking up all sizes and shapes of litter effectively. It was straightforward to change the heads for different types of litter.

Should allow for litter from different scenarios to be removed, making the process easier as I can change the heads and use the head that is appropriate for the situation

Looks fantastic and very aesthetic with a nice contrast between the colours, if tools are able to perform effectively then the tool would then be able to remove litter better

Looks good and would be good if it can function

Commonly, users use a litter picker to remove litter, would you use this tool instead?

Yes, it is a very practical, useful and a unique tool

Yes as it has a litter picker tool and then other tools too so litter will be more effectively removed

Yes, the different heads mean it would be easier to clear up litter in a variety of situations and sometimes more efficiently

Yes as a litter picker is one of the tools so using this product means I'll be able to use the other tools to remove litter

I would as this tool has a combination of tools for different scenarios

Probably as this can remove litter from a wider range of scenarios, however I may only use the litter picker and the others for certain scenarios as the litter picker can basically do it all

Does the design look aesthetic and good looking, what would you change about it?

Yes I think the product is very aesthetic and would be noticeable in a shop, particularly because of the red details on the handle

Nothing I like the black and red on the metal

Yes it does. Everything was in proportion and finished to a high standard. I don't think anything needs changing but personally I may not choose red as the colour

Yes, I like the red and black compared to the shiny look of the metal, I like how the heads are black will be able to see litter

Very aesthetic, wouldn't change the look of it

Yes, like how the parts I will use a red so is clear to see

How much would you expect this tool to cost?

Somewhere around £70 considering the vast adaptations this tool provides

£60 as is 3 different tools into one and would expect one tool to be around £20

I am not sure, about £20

£50 - £100 as it is multi head so is a combination of different tools into one which would cover the cost of multiple tools or be cheaper potentially

Anywhere from £50 to £150

£50

What would you change or add for the design? Suggest any potential improvements or development ideas for the product for it to be improved

Perhaps the metal part of the litter picker could be covered in a material to prevent scratching and improve aestheticism

Make the heads smaller or the metal at the heads smaller so that they can be carried easier

I think the net for the collection of litter could be looked at so that the net stayed open better under the weight of the water which would mean it would be easier to collect rubbish.

Add a brush head so the design can be used as a dustpan and brush as this would be able to remove small litter and dust inside helping keep everything clean.

I think the product is good if the tools can function, potentially add various other tools heads or add other heads such as garden tools like a spade or pitch fork so the tool can be used for litter and gardening, and expand its use

Add more tool heads so that it can be used elsewhere too or remove litter in different scenarios

Why each question was chosen and analysis on responses:

1 – In order to gain knowledge on a user's opinion if the product, by getting their opinion I can find out what they think is good and what is bad about the product so that the product can be improved and result in a better and more developed product with less issues. All users thought that the product was a good idea and would allow for the removal of litter to be made easier, meeting part of the design brief and allowing for a good product

2 – This question asks the user if they would use my product over the most common type of tool to remove litter and is important to see how well my product would do in the market. As all the users said yes this is a good representation of how well my product would do within the market of litter removal tools, as the tool would help ease the process of removing litter, meaning that my product would fill a gap in the market and be commonly used by users

3 – This question asks about the aesthetic which is essential for a good looking and appealing product that the user would feel comfortable and confident using. All of the users said the product looked aesthetic, even if one didn't like the red colour, meaning that my product would look appealing to the user and those who would want to purchase the product. Meaning the tool would look good and professional and could be placed in a shop

4 – This question is important to find out what the user would think of the cost, and the product can then be based around this pricing as it is then appropriate, if the product costs too much then people will not buy it, but if it is sold too cheaply then the profit will not be much.

5 – This question allows for the user to suggest their need and want for the tool, allowing for me to find out what the user would want and be able to change and develop the tool in order to meet the user's needs, the idea should only be chosen if many user's want it as then it is a desired part of the product in order to improve the tool. Many of the users suggested more tool heads which could be an area I could explore, as well as shortening down the tool heads so they are more transportable

From the responses of the potential users of the product, I have been able to gather the benefits of the product that would allow for it to perform effectively, as well as finding out the user's opinion and what they would want to see the product do, or how it would be developed even further. Many of the users wanted more tool heads for other situations, one of the user's wanted one for gardening which is an area I could look into if my development of the product was to continue. This would mean that the tool would not only be able to remove litter but be used elsewhere within a range of environments for different purposes and scenarios that don't involve removing litter. The users also said that the product was aesthetic meaning it would be eye-catching and able to sell as well as being a product that many of the user's who remove litter would want to use.

Commercial evaluation

When manufacturing for an industry the priority is to produce the product or component as fast as can be done as the main aspects of interest for this is profit and money. As I have produced my product, changes in manufacture had to be made and to suit these changes the industrial process may have to change if my design was to be mass or batch produced.

Injection moulding

For my product I suggested that injection moulding was used to produce the complex and unique components as injection moulding would allow for the component to be accurately sized as well, they can be pigmented which would allow for the product to be aesthetically pleasing and easy on the eye of the user. Additionally, injection moulding would allow for a fast-moving process which would result in the components of the tool being able to meet the demands of the industry. This would allow for the components to be easily produced and meet the requirements in order for the product to be produced. The accuracy that this process provides will allow for a good and functional product that can be easily used.

For my prototype, I used inter-screws which were able to work very well, and for all the pieces to connect properly. By using these I can make the packaging of the product small and more secure which would be ideal for an industry product, as the product wouldn't be set up in the factory and would be done by the user, a ready-to-assemble product, that reduces the set up time of the product. As these were able to work well, I could then injection mould the plastic shell that the screw had to connect to the design.

When manufacturing my prototype, I cut out a piece of acrylic on the laser cutter to fit into my scraper head tool design so that it can function effectively, as I will need to produce this piece, and that I want it removable and held in place by friction, creating a push fit mechanism, so that the mesh is replaceable if it breaks. This piece could then be injection moulded which would allow for the accuracy of the component to be reached and to have the same red colour as the other pieces. Injection moulding the scraper head would also allow for the complex shape of the tubing connection to be achieved, and this would result in the tubing being able to connect easily.

Laser cutting ABS

If I were to laser cut the acrylic piece that is meant for the scraper head tool, I could then use the left-over sheets from the laser cut, grind them down and use them for my injection moulding process, I would be able to use ABS sheets for the laser cutting and for the piece that goes onto the scraper head tool. This would also mean that the colour of the scraper head tool piece and the other components which aren't black will be the same colour as they will be produced from the same pigmented ABS sheet.

Laser cutting or CNC milling tubing

Both of these methods of industrial manufacture would still be appropriate if I were to use them within my production process, the laser cutting allows for a slightly easier process too as it is able to cut the tubing into smaller pieces, which will be straight however would need to be turned to cut holes in all sides of the product

TIG and laser welding

Due to me no longer requiring the rods or having to weld the scraper head tool together due to it now being produced from ABS for the final, batch or mass-produced product, then TIG and laser welding is no longer a required process in order for the design to be produced

Screw manufacturing

In order to meet the industrial demands and produce the screws for the inter screws, making an 'in-house' production would allow for cheaper manufacturing of the screws for the inter screws.

Due to some changes that occurred within my manufacture, some changes to the industrial manufacture processes have been changed to allow for new components or mechanisms to function effectively. The manufacture would then build up to be 'in house' to be sustainable.

Modifications

Modifications taken place during manufacturing

When manufacturing the product, certain aspects and designs had to be altered or developed in order for the product to function effectively when testing and using the prototype, the changes were done when designs during testing weren't efficient or able to work.

These changes were:

- Adding two small pieces of cut aluminium tubing above the back end of the wedge, the side of the wedge which is closest to the trigger (the wedge is the component that pushes the rope down so that the point of tension will be above the trigger, so that the trigger can fully return) prevents the back end of the wedge coming or being pinged/flying out, so that the wedge is able to stay in its required position, and allow for the litter picker head tool to be used repeatedly without a disturbance when using it. As found when manufacturing the force exerted on the back end is too great due to it being the point of tension and as a result causes the design to not function. The design is two pieces so that the rope is able to be passed through the gap between them and not be difficult to set up the design, allowing for it to be done in one motion. As well, as the design now has these parts which stop the wedge from moving up, the wedge now must be inserted and removed at an angle so that it is able to fit in place correctly.
- The other change that I made during the manufacture was the method to connect the mesh to the scraper tool head, initially I intended for the mesh to be stuck onto the top of the tool, however adding a laser cut piece of acrylic on top of the mesh, and making the hole where the mesh is located deeper would allow for the mesh to be more protected and therefore have a longer lifespan. When manufacturing I wanted for this piece of acrylic to be removable in case the mesh got damaged such as one of the wires breaking, or a hole becoming larger so that the mesh had to be replaced, this mechanism worked when paint wasn't added however when I painted the design the added thickness to the walls of the design prevented the laser cut piece from staying inside and instead I had to glue the mesh and laser cut piece to the scraper and maintain a good aesthetic.



Suggested modifications and improvements to take place if the development was to continue:

As my manufactured prototype has problems that I and my client have identified throughout testing, suggesting potential redesigns and solutions to these problems would allow for a starting point if the design was to be continued being developed and would hopefully provide solutions to these problems so that the product would be able to effectively function without these drawbacks.

Solution to the problems:

Problem 1 – As the locking mechanism wasn't able to function to its optimal performance and then connected the body and tool head together due to friction, a solution to this would be to extend the length of the button lock so that the button can fit inside snugly and be able to return back to its original size when the pressure from the front is passed, and therefore lock together the tool head and the body unless pressure is applied and the tool head is pulled out and removed. Within my testing, this mechanism was able to function effectively however a reason that it didn't function effectively for my final product was due to either the spray paint having an extra layer which then decreased the length of the lock or the tubing was placed too far and glued in so that the tubing

for the tool head wasn't able to go inside, meaning the button wouldn't be able to go inside as it was blocked by the tubing to enter fully. Another reason is that when the adhesive set, it wasn't removed fully so it set and then narrowed down the length of the button lock so the button couldn't be fully inserted and lock together the two designs.

Problem 2 – As the net head tool contains two parts which use polypropylene rope, the net and hoop, it is difficult to then remove litter from a watered system as the polypropylene rope is lighter than water so will float. A solution to this is to use a material for the net and hoop which is denser than water so it will be able to engulf the litter, the net could therefore be made from nylon or polyester, and the hoop could be made from a stainless-steel cable which will help keep the net remaining open whilst using. This cable would be placed through the net and could be connected by a mechanical fastener such as a U-bolt or by a selection of carabiners connected to the side.

Problem 3 and 4 – As when testing the scraper head tool, it was sometimes difficult to remove larger objects that were on the floor due to the scraper pushing the litter forwards and to remove litter without a surface to push against, however I believe the solution to this is for the user to use the tool and understand and develop their technique when removing the litter. By using this tool head more, they will be more experienced and be able to understand what angles work and what angles do not, as well as if a surface needed to push against or they can quickly push the tool head under. By using the tool head, they can experience what works and what doesn't which will allow for the user to become more experienced with the tool and use it more effectively.

Problem 5 – As the wedge is separate from the litter picker tool and is required for the litter picker tool head to function, and the user may forget the small piece, a solution would be for the wedge to remain connected to the body, whether this be on a hinge or the wedge remains in the same place by the trigger, or the user passes the rope through a different section. It will allow for all the pieces to remain as one and for the wedge to not be forgotten and kept with the components.

Problem 6 – Initially I intended for the piece above the mesh to be removable, however I had to connect this to the scraper and mesh with adhesive as it would pop up, due to the thickness of the paint. The solution for this is that if my design were to be mass or batch produced, injection moulding would be able to produce this piece without any added size as it can be pigmented, or to use an acrylic piece which would be the same colour as the other components and have a professional looking design. As well, easily able to clean or replace if needed.

Problem 1:



Problem 2:



Problem 3 and 4:



Problem 5:



Problem 6:



If I were to continue my development, these are the ideas in which I would start with, develop and test in order for the product to remove the problems that come with it and be an effective and functional product that allows for an easier experience of removing litter from the environment. Additionally, I think developing the bin bag holder, Design 3 of my designs, would help for the sorting process of the litter, which was a problem that my client found, and could be partnered up and sold to users with my multi-head litter removal tool to allow for simple and organised capture. Another idea would be a waterproof bag that goes over the net to stop things from getting wet.

Specification and brief evaluation

As I have manufactured my final design and prototype, I will compare how well my design has met the original design specification and the design brief. By how much my product meets the specification and design brief is not a tell of how good the product is but is a tell of how well the product has been designed and developed to meet the problem that I identified during my research of potential problems that I could investigate further into.

Function of the product

My original specification states that the function of the product is help the user remove litter and reduce the effects of pollution and the damage that litter is able to cause to the environment and animals, as well as creating an unpleasant area. The tool was mean to reduce the difficulty of the litter removal process so that the litter could be easily removed, and therefore the time taken to remove the litter would be less, and then the user would have to spend less time removing litter or would be able to spend the same amount of time removing a larger quantity of litter from the environment. Initially, removing small pieces of litter was very difficult due to the litter picker, being the tool that most users use to remove litter from the environment, this was found out from my litter count investigation where cigarettes were most of the litter found and further backed up by my expert opinion and client, who found the removal difficult and would often have to hand pick the cigarette butts. As well litter would often get stuck to the ground and in water which would then be difficult to grip with the litter picker effectively. The tools could often be straining on the user which would cause discomfort to the user. Essentially, the function of the tool intended was to remove litter effectively from a range of environments without causing or discomfort for the user, so that litter removal is easy and simple.

When testing my manufactured product with my client, the development and investigation that took place to identify these problems was essential as the litter picker head tool and scraper head tool were able to function effectively once the technique of the tools had been learnt. Both heads were able to efficiently pick up the cigarette butts that were found on the floor, I believe my version of the litter picker was able to retrieve these cigarette butts more easily due to the end of the head not being bent like some litter pickers often are, which allowed for my tool head to be positioned at many angles and still remove the litter effectively. The scraper head tool was also able to remove litter stuck to the ground and small pieces of litter such as cigarette butts effectively, the initial concern when testing this product was that it would sometimes push the litter instead of picking it up however, upon further thought and testing it is the technique used. If the scraper tool is used with the correct technique, then it is able to be useful and effective. The tool also causes a limited amount of stain on the user as it is relatively lightweight due to it being produced from hollow aluminium tubing, as well an ergonomic handle was manufactured that allowed for an ergonomic and comfortable grip for users in the 5th – 95th percentile. The tool head that hasn't met the original function that well is the net head tool due to it being manufactured from polypropylene rope which floated on the water and did not maintain the hoop shape. This problem would be resolved if the development of the design was to continue due to the hoop being produced from a stainless-steel cable which will not float and be able to maintain its shape effectively.

Performance requirements of my product

The intention of my product was to simplify the removal of litter from various environments to allow for an easier and simple removal process, and I believe this was done with the litter picker head tool and scraper head tool as they were both able to remove all litter found when me and my client tested the product. They were both able to efficiently remove the major problem of cigarette butts that I found in my litter count investigation page effectively, meaning it solved one of the major problems that was occurring with the removal of litter when investigating problems with the removal of litter. The only head that was unable to meet the requirements effectively was the net tool head due to the hoop floating and not being able to engulf the litter. This will change if the development was to continue as the hoop would be made from stainless steel which is heavier than

water so will be able to sink and engulf the litter that the user intends to capture within a watered environment.

Material and product features:

As my tool is being carried around in their hand by the user it should be lightweight and easy to carry and not cause strain on the user. As well, as the product will be used outside it will be dealing with 'wear and tear' such as bashes and collisions so will need to be strong to resist the product from breaking and maintain a long lifespan. This has been achieved by using aluminium as the body of the tools and tool heads, aluminium is a durable material and is 'self-healing' due to it forming an aluminium-oxide layer on the outside which helps prevent the tool from becoming damaged further. This material will be able to deal with all ranges of use faced and maintain its high quality and performance when removing litter.

User requirements

The requirements that are required for the user is the tool to be ergonomic and comfortable for the user, by being able to accommodate hand sizes between the 5th and 95th percentile the product will be able to meet the ergonomic needs of 90% of the users. This was met by producing an ergonomic handle which was modelled on clay and on foam in order to maintain its ergonomic size, when testing the product with my user they found it comfortable and though that it was able to suit her ergonomic needs. This would then allow for the product to remove litter from the environment more effectively and reduce the amount of strain the user would feel when having to hold and carry the design. Another way the ergonomic requirements for the user was met, was by sizing the tools to meet the ergonomic needs. By sizing the body at just over 600mm, the tool heads when connected would result in the user being able to remove litter at a comfortable distance. This then resulted in the litter being able to be removed comfortably and easily.

Sustainability

My product has been able to meet the sustainable needs by being produced from aluminium and PLA (would be ABS if the product was to be mass or batch produced), which are recyclable materials so when the product meets the end of its life, the product can be appropriately disposed off and then will be recycled. This would then help promote and sustain the green values and 'environmentally-friendly' product that my design incorporates. I and the user and unable to attempt to do further action as we have to trust those who remove and dispose of the product that it is done with care and recycled and not put into landfill.

Scale of production

If my product was to be produced within the industry, it would be mass or batch produced, and in order to meet this demands, the framing and components would be sent to the user for them to connect the design themselves. The components that they connect would be the inter screws and the components that are attached to them as well as potentially the net. This would be done as this part can often be time consuming so allowing the user to do it will result in time saved when manufacturing the product to meet with the demands of the industry. As well, the price of the tool needs to be accessible for many, and from my evaluation questionnaire I believe a price of £60 would be appropriate, as the tool cost £24.61 to make, the price will decrease if bought in bulk for industrial manufacture, however this price did not take into account the price of the plastic components, which would then be more. By placing the price at £60 the user is able to essentially get 3 different tools in one product, allowing for saved space, room and being convenient.

Design a product that can help reduce pollution and litter in a wide range of environments and terrains so that the removal of litter can be done more effectively and be easier for the user. The product should not cause strain on the user or be difficult to use and should provide the user with a simple and easy experience so that litter can be removed more effectively.

Overall, I believe that the product has mostly met the design brief and specification as the litter picker head tool and the scraper can effectively remove litter without causing strain on the user, and allow for an easy experience, the net head tool is the product that doesn't meet the design brief or specification that well due to the litter being difficult to remove, however I believe that the redesign which would use a nylon or polyester net with a stainless steel hoop would function meet them.

Conclusion

As I have come to the end of the evaluation, I can now summarise all of my findings from my product, I have found the advantages and disadvantages of the product and the solutions that will then meet the problems found so that if I was to continue the development of my product then I would be able to minimise the problems so that my product would be more effective at removing litter, allowing for an easier experience for the user and meeting the design brief and specification more appropriately and accurately.



Advantages found with my product:

- Litter picker and scraper tool heads were able to function effectively and help to remove the most common type of litter and biggest problem found when investigating, cigarette butts
- Both litter picker and scraper tool were able to function how designed so that litter was able to be effectively removed from the environment and less damage will be caused
- Connection between the tool heads and the body was secure, even if the button lock didn't fully work, the pieces were able to remain together due to friction and the button being slightly, not fully, locked into place
- Tool heads were simple to attach and remove, push tool head into the body to connect, and push the button down and pull to remove
- Client found that the handle was very comfortable and ergonomic, which would result in minimal strain or stress being applied to the user when using the product
- Stones, sand and water would be able to pass through the scraper head tool, making it appropriate to remove litter by scraping it up. Would work very effectively in a sandy environment such as a beach
- Litter was able to be removed in all environments apart from watered due to the net tool not functioning effectively, meaning that all litter found could be easily removed from the environment, and directly meets the design brief
- Once user understands how to use the scraper head tool effectively, it can be very useful at removing litter, the correct technique has to be used which the user can develop from using
- Litter picker was able to easily pick up butts which was a problem for my client before my tool

Disadvantages found with my product:

- Locking mechanism didn't lock fully into place due to the length of the locking mechanism being too short, this may have been due to errors in manufacture such as the body being pushed too far in or there being dried adhesive in the way of the connection
- Net tool didn't function fully due to being made from polypropylene which floats, this meant the net couldn't go underneath the water and engulf the captured litter
- The wedge component is not attached to the body so will have to be remembered to be brought when removing litter with the litter picker head tool
- Mesh for the scraper head tool wasn't removable so if it got damaged then the tool wouldn't function to its maximum ability

Solutions to the problems found and modifications that would take place if I were to continue my development of my product:

- **Extend the length of the locking mechanism** so the button can snugly fit in and lock into place fully which would result in the tool head and body being strongly connected
- **Make the net tool head hoop from a stainless-steel cable** so that the hoop is able to enter the water and engulf the litter, **use a net made from nylon or polyester** as this is denser than water too, potentially connect this cable with hooks or a carabiner
- **Connect the wedge to the body so it will not get lost**, potentially connect it to the body via hinges and the wedge will be able to rotate in and out to allow the rope to be passed through
- **Injection mould or laser cut the scraper head tool piece** that lays over the mesh so that it will be connected by a push fit mechanism, which will allow for the mesh to be replaced if it breaks

How well my product has met the design brief:

Design a product that can help reduce pollution and litter in a wide range of environments and terrains so that the removal of litter can be done more effectively and be easier for the user. The product should not cause strain on the user or be difficult to use and should provide the user with a simple and easy experience so that litter can be removed more effectively.

Through testing of my product with my client, I was able to find out how effectively the product can meet the design brief. The manufactured design is able to remove litter in nearly all environments apart from a watered environment, such as a pond or stream, as the litter picker tool head and the scraper tool head function effectively, and when tested was able to remove all litter that was found within the environments and terrains. As all litter was able to be removed from the environment, the litter picker and scraper head tools were effective in the function which would allow for a simple and easier removal of litter for the user.

The product has a handle which has been ergonomically designed in order to accommodate 90% of all users as the design is based for users between the 5th and 95th percentile and hand sizes. As well the length of the tools is ergonomic and taken into account anthropometric data which has led to the product being appropriately sized so that you use the tools, all litter is removed from an appropriate and comfortable distance. If the development of my design was to be continued then I would develop on the disadvantages identified with the product, most specifically being the net head tool not functioning, where the solution to this problem would be for the hoop to be produced from a stainless steel cable as this is denser than water and able to sink. Once this has been developed, so that the net is able to engulf and capture litter, the other aspects such as increasing the locking mechanism length so the button can fully lock into place can be done. The product would then be tested again to test if there were any new problems identified, and this iteration and development process would continue until no more problems occur with the product, leaving the user with a product that can effectively meet the design brief and specification and remove litter from a range of environments effectively.

In conclusion, I believe that the product manufactured has been able to remove litter effectively from a range of environments, excluding the net head tool. I believe the scraper and litter picker head tools manufactured have strongly met the design brief and specification which has allowed for them to be functional and effective products at removing litter and allows the user for a simple and easy experience. Additionally, with the suggested modifications that would take place if I continued my development for product and the net head tool, I believe the design would directly meet the design brief and allow for litter to be removed from a wide range of environments and can be done efficiently and be an easy process which doesn't cause strain on the user. Furthermore, if development of this product would continue, developing the bin bag holder with two sections, one for recycling and one for refuse would result in the whole process of removing litter to be easier as now the litter doesn't have to be sorted at the end. I would also be interested in developing and adding new tool heads that could fit into the design, so the multi-head tool will be more effective at removing litter and make the process simpler for the user so that the process is easier.

Links for information

<https://www.unep.org/topics/chemicals-and-pollution-action/pollution-and-health> -UNEP Pollution and health – 22/8/24

<https://playpowercanada.ca/blog/how-commercial-trash-cans-help-prevent-littering/#:~:text=Although%20litter%20behaviour%20relies%20heavily,disposal%2C%20do%20play%20a%20role.> – litter bins – 30/9/24

<https://www.bbc.com/culture/article/20190627-the-people-fighting-the-war-on-waste-at-music-festivals> festival pollution - 17/10/24

<https://www.wastemanaged.co.uk/our-news/festival/festival-waste-guide/#:~:text=The%20clean%20Dup%20of%20festival,a%20dedicated%20team%20of%20workers.> – festival information 2/11/24

<https://www.keepbritaintidy.org/sites/default/files/resources/20200330%20KBT%20Litter%20Composition%20Report%20-%20FINAL.pdf> – Litter composition report

<https://multisite.eos.ncsu.edu/www-ergocenter-ncsu-edu/wp-content/uploads/sites/18/2016/06/Anthropometric-Detailed-Data-Tables.pdf> - anthropometric data

https://www.researchgate.net/publication/307706011_Smoking_Behavior_How_Close_to_the_Tipping_Do_Consumers_Actually_Smoke/fulltext/57cea71608ae83b37461f898/Smoking-Behavior-How-Close-to-the-Tipping-Do-Consumers-Actually-Smoke.pdf?origin=publication_detail&_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6InB1YmxpY2F0aW9uIiwicGFnZSI6InB1YmxpY2F0aW9uRG93bmxxvYWQjLCJwcmV2aW91c1BhZ2UiOiJwdWJsaWNhdGlvbiJ9fQ – cigarette

<http://www.woodcotemillenniumgreen.org.uk/> - Millenium green website